

Overview and Terminology

In this Guide we address the full spectrum of spatial analysis and associated modeling techniques that are provided within currently available and widely used geographic information systems (GIS) and associated software. Collectively such techniques and tools are often now described as *geospatial analysis*, although we use the more common form, *spatial analysis*, in most of our discussions.

The term ‘GIS’ is widely attributed to Roger Tomlinson and colleagues, who used it in 1963 to describe their activities in building a digital natural resource inventory system for Canada (Tomlinson 1967, 1970). The history of the field has been charted in an edited volume by Foresman (1998) containing contributions by many of its early protagonists. A timeline of many of the formative influences upon the field is provided in [Longley et al. \(2015, p20\)](#). The research makes the unassailable point that the success of GIS as an area of activity has been driven by the success of its applications in solving real world problems.

In order to cover such a wide range of topics, this Guide has been divided into a number of main sections or chapters. These are then further subdivided, in part to identify distinct topics as closely as possible, facilitating the creation of a web site from the text of the Guide. Hyperlinks embedded within the document enable users of the web and PDF versions of this document to navigate around the Guide and to external sources of information, data, software, maps, and reading materials.

[Chapter 2](#) provides an introduction to spatial thinking, described by some as “spatial literacy”, and addresses the central issues and problems associated with spatial data that need to be considered in any analytical exercise. Readers who are already familiar with these concepts may wish to skip this chapter and move straight on to [Chapter 3](#).

Real-world problems and applications are typically governed by the organizational *practices* and *procedures* that prevail with respect to particular *places*. Not only are there wide differences in the volume and remit of data that the public sector collects about population characteristics in different parts of the world, but there are differences in the ways in which data are collected, assembled and disseminated (e.g. general purpose censuses versus statistical modeling of social surveys, property registers and tax payments). Data collected by the private sector, often as a result of the use of services that automatically gather information on events, locations and individuals, present a further challenge (see Chapter 9 for an extended discussion of this issue). There are also differences in the ways in which different data holdings can legally be merged and the purposes for which data may be used — particularly with regard to health and law enforcement data. Finally, there are geographical differences in the cost of geographically referenced data. Some organizations, such as the US Geological Survey, are bound by statute to limit charges for data to sundry costs such as media used for delivering data while others, such as most national mapping organizations in Europe, are required to exact much heavier charges in order to recoup much or all of the cost of data creation. Analysts may already be aware of these contextual considerations through local knowledge, and other considerations may become apparent through browsing metadata catalogs. GIS applications must by definition be sensitive to context, since they represent unique locations on the Earth’s surface.

This initial discussion is followed in [Chapter 3](#) by an examination of the methodological background to GIS analysis. Initially we examine a number of formal methodologies and then apply ideas drawn from these to the specific case of spatial analysis. A process known by its initials, PPDAC (Problem, Plan, Data, Analysis, Conclusions) is described as a methodological framework that may be applied to a very wide range of spatial analysis problems and projects. We continue Chapter 3 with a brief look at model-building, with particular reference to the various types of GIS model that can be constructed to address geospatial problems, followed by a discussion of so-called Research-ready data (RRD).

Subsequent Chapters present the various analytical methods supported within widely available software tools. The majority of the methods described in [Chapter 4 \(Building blocks of spatial analysis\)](#) and many of those in [Chapter 6 \(Surface and field analysis\)](#) are implemented as standard facilities in modern commercial GIS packages such as [ArcGIS](#), [MapInfo](#), [Manifold](#), [TNT/Datum Workstation](#) and [Intergraph](#) (now part of Hexagon Geospatial). Many are also provided in more specialized GIS products such as [TerrSet/Idrisi](#), [GRASS](#), [QGIS](#) and [ENVI](#). Note that [GRASS](#) and [QGIS](#) (which includes GRASS in its download kit) are OpenSource. A number of software tools described and illustrated in this book have since been superseded or are no longer available, but their underlying principles and methods remain applicable to the analytical processes examined. Similarly, a number of software tools have been dramatically enhanced, often as applications-specific or sector-specific add-on packages.

In addition we discuss a number of more specialized tools, designed to address the needs of specific sectors or technical problems that are otherwise not well-supported within the core GIS packages at present. [Chapter 5](#), which focuses on statistical methods, and [Chapter 7](#) and [Chapter 8](#) which address Network and Location Analysis, and Geocomputation, are less commonly supported in GIS packages, but may provide loose- or close-coupling with such systems, depending upon the application area. In all instances we provide examples and commentary on software tools that are readily available. The final Chapter ([Chapter 9](#)) addresses issues associated with so-called Big Data and AI - this is clearly an area that is developing extremely rapidly and is expected to result in major changes to the way the GIS industry operates and the forms of Geospatial Analysis that are available.

As noted above, throughout this Guide examples are drawn from and refer to specific products — these have been selected purely as examples and are not intended as recommendations. Extensive use has also been made of tabulated information, providing abbreviated summaries of techniques and formulas for reasons of both compactness and coverage. These tables are designed to provide a quick reference to the various topics covered and are, therefore, not intended as a substitute for fuller details on the various items covered. We provide limited discussion of 2D and 3D mapping facilities, and the support for digital globe formats (e.g. [KML and KMZ](#)), which is increasingly being embedded into general-purpose and specialized data analysis toolsets. These developments confirm the trend towards integration of geospatial data and presentation layers into mainstream software systems and services, both terrestrial and planetary (see, for example, the KML images of Mars DEMs developed by Google as part of the Google Earth project).

Just as all datasets and software packages contain errors, known and unknown, so too do all books and websites, and the authors of this Guide expect that there will be errors despite our

best efforts to remove these! Some may be genuine errors or misprints, whilst others may reflect our use of specific versions of software packages and their documentation. Inevitably with respect to the latter, new versions of the packages that we have used to illustrate this Guide will have appeared even before publication, so specific examples, illustrations and comments on scope or restrictions may have been superseded. In all cases the user should review the documentation provided with the software version they plan to use, check release notes for changes and known bugs, and look at any relevant online services (e.g. user/developer forums and blogs on the web) for additional materials and insights.

The web version of this Guide may be accessed via the associated Internet site: www.spatialanalysisonline.com. The contents and sample sections of the PDF version may also be accessed from this site. In both cases the information is regularly updated. The Internet is now well established as society's principal mode of information exchange and most GIS users are accustomed to searching for material that can easily be customized to specific needs. Our objective for such users is to provide an independent, reliable and authoritative first port of call for conceptual, technical, software and applications material that addresses the panoply of new user requirements.

Spatial analysis, GIS and software tools

Our objective in producing this Guide is to be comprehensive in terms of concepts and techniques (but not necessarily exhaustive), representative and independent in terms of software tools, and above all practical in terms of application and implementation. However, we believe that it is no longer appropriate to think of a standard, discipline-specific textbook as capable of satisfying every kind of new user need. Accordingly, an innovative feature of our approach here is the range of formats and channels through which we disseminate the material.

Given the vast range of spatial analysis techniques that have been developed over the past half century many topics can only be covered to a limited depth, whilst others have been omitted because they are not implemented in current mainstream GIS products. This is a rapidly changing field and increasingly GIS packages are including analytical tools as standard built-in facilities or as optional *toolsets*, *add-ins* or *analysts*. In many instances such facilities are provided by the original software suppliers (commercial vendors or collaborative non-commercial development teams) whilst in other cases facilities have been developed and are provided by third parties. Many products offer software development kits (SDKs), programming languages and language support, scripting facilities and/or special interfaces for developing one's own analytical tools or variants.

In addition, a wide variety of web-based or web-deployed tools have become available, enabling datasets to be analyzed and mapped, including dynamic interaction and drill-down capabilities, without the need for local GIS software installation. These tools include the widespread use of web-based Java, Javascript, AJAX and HTML5 applications, and interactive Virtual Globe explorers, some of which are described in this Guide. They provide an illustration of the direction that many toolset and service providers are taking.

Throughout this Guide there are numerous examples of the use of software tools that facilitate geospatial analysis. In addition, some subsections of the Guide and the software section of the accompanying website, provide summary information about such tools and links to their suppliers. Commercial software products rarely provide access to source code or full details of

the algorithms employed. Typically they provide references to books and articles on which procedures are based, coupled with online help and “white papers” describing their parameters and applications. This means that results produced using one package on a given dataset can rarely be exactly matched to those produced using any other package or through hand-crafted coding. There are many reasons for these inconsistencies including: differences in the software architectures of the various packages and the algorithms used to implement individual methods; errors in the source materials or their interpretation; coding errors; inconsistencies arising out of the ways in which different GIS packages model, store and manipulate information; and differing treatments of special cases (e.g. missing values, boundaries, adjacency, obstacles, distance computations etc.).

Non-commercial packages sometimes provide source code and test data for some or all of the analytical functions provided, although it is important to understand that “non-commercial” often does not mean that users can download the full source code. Source code greatly aids understanding, reproducibility and further development. Such software will often also provide details of known bugs and restrictions associated with functions — although this information may also be provided with commercial products it is generally less transparent. In this respect non-commercial software may meet the requirements of scientific rigor more fully than many commercial offerings, but is often provided with limited documentation, training tools, cross-platform testing and/or technical support, and thus is generally more demanding on the users and system administrators. In many instances open source and similar not-for-profit GIS software may also be less generic, focusing on a particular form of spatial representation (e.g. a grid or raster spatial model). Like some commercial software, it may also be designed with particular application areas in mind, such as addressing problems in hydrology or epidemiology.

The process of selecting software tools encourages us to ask: (i) “what is meant by geospatial analysis techniques?” and (ii) “what should we consider to be GIS software?” To some extent the answer to the second question is the simpler, if we are prepared to be guided by self-selection. For our purposes we focus principally on products that claim to provide geographic information systems capabilities, supporting at least 2D mapping (display and output) of raster (grid based) and/or vector (point/line/polygon based) data, with a minimum of basic map manipulation facilities. We concentrate our review on a number of the products most widely used or with the most readily accessible analytical facilities. This leads us beyond the realm of pure GIS. For example: we use examples drawn from packages that do not directly provide mapping facilities (e.g. the now rather outdated software called [Crimestat](#)) but which provide input and/or output in widely used GIS map-able formats; products that include some mapping facilities but whose primary purpose is spatial or spatio-temporal data exploration and analysis (e.g. GS+, [GeoDa](#), [PySal](#)); and products that are general- or special-purpose analytical engines incorporating mapping capabilities (e.g. [MATLab](#) with the Mapping Toolbox, [WinBUGS](#)) — for more details on these and other example software tools, please see the website page: www.spatialanalysisonline.com/software.html

The more difficult of the two questions above is the first — what should be considered as “geospatial analysis”? In conceptual terms, the phrase identifies the subset of techniques that are applicable when, as a minimum, data can be referenced on a two-dimensional frame and relate to terrestrial activities. The results of geospatial analysis will change if the location or extent of the frame changes, or if objects are repositioned within it: if they do not, then

“everywhere is nowhere”, location is unimportant, and it is simpler and more appropriate to use conventional, *aspatial*, techniques.

Many GIS products apply the term (geo)spatial analysis in a very narrow context. In the case of vector-based GIS this typically means operations such as: map overlay (combining two or more maps or map layers according to predefined rules); simple buffering (identifying regions of a map within a specified distance of one or more features, such as towns, roads or rivers); and similar basic operations. This reflects (and is reflected in) the use of the term *spatial analysis* within the Open Geospatial Consortium ([OGC](#)) “simple feature specifications” (see further [Table 4-2](#)). For raster-based GIS, widely used in the environmental sciences and [remote sensing](#), this typically means a range of actions applied to the grid cells of one or more maps (or images) often involving filtering and/or algebraic operations (*map algebra*). These techniques involve processing one or more raster layers according to simple rules resulting in a new map layer, for example replacing each cell value with some combination of its neighbors’ values, or computing the sum or difference of specific attribute values for each grid cell in two matching raster datasets. Descriptive statistics, such as cell counts, means, variances, maxima, minima, cumulative values, frequencies and a number of other measures and distance computations are also often included in this generic term “spatial analysis”.

However, at this point only the most basic of facilities have been included, albeit those that may be the most frequently used by the greatest number of GIS professionals. To this initial set must be added a large variety of statistical techniques (descriptive, exploratory, explanatory and predictive) that have been designed specifically for spatial and spatio-temporal data. Today such techniques are of great importance in social and political sciences, despite the fact that their origins may often be traced back to problems in the environmental and life sciences, in particular ecology, geology and epidemiology. It is also to be noted that spatial statistics is largely an observational science (like astronomy) rather than an experimental science (like agronomy or pharmaceutical research). This aspect of geospatial science has important implications for analysis, particularly the application of a range of statistical methods to spatial problems.

Limiting the definition of geospatial analysis to 2D mapping operations and spatial statistics remains too restrictive for our purposes. There are other very important areas to be considered. These include: surface analysis —in particular analyzing the properties of physical surfaces, such as gradient, aspect and visibility, and analyzing surface-like data “fields”; network analysis — examining the properties of natural and man-made networks in order to understand the behavior of flows within and around such networks; and locational analysis. GIS-based network analysis may be used to address a wide range of practical problems such as route selection and facility location, and problems involving flows such as those found in hydrology. In many instances location problems relate to networks and as such are often best addressed with tools designed for this purpose, but in others existing networks may have little or no relevance or may be impractical to incorporate within the modeling process. Problems that are not specifically network constrained, such as new road or pipeline routing, regional warehouse location, mobile phone mast positioning, pedestrian movement or the selection of rural community health care sites, may be effectively analyzed (at least initially) without reference to existing physical networks. Locational analysis “in the plane” is also applicable where suitable network datasets are not available, or are too large or expensive to be utilized, or where

the location algorithm is very complex or involves the examination or simulation of a very large number of alternative configurations.

A further important aspect of geospatial analysis is visualization (or *geovisualization*) — the use, creation and manipulation of images, maps, diagrams, charts, 3D static and dynamic views, high resolution satellite imagery and digital globes, and their associated tabular datasets (see further [Slocum et al., 2008](#), [Dodge et al., 2008](#), [Longley et al., 2015](#), Ch12). For early insights into how some of these developments may be applied, see Andrew Hudson-Smith (2008) “Digital Geography: Geographic visualization for urban environments” and Martin Dodge and Rob Kitchin’s earlier “[Atlas of Cyberspace](#)” which is available as a free downloadable document. A Working Paper on this topic from the Centre for Advanced Spatial Analysis ([CASA](#)) can be found here:

https://discovery.ucl.ac.uk/1436767/2/CASAWorkingPaper_paper190.pdf

GIS packages and web-based services increasingly incorporate a range of such tools, providing static or rotating views, draping images over 2.5D surface representations, providing animations and fly-throughs, dynamic linking and brushing and spatio-temporal visualizations. This latter class of tools has been, until recently, the least developed, reflecting in part the limited range of suitable compatible datasets and the limited set of analytical methods available, although this picture is changing rapidly. One recent example is the availability of image time series from [NASA’s Earth Observation](#) Satellites, yielding vast quantities of data on a daily basis (e.g. [Aqua mission](#), commenced 2002; [Terra mission](#), commenced 1999).

Geovisualization is the subject of ongoing research by the International Cartographic Association (ICA), [Commission on Visual Analytics](#), who have organized a series of workshops and publications addressing developments in geovisualization, notably with a cartographic focus.

As datasets, software tools and processing capabilities develop, 3D geometric and photo-realistic visualization are becoming a *sine qua non* of modern geospatial systems and services — see [Andy Hudson-Smith’s “Digital Urban” blog](#) for a regularly updated commentary on this field. We expect to see an explosion of tools and services and datasets in this area over the coming years — many examples are included as illustrations in this Guide. Other examples readers may wish to explore include: the static and dynamic visualizations at [3DNature](#) and similar sites; the 2D and 3D [Atlas of Switzerland](#); Urban 3D modeling programmes such as [CityGML](#); and the integration of GIS technologies and data with digital globe software, e.g. data from [Digital Globe \(Maxar\)](#), and Earth-based frameworks such as Google Earth, Microsoft Virtual Earth, NASA Worldwind and Edushi (Chinese). There are also automated translators between GIS packages such as [ArcGIS](#) and Digital Earth models (see for example the now withdrawn, Arc2Earth).

These novel visualization tools and facilities augment the core tools utilized in spatial analysis throughout many parts of the analytical process: exploration of data; identification of patterns and relationships; construction of models; dynamic interaction with models; and communication of results — see, for example, the work of the city of Portland, Oregon, who have used 3D visualization to communicate the results of zoning, crime analysis and other key local variables to the public; and with Covid at the forefront of much work and reporting from 2020 onward, the US interactive Covid mapping project (now ended). Another example is the

3D visualizations provided as part of the web-accessible [London Air Quality](#) network. These are designed to enable:

- users to visualize air pollution in the areas that they work, live or walk
- transport planners to identify the most polluted parts of London
- urban planners to see how building density affects pollution concentrations in the City and other high density areas, and
- students to understand pollution sources and dispersion characteristics

Physical 3D models and hybrid physical-digital models have also been developed and applied to practical analysis problems. For example: 3D physical models constructed from plaster, wood, paper and plastics have been used for many years in architectural and engineering planning projects; hybrid sand tables are used to help firefighters in California visualize the progress of wildfires (see Figure 1-1A, below and tools from companies like [SimTable](#); very large sculptured solid terrain models are being used for educational purposes, to assist land use modeling programmes, and to facilitate participatory 3D modeling in less-developed communities ([P3DM](#)); and 3D digital printing technology is being used to generate 3D landscapes and cityscapes from GIS, CAD and/or VRML files with planning, security, architectural, archaeological and geological applications (see Figure 1-1B, below and the websites of [3D Systems](#) and [Stratasys](#) for more details). To create large landscape models multiple individual prints, which are typically only around 20cm x 20cm x 5cm, are made, in much the same manner as raster file mosaics.

Figure 1-1A: 3D Physical GIS models: Sand-in-a-box model, Albuquerque, USA



Figure 1-1B: 3D Physical GIS models: 3D GIS printing



GIS software, notably in the commercial sphere, is driven primarily by demand and applicability, as manifest in willingness to pay. Hence, to an extent, the facilities available often reflect commercial and resourcing realities (including the development of improvements in processing and display hardware, and the ready availability of high quality datasets) rather than the status of development in geospatial science. Indeed, there may be many capabilities available in software packages that are provided simply because it is extremely easy for the designers and programmers to implement them, especially those employing object-oriented programming and data models. For example, a given operation may be provided for polygonal features in response to a well-understood application requirement, which is then easily enabled for other features (e.g. point sets, polylines) despite the fact that there may be no known or likely requirement for the facility.

Despite this cautionary note, for specific well-defined or *core* problems, software developers will frequently utilize the most up-to-date research on algorithms in order to improve the quality (accuracy, optimality) and efficiency (speed, memory usage) of their products. For further information on algorithms and data structures, see the online [NIST Dictionary of algorithms and data structures](#).

Furthermore, the quality, variety and efficiency of spatial analysis facilities provide an important discriminator between commercial offerings in an increasingly competitive and open market for software and spatial datasets. However, the ready availability of analysis tools does not imply that one product is necessarily better or more complete than another — it is the selection and application of *appropriate* tools in a manner that is *fit for purpose* that is important. Guidance documents exist in some disciplines that assist users in this process, e.g. Perry *et al.* (2002) dealing with ecological data analysis, and to a significant degree we hope that this Guide will assist users from many disciplines in the selection process.

Intended audience and scope

This Guide has been designed to be accessible to a wide range of readers — from undergraduates and postgraduates studying GIS and spatial analysis, to GIS practitioners and professional analysts. It is intended to be much more than a cookbook of formulas, algorithms and techniques — its aim is to provide an explanation of the key techniques of spatial analysis using examples from widely available software packages. It stops short, however, of attempting a systematic evaluation of competing software products. A substantial range of application examples are provided, but any specific selection inevitably illustrates only a small subset of the huge range of facilities available. Wherever possible, examples have been drawn from non-

academic sources, highlighting the growing understanding and acceptance of GIS technology in the commercial and government sectors.

The scope of this Guide incorporates the various spatial analysis topics included within the seminal NCGIA Core Curriculum (Goodchild and Kemp, 1990, now updated) and as such may provide a useful accompaniment to GIS Analysis courses based closely or loosely on this programme. More recently the Education Committee of the University Consortium for Geographic Information Science ([UCGIS](https://www.ucgis.org)) in conjunction with the Association of American Geographers ([AAG](https://www.aag.org)) has produced a comprehensive “Body of Knowledge” (BoK) document - see the associated website <https://www.ucgis.org/site/gis-t-body-of-knowledge>. This Guide covers materials that primarily relate to the BoK sections **CF**: Conceptual Foundations; **AM**: Analytical Methods and **GC**: Geocomputation. In the general introduction to the **AM** knowledge area the authors of the BoK summarize this component as follows:

“This knowledge area encompasses a wide variety of operations whose objective is to derive analytical results from geospatial data. Data analysis seeks to understand both first-order (environmental) effects and second-order (interaction) effects. Approaches that are both data-driven (exploration of geospatial data) and model-driven (testing hypotheses and creating models) are included. Data-driven techniques derive summary descriptions of data, evoke insights about characteristics of data, contribute to the development of research hypotheses, and lead to the derivation of analytical results. The goal of model-driven analysis is to create and test geospatial process models. In general, model-driven analysis is an advanced knowledge area where previous experience with exploratory spatial data analysis would constitute a desired prerequisite.” (BoK, p83 of the e-book version, first edition).

GIS and related software tools

The GIS software and analysis tools that an individual, group or corporate body chooses to use will depend very much on the purposes to which they will be put. There is an enormous difference between the requirements of academic researchers and educators, and those with responsibility for planning and delivery of emergency control systems or large scale physical infrastructure projects. The spectrum of products that may be described as a GIS includes (amongst others):

- mainstream GIS software packages and services, such as those from [Caliper](https://www.mapititude.com) (Mapititude/TransCAD), Clark Labs ([TerrSet](https://www.terraset.com)/[IDRISI](https://www.idrisi.com)), [ESRI](https://www.esri.com) (ArcGIS Pro and related [toolboxes](#) and services), [Grass](https://grass.osgeo.org) (Opensource), [Manifold](https://www.manifold.com), and Microimages ([TNT/Datum Workstation](https://www.tntmapping.com))
- highly specialized, sector specific packages: for example civil engineering design and costing systems; satellite image processing systems; and utility infrastructure management systems
- transportation and logistics management systems
- civil and military control room systems
- systems for visualizing the built environment for architectural purposes, for public consultation or as part of simulated environments for interactive gaming
- land registration systems

- census data management systems
- commercial location services and Digital Earth models
- Geospatial data visualization tools
- AI-augmented geospatial data services and software tools
- Online versions of offline (desktop) GIS software

The list of software functions and applications is long and in some instances suppliers would not describe their offerings as a GIS. In many cases such systems fulfill specific operational needs, solving a well-defined subset of spatial problems and providing mapped output as an incidental but essential part of their operation. Many of the capabilities may be found in generic GIS products. In other instances a specialized package may utilize a GIS engine for the display and in some cases processing of spatial data (directly, or indirectly through interfacing or file input/output mechanisms). For this reason, and in order to draw a boundary around the present work, reference to application-specific GIS will be limited.

A number of GIS packages and related toolsets have particularly strong facilities for processing and analyzing binary, grayscale and color images. They may have been designed originally for the processing of remote sensed data from satellite and aerial surveys, but many have developed into much more sophisticated and complete GIS tools, e.g. Clark Lab's [Terrset/Idrisi](#) software; MicroImage's [TNT/Datum Workstation](#) product set; the [ERDAS](#) suite of products; and [ENVI](#) with associated packages such as [RiverTools](#). Alternatively, image handling may have been deliberately included within the original design parameters for a generic GIS package (e.g. [Manifold](#)), or simply be toolsets for image processing that may be combined with mapping tools (e.g. the [MATLab](#) Image Processing Toolbox). Whatever their origins, a central purpose of such tools has been the capture, manipulation and interpretation of image data, rather than spatial analysis *per se*, although the latter inevitably follows from the former.

In this Guide we do not provide a separate chapter on image processing, despite its considerable importance in GIS, focusing instead on those areas where image processing tools and concepts are applied for spatial analysis (e.g. surface analysis). We have adopted a similar position with respect to other forms of data capture, such as field and geodetic survey systems and data cleansing software — although these incorporate analytical tools, their primary function remains the recording and georeferencing of datasets, rather than the analysis of such datasets once stored.

For most GIS professionals, spatial analysis and associated modeling is an infrequent activity. Even for those whose job focuses on analysis the range of techniques employed tends to be quite narrow and application specific. GIS consultants, researchers and academics on the other hand are continually exploring and developing analytical techniques. For the first group and for consultants, especially in commercial environments, the imperatives of financial considerations, timeliness and corporate policy loom large, directing attention to: delivery of solutions within well-defined time and cost parameters; working within commercial constraints on the cost and availability of software, datasets and staffing; ensuring that solutions are fit for purpose/meet client and end-user expectations and agreed standards; and in some cases, meeting “political” expectations.

For the second group of users it is common to make use of a variety of tools, data and programming facilities developed in the academic sphere. Increasingly these make use of non-commercial wide-ranging spatial analysis software libraries, such as the [R-Spatial](#) project (in “R”); [PySal](#) (in “Python”); and [Splancs](#) (in “S”). Many of these tools include support for a variety of file types and formats, including those required for mapping the results (e.g. using [ESRI](#)-type shape, shp, file handling) - for more details and worked examples using R, see Lansley and Cheshire's (2016) [Introductory Guide](#) downloadable as a PDF.

Sample software products

The principal products we have included in this latest edition of the Guide are included on the accompanying website's [software page](#). Many of these products are free whilst others are available (at least in some form) for a small fee for all or selected groups of users. Others are licensed at varying per user prices, from a few hundred to over a thousand US dollars per user. Our tests and examples have largely been carried out using desktop/Windows versions of these software products. Different versions that support Unix-based operating systems and more sophisticated back-end database engines have not been utilized. In the context of this Guide we do not believe these selections affect our discussions in any substantial manner, although such issues may have performance and systems architecture implications that are extremely important for many users. Increasingly online services are being offered that provide access to very large datasets, data mining and AI-driven analytical tools; brief discussion of the opportunities and issues relating to some of these services is provided in the final Chapter of this book. [OGC](#) compliant software products are listed on the [OGC](#) resources web page: https://portal.ogc.org/public_ogc/compliance/compliant.php

To quote from the latest [OGC](#) statement: “*The OGC Compliance Program is a certification process that ensures organizations’ solutions are compliant with OGC Standards. It is a universal credential that allows agencies, industry, and academia to better integrate their solutions. OGC compliance provides confidence that a product will seamlessly integrate with other compliant solutions regardless of the vendor that created them*”

Software performance

Suppliers should be able to provide advice on performance issues (e.g. see the [ESRI](#) web site, "Services" area for relevant material relating to their products) and in some cases such information is provided within product Help files (e.g. see the Performance Tips section within the [Manifold](#) GIS support section). Some analytical tasks are very processor- and memory-hungry, particularly as the number of elements involved increases. For example, vector overlay and buffering is relatively fast with a few objects and layers, but slows appreciably as the number of elements involved increases. This increase is generally at least linear with the number of layers and features, but for some problems grows in a highly non-linear (i.e. geometric) manner. Offline (desktop) software like [Manifold](#) has, in recent years, been dramatically improved in terms of data loading and processing speed, notably by using multiple cores and processor units, in particular, NVIDIA GPUs.

Many optimization tasks, such as optimal routing through networks or trip distribution modeling, are known to be extremely hard or impossible to solve optimally and methods to achieve a best solution with a large dataset can take a considerable time to run (see our discussion of [Algorithms and computational complexity theory](#) later in this Guide for a fuller

discussion of this topic). Similar problems exist with the processing and display of raster files, especially large images or sets of images. Geocomputational methods, some of which are beginning to appear within GIS packages and related toolsets, are almost by definition computationally intensive. This certainly applies to large-scale ([Monte Carlo](#)) simulation models, cellular automata and agent-based models and some raster-based optimization techniques, especially where modeling extends into the time domain.

A frequent criticism of GIS software is that it is over-complicated, resource-hungry and requires specialist expertise to understand and use. Such criticisms are often valid and for many problems it may prove simpler, faster and more transparent to utilize specialized tools for the analytical work and draw on the strengths of GIS in data management and mapping to provide input/output and visualization functionality. Example approaches include: (i) using high-level programming facilities within a GIS (e.g. macros, scripts, VBA, [Python](#)) – many add-ins are developed in this way; (ii) using wide-ranging programmable spatial analysis software libraries and toolsets that incorporate GIS file reading, writing and display, such as the [R-Spatial](#) and [PySal](#) projects noted earlier; (iii) using general purpose data processing toolsets, e.g. [MATLab](#), Excel, [Matplotlib](#), Numeric Python ([Numpy](#)), Cython ("an optimizing static compiler for both the [Python](#) programming language and the extended Cython programming language (based on Pyrex). It makes writing C extensions for Python as easy as Python itself" - see further <https://cython.org/>), and other libraries from [Enthought](#); or (iv) directly utilizing mainstream programming languages (e.g. Java, C++). The advantage of these approaches is control and transparency, the disadvantages are that software development is never trivial, is often subject to frustrating and unforeseen delays and errors, and generally requires ongoing maintenance. In some instances analytical applications may be well-suited to parallel or grid-enabled processing – as for example is the case with [GWR](#) (see Harris *et al.*, 2006). In some instances many of these issues are effectively 'outsourced' to a geospatial services provider, with datasets, mapping and analytical toolsets delivered through a single, user-friendly interface, with all the pros and cons of such packaged offerings.

At present there are no standardized tests for the quality, speed and accuracy of GIS procedures. It remains the buyer's and user's responsibility and duty to evaluate the software or service they wish to use for the specific task at hand, and by systematic controlled tests or by other means establish that the product and facility within that product they choose to use is truly fit for purpose — *caveat emptor*! Details of how to obtain many of these products are provided on the [software page](#) of the website that accompanies this book. The list maintained on Wikipedia is also a useful source of information and links, although is far from being complete or independent. A number of trade magazines and websites provide ad hoc reviews of GIS software and service offerings, especially new releases, although coverage of analytical functionality may be limited.

Suggested reading

There are numerous excellent modern books on GIS and spatial analysis, although few address software facilities and developments. Hypertext links are provided here, and throughout the text where they are cited, to the more recent publications and web resources listed.

As a background to this Guide any readers unfamiliar with GIS are encouraged to first tackle "Geographic Information Science and Systems" (GISSc) by [Longley et al.](#) (2015). GISSc

seeks to provide a comprehensive and highly accessible introduction to the subject as a whole. The GB Ordnance Survey's "[GIS pages](#)" also provides an excellent brief introduction to GIS and its applications.

Some of the basic mathematics and statistics of relevance to GIS analysis is covered in Dale (2005) and Allan (2004). For detailed information on datums and map projections, see Iliffe and Lott (2008). Useful online resources for those involved in data analysis, particularly with a statistical content, include the [StatsRef website](#) and the [e-Handbook of Statistical Methods](#) produced by the US National Institute on Standards and Technology, NIST). The more informally produced set of articles on statistical topics provided under the Wikipedia umbrella are also an extremely useful resource. These sites, and the mathematics reference site, [Mathworld](#), are referred to (with hypertext links) at various points throughout this document. For those who find mathematics and statistics something of a mystery, [de Smith \(2018\)](#) and Bluman (2003) provide useful starting points. For guidance on how to avoid the many pitfalls of statistical data analysis readers are recommended the material in the classic work by Huff (1993) "How to lie with statistics", and the 2008 book by Blastland and Dilnot "The tiger that isn't".

A relatively new development has been the increasing availability of out-of-print published books, articles and guides as free downloads in PDF format. These include: the series of 59 short guides published under the [CATMOG](#) umbrella (Concepts and Methods in Modern Geography), published between 1975 and 1995, most of which are now available at the [QMRG](#) website (a full list of all the guides is provided at the end of this book); the [Atlas of Cyberspace](#) by Dodge and Kitchin; and [Fractal Cities](#), by Batty and Longley.

Undergraduates and MSc programme students will find Burrough and McDonnell (1998, 2015) provides excellent coverage of many aspects of geospatial analysis, especially from an environmental sciences perspective. Valuable guidance on the relationship between spatial process and spatial modeling may be found in Cliff and Ord (1981) and Bailey and Gatrell (1995). The latter provides an excellent introduction to the application of statistical methods to spatial data analysis. O'Sullivan and Unwin (2010, 2nd ed.) is a more broad-ranging book covering the topic the authors describe as "Geographic Information Analysis". This work is best suited to advanced undergraduates and first year postgraduate students. In many respects a deeper and more challenging work is Haining's (2003) "Spatial Data Analysis — Theory and Practice". This book is strongly recommended as a companion to the present Guide for postgraduate researchers and professional analysts involved in using GIS in conjunction with statistical analysis.

However, these authors do not address the broader spectrum of geospatial analysis and associated modeling as we have defined it. For example, problems relating to networks and location are often not covered and the literature relating to this area is scattered across many disciplines, being founded upon the mathematics of graph theory, with applications ranging from electronic circuit design to computer networking and from transport planning to the design of complex molecular structures. Useful books addressing this field include Miller and Shaw (2001) "Geographic Information Systems for Transportation" (especially Chapters 3, 5 and 6), and Rodrigue *et al.* (2006) "The geography of transport systems" (see further: <https://transportgeography.org/>).

As companion reading on these topics for the present Guide we suggest the two volumes from the Handbooks in Operations Research and Management Science series by Ball *et al.* (1995): “Network Models”, and “Network Routing”. These rather expensive volumes provide collections of reviews covering many classes of network problems, from the core optimization problems of shortest paths and arc routing (e.g. street cleaning), to the complex problems of dynamic routing in variable networks, and a great deal more besides. This is challenging material and many readers may prefer to seek out more approachable material, available in a number of other books and articles, e.g. Ahuja *et al.* (1993), Mark Daskin’s excellent book “Network and Discrete Location” (1995) and the earlier seminal works by Haggett and Chorley (1969), and Scott (1971), together with the widely available online materials accessible via the Internet. Final recommendations here are Stephen Wise’s excellent GIS Basics (2002) and Worboys and Duckham (2004) which address GIS from a computing perspective. Both these volumes covers many topics, including the central issues of data modeling and data structures, key algorithms, system architectures and interfaces.

Many recent books described as covering (geo)spatial analysis are essentially edited collections of papers or brief articles. As such most do not seek to provide comprehensive coverage of the field, but tend to cover information on recent developments, often with a specific application focus (e.g. health, transport, archaeology). The latter is particularly common where these works are selections from sector- or discipline-specific conference proceedings, whilst in other cases they are carefully chosen or specially written papers. Classic amongst these is Berry and Marble (1968) “Spatial Analysis: A reader in statistical geography”. More recent examples include “GIS, Spatial Analysis and Modeling” edited by Maguire, Batty and Goodchild (2005), and the excellent (but costly) compendium work “The SAGE handbook of Spatial Analysis” edited by Fotheringham and Rogerson (2008).

A second category of companion materials to the present work is the extensive product-specific documentation available from software suppliers. Some of the online help files and product manuals are excellent, as are associated example data files, tutorials, worked examples and white papers (see for example, [ESRI’s What is GIS?](#)), which provides a wide-ranging guide to GIS. In many instances we utilize these to illustrate the capabilities of specific pieces of software and to enable readers to replicate our results using readily available materials. In addition some suppliers, notably [ESRI](#), have a substantial publishing operation, including more general (i.e. not product specific) books of relevance to the present work. Amongst their publications we strongly recommend the “ESRI Guide to GIS Analysis Volume 1: Geographic patterns and relationships” (1999) by Andy Mitchell, which is full of valuable tips and examples. This is a basic introduction to GIS Analysis, which he defines in this context as “a process for looking at geographic patterns and relationships between features”. Mitchell’s Volume 2 (July 2005) covers more advanced techniques of data analysis, notably some of the more accessible and widely supported methods of spatial statistics, and is equally highly recommended. A number of the topics covered in his Volume 2 also appear in this Guide. David Allen has produced a tutorial book and DVD (GIS Tutorial II: Spatial Analysis Workbook) to go alongside Mitchell’s volumes, and these are obtainable from ESRI Press. For a perspective on modern applications of GIS and geospatial analysis see Esri’s [GIS For Science series](#) (Volume 3 is the latest and Chapter 7 focuses on geospatial analysis and AI, also referred to now as GeoAI) and the recently published collection of edited chapters brought together as the “Handbook of Geospatial Artificial Intelligence” (2024, Song Gao *et al.*, with a Foreword

by Prof Mike Goodchild). Those considering using Open Source software may wish to investigate the books by Neteler and Mitasova (2008), Tyler Mitchell (2005) and Sherman (2008).

In parallel with the increasing range and sophistication of spatial analysis facilities to be found within GIS packages, there has been a major change in spatial analytical techniques. In large measure this has come about as a result of technological developments and the related availability of software tools and detailed publicly available datasets. One aspect of this has been noted already — the move towards network-based location modeling where in the past this would have been unfeasible. More general shifts can be seen in the move towards local rather than simply global analysis, for example in the field of exploratory data analysis; in the increasing use of advanced forms of visualization as an aid to analysis and communication; in the growth of online spatial data, mapping and analysis services; in the development of a wide range of computationally intensive and simulation methods that address problems through micro-scale processes (geocomputational methods); and finally, in the recent development of AI-augmented services, many of which fit within the somewhat portmanteau term, GeoAI. These trends are addressed at many points throughout this Guide.

Geospatial analysis involves a wide range of terms used in various contexts, often borrowed from mathematics, statistics, and geography, where meanings may vary depending on the source or software. Below is a comprehensive paragraph-style explanation of selected key terminologies used in geospatial analysis, based on widely recognized sources like the Association for Geographic Information (AGI) and the Open Geospatial Consortium (OGC).

Adjacency is when two or more polygons share a common boundary or side, although this concept also applies to features on opposite sides of a shared boundary, not limited to polygons. An arc typically refers to a straight line segment connecting two nodes in a polyline or polygon but may also include curves or splines, with directed or undirected types having attributes like cost or capacity in network analysis. Artifacts are unusual, insignificant observations such as spikes in 3D plots, often caused by data collection or computation processes, sometimes appearing as ghost lines.

Aspect indicates the direction where the slope of a surface is steepest, relating directly to slope and gradient. An attribute is a data item linked with an object in a spatial database, which can be explicit (stored in tables) or implicit (computed on demand), and raster datasets usually have one attribute value per cell. Azimuth measures the horizontal direction of a vector clockwise in degrees from the positive Y-axis, akin to compass bearings, while azimuthal projection maps areas as if projecting onto a tangent plane, preserving true compass bearings.

Spatial autocorrelation describes the degree to which spatial variables vary together, positively or negatively, commonly seen in ordered datasets like time series or spatial data ordered by distance bands; although it implies association, it does not prove causality. A cartogram distorts map geometry to represent data like population instead of land area, while choropleth maps visualize data categories across areas using color or shading.

Conflation merges data sources into one by reconciling differences, distinct from concatenation where sources overlay but remain separate. Contiguity represents the topological relationship by recording adjacent polygons by arcs, focusing on relative rather than exact geographic

locations. A curve is a one-dimensional geometric object defined by point sequences with interpolation, where LineString or polyline is a common linear subtype.

A datum refers to a reference surface model (e.g., WGS84) used for elevation or horizontal geographic referencing. DEM and DTM represent digital elevation and terrain models, respectively, while EDM denotes electronic distance measurement. EDA and ESDA refer to exploratory data analysis and exploratory spatial data analysis. Ellipsoid or spheroid describes the Earth's shape as an ellipse rotated about its minor axis.

A feature is a GIS object such as a point, line, polygon, or annotation, with geoid representing Earth's mean sea level extended under continents as a gravitational surface. Geodemographics analyze populations by locality or neighborhood type. Geospatial pertains to location relative to Earth's surface and often involves spatial data representations beyond simple maps. Geostatistics are specialized statistical methods dealing with spatial data, accounting for issues like autocorrelation, often applied in environmental and social sciences.

Geovisualization includes techniques that visualize spatial and spatio-temporal data, from 2D maps to dynamic 3D models and immersive systems. GIS-T focuses on GIS applications in transportation. GPS and DGPS are satellite-based positioning systems with DGPS providing enhanced accuracy through fixed reference stations. Gradient is a vector field of slope magnitude and aspect direction across surfaces.

Graphs are mathematical collections of vertices and edges studied in graph theory. Heuristics are problem-solving approaches aimed at finding near-optimal solutions efficiently. IID is an abbreviation for independently and identically distributed data samples. Invariance refers to spatial features maintaining properties under certain transformations. Kernel methods involve localized calculations within defined neighborhoods.

Layers are collections of geographic entities of a single type, which can be combined across types. Map algebra involves cell-based algebraic operations on raster data to produce new layers. A mashup refers to web content created by integrating multiple data sources via programmable interfaces. The minimum bounding rectangle (MBR) or minimum enclosing rectangle (MER) defines the smallest enclosing rectangular boundary for feature sets.

Planar and non-planar refer to geometries lying in a single plane or not, with planar enforced polygons having no overlaps except at boundaries; planar graphs can be drawn without edge intersections outside nodes. Pixels are the smallest image units carrying color or intensity values. A polygon is a closed planar figure defined by connected vertices and edges; if simple, it does not self-intersect, and convex if all internal lines lie within it.

Polyhedral surfaces consist of connected polygons sharing boundaries, while polylines are vertex-ordered connected line segments, with LineStrings a GIS standard curve subtype. Raster/grid data represent geographic features as a grid of discrete cells, each holding values, often referenced from the lower-left corner. Resampling adjusts raster resolutions or reduces dataset size by aggregating pixels, with coarse images containing less detail.

Rubber sheeting adjusts dataset coordinates for alignment by stretching or shrinking while preserving topology. Slope measures elevation change over horizontal distance, with the maximum slope direction called aspect; combined, they form the gradient vector. Spatial

econometrics applies econometric methods to spatial data focusing on spatial dependence and heterogeneity across locations and time.

A spheroid models Earth's shape as a slightly flattened sphere (oblate ellipsoid). SQL in GIS supports spatial queries based on relationships rather than just attributes. Surfaces are 2D geometric objects possibly with holes, while tessellations divide planes into polygons like raster squares or triangles (TIN). Triangulated irregular networks (TINs) model surfaces by connecting irregularly spaced points into non-overlapping triangles.

Topology captures spatial relationships such as connectivity and adjacency independent of exact coordinates, represented in digital data by nodes and links. Transformations convert coordinate data between systems, including affine transformations preserving parallel lines. Data transformations apply mathematical functions on datasets, with back transformations reverting to original data. Vectors represent GIS data as lines or arcs with explicit node storage.

Viewshed maps visibility regions from observation points, indicating visible raster cells. WGS84 is a major global reference system modeling Earth as a spheroid with precise flattening at poles.

Common Measures and Notation

Throughout this Guide a number of terms and associated formulas are used that are common to many analytical procedures. In this section we provide a brief summary of those that fall into this category. Others, that are more specific to a particular field of analysis, are treated within the section to which they primarily apply. Many of the measures we list will be familiar to readers, since they originate from standard single variable (univariate) statistics. For brevity we provide details of these in tabular form. In order to clarify the expressions used here and elsewhere in the text, we use the notation shown in Table 1-2. Italics are used within the text and formulas to denote variables and parameters, as well as selected terms.

The notation and symbology used in geospatial analysis and related mathematical contexts include several fundamental symbols and their meanings. The closed interval $[a,b]$ on the real line includes all values between a and b , including the endpoints. In contrast, (a,b) denotes an open interval, containing all values between a and b but excluding the endpoints. This open interval symbol should not be confused with coordinate pairs like (x,y) or function inputs like $f(x,y)$, where the meaning depends on context. In graph theory, (i,j) represents an edge connecting vertices i and j . The (x,y) pair typically represents a two-dimensional spatial coordinate in Cartesian or spherical form, with latitude and longitude ranges commonly from -90 to $+90$ and -180 to $+180$ degrees respectively. Adding a third dimension as (x,y,z) includes height, depth, or an attribute value associated with the spatial location.

A set of n values $x_1, x_2, \dots, x_n$, denoted $\{x_i\}$, represents data points or measurements, possibly continuous or ratio-scaled. An ordered set $\{X_i\}$ arranges these values so that $x_i \leq x_{i+1}$ for all i , useful in statistics. Bold uppercase letters (X) denote matrices, while lowercase bold letters (x) indicate vectors. Frequency counts of data values or classes are represented by $\{f_i\}$, with the sum of all f_i equal to the number of observations n . Probabilities $\{p_i\}$ derived from frequencies satisfy the summation to unity, $\sum p_i = 1$.

Symbol Σ denotes summation (e.g., $x_1 + x_2 + \dots + x_n$), while Π indicates multiplication across elements. The caret symbol $\wedge$ marks an estimate of a population parameter, often called a "hat".

The tilde $\sim$ signifies a variable's distribution, for example $y \sim N(0,1)$ indicates y is normally distributed with mean 0 and standard deviation 1. The factorial symbol $!$ defines multiplicative sequences for integers, such as $x! = x(x-1)(x-2)\dots 1$. The symbol $\equiv$ means "equivalent to," while $\approx$ means "approximately equal to." The symbol $\in$ is used for set membership, expressing that x belongs to a set or interval, e.g., $x \in$. Inequality symbols $\leq$ (less than or equal to) and $\geq$ (greater than or equal to) are sometimes represented as $\leq$ and $\geq$ for compatibility with text displays.

These symbols form the concise notation backbone for representing intervals, sets, coordinate pairs, vectors, matrices, probabilities, estimations, distributions, and mathematical relationships widely used in geospatial data analysis and statistics.

Statistical measures and related formulas

Following provides common univariate statistical measures and formulas used to analyze datasets. These measures fall into several categories:

Counts and Specific Values include the total count of data points, distinct values (variety), most/least frequent values (majority/minority), and extremes (maximum/minimum).

Measures of Centrality describe the center of a dataset through various means (arithmetic, geometric, harmonic, power), the mode (most frequent value), median (middle value), and mid-range. These provide different perspectives on the typical or central value depending on data distribution and context.

Measures of Spread quantify how data varies around the center, including range, quartiles, inter-quartile range, variance, and standard deviation. These indicate whether data points cluster tightly or scatter widely.

Measures of Distribution Shape use moments about the mean to describe asymmetry (skewness) and peakedness (kurtosis), revealing whether distributions are symmetric or skewed.

Measures of Complexity include information-theoretic entropy and fractal/topological dimensions, capturing data disorder and geometric complexity.

Common Distributions (Uniform, Binomial, Poisson, Normal) describe theoretical patterns from which sample data may originate, essential for statistical inference.

Data Transforms (logarithmic, square root, logit, Box-Cox, angular) are applied to normalize or stabilize variance in skewed or heterogeneous datasets.

Matrix Expressions cover determinants, inverses, transposes, eigenvalues, and eigenvectors—fundamental for multivariate and spatial statistical operations.

Many of these measures extend naturally to two dimensions, providing the foundation for spatial statistics and analysis of geographic data.

Counts and Specific Values

- **Count:** The number of data values in a set, given by $\text{Count}(\{x_i\}) = n$.
- **Top m, Bottom m:** The set of the largest (smallest) m values from a set. For ordered data: $\text{Top}_m\{x_i\} = \{X_{n-m+1}, \dots, X_{n-1}, X_n\}$; $\text{Bot}_m\{x_i\} = \{X_1, X_2, \dots, X_m\}$.

- **Variety:** Number of distinct (different) data values in a set; sometimes called diversity (not to be confused with entropy-based diversity measures).
- **Majority:** The most common/frequent value(s) in a set; similar to mode, often used for raster data where a class is more than 50% of the total.
- **Minority:** The least frequently occurring value(s) in a set; often referenced for raster datasets.
- **Maximum, Max:** The highest value in a set (may not be unique): $\text{Max}\{x_i\} = X_n$.
- **Minimum, Min:** The lowest value in a set (may not be unique): $\text{Min}\{x_i\} = X_1$.
- **Sum:** The total sum of data values in a set.

Measures of Centrality

- **Mean (Arithmetic):** The average of all values, also known as the sample mean. If frequencies $\{f_i\}$ are treated as weights, the result is a weighted mean. Population mean is usually μ , estimate shown as $\hat{\mu}$ or $\bar{x}$.
- **Mean (Harmonic):** The mean of reciprocals of the data values, then take the reciprocal of the result. Harmonic mean $\leq$ geometric mean $\leq$ arithmetic mean.
- **Mean (Geometric):** Calculated by multiplying the data values together and taking the n th root. Geometric mean is $\geq$ harmonic mean but $\leq$ arithmetic mean.
- **Mean (Power):** Generalized mean; for $p = 1$ it's arithmetic mean, $p = 2$ is root mean square, $p = -1$ harmonic mean, and in limits $p = 0$ is geometric mean, $p = -\infty$ minimum, $p = +\infty$ maximum.
- **Trim-mean, TM, t (Olympic mean):** The mean after trimming a percentage $t/2$ of high and low values to remove outliers/extremes. For small samples, a specific value is ignored. Olympic mean used in scoring gymnastics.
- **Mode:** The most frequently occurring value(s). Unimodal sets have a single mode, multimodal have several. Mean – Mode $\approx 3(\text{Mean} - \text{Median})$ for some distributions.
- **Median, Med:** The middle value in an ordered set. For odd n : $\text{Med}\{x_i\} = X_{(n+1)/2}$. For even n : $\text{Med}\{x_i\} = (X_{n/2} + X_{n/2+1})/2$.
- **Mid-range, MR:** Middle value of the range: $\text{MR}\{x_i\} = \text{Range}/2$.
- **Root Mean Square (RMS):** The square root of the mean of squared values—removes negatives.

Measures of Spread

- **Range:** Difference between maximum and minimum: $\text{Range}\{x_i\} = X_n - X_1$.
- **Lower Quartile (25%), LQ:** 25% of ordered data are $\leq$ the upper bound of LQ. For continuous data, it's the cumulative distribution up to 0.25.
- **Upper Quartile (75%), UQ:** 75% of ordered data are $\leq$ the upper bound of UQ. For continuous data, it's the cumulative distribution from 0.75 to 1.

- **Inter-quartile Range (IQR):** Difference between UQ and LQ, covering middle 50%: $IQR = UQ - LQ$.
- **Trim-range, TR, t:** Range computed after removing a percentage $t/2$ of high and low values. For small samples, a specific number may be ignored. $TR_t = X_{n(1-t/2)} - X_{nt/2}$, $t \in$, $TR_{50\%} = IQR$.
- **Variance, Var, σ^2, s^2, μ_2 :** The average squared difference from the mean; population or sample formula. Population variance denoted μ_2 or σ^2 ; sample estimate often s^2 .
- **Standard Deviation, SD, s, RMSD:** Square root of variance (RMSD). Population SD is σ , estimate shown as $\hat{\sigma}$ or s .
- **Standard Error of the Mean, SE:** Estimated SD of sample means from the population, reduced by $\sqrt{n}$.
- **Root Mean Squared Error, RMSE:** Standard deviation from known true values x_i^* ; if $\bar{x}_i$ is the mean of sample values, $RMSE = RMSD$.
- **Mean Deviation/Error, MD (ME):** Mean deviation from known true values x_i^* .
- **Mean Absolute Deviation/Error, MAD (MAE):** Mean absolute deviation from true values x_i^* .
- **Covariance, Cov:** Common (co-)variation across two or more datasets; for identical sets, covariance equals variance.
- **Correlation/Product Moment (Pearson's r):** Similarity between paired datasets; ratio of covariance to the product of standard deviations. Value of 1 means perfect correlation.
- **Coefficient of Variation, CV:** Ratio of SD to mean (may be percent). Ratio ≈ 1 with left skew often suggests exponential distribution. Near-zero mean makes CV unstable.
- **Variance Mean Ratio, VMR:** Variance divided by mean (may be percent). Ratio ≈ 1 with unimodal, count data often suggests Poisson; near-zero mean makes it unstable.

Measures of Distribution Shape

- **Skewness, α_3 :** Measures asymmetry of the distribution. Skewness = 0 for symmetric, unimodal distributions; >0 means right-skewed, <0 left-skewed. Based on the third moment about the mean.
- **Kurtosis, α_4 :** Measures peakedness of frequency distribution; high kurtosis implies a sharper peak. Fourth moment about the mean, adjusted so Normal kurtosis = 0 by subtracting 3.

Measures of Complexity and Dimensionality

- **Information Statistic (Shannon's Entropy), I:** Measures amount of disorder or information in $\{x_i\}$, with p_i the proportion in class i . $I \in [0, \log_2(k)]$. Zero means all data in one category; max means even spread.

- **Information Statistic (Diversity), Div:** Shannon's entropy divided by number of classes, k , ranges from 0 to 1.
- **Dimension (Topological), D_T :** Number of intrinsic coordinates to refer to a point. Point = 0, line = 1, surface = 2, solid = 3. In GIS, 2.5D denotes a planar region with an attribute at each point. Topological dimension similar to cover dimension.
- **Dimension (Capacity, Cover, Fractal), D_c :** For an object covered by $N(h)$ elements of edge h : For a line, $N(h) = 1/h$; surface $N(h) = 1/h^2$; volume $N(h) = 1/h^3$. General: $N(h) = 1/h^D$, so $D_c = -\log(N(h))/\log(h)$, possibly fractional (fractal).

Common Distributions

- **Uniform (Continuous):** All values equally likely within range. Mean = $a/2$, variance = $a^2/12$. Density function $f(x)$ for continuous variables.
- **Binomial (Discrete):** Probability of x successes out of n trials, p = probability of success, $q = 1 - p$ (failure). Mean $m = np$, variance npq . Probability function $p(x)$ for discrete variables.
- **Poisson (Discrete):** Approximation to Binomial with small p and large n ($n > 100$), with fixed mean $m = np$. Mean = variance = m .
- **Normal (Continuous):** Describes measurements with independent, random, additive errors. Approximates Binomial with larger p ($\approx 1/2$), large n . If mean μ and SD σ , Normal is $N(\mu, \sigma)$. Normalization: $z = (x - \mu)/\sigma$ gives $N(0,1)$. By Central Limit Theorem, mean of n random variables is Normal.

Data Transforms and Back Transforms

- **Log:** Log transform (base e or natural log) used for unimodal, left-skewed data, often with +1 added. To reverse, use $\exp()$. $z = \ln(x)$ or $z = \ln(x + 1)$; $x = \exp(z)$ or $x = \exp(z) - 1$.
- **Square Root (Freeman-Tukey):** Used to make data more Normal-like. For rate data ($0 \leq x \leq 1$), often uses the Freeman-Tukey combined form.
- **Logit:** Transform for binary data (e.g., survival), range $(-\infty, \infty)$, p is proportion of ones. Inverse shows p in terms of z . Modified for samples $p = 0$ or $p = 1$ by adjusting the fraction. Used to address logistic/S-shaped relationships.
- **Normal, z-transform:** Standardizes distribution to zero mean, unit variance. z_2 distributed as $N(0,1)$ for large n . Divisor is standard error; SD must be non-zero.
- **Box-Cox, Power Transforms:** Family of transforms for positive values, can make data more Normal-like; k is the parameter. Inverse/back-transform provided.
- **Angular Transforms (Freeman-Tukey):** For proportions p , spreads values near ends of the range. k commonly 0.5. FT version averages the transform, stabilizing variance.

Selected Functions

- **Bessel Functions (first kind):** Solutions to certain differential equations; defined by order (subscript). Can be represented by infinite series for non-negative real orders. Used in spatial analysis, especially for directional statistics and splines.
- **Exponential Integral Function, $E_1(x)$:** Definite integral function; used in spline curve fitting.
- **Gamma Function, Γ :** Definite integral widely used: For integer x : $\Gamma(x) = (x - 1)!$, $\Gamma(x/2) = (x/2 - 1)!$, $\Gamma(3/2) = (1/2)!/2 = (\sqrt{\pi})/2$.

Matrix Expressions

- **Identity:** A diagonal matrix with 1s on the diagonal, 0s elsewhere.
- **Determinant:** Only defined for square matrices. For A (size $n \times n$), $\text{Det} = 0$ means matrix is singular (no inverse); near-zero Det means ill-conditioned. Notation: $|A|$ or $\text{Det}(A)$.
- **Inverse:** Matrix equivalent to algebraic division. Matrix A is invertible if $\text{Det} \neq 0$. A symmetric and positive definite if all eigenvalues are positive; positive semi-definite permits some zero eigenvalues. If A is invertible, $AA^{-1} = I$.
- **Transpose:** Rows and columns swapped; elements a_{ij} become a_{ji} for all i, j . The inverse of the transpose equals the transpose of the inverse: $(A^T)^{-1} = (A^{-1})^T$.
- **Symmetric:** $a_{ij} = a_{ji}$ for all i, j ; notation $A = A^T$.
- **Trace:** Sum of the diagonal elements a_{ii} ; also equals sum of eigenvalues. $\text{Tr}(A)$.
- **Eigenvalue, Eigenvector:** For A (size $k \times k$), and vector x , scalar λ satisfies $(A - \lambda I)x = 0$. A has k eigenvalues and corresponding eigenvectors. A can be decomposed as $A = EDE^{-1}$, where E contains eigenvectors and D is diagonal with eigenvalues (diagonalization).

Conceptual Frameworks for Spatial Analysis

Geospatial analysis provides a distinct perspective on the world, a unique lens through which to examine events, patterns, and processes that operate on or near the surface of our planet. It makes sense, then, to introduce the main elements of this perspective, the conceptual framework that provides the background to spatial analysis, as a preliminary to the main body of this Guide's material. This chapter provides that introduction. It is divided into four main sections. The first, [Basic Primitives](#), describes the basic components of this view of the world — the classes of things that a spatial analyst recognizes in the world, and the beginnings of a system of organization of geographic knowledge. The second section, [Spatial Relationships](#), describes some of the structures that are built with these basic components and the relationships between them that interest geographers and others. The third section, [Spatial Statistics](#), introduces the concepts of spatial statistics, including probability, that provide perhaps the most sophisticated elements of the conceptual framework. Finally, the fourth section, [Spatial Data Infrastructure](#), discusses some of the basic components of the data infrastructure that increasingly provides the essential facilities for spatial analysis.

The domain of geospatial analysis is the surface of the Earth, extending upwards in the analysis of topography and the atmosphere, and downwards in the analysis of groundwater and geology. In scale it extends from the most local, when archaeologists record the locations of pieces of pottery to the nearest centimeter or property boundaries are surveyed to the nearest millimeter, to the global, in the analysis of sea surface temperatures or global warming. In time it extends backwards from the present into the analysis of historical population migrations, the discovery of patterns in archaeological sites, or the detailed mapping of the movement of continents, and into the future in attempts to predict the tracks of hurricanes, the melting of the Greenland ice-cap, or the likely growth of urban areas. Methods of spatial analysis are robust and capable of operating over a range of spatial and temporal scales.

Ultimately, geospatial analysis concerns *what* happens *where*, and makes use of geographic information that links features and phenomena on the Earth's surface to their locations. This sounds very simple and straightforward, and it is not so much the basic information as the structures and arguments that can be built on it that provide the richness of spatial analysis. In principle there is no limit to the complexity of spatial analytic techniques that might find some application in the world, and might be used to tease out interesting insights and support practical actions and decisions. In reality, some techniques are simpler, more useful, or more insightful than others, and the contents of this Guide reflect that reality. This chapter is about the underlying concepts that are employed, whether it be in simple, intuitive techniques or in advanced, complex mathematical or computational ones.

Spatial analysis exists at the interface between the human and the computer, and both play important roles. The concepts that humans use to understand, navigate, and exploit the world around them are mirrored in the concepts of spatial analysis. So the discussion that follows will often appear to be following parallel tracks — the track of human intuition on the one hand, with all its vagueness and informality, and the track of the formal, precise world of spatial analysis on the other. The relationship between these two tracks forms one of the recurring themes of this Guide

Basic Primitives

The building blocks for any form of spatial analysis are a set of basic primitives that refer to the place or places of interest, their attributes and their arrangement. These basic primitives are discussed in the following subsections.

Place

At the center of all spatial analysis is the concept of *place*. The Earth's surface comprises some 500,000,000 sq km, so there would be room to pack half a billion industrial sites of 1 sq km each (assuming that nothing else required space, and that the two-thirds of the Earth's surface that is covered by water was as acceptable as the one-third that is land); and 500 trillion sites of 1 sq m each (roughly the space occupied by a sleeping human). People identify with places of various sizes and shapes, from the room to the parcel of land, to the neighborhood, the city, the county, the state or province, or the nation-state. Places may overlap, as when a watershed spans the boundary of two counties, and places may be nested hierarchically, as when counties combine to form a state or province.

Places often have names, and people use these to talk about and distinguish between places. Some names are official, having been recognized by national or state agencies charged with

bringing order to geographic names. In the U.S., for example, the [Board on Geographic Names](#) exists to ensure that all agencies of the federal government use the same name in referring to a place, and to ensure as far as possible that duplicate names are removed from the landscape. A list of officially sanctioned names is termed a *gazetteer*, though that word has come to be used for any list of geographic names.

Places change continually, as people move, climate changes, cities expand, and a myriad of social and physical processes affect virtually every spot on the Earth's surface. For some purposes it is sufficient to treat places as if they were static, especially if the processes that affect them are comparatively slow to operate. It is difficult, for example, to come up with instances of the need to modify maps as continents move and mountains grow or shrink in response to earthquakes and erosion. On the other hand it would be foolish to ignore the rapid changes that occur in the social and economic makeup of cities, or the constant movement that characterizes modern life. Throughout this Guide, it will be important to distinguish between these two cases, and to judge whether time is or is not important.

People associate a vast amount of information with places. Three Mile Island, Sellafield, and Chernobyl are associated with nuclear reactors and accidents, while Tahiti and Waikiki conjure images of (perhaps somewhat faded) tropical paradise. One of the roles of places and their names is to link together what is known in useful ways. So for example the statements "I am going to London next week" and "There's always something going on in London" imply that I will be having an exciting time next week. But while "London" plays a useful role, it is nevertheless vague, since it might refer to the area administered by the Greater London Authority, the area inside the M25 motorway, or something even less precise and determined by the context in which the name is used. Science clearly needs something better if information is to be linked exactly to places, and if places are to be matched, measured, and subjected to the rigors of spatial analysis.

The basis of rigorous and precise definition of place is a *coordinate system*, a set of measurements that allows place to be specified unambiguously and in a way that is meaningful to everyone. The Meridian Convention of 1884 established the Greenwich Observatory in London as the basis of longitude, replacing a confusing multitude of earlier systems. Today, the World Geodetic System (WGS84) of 1984 and subsequent adjustments provide a highly accurate pair of coordinates for every location on the Earth's surface (and incidentally place the line of zero longitude about 100m east of the Greenwich Observatory). Elevation continues to be problematic, however, since countries and even agencies within countries insist on their own definitions of what marks zero elevation, or exactly how to define "sea level". Many other coordinate systems are in use, but most are easily converted to and from latitude/longitude. Today it is possible to measure location directly, using the Global Positioning System (GPS) or its Russian counterpart [GLONASS](#) (and in future its European counterpart [Galileo](#)). Spatial analysis is most often applied in a two-dimensional space. But applications that extend above or below the surface of the Earth must often be handled as three-dimensional. Time sometimes adds a fourth dimension, particularly in studies that examine the dynamic nature of phenomena.

Attributes

Attribute has become the preferred term for any recorded characteristic or property of a place. A place's name is an obvious example of an attribute, but a vast array of other options has proven useful for various purposes. Some are measured, including elevation, temperature, or

rainfall. Others are the result of classification, including soil type, land-use or land cover type, or rock type. Government agencies provide a host of attributes in the form of statistics, for places ranging in size from countries all the way down to neighborhoods and streets. The characteristics that people assign rightly or mistakenly to places, such as “expensive”, “exciting”, “smelly”, or “dangerous” are also examples of attributes. Attributes can be more than simple values or terms, and today it is possible to construct information systems that contain entire collections of images as attributes of hotels, or recordings of birdsong as attributes of natural areas. But while these are certainly feasible, they are beyond the bounds of most techniques of spatial analysis.

Within GIS the term *attribute* usually refers to records in a data table associated with individual features in a vector map or cells in a grid (*raster* or image file). Most GIS packages do not display a separate attribute table for raster data, since each grid cell contains a single data item, which is the value at that point and can be readily examined. [ArcGIS](#) is somewhat unusual in that it provides an attribute table for raster data

Rows in this raster attribute table provide a count of the number of grid cells (pixels) in the raster that have a given value, e.g. 144 cells have a value of 453 meters. Furthermore, the linking between the attribute table visualization and mapped data enables all cells with elevation=453 to be selected and highlighted on the map. Many terms have been adopted to describe attributes. From the perspective of spatial analysis the most useful divides attributes into scales or levels of measurement, as follows:

- Nominal.** An attribute is nominal if it successfully distinguishes between locations, but without any implied ranking or potential for arithmetic. For example, a telephone number can be a useful attribute of a place, but the number itself generally has no numeric meaning. It would make no sense to add or divide telephone numbers, and there is no sense in which the number 9680244 is more or better than the number 8938049. Likewise, assigning arbitrary numerical values to classes of land type, e.g. 1=arable, 2=woodland, 3=marsh, 4=other is simply a convenient form of naming (the values are nominal). SITENAME in Figure 2-1A is an example of a nominal attribute, as is OBJECTID, even though both happen to be numeric

- Ordinal.** An attribute is ordinal if it implies a ranking, in the sense that Class 1 may be better than Class 2, but as with nominal attributes no arithmetic operations make sense, and there is no implication that Class 3 is worse than Class 2 by the precise amount by which Class 2 is worse than Class 1. An example of an ordinal scale might be preferred locations for residences — an individual may prefer some areas of a city to others, but such differences between areas may be barely noticeable or quite profound. Note that although OBJECTID in Figure 2-1A appears to be an ordinal variable it is not, because the IDs are provided as unique names only, and could equally well be in any order and use any values that provided uniqueness (and typically, in this example, are required to be integers)

- Interval.** The remaining three types of attributes are all quantitative, representing various types of measurements. Attributes are interval if differences make sense, as they do for example with measurements of temperature on the Celsius or Fahrenheit scales, or for measurements of elevation above sea level

- Ratio.** Attributes are ratio if it makes sense to divide one measurement by another. For example, it makes sense to say that one person weighs twice as much as another person, but it

makes no sense to say that a temperature of 20 Celsius is twice as warm as a temperature of 10 Celsius, because while weight has an absolute zero Celsius temperature does not (but on an absolute scale of temperature, such as the Kelvin scale, 200 degrees can indeed be said to be twice as warm as 100 degrees). It follows that negative values cannot exist on a ratio scale. HA_BURNED and ACRES_BURN in Figure 2-1A are examples of ratio attributes. Note that only one of these two attribute columns is required, since they are simple multiples of one another

•**Cyclic.** Finally, it is not uncommon to encounter measurements of attributes that represent directions or cyclic phenomena, and to encounter the awkward property that two distinct points on the scale can be equal — for example, 0 and 360 degrees are equal. Directional data are cyclic (Figure 2-2), as are calendar dates. Arithmetic operations are problematic with cyclic data, and special techniques are needed, such as the techniques used to overcome the Y2K problem, when the year after (19)99 was (20)00. For example, it makes no sense to average 1 degree and 359 degrees to get 180 degrees, since the average of two directions close to north clearly is not south. Mardia and Jupp (1999) provide a comprehensive review of the analysis of directional or cyclic data (see further, Section 4.5.1, [Directional analysis of linear datasets](#))

While this terminology of measurement types is standard, spatial analysts find that another distinction is particularly important. This is the distinction between attributes that are termed *spatially intensive* and *spatially extensive*. Spatially extensive attributes include total population, measures of a place's area or perimeter length, and total income — they are true *only of the place as a whole*. Spatially intensive attributes include population density, average income, and percent unemployed, and if the place is homogeneous they will be true of any part of the place as well as of the whole. For many purposes it is necessary to keep spatially intensive and spatially extensive attributes apart, because they respond very differently when places are merged or split, and when many types of spatial analysis are conducted.

Since attributes are essentially measured or computed data items associated with a given location or set of locations, they are subject to the same issues as any conventional dataset: sampling error; measurement errors and limitations; mistakes and miscalculations; missing values; temporal and thematic errors and similar issues. Metadata accompanying spatial datasets should assist in assessing the quality of such attribute data, but at least the same level of caution should be applied to spatial attribute data as with any other form of data that one might wish to use or analyze.

Objects

The places discussed in Section 2.1.1, [Place](#), vary enormously in size and shape. Weather observations are obtained from stations that may occupy only a few square meters of the Earth's surface (from instruments that occupy only a small fraction of the station's area), whereas statistics published for Russia are based on a land area of more than 17 million sq km. In spatial analysis it is customary to refer to places as *objects*. In studies of roads or rivers the objects of interest are long and thin, and will often be represented as lines of zero width. In studies of climate the objects of interest may be weather stations of minimal extent, and will often be represented as points. On the other hand many studies of social or economic patterns may need to consider the two-dimensional extent of places, which will therefore be represented as areas, and in some studies where elevations or depths are important it may be appropriate to represent

places as volumes. To a spatial statistician, these points, lines, areas, or volumes are known as the attributes' *spatial support*.

Each of these four classes of objects has its own techniques of representation in digital systems. The software for capturing and storing spatial data, analyzing and visualizing them, and reporting the results of analysis must recognize and handle each of these classes. But digital systems must ultimately represent everything in a language of just two characters, 0 and 1 or “off” and “on”, and special techniques are required to represent complex objects in this way. In practice, points, lines, and areas are most often represented in the following standard forms:

- **Points** as pairs of coordinates, in latitude/longitude or some other standard system
- **Lines** as ordered sequences of points connected by straight lines
- **Areas** as ordered rings of points, also connected by straight lines to form polygons. In some cases areas may contain holes, and may include separate islands, such as in representing the State of Michigan with its separate Upper Peninsula, or the State of Georgia with its offshore islands. This use of polygons to represent areas is so pervasive that many spatial analysts refer to all areas as polygons, whether or not their edges are actually straight

Lines represented in this way are often termed *polylines*, by analogy to polygons (see [Table 1-1](#) for a more formal definition). Three-dimensional volumes are represented in several different ways, and as yet no one method has become widely adopted as a standard. The related term *edge* is used in several ways within GIS. These include: to denote the border of polygonal regions; to identify the individual links connecting nodes or vertices in a network; and as a general term relating to the distinct or indistinct boundary of areas or zones. In many parts of spatial analysis the related term, *edge effect* is applied. This refers to possible bias in the analysis which arises specifically due to proximity of features to one or more edges. For example, in point pattern analysis computation of distances to the nearest neighboring point, or calculation of the density of points per unit area, may both be subject to edge effects.

Figure 2-3, below, shows a simple example of points, lines, and areas, as represented in a typical map display. The hospital, boat ramp, and swimming area will be stored in the database as points with associated attributes, and symbolized for display. The roads will be stored as polylines, and the road type symbols (U.S. Highway, Interstate Highway) generated from the attributes when each object is displayed. The lake will be stored as two polygons with appropriate attributes. Note how the lake consists of two geometrically disconnected pieces, linked in the database to a single set of attributes — objects in a GIS may consist of multiple parts, as long as each part is of the same type.

It can be expensive and time-consuming to create the polygon representations of complex area objects, and so analysts often resort to simpler approaches, such as choosing a single *representative point*. But while this may be satisfactory for some purposes, there are obvious problems with representing the entirety of a large country such as Russia as a single point. For example, the distance from Canada to the U.S. computed between representative points in this way would be very misleading, given that they share a very long common boundary.

Maps

Historically, maps have been the primary means to store and communicate spatial data. Objects and their attributes can be readily depicted, and the human eye can quickly discern patterns and anomalies in a well-designed map. Points can be shown as symbols of various kinds, depicting anything from a windmill to a church; lines can be symbolized to distinguish between major roads, minor roads, and rivers; and areas can be symbolized with color, shading, or annotation.

Maps have traditionally existed on paper, as individual sheets or bound into atlases (a term that originated with Mercator, who produced one of the first atlases in the late 16th century). The advent of digital computers has broadened the concept of a map substantially, however. Maps can now take the form of images displayed on the screens of computers, mobile phones or in-vehicle navigation systems. They can be dynamic, showing the Earth spinning on its axis, tracking the movement of migrating birds, ships or aircraft. Their designs can now go far beyond what was traditionally possible when maps had to be drawn by hand, incorporating a far greater range of color and texture, and even integrating audio and video elements.

Multiple properties of places

Dividing objects into points, lines, areas, and volumes is only the crudest of the many ways in which humans classify the things they find surrounding them in their environment. A topographic map may show areas of woodland, rivers, roads, contours, windmills, and several hundred other kinds of objects, each of them classifiable as points, lines, areas, or volumes. Objects may overlap — a river may pass through a wood, a road may cross a river, and all three may lie inside the area shown as “Montara, California”. Spatial analysts think of multiple classes of objects as forming *layers* of information superimposed on each other. A study might be conducted on just one layer, analyzing the patterns formed by the objects in that layer and looking for clues as to the processes that created the pattern. In other cases the patterns of one layer might be analyzed in relation to other layers. For example, the pattern of cases of cancer in an area might be studied to see if it bears any relationship to the pattern of roads or electrical transmission lines in the area, or to the chemistry of drinking water.

Sometimes many different attributes are collected and published for the same sets of places. Census agencies do this when they compile hundreds of attributes for census tracts, counties, or states. In such cases comparing layers is a simple matter of comparing the different attributes assigned to the same places. But more generally the objects in two layers may not be the same, creating difficulties for analysis. For example, suppose a study is being conducted of people’s perception of noise in a neighborhood near an airport. Monitoring devices have been positioned at points around the neighborhood, providing a layer of noise measurements. Residents have been interviewed, providing another layer of perceptual data. The analysis now moves to a comparison of the contents of the two layers, but it is hampered by the fact that the objects in the two layers are in different locations.

The idea that the Earth’s surface is a space littered with various kinds of objects certainly matches the ways humans tend to think about the world around them. Scenes perceived by the eye are immediately parsed by the brain into “tree”, “car”, “house”, and other discrete objects. But in other cases the world may appear more continuous than discrete. Ground elevation (or ocean depth), for example, is in broad terms a continuously varying property that can be measured anywhere on the Earth’s surface, as is atmospheric temperature, or, within an

urbanized area, average level of noise or pollution (Figure 2-5, below — each pixel of this image indicates the measured or modeled level of traffic-related noise in decibels – reds are the highest levels, gray indicates buildings).

In essence there are two ways of thinking about phenomena on the Earth's surface. In the first, the *discrete-object* view, reality is like an empty table-top littered with discrete, countable objects that can be assigned to different classes. In the second, the *continuous-field* view, reality is a collection of continuous surfaces, each representing the variation of one property, such as elevation, over the Earth's surface. When it is necessary to differentiate by height, the field becomes three-dimensional rather than two-dimensional, and time may add a fourth dimension. Mathematically, a field is a continuous function mapping every location to the value of some property of interest.

Fields

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The distinction between continuous fields and discrete objects is merely conceptual, and there is no test that can be applied to the contents of a database to determine which view is operative. But it is a very important distinction that underlies the choices that analysts make in selecting appropriate techniques. Points, polylines, and polygons arise in both cases, and yet their meanings are very different. For example, points might represent the locations where a continuous field of temperature is measured, or they might represent isolated instances of a disease — in the latter case, which falls within the discrete-object view, there is no implication that anything happens in between the cases, whereas in the former case one would expect temperature to vary smoothly between measurement sites. Polylines might represent the connected pieces of a stream, or they might represent the contours of an elevation surface — in the latter case, the polylines would collectively represent a field, and would consequently not be allowed to cross each other. If filled contours rather than separate contour lines are generated then polygons rather than polylines define the field values

Networks

Mention has already been made of the lines that often constitute the objects of spatial data. They include streets, roads, railroads, and highways; rivers, canals, and streams; and the more

abstract interpersonal networks studied by sociologists. Networks constitute one-dimensional structures embedded in two or three dimensions. Discrete point objects may be distributed on the network, representing such phenomena as landmarks or bridges on road networks, or observation points on rivers. Discrete line objects may also be found, in the form of tunnels, stretches of highway with a constant number of lanes, or river reaches. Continuous fields may also be distributed over the one-dimensional network, defining such variables as travel speed, railroad gradient, traffic density, or stream velocity. Mathematically, a network forms a *graph*, and many techniques developed for graphs have applications to networks. These include various ways of measuring a network's connectivity, or of finding shortest paths between pairs of points on a network. These methods are explored in detail in Chapter 7, [Network and Location Analysis](#).

Density estimation

One of the more useful concepts in spatial analysis is *density* — the density of humans in a crowded city, or the density of tracks across a stretch of desert, or the density of retail stores in a shopping center. Density provides an effective link between the discrete-object and continuous-field conceptualizations, since density expresses the number of discrete objects per unit of area, and is itself a continuous field. Mathematically, the density of some kind of object is calculated by counting the number of such objects in an area, and dividing by the size of the area. This is easily done when calculating population density from census data, for example, by dividing the population of each census tract or county by its physical area. But this leads to multiple values of density, depending on the objects selected and the definition of the areas used to make the calculation. Techniques of *density estimation* try to avoid this problem and are discussed in detail in Section 4.3.4, [Density, kernels and occupancy](#). The book by Silverman (1998) provides additional background reading on this issue.

Detail, resolution, and scale

The surface of the planet is almost infinitely complex, revealing more and more detail the closer one looks. An apparently simple concept such as a coastline turns out to be problematic in practice — should its representation as a polyline include every indentation, every protruding rock, even every grain of sand? It is clearly impossible to build a representation that includes every detail, and in reality decisions must be made about the amount of detail to include. *Spatial resolution* is the term given to a threshold distance below which the analyst has decided that detail is unnecessary or irrelevant. For example, one might decide to represent a coastline at a spatial resolution of 100m, meaning that all crenulations, indentations, and protuberances that are no wider than 100m are omitted from the representation. In other words, the representation of the phenomenon appears simpler and smoother than it really is.

Cartographers use a somewhat different approach to the omission of detail from maps. A map is said to have a scale or representative fraction, being the ratio of distances on the map to their corresponding distances in the real world. For example, if a map shows Los Angeles and Santa Barbara as 10cm apart, and in reality they are 100km apart as the crow flies, then the map's scale is 1:1,000,000 (note that in principle, because of projection from a curved to a flat surface, no map's scale can be exactly uniform although this clearly does not apply to globes). This approach to defining level of detail is problematic, however, if the dataset in question is not on paper. Data that are born digital do not have a representative fraction, and the representative fraction for data displayed on a computer screen depends on the dimensions of the screen.

Project onto a larger screen and the representative fraction will change, irrespective of the value shown on the screen. So it is better to use spatial resolution to describe the level of detail of digital data. As a general rule of thumb the spatial resolution of a paper map is roughly 0.5mm at the scale of the map (500m on a 1:1,000,000 map, or 5m on a 1:10,000 map), a figure that takes into account the widths of lines on maps and the cartographic practice of smoothing what is perceived as excessive detail.

When digital map data are displayed decisions have to be made, manually and/or automatically, as to how much detail should be displayed, and how feature data should be *generalized* in order to produce a visually acceptable and meaningful result. This has been the subject of much research, some of which continues to the present day. For a notable book (of collected papers) on this subject, see Mackaness *et al.* (2007). In a white paper produced by [ESRI \(1996\)](#) they summarized the major generalization operators that they planned to implement at that time, as follows:

- Preselection:** deciding on which features to include and which to exclude (e.g. major roads only, ignore footpaths etc.)
- Elimination:** removal of features that are too small/short (below some threshold)
- Simplification:** simplifying (smoothing, straightening) linear or polygon boundaries whilst retaining their basic form (see also, Section 4.2.4, [Line Smoothing and point-weeding](#))
- Aggregation:** combining distinct features into a larger composite object (e.g. a cluster of houses into a polygonal ‘urbanized area’)
- Collapse:** reduction of feature dimensionality, as for example when a town is represented by a point or a broad river by a line
- Typification:** reduction of the level of detail by replacing multiple objects by a smaller number of the same objects occupying broadly the same locations
- Exaggeration:** emphasizing important features that might otherwise be removed
- Classification and symbolization:** grouping together similar features and using different symbology to represent the new arrangement
- Conflict resolution (displacement):** identifying and resolving feature conflicts (e.g. overlaps, labeling), and
- Refinement:** altering feature geometry and alignment to improve aesthetics

Topology

In mathematics, a property is said to be *topological* if it survives stretching and distorting of space. Draw a circle on a rubber sheet, and no amount of stretching or distorting of the sheet will make it into a line or a point. Draw two touching circles, and no amount of stretching or distorting will pull them apart or make them overlap. It turns out that many properties of importance to spatial analysis are topological, including:

- Dimensionality**: the distinction between point, line, area, and volume, which are said to have topological dimensions of 0, 1, 2, and 3 respectively
- Adjacency**: including the touching of land parcels, counties, and nation-states
- Connectivity**: including junctions between streets, roads, railroads, and rivers
- Containment**: when a point lies inside rather than outside an area

Many phenomena are subject to topological constraints: for example, two counties cannot overlap, two contours cannot cross, and the boundary of an area cannot cross itself. Topology is an important concept therefore in constructing and editing spatial databases. Figure 2-7, below, illustrates the concept of topology, using the example of two areas that share a common boundary. While stretching of the space can change the shapes of the areas, they will remain in contact (retain their adjacency) however much stretching is applied.

Spatial Relationships

Co-location

Many useful insights can come from comparing the attributes of co-located places. For example, a cluster of cases of lung cancer in a port city might be explained by knowing that asbestos was widely used in the construction of ships in that city in the Second World War. There is also merit in examining occurrences of co-location in point datasets — in many instances the same coordinates are assigned to multiple points in a sample, which only becomes apparent when the attribute table is examined since mapping will obscure the existence of multiple points at the same location. In Section 2.1.5 a distinction was made between cases where multiple attributes are compared for the same set of objects, and cases where the attributes to be compared are attached to distinct, overlapping objects. In the latter case analysis will require the use of special techniques, such as those discussed in Section 2.2.8. Analysts often use the term *overlay* to refer to the superimposition and analysis of layers of geographic data about the same place. Overlay may require the superimposition of area on area, or line on area, or point on area, or line on line, or point on line, depending on the nature of the objects in the database.

Distance, direction and spatial weights matrices

Knowledge of location also allows the analyst to determine the distance and direction between objects. Distance between points is easily calculated using formulas for straight-line distance on the plane or on the curved surface of the Earth, and with a little more effort it is possible to determine the actual distance that would be traveled through the road or street network, and even to predict the time that it would take to make the journey. Distance and direction between lines or areas are often calculated by using representative points, but this can be misleading (as noted in Section 2.1.3, [Objects](#)). They can also be calculated using the nearest pair of points in the two objects, or as the average distance and direction between all pairs of points (note the difficulties of averaging direction, however, as mentioned in Section 2.1.2, [Attributes](#)). Note that whilst for many applications, Euclidean distance is a suitable metric, for larger distances spherical, geoidal or other metrics may be more appropriate - see further, [Distance Operations](#).

Many types of spatial analysis require the calculation of a table or matrix expressing the relative proximity of pairs of places, often denoted by **W** (a *spatial weights matrix*). Proximity can be

a powerful explanatory factor in accounting for variation in a host of phenomena, including flows of migrants, intensity of social interaction, or the speed of diffusion of an epidemic. Software packages will commonly provide several alternative ways of determining the elements of **W**, including:

- 1 if the places share a common boundary, else 0
- the length of any common boundary between the places, else 0
- a decreasing function of the distance between the places, or between their representative points, or their k^{th} nearest neighbors

In Figure 2-8, below, we show a section of the map of counties in North Carolina, USA. Two of the 100 counties have been highlighted, counties 1 and 3. An extract from a simple binary spatial weights matrix generated by the [GeoDa](#) software package based on ‘Queen’s move adjacency’ (by analogy with the game of chess) is shown below for each of these counties (represented in this instance as a text list for compactness, rather than a sparse binary matrix):

```
1 3
19 18 2
3 5
25 23 18 10 2
```

Here, the first line indicates that county 1 (the most westerly of the two highlighted) has 3 neighbors. Line two shows that their IDs are 19 18 and 2. County 3 is shown as having 5 neighbors, 4 of which share a boundary with it (Rook’s move) and one of which (25) appears to share a corner (Bishop’s move: Rook’s move plus Bishop’s move equal Queen’s move) but in fact shares a very short boundary. Instead of pure adjacency a set of spatial weights could have been computed based on the Euclidean distance separating the centroids of the various counties. In this case the matching entries for the weights matrix would have the form:

```
1 19 0.3290
1 2 0.3138
1 18 0.2535
3 18 0.4985
3 2 0.4077
3 23 0.1738
3 25 0.4475
3 40 0.5202
```

Here the pairwise connections are listed with the distances between the centroids, and entries may include counties that are not adjacent but perhaps lie within some threshold distance (n.b.

distances computed by [GeoDa](#) in this example are not in meaningful units, i.e. *not* in miles; *arc distance* must be specified to produce distances in miles).

In such analyses it will be **W** that captures the spatial aspects of the problem, and the actual coordinates become irrelevant once the matrix is calculated. Note that under Euclidean measure **W** will be invariant with respect to displacement (translation), rotation, and mirror imaging (reflection). Also note that edge effects may be significant – North Carolina is not an isolated State but has the counties of the State of Virginia to its north. Thus there are counties adjacent to those we have highlighted that have been excluded from our analysis purely because of the way we have selected our study area. Likewise, adjacency models based on immediate neighbors (e.g. rook's move) have a substantial effect on resulting computations, for example on computed correlation coefficients (see further Dubin, 2008, for a fuller discussion of these effects).

Multidimensional scaling

Multidimensional scaling (MDS) is the general term given to the problem of reconstructing locations from knowledge of proximities. Tobler and Wineberg (1971) provided an excellent example in which the unknown locations of 33 ancient pre-Hittite settlements in Cappadocia were inferred from knowledge of their interactions with other settlements whose locations were known, based on the assumption that interaction declined systematically with distance (see further, Section 4.4.5, [Distance decay models](#)). Scaling techniques have been used to create a wide range of specialized maps, including maps on which distances reflect travel times rather than actual distances, or the similarity of the species found in each place, or the perceptions of relative proximities in the minds of local residents (Golledge and Rushton, 1972).

Spatial Context

Much useful insight can often be gained by comparing the attributes of objects with those of other objects in close proximity. The behavior of a person on a crowded street might be explained in terms of the proximity of other people; the price of a house might be due in part to the existence of expensive homes in the immediate vicinity; and an area might find its homes losing value because of proximity to a polluting industrial plant. Location establishes context, by allowing distances between objects to be determined, and by providing information on their relevant attributes.

Neighborhood

People think of themselves as living in neighborhoods, or places that are sufficiently close to be experienced on a day-to-day basis. Very often neighborhood is the basis of spatial context, characterizing the nature of a person's surroundings. Neighborhoods are often conceived as partitioning an urban space, such that every point lies in exactly one neighborhood, but this may conflict with individual perceptions of neighborhood, and by the expectation that neighborhood extends in all directions around every individual's location. Figure 2-9, below, shows three examples of possible neighborhood definitions.

In Figure 2-9A the neighborhood is defined as a circle centered on the house, extending equally in all directions. In Figure 2-9B neighborhood is equated with an existing zone, such as a census tract or precinct, reflecting the common strategy of using existing aggregated data to characterize a household's surroundings. In Figure 2-9C weights are applied to surroundings

based on distance, allowing neighborhood to be defined as a *convolution* (see further, Section 2.2.10, [Smoothing and sharpening](#)) with weight decreasing as a simple function of distance.

In many geospatial analysis applications the term neighborhood is used to refer to the set of zones or cells that are immediately adjacent to a selected zone or cell. For example, a raster model is generally defined using a regular grid of square cells. The set of 8 cells that surround each cell (omitting the grid's edge cells) is often referred to as the 'Moore' neighborhood (see further, Section 8.2.1, [Cellular automata \(CA\)](#)) or 'Queen's move' neighborhood (by analogy with the game of chess). Likewise the subset that consists of just the direct North, South, East and West neighbors is sometime called the 'von Neumann' or 'Rook's move' neighborhood. Note that whilst such neighborhoods apply to regular grids, irregular lattices (e.g. administrative areas) also have neighborhood structures of that may be described in this or a similar manner.

Spatial heterogeneity

The Earth's surface displays almost incredible variety, from the landscapes of the Tibetan plateau to the deserts of Australia and the urban complexity of London or Tokyo. Nowhere can be reasonably described as an average place and it is difficult to imagine any subset of the Earth's surface being a representative sample of the whole. The results of any analysis over a limited area can be expected to change as that limited area is relocated, and to be different from the results that would be obtained for the surface of the Earth as a whole. These concepts are collectively described as spatial heterogeneity, and they tend to affect almost any kind of spatial analysis conducted on geographic data. Many techniques such as Geographically Weighted Regression (Fotheringham, Brunsdon, and Charlton, 2002, discussed in Section 5.6.3, [Geographically Weighted Regression \(GWR\)](#), of this Guide) take spatial heterogeneity as given — as a universally observed property of the Earth's surface — and focus on providing results that are specific to each area, and can be used as evidence in support of local policies. Such techniques are often termed *place-based* or *local*.

Spatial dependence

Notwithstanding the comments in the previous section about spatial heterogeneity, anyone examining the Earth's surface in detail would be struck by how conditions tend to persist locally, and how it is possible to divide the surface into regions that exhibit substantial internal similarity. For example, the desert regions are characterized by lack of rainfall, the tropical rainforests by abundant rainfall and dense vegetation, and the Polar Regions by extreme cold. Conditions at nearby points are not sampled independently and randomly from global distributions, but instead show remarkable levels of interdependence. Of course there are exceptions, where conditions change very rapidly over short distances, for example between the plains of India and the high Himalayas, or between coastal plains and adjacent ocean.

The general term for this phenomenon is spatial dependence. Its pervasiveness was aptly captured by Tobler in what has become known as his First Law of Geography: "All things are related, but nearby things are more related than distant things". The magnitude of the effect can be measured using a number of statistics of spatial autocorrelation (Cliff and Ord, 1981, Goodchild [CATMOG](#) 47). It also underlies the discipline known as geostatistics, which describes spatial variation in terms of a function (known as a correlogram) that shows how spatial autocorrelation decreases with increasing distance (for a general overview of

geostatistics see Section 6.7.1, [Core concepts in Geostatistics](#), and the book by Isaaks and Srivastava, 1989). This correlogram reaches zero, or independence, at a distance known as the range. The implications of Tobler's observation are profound. If it were not true, the full range of conditions anywhere on the Earth's surface could in principle be found packed within any small area. There would be no regions of approximately homogeneous conditions to be described by giving attributes to area objects. Topographic surfaces would vary chaotically, with slopes that were everywhere infinite, and the contours of such surfaces would be infinitely dense and contorted. Spatial analysis, and indeed life itself, would be impossible.

Spatial sampling

One of the implications of Tobler's First Law and spatial dependence is that it is possible to capture a reasonably accurate description of the Earth's surface with a few well-placed samples. Meteorologists capture weather patterns by sampling at measuring stations, knowing that conditions between these sample locations will vary systematically and smoothly. Geographers describe the contents of entire regions with a few well-chosen statements, provided that the regions have been defined to enclose approximately uniform conditions. Census data are published in spatially aggregated form to protect the confidentiality of individual returns, and Tobler's First Law ensures that the variance within each aggregate unit is not so large as to render the aggregated data meaningless.

Numerous approaches to spatial sampling exist, depending on the nature of the questions being asked. Perhaps the simplest occurs when sample observations are made in order to obtain a best estimate of average conditions within an area. For example, to show that average temperatures across the globe are increasing due to global warming, one would want to ensure that every location on the globe had an equal chance of being sampled (in reality weather stations are distributed far from randomly, often co-located with airports, and more common on land and in heavily populated areas). In other cases it might be more important to characterize various regions, by placing an equal number of sample points in each region, or to characterize each of a number of vegetation or soil types. If the objective is to characterize the pattern of spatial dependence in an area, then it may be more important to place samples so as to obtain equally reliable estimates of similarity in every range of distances. A discussion of spatial sampling can be found in the book by [Longley et al.](#) (2015, Ch2.4) and in Section 5.1.2, [Spatial sampling](#), of this Guide.

Spatial interpolation

If spatial sampling is an efficient way of capturing knowledge of spatial variation, then there must be reliable ways of filling in the unknown variation between sample points. Spatial interpolation attempts to do this, by providing a method of estimating the value of a field anywhere from a limited number of sample points. Spatial interpolation is no more and no less than intelligent guesswork, supported by Tobler's First Law. For example, in Figure 2-10, below, we may wish to estimate the value at location (5,5) utilizing knowledge of the values of nearby locations.

In this example a total of 5 nearby points have been selected and the value at location (5,5) will be estimated as some combination of the known values at other locations, for example as a weighted average of these values. Typically weights are chosen to be larger for nearby values (e.g. point 4, with value 4 and point 5 with value 6) than more distant sites, such as location 2

with value 4. For example, weights could be chosen as a simple inverse function of the distance between the point to be evaluated and the measured locations – the further away a known location is the less its contribution (or weight) will be to the estimated value at (5,5). For the example shown this yields an estimated value of 4.18 at location (5,5), with point 4 contributing 44% to this estimate, whilst point 3 contributes only 8%.

Many methods of spatial interpolation exist, all of them based to some extent on the principle that conditions vary smoothly over the Earth's surface. Methods have even been devised to cope with exceptions to Tobler's First Law, by recognizing the discontinuities that exist in some geographic phenomena, such as faults in geologic structures, or the barriers to human interaction created by international boundaries.

Arguably the most rigorous approach to interpolation is Kriging, a family of techniques based on the theory of geostatistics (Isaaks and Srivastava, 1989). Kriging and a wide range of other interpolation methods are described in Chapter 6 ([Surface and Field Analysis](#)) of this Guide.

Smoothing and sharpening

The Earth's surface is very dynamic, particularly in those aspects that fall in the social domain. People move, houses are constructed, areas are devastated by war, and severe storms modify coastlines. Patterns of phenomena on the Earth's surface reflect these processes, and each process tends to leave a distinctive footprint. Interpreting those footprints in order to gain insight into process is one of the most important motivations for spatial analysis.

The pervasiveness of Tobler's First Law suggests that processes that leave a smooth pattern on the Earth's surface are generally more prevalent than those that cause sharp discontinuities, but spatial analysts have developed many ideas in both categories. Many processes involve some form of *convolution*, when the outcome at a point is determined by the conditions in some immediate neighborhood. For example, many social outcomes are responses to neighborhood conditions — individual obesity may result in part from a physical design of the neighborhood that discourages walking, alcoholism may be due in part to the availability of alcohol in the neighborhood, and asthma incidence rates may be raised due to pollution caused by high local density of traffic. Mathematically, a convolution is a weighted average of a point's neighborhood, the weights decreasing with distance from the point, and bears a strong technical relationship to density estimation. As long as the weights are positive, the resulting pattern will be smoother than the inputs. The blurring of an out-of-focus image is a form of convolution.

First- and second-order processes

In the analysis of point patterns, a *first-order process* is one that produces a variation in point density in response to some causal variable. For example, the density of cases of malaria echoes the density of particular species of mosquito, which in turn reflects the abundance of warm stagnant water for breeding. Because both mosquitoes and people move around, the pattern of malaria density can be expected to be smoother than the patterns of either mosquito density or stagnant-water abundance.

A *second-order process* results from interactions, when the presence of one point makes others more likely in the immediate vicinity. Patterns of contagious disease reflect second-order processes, when the disease is passed from an initial carrier to family members, co-workers, sexual partners, and others who come into contact with infectious carriers. The result will be a

pattern that clusters around carriers (Figure 2-11B and D, below). Both first- and second-order processes produce clustering and smooth variations in density, but the mechanisms involved and the relationships to other variables are very different. Competition for space provides a familiar example of a form of second-order process that results in an exception to Tobler's First Law. The presence of a shopping center in an area generally discourages other shopping centers from locating nearby, particularly when the focus is on convenience goods and services. It is unusual, for example, to find two supermarkets next to each other, or two outlets of the same fast-food chain. These are examples of *dispersed* rather than clustered point patterns (Figure 2-11C), in which the presence of one point makes others less rather than more likely in the immediate vicinity.

Spatial Statistics

The term *spatial statistics* refers to the application of statistical concepts and methods to data that has an explicit spatial structure that is important to understanding those data. Where data are available, with associated spatial information, and the latter adds nothing to the understanding of the data, then analysis can proceed as if the data had no spatial context — i.e. as if the data was *aspatial*. Spatial statistics is a discipline in its own right, closely associated with traditional statistical methods and more modern computational statistics, but with the added complexity of spatial dependence — typically spatial data samples are not independent as is usually required in traditional statistical methods. Key aspects of spatial statistics are briefly discussed in the sections that follow, with more detailed specific procedures being covered in [Chapter 5: Data exploration and spatial statistics](#).

Spatial probability

Humans will never have a complete understanding of everything that happens on the Earth's surface, and so it is often convenient to resort to thinking in terms of probabilities. In principle one could completely characterize the physics of a human hand and a coin, but in practice it is much more productive to assign probabilities to the outcomes of a coin toss. In similar fashion spatial analysts may avoid the virtual impossibility of predicting exactly where landslides will occur by assigning them probabilities based on patterns of known causes, such as clay soils, rainfall, and earthquakes. A map of probabilities assigns each location a value between 0 and 1, forming a probability field.

Such a map considers only the probability of a single, isolated event, however. The probability that two points a short distance apart will both be subject to landslide is not simply the product of the two probabilities, as it would be if the two outcomes were independent, a conclusion that can be seen as another manifestation of Tobler's First Law. For example, if the probability of a landslide at Point A is $\frac{1}{2}$ and at Point B a short distance away is also $\frac{1}{2}$, the probability that both will be affected is more than $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$, and possibly even as much as $\frac{1}{2}$. Technically, the *marginal* probabilities of isolated events may not be as useful as the *joint* probabilities of related events — and joint probabilities are properties of pairs of points and thus impossible to display in map form unless the number of points is very small.

Probability density

One of the most useful applications of probability to the Earth's surface concerns uncertainty about location. Suppose the location of a point has been measured using GPS, but inevitably the measurements are subject to uncertainty, in this case amounting to an average error of 5m

in both the east-west and north-south directions. Standard methods exist for analyzing measurement error, based on the assumption, well justified by theory, that errors in a measurement form a bell curve or [Gaussian distribution](#). Spatially, one can think of the east-west and north-south bell curves as combining to form a bell. But the surface formed by the bell is not a surface of probability in the sense of Section 2.3.1, [Spatial probability](#) — it does not vary between 0 and 1, and it does not give the marginal probability of the presence of the point. Instead, the bell is a surface of *probability density*, and the probability that the point lies within any defined area is equal to the volume of the bell's surface over that area. The volume of the entire bell is exactly 1, reflecting the fact that the point is certain to lie somewhere.

It is easy to confuse probability density with spatial probability, since both are fields. But they have very different purposes and contexts. Probability density is most often encountered in analyses of positional uncertainty, including uncertainty over the locations of points and lines.

Uncertainty

Any geographic dataset is only a representation of reality, and it inevitably leaves its user with uncertainty about the nature of the real world that is being represented. This uncertainty may concern positions, as discussed in the Section 2.3.2, [Probability density](#), but it may also concern attributes, and even topological relationships. Uncertainty in data will propagate into uncertainty about conclusions derived from data. For example, uncertainty in positions will cause uncertainty in distances computed from those positions, in the elements of a **W** matrix, and in the results of analyses based on that matrix.

Uncertainty can be due to the inaccuracy or limitations of measuring instruments, since an instrument that measures a property to limited accuracy leaves its user uncertain about the true value of the property. It can be due to vagueness in definitions, when land is assigned to classes that are not rigorously defined, so that different observers may classify the same land differently. Uncertainty can also be due to missing or inadequate documentation, when the user is left to guess as to the meaning or definitions of certain aspects of the data. Clearly it is important to spatial analysts to know about the uncertainties present in data, and to investigate how those uncertainties impact the results of analysis. A range of techniques have been developed, and there is a rich literature on uncertainty in spatial data and its impacts (see further, Zhang and Goodchild, 2002 and [Longley et al.](#) (2015, Ch5).

Statistical inference

One of the most important tools of science is statistical inference, the practice of reasoning from the analysis of samples to conclusions about the larger populations from which the samples were drawn. Entire courses are devoted to the topic, and to its detailed techniques — the *t*, *F*, and [Chi-Squared tests](#), linear modeling, and many more. Today it is generally accepted that any result obtained from an experiment, through the analysis of a sample of measurements or responses to a survey, will be subjected to a *significance test* to determine what conclusions can reasonably be drawn about the larger world that is represented by the measurements or responses.

The earliest developments in statistical inference were made in the context of controlled experiments. For example, much was learned about agriculture by sowing test plots with new kinds of seeds, or submitting test plots to specific treatments. In such experiments it is reasonable to assume that every sample was created through a random process, and that the

samples collectively represent what might have happened had the sample been much larger — in other words, what might be expected to happen under similar conditions in the universe of all similar experiments, whether conducted by the experimenter or by a farmer. Because there is variation in the experiment, it is important to know whether the variation observed in the sample is sufficiently large to reach conclusions about variation in the universe as a whole. Figure 2-12, below, illustrates this process of statistical inference. A sample is drawn from the population by a random process. Data are then collected about the sample, and analyzed. Finally, inferences are made about the characteristics of the population, within the bounds of uncertainty inherent in the sampling process.

These techniques have become part of the standard apparatus of science, and it is unusual for scientists to question the assumptions that underlie them. But the techniques of spatial analysis are applied in very different circumstances from the controlled experiments of the agricultural scientist or psychologist. Rather than creating a sample by laying out experimental plots or recruiting participants in a survey, the spatial analyst typically has to rely on so-called *natural* experiments, in which the variation among samples is the result of circumstances beyond the analyst's control.

In this context the two fundamental principles of statistical inference raise important questions: (i) were the members of the sample selected randomly and independently from a larger population, or did Tobler's First Law virtually ensure lack of independence, and/or did the basic heterogeneity of the Earth's surface virtually ensure that samples drawn in another location would be different? (ii) what universe is represented by the samples? and (iii) is it possible to reason from the results of the analysis to conclusions about the universe?

All too often the answers to these questions are negative. Spatial analysis is often conducted on all of the available data, so there is no concept of a universe from which the data were drawn, and about which inferences are to be made. It is rarely possible to argue that sample observations were drawn independently, unless they are spaced far apart. Specialized methods have been devised that circumvent these problems to some degree, and they will be discussed at various points in the book. More often, however, the analyst must be content with results that apply only to the sample under analysis, and cannot be generalized to some larger universe.

Spatial Data Infrastructure

This final section introduces some of the basic concepts associated with what has become known as *spatial data infrastructure*, the set of representations created using recognized standards and other arrangements that provide sources of spatial data and tools, in ways that support spatial analysis. Spatial data infrastructures provide some of the raw ingredients to spatial analysis, as well as a framework for the display of results that enable ready interpretation by users.

Until the 1980s it was generally accepted that responsibility for the acquisition and dissemination of spatial data lay with national mapping agencies, which produced topographic maps and other information at public expense. Parts of this system began to degrade and decentralize around 1990, however, for a number of reasons. On the one hand, governments became less and less willing to foot the steadily increasing bill; particularly as new technologies created demands for increasingly high value information products. On the other hand, some of the same developments in computing technology and GPS had made it possible for virtually

anyone to become a creator of spatial data at low cost; and the ‘creative commons of the Internet allowed the creation of spatial data infrastructures by communities of volunteers. For example, [Figure 2-13](#) shows the results of calculating a school catchment area and an analysis of pupil educational attainment in and around the area, mapped against a backcloth of data from Open Streetmap (www.openstreetmap.org). Related to this second trend has been the commercial sector lobby for new markets for its products, arguing that subsidized government production of spatial data constituted unfair competition. The movement of the search engines of Google and Microsoft into Internet mapping and more recently, geospatial artificial intelligence (GeoAI), has resulted in spatial data infrastructures becoming available to any non commercial user at zero cost, on the presumption that the supply of such data can be funded through an advertising or subscription driven business model. Today, many countries now have extensive spatial data infrastructures, and responsibility for the acquisition and dissemination of spatial data is now shared between numerous agencies, companies, and individuals.

Geoportals

A *geoportal* is a Web site providing a single point of access to spatial data, allowing its users to browse through extensive collections of data and related materials, evaluating their potential fitness for use. Geoportals may include data that are in the public domain or are otherwise available free of charge, and may also support licensing and purchasing of proprietary data. They include the ability to search through catalogs, examining the contents and quality of data.

The screenshot shown in [Figure 2-13](#), below, is a UK example of a geoportal that brings together datasets concerned with pupil and student education, along with associated population characteristics and framework data. Geoportals include standard layers of data on topography, roads and streets, rivers and streams, named places, imagery, and political boundaries; and may also provide access to more specialized data. A number of geoportals are listed in the Web links: “Selected national and international data and information sources” Annex to this Guide.

Metadata

Fundamental to the operation of a geoportal are *metadata*, the catalog entries that allow users to search for datasets and to assess their contents. Metadata may include information on contents, quality, the data production process, details of formats and coordinate systems, and many other topics. For more details see the US National Spatial Data Infrastructure site: <https://www.fgdc.gov/nsdi/nsdi.html>

Interoperability

Over the past four decades many hundreds of formats have been devised and implemented for spatial data. Some are open, while others are proprietary. The development of an effective spatial data infrastructure has required extensive efforts to achieve *interoperability*, in other words the ability of systems to exchange and use data without major effort. The standards that make this possible have been devised and promulgated by several organizations, including national mapping agencies, international standards bodies, and the Open Geospatial Consortium (OGC): <https://www.ogc.org/>

Conclusion

This chapter has provided a brief overview of some of the more important and fundamental concepts that underpin spatial analysis. Many more will be encountered at various points in

this book, and many more can be found in the background readings cited in Section 1.3.2, [Suggested reading](#). These concepts and others together frame the theory of (geo)spatial analysis, just as the field of statistics is framed by the concepts of statistical analysis. While much of spatial analysis will appear intuitive, it is important to recognize that it is underpinned by concepts and terms whose meanings are often far from intuitive, and require explicit recognition.

Geospatial Project Design and Research Practice

Geospatial analysis is not simply a collection of techniques that may be applied in certain specific situations, assuming that the relevant data are available. It is very much a part of a larger process, and is often a relatively small element, albeit vitally important and at times, relatively complex part. In the following sections a number of analytical methodologies are described, with a particular focus on our spatial version of a methodology known as PPDAC: Problem, Plan, Data, Analysis and Conclusions.

As with all scientific disciplines, spatial analysis has its fair share of terminology, abbreviations, methods and tools. In this subsection we start by examining spatial analysis in the broader context of analytical methodologies, and attempt to highlight the similarities with methods applied in a range of disciplines. We then extend the discussion to consider some of the distinctive aspects of spatial and GIS analysis.

Earlier we suggested that the process of spatial analysis often proceeds in a simple sequence from problem specification to outcome. In reality this is an over-simplification — the analytical process is far more complex and iterative than this summary of steps suggests. So, for example, Mitchell (2005) suggests the sequence for spatial data analysis shown in Figure 3-1, below. Here it appears that there is a natural flow from start to finish, but this is rarely the case. Not only is the process iterative, but at each stage one often looks back to the previous step and re-evaluates the validity of the decisions made. Explicit recognition of this need is incorporated in the approach of Haining (2003, p.359) to “Data-driven statistical modeling”, which is otherwise very similar to Mitchell’s model.

The need for a more iterative approach is partly a reflection of adopting scientific methods, but also a recognition that most analytical tasks take place within a much broader context. For example, pragmatic decisions often have to be made based on a series of common questions:

- how well defined is the problem I am seeking to address?
- how much time, money and resource can I afford to apply to this problem?
- what research has previously been carried out on problems of this type, and what strengths and weaknesses have these shown?
- who will be the recipient of the results, and what are their expectations and requirements?
- if I commence by examining the requirements of the outcome closely, what implications does this have on my selection of techniques and data, and what caveats should I apply to the scope and validity of my analyses — e.g. in relation to sample/region size, timespan, attribute set, validation etc.?
- how will I deal with data inadequacies — e.g. missing data, unsuitable data, delays in receiving or obtaining access to key datasets?
- how will I deal with limitations and errors in the software I have chosen to use?

- what are the implications of producing wrong or misleading results?
- are there independent and verifiable means for validating results obtained?

This list is by no means exhaustive but serves to illustrate the kind of questioning that should form part of any analytical exercise, in GIS and many other fields. In the context of GIS, readers are recommended to study the “Framework for theory” section of the paper by Steinitz (1993), which proposes a six step approach to landscape planning, with many resonances in our own methodological discussions, below. Also recommended is the technical report by Stratton (2006) “Guidance on spatial wildfire analysis”, which again echoes many of the observations we make in the following subsections.

In supplier-client relationships it is often useful to start with the customer requirements and expectations, including the form of presentation of results (visualizations, data analyses, reports, public participation exercises etc.). From a clear understanding of the problem and expectations, one can then work *back* through the process stages identifying what needs to be included and might be excluded in order to have the best chance of meeting those expectations and requirements within the timescale and resourcing available. Echoes of these ideas are found in the commentary on cartographic modeling provided with the [TerrSet/Idrisi](#) GIS package:

“In developing a cartographic model we find it most useful to begin with the final product and proceed backwards in a step-by-step manner towards the existing data. This process guards against the tendency to let the available data shape the final product.”

An interesting example of the modern analytical process at work is a study of the incidence of childhood cancer in relation to distance from high voltage power lines by Draper *et al.* (2005). The client in this instance was not a single person or organization, but health policy makers and practitioners, together with academic and medical specialists in the field. With such scrutiny, political sensitivity and implications for the health of many children, great care in defining the problem to be addressed, selecting the data, conducting the analysis and reporting the findings was essential. In this instance some of their initial findings were reported by the press and radio prior to the intended publication date, so the authors took the decision to publish ahead of schedule despite the issues this might raise

Each step of this process is described clearly, qualifying the information and referring to earlier work where appropriate. In addition the paper clarifies the roles of the various participants in the research, addresses issues relating to competing interests (one of the researchers involved was an employee of the main power grid company), and deals with issues of funding and ethical approval. All of these elements provide transparency and enable “clients” to evaluate the quality and importance of the research. The form of publication (via the British Medical Journal) also facilitated extensive online discussion, with comments on the paper by acknowledged experts from around the world being published electronically, as well as the author’s responses to these comments.

An interesting aspect of this research, and a significant number of similar studies, is that it deals with rare events. In general it is far more difficult to be confident about drawing definitive conclusions about such events, but this does not detract from the importance of attempting to carry out research of this kind. What it does highlight is the need to identify and report on the scope of the study, and attempt to identify and highlight any obvious weaknesses in the data or methodology. In the cancer study just cited, for example, the datasets used were details of the

location of power grid lines (by type, including any changes that had occurred during the 30+ year study period) together with around 30,000 records stored in the national cancer registry. No active participation from patients or their families was involved, and homes were not visited to measure actual levels of Electro-Magnetic (EM) radiation. These observations raise important but unstated questions: how accurate are the grid line datasets and the cancer registry records? is home address at birth an appropriate measure (surrogate for exposure to EM radiation)? is vertical as well as horizontal proximity to high voltage lines of importance? is proximity to pylons carrying insulators and junction equipment rather than just the lines an important factor? is voltage variability important? were the controls similarly located relative to the power lines as the cases — if not, could this account for some aspects of the results reported? given that the research findings identified a pattern of increased risk for certain cancers that was not a monotonic function of distance, can this result be explained, are the observations purely chance occurrences, or are there some other factors at work?

The latter issue is often described as confounding, i.e. identifying a result which is thought to be explained by a factor that has been identified, but is in reality related to one or more other factors that did not form part of the analysis. For example, in this instance one might also consider that: (i) socio-economic status and lifestyle factors may be associated with proximity to overhead power lines, pylons and transformers; (ii) population densities may vary with distance from power lines; (iii) pre-natal exposure may be important; and (iv) the location of nurseries and playgroups may be relevant. Any of these factors may be important to incorporate in the research. In this instance the authors did analyze some of these points but did not include them in their reporting, whilst other points remain for them to consider further and comment on in a future publication. The “distance from power lines” study is an example of modern scientific research in action — indeed, it has provided an important input to the debate on health and safety, leading to national recommendations affecting future residential building programmes.

[Mackay and Oldford \(2000\)](#) have produced an excellent discussion of the context and role of (statistical) analysis within the broader framework of scientific research methods: “Scientific method, statistical method and the speed of light”. Their paper studies the work carried out in 1879 by A A Michelson, a 24-year old Naval ensign, and his (largely successful) efforts to obtain an accurate estimate of the speed of light. The paper is both an exercise in understanding the role of statistical analysis within scientific research, and a fascinating perspective on a cornerstone of spatial analysis, that of distance determination. They conclude that statistical analysis is defined in large measure by its methodology, in particular its focus on seeking an understanding of a population from sample data. In a similar way spatial analysis and GIS analysis are defined by the methods they adopt, focusing on the investigation of spatial patterns and relationships as a guide to a broader understanding of spatial processes. As such spatial analysis is defined by both its material (spatial datasets) and its methods. Within this set of methods are the full range of univariate and multivariate statistical methods, coupled with a wide range of specifically spatial tools, and a broad mix of modeling and optimization techniques.

[Mackay and Oldford](#) describe the statistical method in terms of a sequence of steps that they label PPDAC: Problem; Plan; Data; Analysis; and Conclusions. This methodology is very similar to those we have already described, although a little broader in scope. They then expand upon each step and in the paper relate each step to the process adopted by Michelson. This

approach emphasizes the place of formal analysis as very much a part of a process, rather than a distinct discipline that can be considered in isolation. Daskin (1995, Ch. 9) makes a very similar case within the context of planning facility locations, and sees this as an iterative process culminating in implementation and ongoing monitoring. Spatial analysis as a whole may be considered to follow very similar lines — however the steps need clarification and some amendment to Mackay and Oldford’s formulation.

A summary of a revised PPDAC approach is shown in [Figure 3-3](#), below. As can be seen from the diagram, although the clockwise sequence (1→5) applies as the principal flow, each stage may and often will feed back to the previous stage. In addition, it may well be beneficial to examine the process in the reverse direction, starting with Problem definition and then examining expectations as to the format and structure of the Conclusions (without pre-judging the outcomes!). This procedure then continues, step-by-step, in an anti-clockwise manner (e→a) determining the implications of these expectations for each stage of the process. A more detailed commentary on the PPDAC stages, particularly as they relate to problems in spatial analysis, is provided in Section 3.3, [Spatial analysis and the PPDAC model](#). In addition, an example application of the methodology to noise mapping is provided on the accompanying website in the [RESOURCES](#) section.

Spatial analysis as a process

In many instances the process of spatial analysis follows a number of well-defined (often iterative) stages: problem formulation; planning; data gathering; exploratory analysis; hypothesis formulation; modeling and testing; consultation and review; and ultimately final reporting and/or implementation of the findings. GIS and related software tools that perform analytical functions only address the middle sections of this process, and even then may be relatively “blunt” instruments.

Having first identified and formulated the problem to be addressed (often a significant task in itself), and developed an outline plan, the first task typically involves obtaining the data which are to be the subject of analysis. This immediately raises many questions that have an important bearing on subsequent stages: what assumptions have been invoked in order to represent the “real-world” and what are the implications of this for subsequent analysis? how complete are the data — spatially and temporally? how accurate are the data (spatially, temporally and in terms of measured attributes)? are all of the datasets compatible and consistent with one another — what sources are they drawn from and how do they compare in terms of scale, projection, accuracy, modeling, orientation, date of capture and attribute definition? are the data adequate to address the problem at hand? Can the available data be combined and processed with the resources available?

The second stage, once the data have been obtained and accepted as fit for purpose (and/or as the best available), is often exploratory. This may involve: simple mapping of the data, points, lines, regions, grids, surfaces, computation of rates, indices, densities, slopes, directional trends, level sets, classifications etc.; or more complex and dynamic exploration of the data, such as brushing and linking (see further, Section 5.2.1). One or more analytical techniques and tools may be utilized at this and subsequent stages.

The third stage will depend upon the objective of the analysis. In many instances presentation of the results of exploratory analysis in the form of commentary, maps, descriptive statistics

and associated documents will complete the process. In others it will involve the development and testing of hypotheses about the patterns observed, and/or modeling of the data in order to undertake some predictive or optimization exercise. Frequently the result of this process is a series of possible outcomes (scenarios) which then need to be summarized and presented for final analysis and decision-making by stakeholders, interest groups, policy makers or entrepreneurs. The process may then iterate until an agreed or stable and robust flow is achieved, from problem specification to data selection, and thence to analysis and outcomes.

This kind of process can be formalized and may be implemented as a standard procedure in operational systems or as part of a planning process. Such procedures may involve relatively lengthy decision cycles (e.g. identifying the next location for a new warehouse as customer demand grows) or highly dynamic environments, for example controlling road traffic lights and routing to reflect the type and density of traffic in real time.

To a substantial degree spatial analysis can be seen as part of a decision support infrastructure, whether such decision-making reflects purely academic interest or commercial, governmental or community interests (see further Batty and Densham, 1996; Miller and Shaw, 2001, Chapter 8). Increasingly, to reflect this trend, GIS vendors have developed tools to facilitate such processes. This has been through the provision of Internet-enabled input, output and processing facilities, and through improved usability, speed, interactivity, spatial decision support systems (SDSS) and visualization techniques. Increasingly these have involved the use of formal decision-support systems, from simple cost-benefit analysis and Multi-Criteria Evaluation (MCE) techniques to the more sophisticated and formalized multi-criteria process such as Saaty's Analytical Hierarchical Process ([AHP](#)) and his more recent Analytical Network Process (ANP) models — see Saaty (1980, 1999), commercialized as the [Decision Lens](#) and the [Superdecisions](#) suites of software.

With these improvements in data and toolsets the range and experience of users has grown and will continue to develop. For this reason it is essential that the underlying principles are well-understood and not misused, accidentally or deliberately, leading to ill-informed or invalid decision-making. All providers and users of such systems have a responsibility to ensure that information generated is communicated in an unambiguous and unbiased manner, and to ensure that limitations and assumptions made are transparent and well-understood. These concerns are of particular importance and relevance to those in the print, internet and broadcast media, especially television, where visualization is so important and timeslots for presentation so brief.

Spatial analysis and the PPDAC model

[Mackay and Oldford](#)'s perspective, discussed in the preceding section, is a little too narrow for many complex real-world situations. It also tends to focus on problems that have scientific research focus rather than the broader range of practical problems that the GIS analyst is likely to encounter.

The approach we describe in the following subsections is intended to provide a strong framework for undertaking projects that have a significant geospatial component. For example, the framework might be used to guide the process of preparing a response to an initial government-initiated RFI (request for information). It might subsequently be used to help guide the development of a full-scale costed proposal in response to an RFP (request for proposals), or to address a subset of a larger project within which the geospatial element is a small element.

Alternatively, the PPDAC methodology can be applied to problems in which the collection and analysis of particular datasets is the central task, as may be the case with primary environmental, socio-economic or epidemiological research. PPDAC is a flexible and dynamic methodology, not a rigid set of procedures or forms, and thus may be applied at several stages of a project. An example ‘case study’ of the use of the PPDAC approach is provided on the accompanying website under the section entitled RESOURCES. This case study examines the hypothetical response to an RFP covering noise mapping of urban areas (in order to produce maps of the type illustrated in Figure 3-4 and in the 3D version included at the front of this Guide, showing noise levels in Augsburg, Germany). The example is illustrative only and is not based on actual application of the PPDAC methodology to a specific project that has been implemented. However, it draws on a wide range of materials relating to this topic that have been developed at a national and international level

Problem: Framing the question

Understanding and defining the problem to be studied is often a substantial part of the overall analytical process — clarity at the start is obviously a key factor in determining whether a programme of analysis is a success or a failure. Success here is defined in terms of outcomes rather than methods. And outcomes are typically judged and evaluated by third parties — customers, supervisors, employers — so their active involvement in problem specification and sometimes throughout the entire process is essential. Breaking problems down into key components, and simplifying problems to focus on their essential and most important and relevant components, are often very effective first steps. This not only helps identify many of the issues to be addressed, likely data requirements, tools and procedures, but also can be used within the iterative process of clarifying the customer’s requirements and expectations. Problems that involve a large number of key components tend to be more complex and take more time than problems which involve a more limited set. This is fairly obvious — but perhaps less obvious is the need to examine the interactions and dependencies between these key components. The greater the number of such interactions and dependencies the more complex the problem will be to address, and as the numbers increase complexity tends to grow exponentially.

Analysis of existing information, traditionally described as “desk research”, is an essential part of this process and far more straightforward now with the advantage of online/Internet-based resources. Obtaining relevant information from the client/sponsor (if any), interested third parties, information gatekeepers and any regulatory authorities, forms a further and fundamental aspect to problem formulation and specification.

Spatial analysts also need to be particularly aware of a number of well-known problems associated with grouped data in general, and spatial data in particular. This applies to both problems in which the analyst contributes to the entire PPDAC process as well as the common situation in which the analyst is presented with the problem or task and maybe the (broad) approach to be taken by another party (for example an academic or business supervisor, a technical committee, or even a client). Issues requiring particular attention in the case of spatial problems include:

- spatial scale factors: over what study region is the work to be carried out, and what are the implications of altering this for some or all datasets? Do the same scale factors apply for all the data of interest?

- statistical scale factors: at what levels of grouping are data to be analyzed and reported?
- spatial arrangement factors: does the specification of the spatial arrangement or re-arrangement of subsections of the study area have an impact on the analysis?
- does the problem formulation require data of types, sizes or quality standards that are available, within the time, budget and resources available? if not, compromises will be necessary
- are conclusions regarding spatially grouped data being sought that imply the grouping (e.g. at the county level, at the farm level) is truly representative of all the components in the group (e.g. individuals living within the county, fields within the farm)? If so, the grouped regions must be entirely or largely homogeneous in order to avoid the so-called *ecological fallacy* — ascribing characteristics to members of a group when only the overall group characteristics are known (special statistical techniques have been devised to address certain problems of this type, for example as discussed in King *et al.*, 2004)
- are conclusions regarding spatial grouped data being sought based on the measured characteristics of sampled individuals? If so, the sample must be entirely or highly representative of the grouping in order to avoid the so-called *atomistic fallacy* — ascribing characteristics to members of a group based on a potentially unrepresentative sample of members

Framing the question is not always a one-time exercise. Once an initial problem specification has been drafted, it may be altered in the light of preliminary investigations, technical or commercial considerations, or unforeseen events ([Figure 3-3](#), feedback loops). As far as possible, however, only essential changes should be made (and documented as such) once problem formulation has been completed and documented, and all interested parties have agreed on problem content. GIS has a particular role to play here, in providing tools (principally mapping related) for storing and visualizing existing data and facilitating discussion of aspects of the problem prior to subsequent stages in the process. This role may continue throughout the various stages of a project, assisting in the interpretation, analysis and presentation of results. Careful consideration should be given to change control management, documentation and reporting, especially in commercial and governmental environments.

Note that in many instances it will be difficult if not impossible to compute the timescale and budget required to address a particular problem without a preliminary study or project. For example, it is quite common for there to be preliminary project involving the preparation of a detailed Requirements Specification (RS) and/or undertaking a smaller-scale or feasibility project, e.g. confined to a small, but representative, area.

Plan: Formulating the approach

Having agreed on the problem definition the next stage is to formulate an approach that has the best possible chance of addressing the problem and achieving answers (outcomes) that meet expectations. Although the PLAN phase is next in the sequence, the iterative nature of the PPDAC process emphasizes the need to define and then re-visit each component. Thus whilst an outline project plan would be defined at this stage, one would have to consider each of the subsequent stages (DATA, ANALYSIS, CONCLUSIONS) before firming up on the detail of the plan. With projects that are more experimental in nature, drawing up the main elements of the PLAN takes place at this stage. With projects for which pre-existing datasets and analysis

tools are expected to be used, the PLAN stage is much more an integrated part of the whole PPDAC exercise.

The output of the PLAN stage is often formulated as a detailed project plan, with allocation of tasks, resources, timescales, analysis of critical path(s) and activities, and estimated costs of data, equipment, software tools, manpower, services etc. Frequently project plans are produced with the aid of formal tools, which may be paper-based or software assisted. In many instances this will involve determining all the major tasks or task blocks that need to be carried out, identifying the interconnections between these building blocks (and their sequencing), and then examining how each task block is broken down into sub-elements. This process then translates into an initial programme of work once estimated timings and resources are included, which can then be modified and fine-tuned as an improved understanding of the project is developed. In some instances this will be part of the Planning process itself, where a formal functional specification and/or pilot project forms part of the overall plan.

As with other parts of the PPDAC process, the PLAN stage is not a one-shot static component, but typically includes a process of monitoring and re-evaluation of the plan, such that issues of timeliness, budget, resourcing and quality can be monitored and reported in a well-defined manner.

The approach adopted involves consideration of many issues, including:

- the nature of the problem and project — is it purely investigative, or a formal research exercise; is it essentially descriptive, including identification of structures and relationships, or more concerned with processes, in which clearer understanding of causes and effects may be required, especially if predictive models are to be developed and/or prescriptive measures are anticipated as an output?
- does it require commercial costings and/or cost-benefit analysis?
- are particular decision-support tools and procedures needed?
- what level of public involvement and public awareness is involved, if any?
- what particular operational needs and conditions are associated with the exercise?
- what time is available to conduct the research and are there any critical (final or intermediate) deadlines?
- what funds and other resources are available?
- is the project considered technically feasible, what assessable risk is there of failure and how is this affected by problem complexity?
- what are the client (commercial, governmental, academic) expectations?
- are there specifications, standards, quality parameters and/or procedures that must be used (for example to comply with national guidelines)?
- how does the research relate to other studies on the same or similar problems?
- what data components are needed and how will they be obtained (existing sources, collected datasets)?

- are the data to be studied (units) to be selected from the target population, or will the sample be distinct in some way and applied to the population subsequently (in which case one must consider not just sampling error but so-called study error also)?

When deciding upon the design approach and analytical methods/tools it is essential to identify available datasets, examine their quality, strengths and weaknesses, and carry out exploratory work on subsets or samples in order to clarify the kind of approach that will be both practical and effective. There will always be unknowns at this stage, but the aim should be to minimize these at the earliest opportunity, if necessary by working through the entire process, up to and including drafting the presentation of results based on sample, hypothetical or simulated data.

The application of a single analytical technique or software tool is often to be avoided unless one is extremely confident of the outcome, or it is the analytical technique or approach itself that is the subject of investigation, or that this approach or toolset has already been approved for use in such cases. If analysis is not limited to single approaches, and a series of outputs, visualizations, techniques and tests all suggest a similar outcome then confidence in the findings tends to be greatly increased. If such techniques suggest different outcomes the analyst is encouraged to explain the differences, by re-examining the design, the data and/or the analytical techniques and tools applied. Ultimately the original problem definition may have to be reviewed.

The impact on research of exceptions — rare events, spatial outliers, extreme values, unusual clusters — is extremely important in geospatial analysis. Exploratory methods, such as mapping and examining cases and producing box-plots (see further, Section 5.2.2, [Outlier detection](#)), help to determine whether these observations are valid and important, or require removal from the study set.

Some analytical techniques are described as being more robust than others. By this is meant that they are less susceptible to data extremes or unusual datasets — for example the median or middle value of a dataset is generally regarded as more robust than the mean or average value, because it is unaffected by the specific values of the set. However, the spatial mean and median exhibit different properties from those applied to individual tabulated attributes, and other measures of centrality (e.g. the central feature of a set) may be more appropriate in some instances. Likewise, statistical tests that make no assumptions about the underlying distribution of the dataset tend to be more robust than those that assume particular distributional characteristics — for example non-parametric versus parametric tests. However, increasing robustness may result in loss of *power* in the sense that some methods are described as being more powerful than others, i.e. they are less likely to accept hypotheses that are incorrect or reject hypotheses that are correct.

Data: Data acquisition

Spatial analysis is somewhat unusual in that key datasets are often provided by or acquired from third parties rather than being produced as part of the research. Analysis is often of these pre-existing spatial datasets, so understanding their quality and provenance is extremely important. It also means that in many instances this phase of the PPDAC process involves selection of one or more existing datasets from those available. In practice not all such datasets will have the same quality, cost, licensing arrangements, availability, completeness, format, timeliness and detail. Compromises have to be made in most instances, with the over-riding

guideline being fitness for purpose. If the datasets available are unsuitable for addressing the problem in a satisfactory manner, even if these are the only data that one has to work with, then the problem should either not be tackled or must be re-specified in such a way as to ensure it is possible to provide an acceptable process of analysis leading to worthwhile outcomes.

A major issue related to data sourcing is the question of the compatibility of different data sets: in formats and encoding; in temporal, geographic and thematic coverage; in quality and completeness. In general datasets from different sources and/or times will not match precisely, so resolution of mismatches and data linkage issues can become a major task in the data phase of any project. And as part of this process the issue of how and where to store the data arises, which again warrants early consideration, not merely to ensure consistency and retrievability but also for convenient analysis and reporting.

Almost by definition no dataset is perfect. All may contain errors, missing values, have a finite resolution, include distortions as a result modeling the real world with discrete mathematical forms, incorporate measurement errors and uncertainties, and may exhibit deliberate or designed adjustment of positional and/or attribute data (e.g. for privacy reasons, as part of aggregation procedures). Spatial analysis tools may or may not incorporate facilities for explicitly handling some of the more obvious of these factors. For example, special GIS tools exist for handling issues such as:

- boundary definition and density estimation
- coding schemes that provide for missing data and for masking out invalid regions and/or data items
- modeling procedures that automatically adjust faulty topologies and poorly matched datasets, or datasets of varying resolutions and/or projections
- a wide range of procedures exist to handle difficulties in classification
- data transformation, weighting, smoothing and normalization facilities exist to facilitate comparison and combination of datasets of differing data types and extents
- lack of continuity in field data can be explicitly handled via breaklines and similar methods
- a range of techniques exist for modeling data problems and generating error bounds, confidence envelopes and alternative realizations

In other words, many facilities now exist within GIS and related software packages that support analysis even when the datasets available are less than ideal. It is worth emphasizing that here we are referring to the analytical stage, not the data cleansing stage of operations. For many spatial data this cleansing has been conducted by the data supplier, and thus is outside of the analyst's direct control. Of course for data collected or obtained within a project, or supplied in raw form (e.g. original, untouched hyperspectral imagery) data cleansing becomes one element in the overall analytical process.

For many problems important components of the project involve intangible data, such as perceptions or concerns relating to pollution, risk or noise levels. Quantifying such components in a manner that is accepted by the key participants, and in a form suitable for analysis, is a difficult process. This is an area in which a number of formal methodologies have been established and widely tested (including spatial decision support tools and cost-benefit analysis

tools). Such tools are almost always applied separately and then utilized to generate or qualify input to GIS software (e.g. weightings).

An additional issue arises when data are generated as part of a research exercise. This may be as a result of applying a particular procedure to one or more pre-supplied datasets, or as a result of a simulation or randomization exercise. In these latter cases the generated datasets should be subjected to the same critical analysis and inspection as source datasets, and careful consideration should be given to generated and magnified distortions and errors produced by the processing carried out. A brief discussion of such issues is provided in Burrough and McDonnell (1998 Ch.10, and 2015 Ch.4) whilst Zhang and Goodchild (2002) address these questions in considerable detail. For further discussion of the nature of spatial data and some of the key issues raised when undertaking analyses using such datasets, see Haining (2008).

Analysis: Analytical methods and tools

One of the principal purposes of this Guide is to assist readers in selecting appropriate analytical methods, tools and models from those readily available in the GIS and related marketplace. Since analysis takes place well down the chain in what we have described as the analytical methodology, initial selection of methods and tools should have been made well before the analytical stage has been reached. Simplicity and parsimony — using the simplest and clearest tools, models and forms of visualization — and fit to the problem and objectives are the key criteria that should be applied. Other factors include: the availability of appropriate tools; time and cost constraints; the need to provide validity and robustness checks, which could be via internal and/or external checks on consistency, sensitivity and quality; conformance to relevant standards; the use of multiple techniques; and the use of independent and/or additional datasets or sampling.

There are a very large number of software tools available for performing spatial analysis: both for simple data summarizing and exploratory spatial data analysis (ESDA), which will often form the initial stage of analysis; and to assist the development and/or application of specific techniques or models. Frequently spatial model-building and spatial analysis are very closely coupled, with the output of one leading to revisions in the other. This is particularly true of micro-modeling techniques, such as geosimulation, where results often ‘emerge’ from the modeling exercise and reveal unexpected patterns and behaviors which in turn lead to revised ideas and hypotheses (see for example, Section 8.2.2 on [Agents and agent-based models](#)).

All products describing themselves as geographic information systems have a core set of analytical facilities, including many of those we describe in this Guide. We cannot hope to draw examples from even a representative sample of products, so we have made our own selection, which we hope will provide sufficient coverage to illustrate the range of analytical techniques implemented in readily available toolsets. In addition to mainstream GIS products we have included a number of specialist tools that have specific spatial analysis functionality, and which provide either some GIS functionality and/or GIS file format input and/or output. Throughout this Guide there are many examples of the application of tools and models to geospatial problems. In addition, in Section 3.4, we discuss the application of certain types of modeling, particularly high-level process-flow models, to geospatial problems.

A recurrent theme in spatial analysis is the notion of pattern. Frequently the objective of analysis is described as being the identification and description of spatial patterns, leading on

to attempts to understand and model the processes that have given rise to the observed patterns. Unfortunately the word ‘pattern’ has a very wide range of meanings and interpretations. One way of defining whether a particular set of observations constitute a spatial pattern is by attempting to define the opposite, i.e. what arrangements of objects are not considered to constitute a pattern. The generally agreed notion of not a pattern is a set of objects or an arrangement that provides no information to the observer. From a practical point of view no information is often the situation when the arrangement is truly random, or is indistinguishable from a random arrangement. An alternative definition might be an even arrangement of objects, with deviations from this uniformity considered as patterns. Thus spatial pattern is a relative concept, where a model of not a pattern (e.g. Complete Spatial Randomness or CSR) is a pre-condition (see for example, Figure 2-11, [First- and second-order processes](#)).

Observed spatial arrangements are frequently of indirect or mapped data rather than direct observations — the process of data capture and storage (e.g. as points and lines or remote sensed images) has already imposed a model upon the source dataset and to an extent pre-determined aspects of the observable arrangements. The method(s) of visualization and the full and/or sampled extent of the dataset may also impose pre-conditions on the interpretation and investigation of spatial patterns.

Identification of spatial pattern is thus closely associated with a number of assumptions or pre-conditions: (i) the definition of what constitutes not a pattern for the purposes of the investigation; (ii) the definition of the dataset being studied (events/observations) and the spatial (and temporal) extent or scale of the observations; and (iii) the way in which the observations are made, modeled and recorded.

Observed patterns may suggest a causal relationship with one or more principal processes, but they do not provide a secure means of inference about process. For example, consider the case of the distribution of insect larvae. Imagine that an insect lays its eggs at a location in a large region at least 200 meters from the egg sites of other insects of the same species, and then flies away or dies. Other insects of the same species do likewise, at approximately the same time, each laying a random number of eggs, most of which hatch to form larvae that slowly crawl away in random directions from the original site for up to 100 meters. At some point in time shortly thereafter an observer samples a number of sites and records the pattern of all larvae within a given radius, say 10 meters. This pattern for each sampled site is then individually mapped and examined. The observer might find that the mapped patterns appear entirely random or might have a gradient of larvae density across the sampled regions. Zooming out to a 100m radius (i.e. using a larger region for sampling) a different pattern might have been observed, with a distinct center and decreasing numbers of larvae scattered in the sampled region away from this point. However, if observations had been made over 1km squares in a 10kmx10km region it might only have been practical to identify the centers or egg sites. At this scale the pattern may appear regular since we implied that each egg site is not randomly distributed, but is influenced by the location of other sites. However, it could equally well be the case that egg sites are actually random, but only eggs on those sites that are laid on suitable regularly distributed vegetation survive and go on to produce live larvae. Zooming out again to 100kmx100km the observer might find that all egg sites are located in particular sub-regions of the study area, thus appearing strongly clustered, a pattern perhaps related to an attraction factor between insects or uneven large-scale vegetation cover or some other environmental variable. The mapped patterns at any given scale may not be sufficient to enable the analyst to

determine whether a given set of observations are random, evenly spread, clustered or exhibit some specific characteristic such as radial spread, nor to infer with any reliability that some particular process is at work (as discussed in Section 2.1.9: [Detail, resolution, and scale](#) and Section 5.4, [Point Sets and Distance Statistics](#)).

On the other hand it is fairly straightforward to generate particular spatial patterns, in the manner described above, using simple (stochastic) process models (see further, Section 5.4, [Point Sets and Distance Statistics](#)). These patterns are specific realizations of the models used, but there is no guarantee that the same pattern could not have been generated by an entirely different process. A specific mapped dataset may be regarded as a specific outcome of a set of (known and unknown) processes, and as such is but one of many possible outcomes or process realizations. To this extent, and perhaps to this extent only, such datasets can be thought of as samples (see further, Sections 2.2.8, [Spatial sampling](#) and 2.3.4, [Statistical inference](#)).

Thus the ANALYSIS phase can be seen as a multi-part exercise. It commences with the review of data collected and the manipulation of the many inputs to produce consistent and usable data. It then extends into the pure analytical phase, where the data are subjected to study in order to identify patterns of various kinds then help to develop new ideas and hypotheses regarding form and process. And this in turn may lead on to the use of or development of one or more models within a formal build-fit-criticize cycle. Finally the output of the models and analysis is examined, and where necessary the dataset and data gathering is re-visited, working back up the PPDAC model chain, prior to moving on to producing the output from the project and delivering this in the Conclusion stage.

Conclusions: Delivering the results

The last stage of the PPDAC process is that of reaching conclusions based upon the analyses conducted, and communicating these to others. Note that implementation of findings (e.g. actually proceeding with building a bypass, designating an area as unfit for habitation, or implementing a vaccination programme) does not form part of this model process, but lies beyond its confines.

“The purpose of the Conclusion stage is to report the results of the study in the language of the Problem. Concise numerical summaries and presentation graphics [tabulations, maps, geovisualizations] should be used to clarify the discussion. Statistical jargon should be avoided. As well, the Conclusion provides an opportunity to discuss the strengths and weaknesses of the Plan, Data and Analysis especially in regards to possible errors that may have arisen” [Mackay and Oldford](#) (2000)

For many problems treated in spatial analysis this summary is sufficient. For others the conclusions stage will be the start of additional work: re-visiting the problem and iterating the entire process or parts of the process; a new project; implementing proposals; wider consultation, such as public participation GIS ([PPGIS](#), [Communityviz](#)), Geoweb applications (such as Digital globes and [StreetView](#)) and participatory 3D modeling; and/or the development of models that help to understand or explain the observations, guide future data gathering, and predict data values at locations/times not included within the analysis.

Geospatial analysis and model building

Geospatial analysis often involves the design and execution of one or more ‘models’. In many instances pre-existing models or simple variants of these are used. Such models may be provided within standard GIS packages, or they may be offered by third parties as add-ons, or they may have been specially developed for specific applications, such as airborne pollution dispersal modeling, flood modeling, wildfire management or traffic forecasting. In other cases models are designed using a number of approaches and tools, and then applied to address the specific problem at hand. The construction of such models follows the well-trodden procedure known as the “build-fit-criticize” cycle. This envisages the designer building a model on paper or computer or using a physical analogue, and then developing the model by evaluating its behavior under various conditions and datasets. By examining the performance of the model (the criticize part of the cycle) in terms of outputs, performance speed, completeness, robustness, efficiency, quality and ultimately, fitness for purpose, the model can be progressively improved and fine-tuned to meet the requirements.

Elsewhere in this Guide are many examples of specific and generic models that have been developed for use in the geospatial field. These range from simple deterministic models of behavior (distance decay models), to so-called ‘cartographic modeling’ (essentially map algebra and related operations), through to complex micro-simulation modeling frameworks that may be used for a wide range of bottom-up and dynamic modeling problems. Extended discussion of models and modeling in the context of geospatial analysis has been provided by Harvey (1968, Chs. 10-12), Goodchild *et al.* (1993) and [Longley et al.](#) (2015, Ch15) and readers are recommended to review these discussions before embarking upon spatial model building. In particular [Longley et al.](#) (2015) characterize geospatial modeling, as opposed to analysis, as involving: multiple stages (perhaps representing different points in time); implementing ideas and hypotheses about the behavior of the real world; and experimenting with policy options and scenarios.

Unfortunately, as with many other areas of GIS, the term *model* has acquired a wide range of meanings. In a number of GIS packages, such as [ArcGIS](#) and [TerrSet/Idrisi](#), model building refers specifically to the use of graphical tools to design and execute a series of geoprocessing tasks. This provides both a way of designing and communicating the workflow associated with a task, and a means of implementing that workflow in a convenient manner.

In this example two items have been dragged to the [ArcGIS](#) ModelBuilder design window: the first is a point set file (a shapefile of lung cancer cases from Lancashire, UK, which is an X,Y point set); the second is the geoprocessing function, Kernel Density, which generates two linked items in the graphical window, the processing function itself and the output raster. All that remains is to link the input shapefile to the processing function (identifying the input and flow) and then to run the ‘model’. Each of the graphical entities can be clicked to specify attributes and parameters, and the model saved for future use. When the model is run an output raster of density data is generated, according to the parameters set in the model.

Clearly this is a trivial example — it is essentially an example of the simple form of general-purpose model structure:

INPUT → PROCESS → OUTPUT

This form of model can readily be developed in a number of ways. For example, in Figure 3-5 we have a single input, single output and single process. Many single geospatial processing

operations require more than one input – for example, the Overlay and Intersect functions (note that the order in which links are established may be important). Likewise, some processing functions will produce multiple outputs.

Even this reflects a very simple model structure. In many real-world workflow situations a whole series of input → process → output steps are involved, with connections between the various outputs and the next stage of inputs. Such models may be structured into a series of sub-models, nested within a broader overall structure, designed to address a broader range of problems or a complex task. The process of analyzing a large task and breaking it down into smaller, more manageable components, is typical of systems analysis and business analysis exercises, which precede the process of systems design and implementation.

A very wide range of models can be constructed using this kind of approach, and optionally exported to a high-level programming language or script, such as [Python](#) or VBscript. Once in the explicit form of the preferred scripting language the functionality of the model can be examined closely, the ordering of links double-checked, extra functionality added, test datasets used to validate the model and outputs, and the ‘program’ retained for repeated use, perhaps with more complex parameterization of inputs.

The type of model described above is relatively simple, in that it flows from input to output in a well-defined manner, using pre-existing processing functionality. This is a static, deterministic model — the result is a specific output which will always be generated in the same manner from a given input. This is ideal for a range well-defined processing tasks, but is still limited in terms of model functionality.

One way to make high-level models of this type become more like traditional concepts of high-level or ‘top-down’ model-building is to incorporate new, purpose-designed, processing modules within the flow. This could involve entirely new model component design, or the development of components that incorporate pre-existing facilities within the GIS package or from third party toolsets (which may or may not be GIS-package related). Typically such models are designed as block diagrams with flows, rather as with the simple models described earlier, but with a much wider range of operations within and between blocks permitted. In particular, one development of the high-level model framework is to permit iterative behavior, leading to models with a dynamic-like structure. In such models one or more output datasets becomes the input data for the next iteration. A series of iterations are then run, with intermediate outputs providing a time-like profile for the model. Indeed, generating a large number of iterations may produce a series of raster grids which provide a sufficient number of data ‘frames’ to generate a video of the process as it develops.

Another common development for high-level models is to build-in stochastic behavior for some components, thus enabling a range of outcomes to be generated from a single starting set of data. In such models the results are obtained after repeated runs of the model, generating a range of outcomes and a probability ‘envelope’ of results. Non-deterministic models of this type are especially appropriate in forecasting or predictive studies. Figure 3-6 illustrates the kind of model of this type that can be constructed using [TerrSet/Idrisi](#).

In this instance a static model was initially designed to predict areas of likely residential expansion within land use classes designated as ‘forest’. The initial input is the pattern of residential land use in 1991 (RESIDENTIAL91) together with a pre-existing low density

residential suitability raster (LDRESSUIT) which is based on factors such as proximity to roads, terrain etc. Both these input files are grids with matching size (565x452 cells) and resolution. In the model diagram rectangular boxes are grids and processing modules are parallelograms. The details of the model are not especially important here, but one can see that the process flows from top left (inputs) to bottom right) output, and is concluded with an overlay of the original input grid and the model output grid (NEWRESID) to produce a map which highlights the predicted growth.

This static model forms part of the tutorial data distributed with the software, so readers can work through this example themselves if they have access to [TerrSet/Idrisi](#). There are two aspects of this particular model we wish to highlight: the first is the process box marked RANDOM which appears early in the process — this takes as input the LANDUSE91 grid and introduces a small random chance that forested land might convert to residential in any part of the study area; the second aspect is the line at the bottom of the model, which takes the output (NEWRESID) and feeds it back as the input for the next iteration of the model. This feedback loop provides a form of dynamic modeling, and enables the overall model to be run for a number of iterations, with intermediate growth maps and associated data being optionally retained for closer examination. Clearly these kinds of macro-level models can provide a very powerful framework for certain classes of geospatial analysis, especially when addressing complex recurring processing tasks and when examining alternative scenarios in multi-criteria decision making problems.

GIS data and analytical processes often contribute a part of a larger multi-disciplinary modeling process. Consider, for example, the [FlamMap](#) wildfire modeling system. Input to this model includes eight GIS raster layers that describe the ‘fuels’ and topography: elevation, slope, aspect, fuel model, canopy cover, canopy height, crown base height and crown bulk density; these layers are then combined in the model with an Initial Fuel Moistures file, and optional components including the inclusion of a custom Fuel Model, weather, and wind files. Within the model further models are applied in combination: a surface fire model; crown fire initiation model; crown fire spread model and a dead fuel moisture model. This example serves to illustrate how many components make up many real-world operational models. In this instance GIS provides key inputs, and also is used to map key outputs, which are produced in the form of both raster and vector files for subsequent mapping. These outputs are then utilized in a second stage of spatial analysis, covering issues such as risk and impact assessment (Figure 3-7), fire management planning, and comparing modeled minimum travel time (MTT) and travel paths of a fire front to actual data. The [FlamMap](#) input and output data used to create Figure 3-7, with accompanying metadata and output files, was obtained from ForestERA.

Research-ready data (RRD)

Overview

The best geospatial data are often costly to produce - despite huge innovations in sensor technologies and ubiquitous use of digital devices to record the characteristics and activities of citizens. But once created, they are very cheap to reproduce. The onus is on new innovation-led data services to share up-front costs of database development and maintenance between ever-increasing numbers of research users. Just as the sharing of GIS software development costs a generation ago enabled the GIS industry, so generic ‘off-the-peg’ Research Ready Data

(RRD: sometimes called ‘analysis ready data’) are today enabling wide dissemination of exciting new data sources that can enable efficient, effective and safe geospatial analysis.

In this section we describe these developments from the standpoint of socioeconomic data, but the core idea is perhaps best illustrated by remote sensing in environmental science. Classic remotely sensed satellite imagery records the reflectance characteristics of the Earth’s surface, rather than the land cover (and by inference, land use) characteristics that most users require. Surface reflectance turns out to be a direct indicator of land cover, and established and widely accepted procedures exist to assign different reflectance characteristics to land cover classes. These procedures are well-documented, as are caveats to their use that have become evident through particular research case studies. It therefore makes little sense for multiple users of the same data source to repeat the same steps in image preparation: instead, a single ‘research ready’ processed dataset can be made available for wide use, and new research can focus upon innovative analysis and interpretation.

This straightforward example raises several important issues that merit consideration when thinking about RRD for socioeconomic applications:

(a) It is understood that the satellite scene is ‘all that there is’ – that it offers complete coverage of a scene at a level of detail that is understood. If any areas are obscured by cloud, this will be evident and all or part of the image will be replaced by another taken on a cloud free day. Comprehensive coverage means that the data can be construed as data **infrastructure**.

(b) Cloud cover is not randomly scattered above the Earth’s surface, so it may be necessary to pay more attention to scenes for areas that are systematically prone to cloud cover – such as mountainous areas or locales with convectional microclimates.

(c) Inference of land cover from surface reflectance is widely understood, generally accepted and well-documented. It is a geographically invariant procedure that can be applied to any cloud free dataset.

(d) This said, image classification is a probabilistic process that is inherently uncertain. Moreover, assignments to some land use categories are more uncertain than others, and so the property of uncertainty is not geographically invariant.

(e) Once classified, provision of RRD is ‘vertically integrated’ – that is, the organization that collected the satellite data either pre-processes them itself, or has formal data licensing arrangements with a partner organization to process them, render them fit for a prescribed range of purposes, and make them available to an agreed user base.

Here we develop these observations in the context of RRD for socioeconomic GIS. More data are collected about citizens today than at any point in human history, and more of these data can be described as ‘smart’ – that is, created through everyday interactions between humans and digital devices. As yet, only tiny fractions of these data are consolidated into usable datasets, or are made available to researchers through FAIR (Findable, Accessible, Interoperable and Reusable) channels. Even where smart data do become available, uncertainty about their inherent incompleteness and bias can limit their usefulness in socially inclusive research, although researchers may not be aware of deficiencies in data that they have not themselves collected. This chapter: (a) sets out differences between use of smart data in social investigation and conventional survey-based research designs; (b) assesses the prospects for

designing RRD infrastructure that manages the risk of bias in geospatial analysis; and (c) sets out how innovation-led data services can serve smart data to today's mass research culture.

A retrospective on sampling and inference in social investigation

Moser and Kalton's (1971, 1985) *Survey Methods in Social Investigation* remains a definitive statement of the core organizing principles and practices of social survey research design and implementation, from conception of a research problem, through sample design and monitored data collection to robust, transparent inference and interpretation. Social 'investigations' are not carried out in controlled laboratory conditions but typically sample individuals from a known population ('all that there is' that is relevant to the research hypotheses) in the observable world. If a sub-group is deemed worthy of special interest to the research, special efforts can be made to adequately represent them in the sample design. The extent of the population is understood, and each member of it has a known and pre-specified chance of selection, independent of use of extraneous considerations such as social media participation, travel card use or voter registration – considerations that may make some elements of the population easier to engage with in data collection. Research proceeds through drawing samples and monitoring of patterning in non-response. It is then possible to use statistical inference to infer population characteristics with known margins of error. The project culminates with report-writing, after which the data are retired from any use apart from checks for reproducibility by other researchers.

Survey-led investigations remain important today (albeit plagued by falling response rates), but this world has fundamentally changed with the advent of vast assemblages of smart data, arising from consumer transactions and other interactions between humans and smart devices (see Longley *et al.*, 2016). This shift has spurred growing use of smart data to underpin geospatial analysis that is data rich but also data driven. Datasets that are 'found' rather than designed provide no safeguards with respect to content or coverage, and their use to create representations of the world may ultimately prove delusional, and akin to the hallucinatory results of unchecked AI large language models.

What can be done? A first step is to investigate the potential sources and operations of bias in the creation of smart datasets, in order that they might be anticipated, understood and ultimately accommodated in subsequent analysis. Although precise impacts upon specific analyses may only become apparent in specific applications, metadata on general characteristics may enable multiple reuses of a smart dataset, akin to the reuse of pre-processed scenes from remote sensing. Smart data can be similarly rendered 'research ready' by diagnostic checking to documented standards and hence be prepared for multiple purposes. Social investigations present overlapping data requirements, and RRD enables economies of scale in database creation and maintenance to be realized for the wider good. Carefully managed, such RRD can retain similar internal consistency to data collected using the design of surveys for specific investigations – mindful also that non-response rates and weaknesses of some (e.g. clustered) sample designs in the past never truly presented a panacea for analysis.

The design of Smart Data Infrastructure

Like survey investigation, RRD can entail significant up-front costs. Historically, up-front costs of survey-led social investigations have been borne by the funder of a specific project, and are justified by the value of the end-uses to which the data are put. Smart data (and some other

sources, such as administrative data) arise out of unrelated and non-research purposes. They can be acquired for research use only with the consent of the data owner, with any agreement likely subject to legal consents as to third party use. Figure 3-8 illustrates how a syndicated approach to RRD creation can develop and evolve.

Interest in a hitherto unused data source may arise in different parts of the research community, often by serendipity. For example: an employee of an organization that generates smart data may become aware of a potential data use case for a problem that, although not of immediate operational concern, merits a horizon-scanning investigation and may enlist an academic contact in collaboration; or an academic engaged in an industry knowledge exchange project may negotiate personal access to a company's data in fulfillment of project objectives; or a masters student may become aware that data available through a short-term placement may have potential for their dissertation topic. In each instance, the novelty of the data source may motivate a data service to engage with the data owner to propose more broad-based collaboration, albeit that the provenance of the data remains largely unknown. Research findings thus far do not meet open science criteria of reproducibility and this may also motivate early users of the data to lobby for widened access.

Where potential research usage is assessed as being widespread and worthwhile, a research data service may consult with prospective users to establish priorities for sector-wide data acquisition. The next step is then to explore the possibility of acquisition from the owner and, given a reasonable likelihood of success, to negotiate temporary access to a sufficiently large sub-set of the data. This enables critical appraisal of the value of the data for a wide class of research problems using due diligence analysis of content, coverage and provenance. This has two components: internal validation of the ostensible content of the data and their spatial and temporal granularity; and external validation of the content that can be triangulated against external (statistical and administrative) sources or augmentation with other smart data sources that may also provide cross-validatory evidence.

If the over-all outcome of due diligence is positive, a multilateral data licensing agreement (DLA) may then be drawn up with the data owner. This enables sector-wide sub-licensing of raw data and derivative data products, or RRD, alongside metadata for each. RRD may be developed from a single smart dataset, or by pooling multiple smart and other data sources. RRD are designed to be non-disclosive, in order that they may be distributed to the broadest possible constituency of research users. Once established, these procedures are straightforward to repeat by updating the DLA for updates or extensions to the initial data resources.

These processes together enable an industrialization of the survey investigation research design shown in Figure 3-8. Instead of framing a specific research problem with a unique set of hypotheses within a bespoke research design, data are acquired and engineered for use across a wide range of research projects, and the fitness for purpose of the data is documented in generic terms. The life of the dataset is not confined to a single project, terminating upon acceptance of a single final report: rather, the data support (alone or in combination with other sources) multiple projects. The data infrastructure that the smart data provide are extended and updated in continuous time. Sampling and the apparatus of inference are replaced by triangulation with (less detailed and frequently collected) statistical sources or augmentation with administrative or smart data that establish reliability.

Access to RRD

Research data services are becoming increasingly attuned to current and envisaged user requirements. RRD content and geographic coverage are important, since this fundamentally shapes the nature and range of hypotheses that can drive social investigations. A successful data service should develop and engage with a community of practice around existing and prospective data assets, with likely benefits to all stakeholders. Numerically, most benefits will accrue to the many RRD users, but data services can also avail specialist users of the tacit knowledge necessary to hone raw (or less pre-processed) smart data to more specialized research tasks. Additionally, whilst data owners may not have panoptic awareness of the content and coverage of their own data (particularly with respect to ‘less often heard’ groups), knowledge exchange embedded in the data service can signpost potential shortcomings in metadata. McGrath-Lone et al. (2022) set out the requirements for effective provision of RRD from administrative sources. Their core considerations extend from creation through documentation, maintenance and service delivery of RRD under FAIR data principles, as elaborated upon in Figure 3-9. There are four components to service provision.

First, creation of smart RRD requires vision and breadth. The early pilot studies discussed in Section 2 may originate in commonalities of interest in problem-solving that are very restricted in space, time and scope. Such pilots are useful to establish proof of concept but are unlikely to meet the needs of the broader constituency of research users. Community-wide multilateral licenses should seek the widest possible (e.g. national) coverage and include content relevant to the greatest number of users. A strategic view of smart RRD creation and maintenance also provides that acquisitions are amenable to linkage with minimal or no geographic aggregation – for example to an integrative smart data spine that readily enables cross-classification and new or updated RRD following new data acquisitions. Whilst it can be more challenging to secure multiple data licensing agreements for the full extent of datasets, ambition in acquisition has the greatest potential to drive new RRD creation.

Second, smart RRD require adequate documentation. Crucially, many owners of smart data are drawn from the private sector and operate in markets in which they have no monopoly of provision. As a result, data are likely to be biased in content and may be geographically restricted in coverage. This scenario requires documentation in quantitative and qualitative terms – the former with respect to available framework data, such as censuses and property gazetteers, and the latter to confirm how and why the data were generated in terms of the mission statements of the data providers. User experiences can also be added as testimonies of suitability for specific research purposes, and to flag potential RRD extensions or refinements.

This leads to the third consideration of curation of RRD, which extends documentation through dynamic updating of the underpinning data infrastructure of RRD, including but not limited to the addition of new data. It includes procedures of internal validation through longitudinal profiling of underpinning infrastructure, and external validation with respect to framework data and other ancillary sources. These procedures enable improved harmonization of the multiple smart data sources used to create smart RRD and suggests potential new RRD products that the research community might find useful.

Fourth, innovation-led data services must render accessible the pipeline of smart RRD to a large and diverse research user community. Proliferation of data sources creates a requirement for gateways to data catalogs that are accessible to users and that provide clear and streamlined

procedures governing access to smart RRD. Ideally data should be available on open access licenses, but issues of ownership, service funding and licensing will typically necessitate additional access protocols – for example providing for data owner clearance in circumstances in which there are potential commercial conflicts of interest.

The final consideration addresses highly skilled requirements for RRD enhancement. Smart data are at the same time data rich and plagued with detail that most researchers will deem extraneous. Innovation-led data services should selectively discard unnecessary detail to improve database manageability, without unduly restricting the range of research questions that the raw data might be used to address. A successful data center is thus both technocratic and research focused, balancing the opportunity costs and benefits of prioritizing different RRD products.

Creation and maintenance of geospatial RRD

Smart geospatial data are typically recorded as point events with high apparent levels of precision. As with statistical and administrative data, this brings disclosure risks which can most easily be addressed in RRD by geographic aggregation. This approach inevitably brings loss of within-zone geographic detail and, where RRD creation involves multiple variables, creates uncertainty of possible ecological fallacies. Yet the effects of aggregation can be documented by data services if the level of the individual has been retained in the underpinning data infrastructure used in RRD creation. Disclosure risks can be managed by use of secure TREs (Trusted Research Environments) to maintain the underpinning individual level data infrastructure. Data services using such facilities typically have additional up-front costs of ensuring safe researcher training, output checking and other data governance procedures.

The ubiquity of georeferencing technology in many smart data sources makes it possible to retain the locations of individuals in underpinning smart data infrastructure. The reliability of positioning measures is one important aspect of internal and external validation of the quality of smart data infrastructure and should be an important element of derivative RRD metadata. Wide dissemination outside secure data environments means that RRD are unlikely to retain identifiable individual observations: however, individual level georeferencing in underpinning data infrastructure makes it possible to create metadata on the effects of aggregation across a full range of scales, consistent with uncompromised disclosure control. This is important in standalone applications such as residential segregation studies and is of value in addressing challenges of ecological fallacy in multivariate studies.

Georeferenced assemblages of raw smart data rarely link a comprehensive range of individual subject characteristics, leading to what Goodchild (2022) has referred to as ‘Balkanisation of the quantitative self’ if smart data sources are siloed. Different sources may nevertheless be linked at individual level in TREs if the necessary data licensing and data subject consents are in place. Augmentation of multiple datasets with each other is also an important aspect of smart data validation, which is otherwise not possible using pre-aggregated statistical or administrative sources. Retention of the level of the individual can also facilitate greater clarity in representing dynamics through updates over time.

The outcome of triangulation of smart data and their augmentation with other sources is spatial data infrastructure that enables geospatial analysis of linked individual characteristics for any convenient spatial aggregation. This infrastructure can be spliced and diced within a secure

research environment into RRD products of known provenance and in response to user demands. Successful delivery of RRD derived from multiple sources requires devising data licensing agreements that are sufficiently flexible to support individual level data linkage and modeling. Innovations in service delivery for RRD thus require extended legal and ethical support alongside improved scientific and technical capacity.

The changing context of GIScience

The preceding discussion of the PPDAC methodology has illustrated how the data used for decision-making in a wide range of applications areas often have a spatial component. The general context to the supply and provision of spatially referenced research-ready data (RRD) has changed profoundly over the last 20+ years and this has had far-reaching implications for the practice of applications-led science. Specifically:

- Most developed countries now have advanced spatial data infrastructures, comprising ‘framework’ data. Such data typically record the variable nature of the Earth’s surface, landscape characteristics, and a range of artificial structures ranging from roads to mail delivery points. This provides valuable context for decision-making. Even in parts of the world where this is not the case, high resolution [remote sensing](#) imagery or the data sources assembled into the ‘virtual Earths’ of Google and Microsoft can provide an acceptable framework from which environmental, population and infrastructure characteristics can be ascertained
- Advances in data dissemination technologies, such as geoportals, have encouraged governments to disseminate public sector data on social, health, economic, demographic, labor market and business activity to the widest possible audiences (e.g. www.gov.uk/government/statistics; data.gov.uk/). Many geoportals include metadata (‘data about data’) on their holdings, which can be used to establish the provenance of a dataset for a given application. The advent of geoportals is having a catalytic role in stimulating public participation in GIS ([PPGIS](#), [Communityviz](#)), and for example amongst genealogists when historic census data are put online. Of more central importance to professional users, geoportals empower very many more users to assemble datasets across domains and create added value information products for specific uses
- Advances in data capture and collection are leading to conventional data sources being supplemented, and in some cases partially replaced, with additional measures of physical, social and environmental information. Developments in the supply and assembly of volunteered geographic information are creating new spatial data infrastructures such as OpenStreetMap (OSM) (www.openstreetmap.org), that help to contextualize spatial data and enhance geovisualization. Additional framework data sources are being provided by the likes of Microsoft and Google, often underpinned by an advertising led business model, but increasingly offered as more traditional subscription or licensing based commercial arrangements. This is creating challenges for those national mapping agencies that are required to recover most of their costs through user charges and copyright enforcement. One scenario is that the monolithic structure of parts of the data industry may become obsolete. In the socio-economic realm, Longley and Harris (1999) have written about the ways in which routine customer transactions in retailing, for example, are captured and may be used to supplement census-based measures of store catchment characteristics. Although the provenance of such data sources is less certain than that of conventional censuses and other government surveys,

there is nonetheless some movement towards adoption of rolling surveys and other sample-based methods to replace conventional censuses — see further, Chapter 9

- There is now a greater focus upon metadata. As suggested above, the proliferation of data sources from formal and informal sources is leading to greater concern with the quality and provenance of data, and the situations in which such data are fit for purpose. This is particularly important post the innovation of the Internet, since data collectors, custodians and users are often now much more dispersed in space and time, and Internet GIS enables much greater apparent ease in conflating and concatenating remotely located data

- Allied to the previous two points, there is a wider realization that the potential of data collected by public sector agencies at the local level or at fine levels of granularity, remains under-exploited. There exist very good prospects for public sector organizations to ‘pool’ data pertaining to their local operations, to the good of the populations that they serve. Innovations such as Google Earth are making it easy for organizations with no previous experience of GIS to add spatial context to their own data

- The practice of science itself is changing. Goodchild and Longley (1999) have written of the increasingly interdisciplinary setting to the creation of GIS applications, of the creation and maintenance of updates of GIS databases in real time, and of the assembly and disbanding of task-centered GIS teams — most obviously in emergency scenarios, but increasingly across the entire domain of GIS activity

- An increasingly important source of geospatial data is the rise of datasets that are collected by automated systems that incorporate geolocation information (e.g. smart mobile phones, gps navigation devices, retail systems, social media systems). The quality, volume and ‘velocity’ of such datasets present numerous problems — see further, Chapter 9

Each of these developments potentially contributes to the development of GIS applications that are much more data rich than hitherto. Yet such applications may be misguided if the analyst is unaware of the scientific basis to data collection, the assumptions implicitly invoked in conflation and concatenation of different datasets, the prior application of Machine Learning (ML) and regenerative AI (see, for example, the recently developed [Conflex](#) AI-assisted software from Citigate), or general issues of data quality. Additionally, the creation of GIS applications and the dissemination of results may be subject to a range of ethical considerations, and issues of accountability and data ownership. The remaining sections of this Guide give the flavor of these issues, and illustrate the ways in which GIS based representations are sensitive to the properties of software and the data that are used. In an ideal world, all data and software would be freely available, but in practice such a world would be unlikely to prioritize investments for future applications development. In the case of software, public domain sources are by no means all open source, but it is possible to form a view as to the quality of a software product by examining the nature of the results that it yields compared against those obtained using similar software offerings and known results with test datasets. Gauging the provenance of data sources is often an altogether trickier proposition, however. In seeking to validate one dataset with reference to a second, as in the comparison of two digitized street network products or the classification of two remotely sensed data resources, analysis of mismatches is more likely to suggest uncertainties than pinpoint precisely measurable inaccuracies. Discrepancies are only ultimately reconcilable through time-consuming field

measurement, although models of error propagation and understanding of data lineage or classification method may optimize use of scarce validation resources.

While it is possible to validate some of the data used in representations of built or natural environments, validation of social measurements is an altogether more problematic task (Kempf-Leonard, 2005). Repeated measurement designs are usually prohibitively expensive, and some of the attitudes and behaviors that are under investigation may in any case be transient or ephemeral in nature. In general terms it is helpful to distinguish between *direct* versus *indirect* indicators of human populations. A direct indicator bears a clear and identifiable correspondence with a focus of interest: net (earned plus unearned) income, along with measures of wealth accumulated in assets provides a direct indicator of affluence. Direct indicators may be time-consuming or difficult to measure in practice (how affluent are you?), as well as being invasive of personal privacy. Thus many data sources rely upon indirect indicators, which are easier to assemble in practice, but which bear a less direct correspondence with the phenomenon of interest — housing tenure, car ownership and possession of selected consumer durables might be used in a composite indicator of affluence, for example. Many of these issues are crystallized in the current focus upon ‘spatial literacy’, which has been defined by the National Research Council, part of the US [National Academy of Sciences](#), as comprising the following integrated components:

- understanding spatial relationships
- developing knowledge about how geographic space is represented; and
- the ability to reason and make key decisions about spatial concepts

In this Guide we seek to demonstrate how the *practice* of GIS is fundamentally about finding out about the real (geographic) world. To quote the NRC (2006): "*Spatial thinking is a cognitive skill that can be used in everyday life, the workplace, and science to structure problems, find answers, and express solutions using the properties of space*"

Building Blocks of Spatial Analysis

This Chapter embraces a very broad range of tools and techniques, many of which are now incorporated into all modern GIS software packages. Indeed, many form part of the basic set of facilities that must be provided if the GIS software is to be compliant with the Open Geospatial Consortium's ([OGC](#)) specifications. Even if a product is not listed by the OGC as compliant it may be partially or fully compliant but simply has not been submitted to them for testing.

Initially we explain the basic geometric, querying and distance-based operations that form part of the building blocks for GIS software. These operations are often referred to as *spatial analysis* by the software providers, but are in reality little more than the essential components of a usable spatial data handling system. A number of operations, notably those relating to directional analysis and some grid-based operations, are provided in a subset of GIS software packages, but may still be regarded as core building blocks of spatial analysis.

Spatial and Spatio-temporal Data Models and Methods

Spatial Data Models and Methods

Spatial datasets enable the creation of operational models of the real world, relying on the **field** and **object** conceptualizations and using coordinate geometry to represent object classes like discrete points, ordered lines (**polylines**), and closed areas (**simple polygons**). Continuously varying phenomena, often described as surfaces or fields, are frequently represented by a **continuous grid of square cells**, where each cell holds a value (e.g., average height or strength). In most literature and software, the points/lines/areas model is called **vector data**, while the grid model is called **raster** (or image) data.

Geographic Information Systems (GIS) employ various data models, including **Computer-aided design (CAD)** for engineering, **Graphical** for simple mapping, **Image** for processing and simple grid analysis, **Raster/grid** for spatial analysis in environmental applications, and **Vector/Geo-relational topological** for cartography, socio-economic analysis, and modeling operations on geometric features. Specialized models also exist, such as **Network** for transportation and utility analysis, **Triangulated irregular network (TIN)** for surface visualization, and **Object** models for operations across all entity types (Table 4-1).

The distinction between vector and raster data is not absolute; they can be **converted (transformed)** into one another. However, transformation often leads to a **loss of information** (e.g., resolution, topological structure) and may not be reversible. For instance, converting a raster map of soil types to vector polygons involves boundary detection and potential smoothing, which, when reversed back to raster form, generally results in a slightly different output due to factors like the degree of simplification and the algorithms used. Similar issues arise when combining or manipulating datasets through filtering or resampling. The Open Geospatial Consortium (OGC) Simple Features Specification defines various methods for spatial analysis, including **spatial relations** (Equals, Intersects, Within) and **spatial analysis** (Distance, Buffer, Convex Hull, Intersection, Union, Difference) (Table 4-2). These operations, often related to **Computational Geometry**, almost always alter the data, making it standard practice to create new output files or map layers rather than modifying source datasets.

Spatio-Temporal Data Models and Methods

Historically, GIS software focused primarily on **spatial data** and often has weak capabilities for handling **spatio-temporal data**. However, the massive amounts of spatio-temporal data now available are driving the need for extended data models and analytical toolsets. This shift involves viewing the **temporal domain** as increasingly critical. Spatio-temporal data can be categorized and analyzed in several ways:

- **T-mode Data Model: Complete spatial fields** recorded at distinct points in time, viewed as a set of time slices (common in GIS, often suitable for video display). Analysis focuses on differences between slices.
- **S-mode Data Model: Complete spatial fields** viewed as a set of point locations or pixels, each having a **temporal profile** (a space-time cube, common in meteorology/oceanography). Analysis concentrates on patterns detected in the profiles.

- **Incomplete Spatial Fields:** Data recorded at **regular distinct points in space and time** (e.g., environmental monitoring stations). Analysis often focuses on time-series data and spatial field estimation.
- **Mobile Objects (Track Data):** Geospatial coordinates with time-stamps for a moving object, which may have irregular spacing or incompleteness. Additional attributes like velocity and acceleration are common.
- **Network-based Data:** Data related to networks, like traffic monitoring or events (crime, accidents) on or related to network structures, where the network provides critical context for modeling (e.g., diffusion).
- **Patterns of Points (Events) over Time:** Monitoring the evolution of point events like diseases (epidemiology) or crime activity.
- **Patterns of Regions (Zones) over Time:** Data collected regularly by zones, such as census-based socio-economic or health district data. An added complexity is changing zone boundaries over time, which may require common zoning or rasterization.

There is no single standard database model or analytical approach for these complex and large datasets. Specialized techniques have been developed for specific cases, such as the **Land Change Modeler** for land use change. **Time-series analysis** (e.g., ARMA or spatio-temporal STARMA models) and extensions to **spatial statistical models** (like spatio-temporal scan statistical procedures, demonstrated by the **SatScan** analysis for emergent crime clusters in Figure 4-0) are also used. However, the volume and complexity often necessitate the application of methods like **microsimulation** (including **Agent Based Modeling - ABM**), **machine learning** (Artificial Neural Networks - ANN, Support Vector Machines - SVN), and the growing field of **GeoAI**. Data mining, sampling, filtering, and advanced visualization are often the only practical ways to gain a full understanding of these datasets.

Geometric and Related Operations

Individual vector features (e.g. polygons) and groups of grid cells exhibit a range of distinctive spatial attributes — for example the *length* of a polyline or polygon boundary and the *area* of a polygon or grid patch. In this section we examine a number of basic geometric attributes that apply to such features. These basic attributes may be included in the attribute table record associated with each feature, or may be treated as an *intrinsic* attribute that may be computed on demand and optionally added as a field in the associated table. If it is necessary to perform operations on these attributes, for example sorting areas by size, or calculating the ratio of areas to perimeters, it is helpful and often more efficient to add the data as an explicit field if it is not already available.

Length and area for vector data

Summary: GIS Distance and Area Computations

Most GIS mapping uses projected Cartesian (x,y) coordinates, allowing distances and areas to be calculated with Euclidean geometry. Polygon areas are commonly computed using integration methods such as the trapezoidal or Simpson's rule. When polygon vertices are traversed clockwise, consecutive vertex pairs form positive and negative "rectangular" components whose sum yields the polygon's total area.

Reversing the vertex order changes only the sign of the computed area. This property enables exclusion of internal polygons (e.g., lakes or built-up areas) by subtracting holes. The method fails if the polygon is self-crossing or if y-values are negative; in the latter case, a constant offset can temporarily be added to all y-coordinates.

Cartesian methods are sufficiently accurate for small regions (under $\sim 100 \text{ km} \times 100 \text{ km}$). For larger regions, or when data is stored in latitude/longitude, lengths and areas should be computed using spherical or ellipsoidal geometry. Using Cartesian formulas over large regions introduces significant error. Accurate spherical or ellipsoidal area computation requires numerical integration; specialized tools (e.g., MATLAB's AREAS) implement such methods.

Polyline lengths and polygon perimeters are the sum of their segment lengths. The Euclidean distance from a point to a line $ax + by + c = 0$ is given by the standard point-to-line formula. Point-to-polyline distance calculations evaluate (i) perpendicular offsets when applicable, or (ii) distances to the nearest segment endpoints.

For spherical coordinates, great-circle distance can be computed using the cosine formula, but this can be numerically unstable for small angles. The Haversine formula provides a more reliable computation of spherical distance, denoted d_S . Many GIS tools do not explicitly state which formula they use; in some cases distances are approximated using Euclidean computations on geographic coordinates.

Length and area for raster datasets

Length and area computations may also be meaningful for grid datasets, and in this case the base calculations are very simple, reflecting the grid cell size or resolution (summing the number of cells and multiplying by the edge length or edge length squared, as appropriate). The difficulty in this case is in the definition of what comprises a line or "polygon". A set of contiguous cells (a clump or patch) with a common attribute value or classification (e.g. deciduous woodland) has an area determined by the number of cells in the clump and the grid resolution. The perimeter length of the clump is a little less obvious. Taking the perimeter as the length of the external boundary of each cell in the clump appears to be the simplest definition, although this will be much longer than a vectorized (smoothed) equivalent measure (typically around 50% longer). Furthermore, depending on the software used, differences in definition of what a clump is (for example, whether a clump can include holes, open edges that are not part of the perimeter, cells whose attributes do not match those of the remaining cells) will result in variations in both area and perimeter calculation.

Many environmental science datasets (e.g. soil type, land-use, vegetation cover) show a near continuous spread of attribute values across cells, with few if any distinct sharp breaks. In this case boundaries are less definitive, determined more by common consent or other procedures (see further, Section 4.2.9, [Overlay and combination operations](#)) than by clear demarcation of cell values. The determination of areas and perimeters of zones in such cases is less clear, although some progress has been made using the concept of *membership functions* (MFs). A further issue with gridded data is that the gridding model (or imaging process) results in distortions or constraints on the way in which data are processed. The distortions are primarily of three kinds — Orientation, Metrics and Resolution:

Orientation — with a single rectangular grid the allocation of source data values to cells and the calculations of lengths and areas will alter if the grid is first rotated; with two or more grids common orientation and resolution are essential if data are to be combined in any way. This may mean that source data (e.g. remote sensed images) will have been manipulated prior to provision of the gridded data to ensure matching is possible, with each source item receiving separate treatment to achieve this end, and/or that data will require resampling during processing. Of course, with multi-band data relating to a single image the orientation and resolution will always match.

Metrics — in order to compute distances for gridded data it is common to add up the number of cells comprising a line or boundary and multiplying this number by the cell edge length, E . This is essentially a “rook’s move” calculation, zig-zagging across the grid, although it may be adjusted by allowing diagonal or “bishop’s moves” also (see Figure 4-2). Diagonal lengths are then taken as $E\sqrt{2}$.

This 3x3 neighborhood model is used in many areas of geospatial analysis, including distance computations, image processing, cellular automata methods (where the complete 8-cell set that surrounds the central cell is known as the Moore neighborhood (see further, Section 8.1.2, [Cellular automata \(CA\)](#)), surface analysis, and spatial autocorrelation analysis amongst others.

These methods only provide correct values for the distance between cells in 4 or 8 directions, with global distances to all other directions (for example, those highlighted in gray in Figure 4-3) being in error by up to 41% for the 4-point rook’s move model and 7.6% for the 8-point queen’s move model. Each of the squares shown in gray below are 2.24 units from the central cell but would be calculated as either 3 units away (rook’s move) or 2.41 units (queen’s move). The position of these gray squares is sometimes described as Knight’s move, again by reference to the movement of chess pieces.

Similar issues arise in computations of cost distance, path alignment and the calculation of surface gradient, aspect and related measures (e.g. material flows). This is not always a question of what is correct or incorrect in some absolute sense, since this will be determined by the application and adequacy or appropriateness of the approach adopted, but it is a significant issue to be aware of.

Resolution — the cell size (and shape) of a grid or image file and the range of attribute values assigned to cells has an important role to play in spatial measures and analysis. Amongst the more obvious is the fact that finer resolution images/grids provide more detailed representation of points, lines and areas, as well as a finer breakdown of associated attribute data. But there are many additional factors to be considered, including the increase in data and processing overhead with finer resolutions. Another factor is the implication that an attribute value assigned to a cell applies throughout the cell, so larger cells imply a greater degree of averaging, including the possibility of missing important variations within the cell. At one extreme, assuming cells are assigned a single attribute value, a grid consisting of a single huge cell will show no variation in attribute values at all, whilst variation between cells will tend to increase as cell sizes are reduced and the number of cells increases. At the other extreme, with cell sizes that are extremely small (and thus very large in number) every cell may be different and contain a unique value or a single observation (presence/ absence). Likewise, with attributes showing binary presence/ absence all cells will show either presence or absence, whilst with continuous

variables (e.g. terrain height, soil moisture levels) entries may be continuous (e.g. any real positive number) or categorized (e.g. coded as 0-255 in an 8-bit attribute coding). Cell entries may also be counts of events, in which cases the earlier comments on cell size have a direct bearing — with one large cell all N observations will fall in the single grid cell, whilst with smaller cell sizes there will be a range of values $0 \leq n \leq N$ until with very small cell sizes the only values for n will be 0 and 1 unless events are precisely co-located.

Surface area

Projected surfaces

Most GIS packages report planar area, not surface area. Some packages, such as [ArcGIS](#) (3D Analyst toolbox) and [Surfer](#) provide the option to compute surface area. Computation of surface areas will result in a value that is greater than or equal to the equivalent planar projection of the surface. The ratio between the two values provides a crude index of surface roughness.

If a surface is represented in TIN form, the surface area is simply the sum of the areas of the triangular elements that make up the TIN. Let $T_j = \{x_{ij}, y_{ij}, z_{ij}\} \ i=1,2,3$ be the coordinates of the corner points of the j^{th} TIN element. Then the surface area of the planar triangulation (or *planimetric* area) determined when $z=\text{constant}$, can be calculated from a series of simple cross-products

Note that this expression is the same as the polygon area formula we provided earlier, but for a $(n-1)=3$ -vertex polygon.

If $z=0$ for all i , the second and third terms in this expression disappear and we are left with the result for the plane. Suppose that we have a plane right angled triangle with sides 1,1 and $\sqrt{2}$, defined by the 3D coordinates: (0,0,0), (1,0,0) and (1,1,0). This plane triangle has area of one half of a square of side length 1, so its area is 0.5 units, as can be confirmed using the first formula above. If we now set one corner, say the point (1,1,0) to have a z -value (height) of 1, its coordinate becomes (1,1,1) and the area increases to 0.7071 units, i.e. by around 40%.

If the surface representation is not in TIN form, but represented as a grid or DEM, surface area can again be computed by TIN-like computations. In this case the 8-position immediate neighborhood of each cell can be remapped as a set of 8 triangles connecting the center-points of each cell to the target cell. Connecting all cell centers in this way creates a form of TIN and surface computation proceeds as described above – this is the procedure used in [ArcGIS](#) (3D Analyst toolbox). A more sophisticated method involves a similar form of computation but calculated on a cell-by-cell weighted basis. In this procedure the surface area for each cell is taken as a weighted sum of eight small triangular areas. The area of triangles centered on the target cell are computed and 25% of each such area is assigned to the central cell (shaded gray in Figure 4-5B)

Research has suggested that the resulting total surface area for the cell is close to that produced by TIN models for grids of 250+ cells. We can examine this model more closely by taking a specific example. Let us assume that the 3x3 DEM surface segment in Figure 4-5 has cells which are 1 meter square and elevation values given in meters as shown below:

The nine individual cells can be referenced in various ways – below use a simple numerical referencing around the central or target cell.

The planar area of each cell is simply 1×1 , so 1 square meter. To compute the estimated *surface* area we need to calculate the area of each of the 8 sloping triangles and allocate 25% of this area to the central cell. To do this we use the determinant expression shown earlier. For the triangle shown in Figure 4-5A (the triangle connecting cells 1-2-0) this gives an area of $3/2$, so 25% of this gives $3/8$ units as opposed to the $1/8$ units that to a flat triangle such as 0-4-5. Continuing this computation for all 8 triangles gives the overall estimated surface area using this algorithm as 2.5 sq.m. The process is then repeated for every pixel, adjusting for edge effects where necessary as with many other operations of this type (e.g. by adding a temporary row or column to each edge that duplicates the existing values).

This example is unrealistic of course since in most DEM datasets the horizontal and vertical scales are not the same. For example, if the individual cell sizes in the example above are 10 meters x 10 meters and the vertical measurements are in meters, the procedure yields a surface area approximately 3% greater than the planar area. Grid surface areas can also be computed using the slope adjustment method described above assuming that a slope grid has been first been computed, or using grid data with linear interpolation to facilitate simpler 3D quadrilateral computations.

Surface areas are computed relative to a reference plane, usually $z=z_{min}$ or $z=0$. In some instances it may be necessary to specify the value to be associated with the reference plane, and whether the absolute values, positive values and/or negative values are to be computed. [Surfer](#), for example, supports all three options which it describes as positive planar and surface area, negative planar and surface area and total planar and surface area. For example, consider the surface illustrated in Figure 4-6. This shows a surface model of a GB Ordnance Survey DEM for tile TQ81NE, which we shall use as a test surface in various sections of this Guide. This is a 5000mx5000m area provided as a 10mx10m grid of elevations, with an elevation range from around 10m-70m. Using a reference plane of $z=30$ m the surface area above this reference plane is roughly 1.3sq kms and below 30m is roughly 2.1sq kms (3.4sq kms in total compared to 2.5sq kms for the planar area). For details of volume computations, which also use reference surfaces, see Section 6.2.6, [Pit filling](#).

Terrestrial (unprojected) surface area

For large geographic regions, Earth curvature significantly affects surface-area calculations, making spherical trigonometry or numerical integration preferable to Cartesian methods. The surface area north of a given latitude ϕ on a spherical Earth (radius $R \approx 6378$ km) can be computed analytically, and the area of a latitude–longitude quadrangle is obtained from the difference between such zones scaled by the longitude span. Internal angles must be derived from true great-circle azimuths if only latitude/longitude coordinates are known. Tools such as MATLAB's AREAQUAD and AREAINTE, and GRASS GIS modules (e.g., `area_poly1` and `area_poly2`), provide automated spherical or ellipsoidal area calculations, with AREAINTE using numerical integration and additional vertices for improved accuracy.

Line Smoothing and point-weeding

Polylines and polygons that have a large number of segments may be considered over-complex or poor representations of the original features, or unsuitable for display and printing at all map scales (a problem of *map generalization*). A range of procedures exist to achieve simplification of lines in such cases, including those illustrated in Figure 4-7.

A. Point weeding is a procedure largely due to Douglas and Peuker (1973) — for a fuller discussion, see Whyatt and Wade (1988) and Visvalingam and Whyatt (1991). In the example shown in Figure 4-7a the original black line connecting all the vertices 1 to 10 is progressively simplified to form the dotted (green) line shown. The procedure commences by connecting point 1 (the anchor) to point 10 (the floating point) with a temporary line (the initial segment). This enables point 8 to be identified as the vertex that deviates from this initial segment by the greatest distance (and more than a user-defined tolerance level, shown here by the bar symbol on the left of the upper diagram). Euclidean distance squared is sufficient for this algorithm and is significantly faster to compute. Point 10 is then directly connected to point 8 and point 9 is dropped. A new temporary line is drawn between point 8 (the new floating point) and 1 and the process is repeated. This time points 5, 6 and 7 are dropped. Finally point 3 is dropped and the final line contains only 4 segments and 5 nodes (1, 2, 4, 8 and 10).

This process can be iterated by moving the initial anchor point to the previous floating point and repeating the process for further simplification. Some implementations of this algorithm, including those of the original authors, fail to handle exceptions where the furthest point from the initial segment is one end of a line that itself lies close to the initial segment. Ebisch (2002) addresses this error and provides corrected code for the procedure. The procedure is very sensitive to the tolerance value selected, with larger values resulting in elimination of more intermediate points.

A number of GIS products provide a range of simplification methods. The default in [ArcGIS](#) is point-weeding (POINT_REMOVE) but an alternative BEND_SIMPLIFY which removes extraneous bends is also available.

B. Simple smoothing utilizes a family of procedures in which the set of points is replaced by a smooth line consisting of a set of curves and straight line segments, closely following but not necessarily passing through the nodes of the original polyline. This may be achieved in many ways with Figure 4-7b showing one example.

C. Spline smoothing typically involves replacing the original polyline with a curve that passes through the nodes of the polyline but does not necessarily align with it anywhere else, or approximates the point set rather like a regression fit (Figure 4-7c). A range of spline functions and modes of operation are available (including in some cases [Bézier curves](#) and their extension, B-splines), many of which involve using a series of smoothly connected cubic curves. In some cases additional temporary nodes or “knots” are introduced (e.g. half way between each existing vertex) to force the spline function to remain close to the original polyline form.

Each method and implementation (including the selection of parameters and constraints for each variant) will have different results within and across different GIS packages and the process will not generally be reversible. Furthermore, different GIS packages tend to offer only one or two smoothing procedures, so obtaining consistent results across implementations is almost impossible. Note that once a polyline or polygon has been subject to smoothing its length will be altered, becoming shorter in many cases, and enclosed areas will also be altered though generally by a lesser amount in percentage terms. The orientation of the line segments also changes dramatically, but not the relative positions of the two end nodes, so the end-node to end-node orientation is retained. Determination of the length and area of amended features may require numerical integration utilizing fine division of the smoothed objects rather than

the simple procedures we have described in Section 4.2, [Geometric and Related Operations](#). Further discussion of smoothing, including additional approaches, is provided in Chapter 6, [Surface and Field Analysis](#). These methods have been described by McMaster and Shea (1992) as forms of *feature generalization* (see also, Section 2.1.9, [Detail, resolution, and scale](#)). Other forms of generalization include: *amalgamation* (replacing a large number of separate area features or cells by a smaller number, e.g. a single feature); *merging* (replacing several line features with a single line feature); and *collapse* (replacing an area object by a point, cells or a combination of points and lines, or replacing dual lines by a single centerline. For a fuller discussion of generalization issues see the ESRI (1996) white paper, available via the [ESRI](#) web site.

Centroids and centers

The terms center and centroid have a variety of different meanings and formulas, depending on the writer and/or software package in which they are implemented. There are also specific measures used for polygons, point sets and lines so we treat each separately below. Centers or centroids are provided in many GIS packages for multiple polygons and/or lines, and in such cases the combined location is typically computed in the same manner as for point sets, using the centers or centroids of the individual objects as the input points. In almost all cases the metric utilized is d_E or d_S , although other metrics may be more appropriate for some problems. Centroids are often defined as the nominal center of gravity for an object or collection of objects, but many other measures of centrality are commonly employed within GIS and related packages. In these latter instances the term center (qualified where necessary by how this center has been determined) is more appropriate than centroid, the latter being a term that should be reserved for the center of gravity. Note that in this section only geometric centroids are considered, i.e. those based on form rather than process.

Polygon centroids and centers

Polygon centers have many important functions in GIS: they are sometimes used as “handle points”, by which an object may be selected, moved or rotated; they may be the default position for labeling; and for analytical purposes they are often used to “represent” the polygon, for example in distance calculations (zone to zone) and when assigning zone attribute values to a point location (e.g. income per capita, disease incidence, soil composition). The effect of such assignment is to assume that the variable of interest is well approximated by assigning values to a single point, which essentially involves loss of information. This is satisfactory for some problems, especially where the polygons are small, homogeneous and relatively compact (e.g. unit ZIP-code or postcode areas, residential land parcels), but in other cases warrants closer examination.

With a number of raster-based GIS packages the process of vector file importing may utilize polygon centers as an alternative to the polygons themselves, applying interpolation to the values at these centroids in order to create a continuously varying grid rather than the common procedure of creating a gridded version of discretely classified polygonal areas (see further Chapter 6). Similar procedures may be applied in vector-based systems as part of their vector-to-raster conversion algorithms.

The Mean Center is not the same as the center of gravity for a polygon.

An alternative to using the MBR is to find the smallest circle that completely encloses the polygon, taking the center of the circle as the polygon center (Figure 4-11, M4). This is the default in the [Manifold](#) GIS (viewable simply by selecting the menu option to View|Structure|Centroids) and/or using the Transform function “Centroids”. This procedure suffers similar problems to that of using the MBR. Yet another alternative is to find the largest inscribed circle, and take its center as the polygon center (Figure 4-11, M5). This approach has the advantage that the center will definitely be located inside the polygon, although possibly in an odd location, depending on the polygon shape. If the polygon is a triangle then M5 and M2 will coincide. [Manifold](#) supports location of centers of types M2, M3 and M4.

As we can also see in Figure 4-10 multiple polygons may be selected and assigned combined centers. In this example the combined center appears to have been computed from the MBR of the two selected census tracts, but different packages and toolsets may provide alternative procedures and thus positions. In such cases the center may be nowhere near any of the selected features, and if it is important to avoid such a circumstance the GIS may facilitate selection of the most central feature (e.g. the most central point or polygon) if one exists. In some instances the combined center calculation may be weighted by the value of an attribute associated with each polygon. This procedure will result in a weighted center whose location will be pulled towards regions with larger weights. However, in such cases the distribution of polygons used in the calculation may produce unrepresentative results.

This is apparent again in Figure 4-10 where a polygon center calculation for all census tracts in the western half of the region, weighted by farm revenues, would be pulled strongly to the east of the sub-region where many small urban tracts are found, even though these may have relatively low weights associated with them.

Point sets

In this version of the expression for M1 there is an additional (optional) component, w_i , representing weights associated with the points included in the calculation. For points sets the weights might be the number of crimes recorded at a particular location, or the number of beds in a hospital; for multiple polygons, where these are represented by their individual centers or centroids, the weights would typically be an attribute value associated with each polygon. Clearly if all weights=1 the formula simplifies to a calculation that is purely based on the coordinate values, with the number of points, $n=\sum w_i$.

M1 in this case is the center of gravity, or *centroid*, of the point set assuming the locations used in the calculation are treated as point masses. M1 is the location that minimizes the sum of (weighted) *squared distance* to all the points in the set, is simple and fast to compute, and is widely used. However, it is not the point that minimizes the sum of the (weighted) *distance* to each of the points. The latter is known sometimes as the (bivariate) “median center” (although [Crimestat](#), for example, uses the term median center to be simply the middle values of the x - and y -coordinates). It is best described as the MAT point (the center of Minimum Aggregate Travel, M6). This point may be determined using the following iterative procedure, with M1 as the initial pair (x_0, y_0) and $k=0, 1 \dots$:

In this expression $d_{i,k}$ is the distance from the i^{th} point to the k^{th} estimated optimal location. It is usual to adjust distances in this, and similar formulas involving division by interpoint distances, by a small increment and/or to apply code checks to avoid divide-by-zero or close

to zero situations. For this particular problem testing the numerically estimated surface derivative and curvature, in addition for checking for division by zero, has been shown to be an effective approach to avoiding overshooting the true optimum (see Ostresh's WEBER algorithm, in Rushton *et al.*, 1973). Iteration is continued until the change in the objective function (the cumulative distance) or both coordinates is less than some pre-specified small value. These formulas for M1 and M6 can be derived by taking the standard equation for distance, d_E , or distance squared, d_E^2 , and partially differentiating first with respect to x and then with respect to y , and finally equating the results to 0 to determine the minimum value. An extension to this type of weighted mean is provided by the Geographically Weighted Regression ([GWR](#)) software package, and is described further in Section 5.6.3. With the [GWR](#) software it is possible to compute a series of locally defined (geographically weighted) means, and associated variances, which may then be analyzed and/or mapped.

The positions of M1, M3 and M6 are illustrated in Figure 4-12, using the same set of coordinates as for the vertices A-F of the polygon described above.

Note that $M1=M2$ in this case, and for both M1 and M3 (the MBR center) their position is unchanged from the polygonal case. Also note that if we move point B to B' the position of M6 is unchanged (although the cumulative distance will be greater). This observation is true for each of the points A-F — they can be moved away from the MAT point to an arbitrary distance along a line connecting them to M6, and the position of M6 will be unaffected. This would not generally be the case with M1 or M3.

The locations described for different types of center often assume that all points (or polygons) are weighted equally. If the set of weights, w_i , are not all equal, the locations of M1 and M6 will be altered, but M3 will be unchanged. The effect of unequal weights is to “pull” the location of M1 and M6 towards the locations with higher weights. For example, if point B in our previous example had a weight of 3, M1 would be moved to (9.00,5.63) just to the right of M6 in Figure 4-12, M3 would be unaltered, and M6 would be altered to (10.41,4.29) which is very close to point C. Any weights in the dataset that are zero or missing will generally result in the point being removed from the calculation. Such occurrences require checking to ensure valid data are not being discarded due to incomplete information or errors.

There is no difference between weighted calculations where the weights are integer values and the standard calculations for unit weights if some points in a set are co-located. For example, if point B is recorded in a dataset 3 times, its effective weight would be 3, and instead of 6 points there would be 8 to consider. Co-located point recording is very common, especially in crime and medical datasets where each incident is associated with a nominal rather than precise location. Examples of co-located data might be the closest street intersection to an incident, the nominal coordinates of a shopping mall, the location of the doctor's surgery where a patient is registered, or a location that recurs because incidents or cases have been rounded to the nearest 50 meters for data protection reasons. It is often a good idea to execute queries looking for duplicate and unweighted locations before conducting analyses of this type, since mapped datasets may not reveal the true underlying patterns. Likewise it is important to check that co-located data are meaningful for the analysis to be undertaken — surgery location is not generally a substitute for a patient's home address.

The preceding calculations have all been carried out using the standard Euclidean metric, d_E . As previously stated, this is the standard for GIS packages, but other metrics may be more

appropriate depending on the problem at hand (see further, Section 4.4.1). Specialist packages like the now defunct LOLA project and [Crimestat](#) support a range of other metrics. Using the city block metric (L_1) and the minimax metric (L_∞) the MAT point, M6, is no longer guaranteed to be unique — all points within a defined region may be equally close to the input point set ([Crimestat](#) only provides a single location, so LOLA or similar facilities are preferable for such computations). For example, the point set in Figure 4-12 with L_1 metric has an MAT solution point set (a rectangle) bounded by (4,4) and (10,8). Clearly, if the point set lay on a network the location of M6 would again be different.

To add to the confusion of the above [Crimestat](#) provides three further measures of centrality for point sets. Each involves variations in the way the mean is calculated (see further, [Table 1-3](#)): the geometric mean (the x - and y -coordinates are calculated using the sum of the logarithms of the coordinates, averaging and then taking antilogs); the harmonic mean (the x - and y -coordinates are calculated using the reciprocal of the coordinates, averaging and then taking the reciprocal of the result); and the triangulated mean (a [Crimestat](#) “special”). The first two alternative means are less sensitive to outliers (extreme values) than the conventional mean center, whilst the latter measure is claimed to represent the directionality of the data better. The [Crimestat](#) manual, Chapter 4, provides more details of each measure, with examples. Note that the harmonic mean is vulnerable to coordinate values which are 0 or close to 0.

Lines

In the case of a line (single segment, polyline or curve) the common notion of the center is simply the point on the line that is equidistant from the two endpoints. This provides both the (intrinsic) center of gravity and the mean center, although when a polyline is viewed as embedded in a plane its “center” should be considered to lie in the plane and not on the line.

For collections of lines there is no generally applied formula and a common central point might be selected from the centers or centroids of the individual elements, or from the MBR, or utilizing a central feature if one exists. Combinations of points, lines and polygons are treated in a similar manner. Because lines and line segments have a well-defined orientation, labeling tools that utilize line or segment centroids may also use the orientation of that element to align associated labels.

Point (object) in polygon (PIP)

One of the most basic of spatial operations is that of determining whether a given point lies inside a polygon. More generally the problem extends to the case of multiple points and polygons, with the problem being to assign every point to the correct polygon. Related problems include line in polygon and polygon in polygon tests. A fast initial screening approach is to check whether a point (or line, polygon) lies in the polygon’s MBR. However, the standard algorithm for determining PIP in the vector model (known as the semi-line algorithm) is to extend a line vertically upwards (or horizontally, e.g. to the right) and then count the number of times this line crosses the polygon boundary. If the line crosses the boundary an odd number of times it lies inside the polygon. This is true even when the polygon is concave or contains holes. A number of special cases need to be considered, as illustrated in Figure 4-13, below. These include points that lie on the boundary or at a vertex, and points that lie directly below a vertical segment of the polygon boundary. A useful discussion of PIP algorithms and related procedures is provided in Worboys and Duckham (2004, pp 197-202).

If the given point lies on the boundary or at a vertex of the polygon there is a further difficulty, since a unique assignment rule could prevent the point from being allocated to any polygon. Solutions to such cases include: assigning the point to the first polygon tested and not to any subsequent polygons; randomly assigning the point to a unique polygon from the set whose boundary it lies on; reporting an exception (non-assignment); assigning a weight value to the point (e.g. 0.5 on a boundary or $1/n$ at a vertex, where n is the number of polygons meeting at that vertex).

Where the PIP process is one of overlaying, the procedure within a GIS will typically append attribute data from the polygon to the point, although this may be in accordance with user-selectable transfer rules (e.g. sum or average a particular field). [ArcGIS](#) includes two PIP-related topology rules: (i) point must be properly inside polygon; and (ii) point must be covered by boundary of polygon (i.e. must be exactly on the boundary). These are two of 16 main vector topology rules supported within [ArcGIS](#) for combinations of point, line and polygon features.

A second PIP algorithm of similar computational complexity, but requiring trigonometric operations in its standard form (so more processor intensive) is known as the *winding number* method. In this procedure a line is extended to each vertex of the polygon from the sample point in turn (traversing anticlockwise). If the sum of the angles from the point to the vertices, v_i , is 0 the point lies outside the polygon, otherwise it lies inside or on the polygon. To clarify this procedure, let the polygon have n vertices $v_i = v_1, v_2, v_3, \dots, v_n = v_0$ and consider a unit vector from a sample point, p , to each vertex in turn. Define the (signed) angle at p between the vectors from p to v_i and v_{i+1} , as θ_i .

If $\sum \theta_i = 0$ the point lies outside the polygon, otherwise it lies inside or on the polygon boundary. With some adjustment the algorithm can avoid calculating the angle and result in a computation that is as fast, or faster than crossing methods (tests suggest 20-30% faster for many non-convex polygons). It is also more general, in that it applies to cases where polygons are not simple, i.e. self-crossing of the polygon boundary is permitted.

The above addresses the issue for the case of a single polygon and for multiple non-overlapping contiguous polygons. With overlapping polygons, which is supported for example within the [Manifold](#) GIS, then unique allocation is not assured. Furthermore, if the sample region is not completely covered by polygons sample points may fall into no polygon. Similar issues arise with raster representation of points and polygons (as zones of continuous fields) but the problem of locating points inside zones is far simpler, whilst the allocation issues remains.

Polygon decomposition

Polygons may be divided into smaller sections in many different ways, for a wide variety of reasons. Most vector GIS packages support some decomposition procedures, through overlay of a template or *cookie cutter* polygons (e.g. the Clip operation in [ArcGIS](#)) and/or through the provision of explicit decomposition procedures. For example, complex polygons with convoluted boundaries may be divided up into a series of non-overlapping convex polygons. These polygons will have centers (e.g. centers of gravity) that are guaranteed to lie inside the original polygon. The multiple centers thus generated may then be used to assist labeling or in assigning data in spatial overlay operations. Polygons can also be divided into non-overlapping triangles, in a variety of different ways. [Manifold](#), for example, provides commands to perform both such decompositions: Transform|Decompose to convex parts, and Transform|Decompose

to triangles. A (simple) polygon with n vertices or nodes can be triangulated into $n-2$ triangles using a procedure known as diagonal triangulation. This form of triangulation utilizes existing polygon vertices, but other triangulations are possible if additional internal vertices are permitted. With m such internal vertices (or Steiner points) a total of $n+2m-2$ triangles will be generated.

A polygon can also be skeletonized, providing an additional view of its form and key internal points. In Figure 4-14, below, bands of increasing distance in constant steps from the external boundary are shown. These terminate in a single central point and generate one intermediate point. Connecting these internal points to the polygon nodes creates a skeleton of lines known as the medial axis of the polygon. The medial axis is not necessarily comprised of straight line segments. In practice skeletonization is carried out using algorithms known generically as medial axis transforms, and may be computed in linear time (i.e. as a linear function of the number of polygon vertices, see Chin *et al.*, 1999). The single central point in the example shown in Figure 4-14 is the center of the largest inscribed circle.

Applications of skeletonization include label placement, especially useful for complex polygons which are concave and/or contain holes, and polygon generalization/smoothing, which has been applied to closed polygons and to open regions such as contour maps — see for example Gold and Thibault (2001). The reverse process, “given a skeleton (e.g. stream pattern) reconstruct a landscape”, has also been the subject of much research. Furthermore, the dual of skeletonization is one form of polygon decomposition, although not generally the version implemented within currently available GIS software. Neither does such software tend to provide skeletonization facilities, but asymmetric buffering (see Section 4.4.4, [Buffering](#)), which is a commonly available feature, enables first-stage skeletonization to be carried out.

Shape

Here is a clean, concise **multi-paragraph summary** suitable for inclusion in a RAG-ready PDF.

Polygon shape plays an important role in GIS analyses, influencing center calculations, compactness assessments, and decomposition into simpler elements. Numerous shape indices have been proposed, but defining a universally satisfactory measure is difficult. Shape metrics often mirror those used for grid-based landscape patches, where complexity is commonly expressed using perimeter-to-area ratios, fractal dimensions, or comparisons to simple Euclidean shapes. In general, higher values indicate more irregular or complex forms. Vector-based GIS systems typically do not compute such metrics automatically, so users must derive them—usually from polygon area (A_i) and perimeter (L_i).

Basic shape measures include the perimeter-to-area ratio (P1A), which is sensitive to size and therefore unsuitable for many applications, and the perimeter-squared-to-area ratio (P2A), used for example in UK census output area design. More robust is the compactness ratio (C), which is dimensionless, ranges from 0 to 1, and equals 1 only for a perfect circle; variations invert the index to express irregularity. Another class of measures compares the polygon to an associated bounding figure (RBF), such as the minimum bounding circle, convex hull, or minimum bounding rectangle. The resulting indices, ranging from 0 to 1, can be computed for many polygons and analyzed using conventional statistical summaries or histograms.

More complex measures involve characteristics such as major/minor internal axes or sets of straight lines within a polygon, but these usually require custom programming. Shape metrics can also extend to point sets—commonly through convex hulls or the standard deviational ellipse, which describes orientation and dispersion. For linear objects, measures such as stream-length-to-basin-area ratios are often built into GIS tools, while polyline complexity can be expressed as the ratio of actual polyline length to Euclidean (or spherical) straight-line distance. Network-level measures include Karsky's circuitity index, which compares network distances with Euclidean distances across all vertex pairs and can be disaggregated to assess individual vertices; weighted versions may incorporate travel preferences or "desire lines." Many such indices require custom calculation either inside the GIS or using external tools.

Overlay and combination operations

Overlay operations involve the placement of one map layer (set of features) *A*, on top of a second map layer, *B*, to create a map layer, *C*, that is some combination of *A* and *B*. *C* is normally a new layer, but may be a modification of *B*. Layer *A* in a vector GIS will consist of points, lines and/or polygons, whilst layer *B* will normally consist of polygons. All objects are generally assumed to have planar enforcement, and the resulting object set or layer must also have planar enforcement. The general term for such operations is topological overlay, although a variety of terminology is used by different GIS suppliers, as we shall see below. In raster GIS layers *A* and *B* are both grids, which should have a common origin and orientation — if not, resampling is required.

The process of overlaying map layers has some similarity with point set theory, but a large number of variations have been devised and implemented in different GIS packages. The principal operations have previously been outlined as the spatial analysis component of the [OGC](#) simple features specification (Table 4-2). The open source package, [GRASS](#), is a typical example of a GIS that provides an implementation of polygon overlay which is very similar to conventional point set theory (Figure 4-15). Functions provided by [GRASS](#) include:

- *Intersection*, where the result includes all those polygon parts that occur in both *A* and *B*
- *Union*, where the result includes all those polygon parts that occur in either *A* or *B*, so is the sum of all the parts of both *A* and *B*
- *Not*, where the result includes only those polygon parts that occur in *A* but not in *B* (sometimes described as a Difference operation), and
- *Exclusive or* (XOR), which includes polygons that occur in *A* or *B* but not both, so is the same as (*A* Union *B*) minus (*A* Intersection *B*)

[TNT/Datum Workstation](#) provides similar functionality and uses much the same terminology as [GRASS](#) (AND, OR, SUBTRACT, XOR) under the heading of vector combinations rather than overlay operations, and permits lines as well as polygons as the "operator" layer.

Note: (a) these operations may split up many of the original polygons (and/or lines) to create a new set of features, each with their own (inherited and derived) attributes (lengths, areas, perimeters, centroids, and associated attributes from *A* and/or *B*). New composite regions (with associated combined attribute datasets) are often the desired end result, in which case borders between adjacent polygons may be dissolved and datasets merged in order to achieve this end result; (b) operations are not generally symmetric, i.e. *A*<operation>*B* is not necessarily the

same as $B \langle \text{operation} \rangle A$, both from a point of view of the geometry and associated attribute data held in tables.

In [ArcGIS](#) operations of this type are known by the general term “Overlay”, whilst in [Manifold](#) the [ArcGIS](#) Overlay operations are described as *Topological Overlays* to distinguish these composite operations from the more component based operations that [Manifold](#) describes as *Spatial Overlays*. Other vector GIS packages use descriptions such as *cookie cutters* (or EXTRACT operations) and *spatial joins*, and without doubt the terminology is confusing — the exact consequences of such operations applied using any particular GIS package will be different and should be examined very carefully. The [ArcGIS](#) implementation provides four main Overlay functions: Intersection and Union, which are essentially similar to those illustrated in Figure 4-15, and two additional operations: Identity, which splits up the elements in B based on the position of elements in A ; and Update, which amends those parts of B that intersect with A such that in these regions B 's geometry and associated data reflects those of A .

The [Manifold](#) GIS generalizes the notion of overlay to include spatial proximity, for example lines that touch areas, and areas that are adjacent to (immediate neighbors of) other areas..

[Manifold](#) also permits polygons within a layer to be overlapping, which can result in more complex output patterns. In all cases the [Manifold](#) Spatial Overlay process transfers data (fields) from a source object (set of features of the same type) to a target object (also a set of features of the same type) according to transfer rules. These rules specify how the data fields are to be transferred (e.g. copied, summed) conditional upon the spatial overlay method. As such they are rather like an SQL "Update table" or "Create table" command, subject to "Where" conditions. Polygon-on-polygon overlay, which is a relatively common procedure, frequently results in the creation of very small thin polygons known as *slivers*. These may be the genuine result of an overlay operation (i.e. valid objects and data) or they may be artifacts, created as a result of differences in the original data capture, manipulation or storage process associated with A and B , even though they should exactly match in places. Such slivers may be removed automatically during the overlay operation, usually by setting tolerance levels, or by post-processing, e.g. removing all polygons with a width less than a tolerance value and replacing these with an arc along the sliver centerline or assigning them to the largest adjacent polygon (e.g. in [ArcGIS](#) use the Eliminate function; in [Manifold](#) use the Normalize Topology function).

Overlays involve operations where two distinct sets of spatial object are combined in particular manner, visualized as overlaying one upon the other. Many other forms of combination are possible. For example, with a single layer of polygons, boundaries between polygons could be removed (*dissolved*) if the polygons are adjacent/share a common boundary and meet certain criteria. The attributes of the newly formed polygon will be the sum or some other measure of the component parts. If a new map layer is created by this process rather than an existing layer modified the process is sometimes described as *merging* rather than dissolving. [Manifold](#) permits the use of the Dissolve operation on lines and points as well as polygons. In the case of lines, multiple line segments are replaced with a single line segment, whilst with points multiple points are replaced by a single center point. The attribute used to guide the dissolve process may be transferred to the new object(s) as: Blank (i.e. do not transfer values for this field); Average; Count; Maximum; Minimum; Sample (randomly chosen value from the original object); or Sum. Other fields are transferred according to transfer rules set for

each field. [ArcGIS](#) supports two additional composite operations: Erase and Symmetric difference. In the Erase operation objects in A overlaid on B result in the removal of those parts of B that intersect with A (essentially the NOT operation of [GRASS](#)). Symmetric difference is similar to Erase, but those parts of A that do not intersect with B are included in the output set also (essentially the XOR operation we described earlier for [GRASS](#)). In addition [ArcGIS](#) supports operations such as Split, Append (a merge operation) and Integrate, which combine or separate features according to well-defined rules. Collectively within [ArcGIS](#) these various facilities are described as Geoprocessing Tools, and in [ArcGIS](#) V8 were available via a “Geoprocessing Wizard” but in [ArcGIS](#) V9 and later editions (ArcGIS Pro) are provided in Geoprocessing Tools as separately identified functions.

Areal interpolation

The polygon-on-polygon overlay process highlights an important problem: “how should attributes be assigned from input polygons to newly created polygons whose size and shape differ from the original set?” This is a specific example of a closely related problem, that of re-districting. Suppose that we have a set of non-overlapping polygons that represent census tracts, and a second set of polygons that represent hospital service districts, generated (for example) as simple Voronoi polygons around each hospital (see further, Section 4.2.14, [Tessellations and triangulations](#)). We wish to estimate the population numbers and age/sex mix for the service districts. Based on this information we might assign budgets to the hospitals or iteratively adjust the service districts to provide a more even set of service areas (see further, Section 4.2.11, [Districting and re-districting](#)). In a simple overlay procedure we calculate the proportion of each census district that intersects with each separate service zone (by area) and then assign this proportion of the relevant attribute to the hospital in question. This assumes that the distribution of each attribute is constant, or uniform, within each census tract while also ensuring that the total counts for each attribute remain consistent with the census figures (so-called volume-preserving or *pyncophylactic* assignment). The standard intersection operator in many GIS packages will not carry out such proportional assignment, but simply carry over the source attributes to the target polygons. Assuming the source attribute table includes an explicit area value, and the intersected target provides an explicit or intrinsic measure of the intersection areas, then the proportional allocation procedure may be carried out by adding a calculated field containing the necessary adjustments. Note that this discussion assumes spatially extensive attributes (see Section 2.1.2, [Attributes](#)) whereas with spatially intensive attributes alternative procedures must be applied (for example, initially using kernel density estimation — see Section 4.3.4, [Density, kernels and occupancy](#), for more details). Figure 4-16 and Figure 4-17 illustrate this process for test UK census Output Areas in part of Manchester. The population totals for each source area are shown in Figure 4-16, together with the area of overlap with a sample 1 km square region and a selected source polygon intersection (highlighted in red/darker gray).

Figure 4-17 shows the result of applying area-based proportional assignment to the square region in question. The region highlighted in red/darker gray in the Figure 4-16 has its population estimate of 173 reduced to 32 in this case, with the sum of these proportional assignments (1632) providing the estimated total population for the entire square. Variants of this procedure attempt to correct for the often unrealistic assumption of uniformity in the spatial distribution within polygons. If data are available for smaller sample regions (e.g. unit postcode areas in the above example) then this may be utilized in a similar manner to that already

described. Another alternative is to model the distribution of the variable of interest (e.g. population) as a continuous surface. Essentially this procedure assigns the attribute value of interest to a suitable polygon center and then calculates the attribute value over a fine grid within the polygon by taking into account the values of the attribute at adjacent or nearby polygon centers.

For example, if the source attribute values were much higher east of the target polygon than west, the values assigned within the source polygon might be assumed to show a slope from east to west rather than be distributed uniformly. Having estimated values for the *source* polygons on the fine grid, these values are then summed for each target polygon and adjusted to make sure that the totals match those for the original polygon. The individual grid cells may then be re-assigned to the *target* polygons (the hospital service areas in our example) giving a hopefully more accurate picture of the attribute values for these service areas. Current mainstream GIS packages do not tend to support the latter procedure, although simple sequences of operations, scripts or programs can be written to make such field assignments.

The above, adjusted volume-preserving assignment approaches, have significant drawbacks where the geography of the study region is highly variable and/or where attributes do not vary in a similar manner (do not exhibit strong positive covariance). For example, population is typically concentrated in urban areas and not spread evenly across arbitrary zones. In urban-rural borders and rural areas with distinct settlements a far better approach would be to utilize ancillary information, such as land-use, road network density (by type) or remote sensed imagery, to adjust attribute allocation. This ancillary data, which is often selected on the basis that it is readily available in a convenient form for use within GIS packages, can then be used as a form of weighting. A number of tests have shown that the quality of areal interpolation can be substantially improved by such methods (see for example, Hawley and Moellering, 2005).

Districting and re-districting

A common analysis problem involves the combination of many small zones (typically stored as polygons) into a smaller number of merged larger zones or districts. This merging process is usually subject to a set of spatial and attribute-related constraints. Spatial constraints might be:

- districts must be comprised of adjacent (coterminous) regions
- districts must be sensible shapes e.g. reasonably compact

Attribute constraints might be:

- no district may have less than 100 people
- all districts must have a similar number of people (e.g. within a target range)

This kind of regionalization problem comes in many forms, depending on the precise nature of the problem to be tackled and constraints applied. For example, problems may seek solutions for which a given attribute exceeds some pre-specified threshold value (see further, Duque *et al.*, 2012) or may seek to define zones that maximize within-region flows, often referred to as *functional regions*. In general the problems can only be solved for non-trivial numbers of source zones using heuristic algorithms. Whilst these will not produce provably optimal solutions (even if such solutions exist) they may be able to produce good quality solutions

(substantial improvements over random or manual solutions) within a reasonable amount of processing time. In some instances solutions can be compared with known optimal solutions, enabling the quality of alternative solution procedures to be assessed formally.

Several GIS packages now facilitate the automatic or semi-automatic creation of districts, or the re-organization of existing districts. These operations have many applications, from the designation of service areas to the definition (and re-definition) of census and electoral districts. For example, within [Manifold](#) (with the Business Tools extension) there are facilities to enable an existing set of districts to be re-assigned automatically to one of N new districts, such that each new district has a similar (balanced) value by area (an intrinsic attribute) or by a selected attribute (optionally weighted). This automated process may yield unsatisfactory results (for example, very convoluted districts) and an alternative, visual interface facility, is provided to enable the user to make manual alterations and see the results in terms of the attribute or attributes of interest. In the case of [ArcGIS Districting](#) is provided as a free downloadable add-in, and is essentially a manual operation supported by map and statistical windows (CitygateGIS's [autoboundEdge](#) provides similar functionality as a standalone product, i.e. without the need for a separate GIS application).

Districting and re-districting are generally processes of agglomeration or construction. The initial set of regions is reduced to a smaller set, according to selected rules. Automating this process involves a series of initial allocations, comparison of the results with the constraints and targets, and then re-allocation of selected smaller regions until the targets are met as closely as possible. As with many such problems, it may not be possible to prove that a given solution is the best possible (i.e. is optimal), but only that it is the best that has been found using the procedures adopted. Creating new districts can be a confusing process. In addition to the kinds of issue we have already discussed there are two important affects to be aware of: scale (*grouping* or *statistical*) effects; and zoning (*arrangement*) effects. The first of these issues, scale effects, is best understood through the use of a simple example. Consider the employment statistics shown in Table 4-4. Areas A and B both contain a total of 100,000 people who are classified as either employed or not. In area A 10% of both Europeans and Asians are unemployed (i.e. equal proportions), and likewise in Area B we have equal proportions (this time 20% unemployed). So we expect that combining areas A and B will give us 200,000 people, with an equal proportion of Europeans and Asians unemployed (we would guess this to be 15%), but it is not the case — 13.6% of Europeans and 18.3% of Asians are seen to be unemployed! The reason for this unexpected result is that in Area A there are many more Europeans than Asians, so we are working from different total populations. This kind of problem is widespread in spatial datasets and is best addressed by the use of multiple scales of analysis, if possible and appropriate, as part of the initial research process (e.g. using variance/semi-variance analysis techniques, fractal analysis, or creating optimized districts with specified attributes as discussed above).

The second issue is due to the way in which voting and census areas are defined — their shape and the way in which they are aggregated affects the results and can even change which party is elected. This is not to say that a particular arrangement *will* have such effects, but that it is possible to deliberately produce a districting plan that meets specific criteria, at least for a single attribute. This question (with or without reference to scale effects) is often referred to by the awkward name Modifiable Areal Unit Problem, or MAUP, as described by Openshaw (1997, 1984) and more recently discussed in some detail by Wong (2008). Figure 4-18

illustrates this issue for an idealized region consisting of 9 small voting districts. The individual zone, row, column and overall total number of voters are shown in diagram A, with a total of 1420 voters of whom roughly 56% will vote for the first party listed/the red party (**R**) and 44% for the second party listed/ the blue party (**B**). With 9 voting districts we expect roughly 5 to be won by the reds and 4 by the blues on a “first past the post” voting system (majority in a voting district wins the district), as is indeed the case in this example. However, if these zones are actually not the voting districts themselves, but combinations of adjacent zones are used to define the voting areas, then the results may be quite different. As diagrams B to F show, with a first-past-the-post voting system we could have a result in which every district was won by the reds (case C), to one in which 75% of the districts were won by the blues (case F). Note that the solutions shown are not unique, and several arrangements of adjacent zones will give the same voting results.

So it is not just the *process* of grouping that generates confusing results, but also the *pattern* of grouping, which is of great interest to those responsible for defining and revising electoral district boundaries. And this is not just a problem confined to voting patterns. For example, if the information being gathered relates to the proportions of trace metals (for example lead and zinc) in the soil, similar issues arise. Samples based on different field boundaries would show that in some arrangements the proportion of lead exceeded that of zinc, whilst other arrangements would show the opposite results. Similarly, measures of inter-zonal flows and spatial autocorrelation statistics (see further, Section 5.5.2, [Global spatial autocorrelation](#)) are significantly affected by scale and pattern aspects of grouping zones.

In practice the problem of zones arrangements may not be as significant an issue as it might appear. As Professor Richard Webber, an expert on geodemographics has remarked (private communication): “I have yet to come across any real world example of a conclusion being invalidly reached as a result of this hypothetical possibility”. A wise strategy is to examine the data at various levels of aggregation, including the lowest (finest level) possible, and check that scale and zoning effects are understood and are unlikely to distort interpretations. Automated zoning was used to create the UK 2001 Census of Population areas — see further Martin (2000) and Openshaw (1977). The procedure (known as AZP) was based on a 7-step approach, as follows:

- Step 1:** Start by generating a random zoning system of N small zones into M regions, $M < N$
- Step 2:** Make a list of the M regions.
- Step 3:** Select and remove any region K at random from this list
- Step 4:** Identify a set of zones bordering on members of region K that could be moved into region K without destroying the internal contiguity of the donor region(s)
- Step 5:** Randomly select zones from this list until either there is a local improvement in the current value of the objective function or a move that is equivalently as good as the current best. Then make the move, update the list of candidate zones, and return to step 4 or else repeat step 5 until the list is exhausted
- Step 6:** When the list for region K is exhausted return to step 3, select another region, and repeat steps 4 to 6
- Step 7:** Repeat steps 2-6 until no further improving moves are made

In the UK case the initial small zones were taken as unit postcodes, which identify roughly 10-14 addresses. Unfortunately these individual addresses are stored as a list of georeferenced point data (nominal centers) rather than postcode areas, so the first stage of the analysis involved generating Voronoi polygons from these points, for the whole of the UK. These initial polygons were then merged to form unit postcode areas ([Figure 4-19](#)), based on the generated polygons and road alignments. Once these basic building blocks had been constructed the automated zoning process could begin. This is illustrated as a series of assignments in [Figure 4-20](#), starting with an arbitrary assignment of contiguous unit postcode areas ([Figure 4-20a](#)), and then progressively selecting such areas for possible assignment to an adjacent zone (e.g. the pink regions in [Figure 4-20b](#) and [Figure 4-20e](#)). Assignment rules included target population size and a measure of social homogeneity, coupled with a measure of shape compactness and enforced contiguity. This process was then applied to a number of test areas across the UK, as is illustrated in the two scenarios, A and B, for part of Manchester, shown in [Figure 4-21](#) (computed zones are shown with colors assigned on a 'unique values' basis). These scenarios were as follows:

- A: Threshold population=100; Target population=150; Shape constraint $P2A = (\text{perimeter squared})/\text{area}$; no social homogeneity constraint applied
- B: Threshold population=100; Target population=250; Shape constraint $P2A = (\text{perimeter squared})/\text{area}$; social homogeneity constraint = tenure and dwelling type intra-area correlations

If scenario B is modified to exclude the shape constraint the assignment of areas is broadly similar, with detailed variations resulting in less compact final areas. and just 5% containing 40-99 households. Based on this methodology the Great Britain Ordnance Survey developed a full set of Output Areas (OAs) with a target population of 125 households with a minimum of 40. These OAs were used for the 2001 census enabling OA data and postcodes to be linked, and providing a set of OAs that can be grouped into higher level units, such as wards and counties, as required. A total of 175,434 OAs were created in this manner, with 37.5% containing 120-129 households, 79.6% containing 110-139 households,

Classification and clustering

“Classification is, perhaps, *the* basic procedure by which we impose some sort of order and coherence upon the vast inflow of information from the real world.” Harvey (1969)

Harvey provides a very useful background to the field of classification, especially in a spatial context, but for the basics of classification within a GIS context Mitchell (1999, pp 46-55) and [Longley et al. \(2015\)](#) are probably more useful and up-to-date starting points. These latter works address the question of how to classify and map attribute data associated with polygonal regions or provided as continuous fields. The manuals and online help facilities of the main GIS and image processing packages are also a good source of material for deciding which scheme might be the most appropriate for the dataset you have. Most of these schemes provide a range of standard alternatives for classifying univariate attributes, for example mapping population density by census tract. The attributes selected may be transformed in advance, for example: by producing spatially intensive variables such as densities rather than population totals; and/or by mathematically transforming data, e.g. by classifying log transformed data rather than the original dataset; and/or by normalizing datasets (as discussed in Section 4.3.3, [Ratios, indices, normalization, standardization and rate smoothing](#)). Alternatively the

classification procedure selected may impose some form of normalization on the untransformed data (e.g. standard deviation based classification).

Classification needs to be seen in terms of purpose as well as method. However, GIS software places almost no constraints upon a user's selection of classification method, so it is the responsibility of the analyst to choose appropriate procedures and settings that are fit for purpose. Classification is by no means a peculiarly spatial problem, and most methods apply to almost any kind of data, i.e. there is rarely a specific spatial aspect, such as a contiguity condition, applied. Such facilities are more likely to be found in the spatial autocorrelation and pattern analysis features of GIS toolsets.

Univariate classification schemes

Within GIS software, univariate classification facilities are found as tools to: aid in the production of choropleth or thematic maps; explore untransformed or transformed datasets; analyze (classify and re-classify) image data (see further, Section 4.2.12, [Classification and clustering](#)); and display continuous field data. In the majority of cases these procedures perform classification based purely on the input dataset, without reference to separate external evaluation criteria.

In almost all instances the objects to be classified are regarded as discrete, distinct items that can only reside in one class at a time (sometimes referred to as *hard* classification). Separate schemes exist for classifying objects that have uncertain class membership (*soft* or *fuzzy* classification) and/or unclear boundaries (as discussed briefly in Section 4.2.13, [Boundaries and zone membership](#)), or which require classification on the basis of multiple attributes (see Section 4.2.12, [Classification and clustering](#)). Typically the attributes used in classification have numerical values that are real or integer type. In most instances these numeric values represent interval or ratio-scaled variables. Purely nominal data are effectively already classified (see, for example, Figure 4-21 in which each zone has been assigned a unique color from a pre-selected palette; and see Section 2.1.2, [Attributes](#), for a brief discussion of commonly used scales). Of course, there is always the option to apply no predefined classification procedure, but instead when mapping such data to apply a continuous gradation of gray tones or colors, using a linear or non-linear (e.g. ogive) scaling of the data values (see further Tobler, 1973 and Cromley and Ye, 2006).

Table 4-5 provides details of a number of univariate classification schemes together with comments on their use. Most of the main GIS packages provide classification options of the types listed, although some (such as the box and percentile methods) are only available in a limited number of software tools (e.g. [GeoDa](#)). A useful variant of the box method, known as hybrid equal interval, in which the inter-quartile range is itself divided into equal intervals, does not appear to be implemented in mainstream GIS packages. Schemes that take into account spatial contiguity, such as the so-called minimum boundary error method described by Cromley (1996), also do not appear to be readily available as a standard means of classification.

Brewer and Pickle (2002) examined many of these methods, with particular attention being paid to ease of interpretation and comparison of map series, e.g. mapping time series data, for example of disease incidence by health service area over a number of years. They concluded that simple quantile methods were amongst the most effective, and had the great advantage of consistent legends for each map in the series, despite the fact that the class interval values often

vary widely. They also found the schemes that resulted in a very large central class (e.g. 40%+) were more difficult to interpret.

In addition to selection of the scheme to use, the number of breaks or intervals and the positioning of these breaks are fundamental decisions. The number of breaks is often selected as an odd value: 5, 7 or 9. With an even number of classes there is no central class, and with a number of classes less than 4 or 5 the level of detail obtained may be too limited. With more than 9 classes gradations may be too fine to distinguish key differences between zones, but this will depend a great deal on the data and the purpose of the classification being selected. In a number of cases the GIS package provides linked frequency diagrams with breaks identified and in some cases interactively adjustable. In other cases generation of frequency diagrams should be conducted in advance to help determine the ideal number of classes and type of classification to be used.

GIS attribute classification schemes vary widely in purpose and method. **Unique values** classification treats every distinct attribute value as its own class, often mapping each value with a separate color. **Manual classification** allows the analyst to define class boundaries directly—either by specifying individual breakpoints, defining a lower bound and fixed interval, or using both a lower and upper bound with a specified number of intervals. **Equal interval** (or *slice* in raster processing) divides the full attribute range into n classes of identical width, defined as $Range/n$, while **defined interval** is a related method in which users explicitly set the width of each class.

Exponential interval schemes create class boundaries so that observations increase or decrease exponentially across classes. **Equal count** (or **quantile**) classification assigns the same number of observations to each class; for example, four classes each containing 25% of values produce quartiles, though actual class sizes are rarely perfectly equal. **Percentile** classification is a variant of quantiles; in GeoDa's ESDA-focused implementation, classes use uneven percentile ranges to highlight distributional structure, specifically: $\leq 1\%$, $1 < 10\%$, $10 < 50\%$, $50 < 90\%$, $90 < 99\%$, and $\geq 99\%$.

Natural breaks (Jenks) classification is widely used in GIS software and identifies class boundaries that minimize internal variance while maximizing differences between classes. Class widths are irregular, the results depend heavily on the number of classes chosen, and boundaries may appear unintuitive; the method follows Jenks and Caspall (1971) and Fisher (1958) and is generally unsuitable for comparing multiple maps. **Standard deviation** classification differs by transforming attribute values into z-scores and grouping them according to distance from the mean, typically at 0.5 or 1.0 SD intervals; standard versions lack a central class, though variants include a mean-centered class defined as $mean \pm 0.5 SD$.

Finally, **Box** classification, based on Tukey's boxplot method, is a quartile-based approach designed to emphasize outliers. It produces six classes: the four quartiles plus two outlier classes for values more than 1.5 times the interquartile range (IQR) from the median, with a stricter outer threshold at $3.0 \times IQR$. Box-based schemes appear in statistical software such as MATLAB or exploratory spatial data analysis tools like GeoDa and STARS but are uncommon in mainstream GIS platforms.

For data with a very large range of values, smoothly varying across a range, a graduated set of classes and coloring may be entirely appropriate. Such classification schemes are common with

field-like data and raster files. Continuous graded shading of thematic maps is also possible, with some packages, such as [Manifold](#)'s Thematic Formatting facilities providing a wide range of options that reduce or remove the dependence on formal class boundaries. The Jenks method, as implemented in a number of GIS packages such as [ArcGIS](#), is not always well-documented so a brief description of the algorithm is provided in Figure 4-22. This description also provides an initial model for some of the multivariate methods described in subsection 4.2.12, [Classification and clustering](#).

Figure 4-22 Jenks Natural Breaks algorithm

Step 1: The user selects the attribute, x , to be classified and specifies the number of classes required, k

Step 2: A set of $k-1$ random or uniform values are generated in the range $[\min\{x\}, \max\{x\}]$. These are used as initial class boundaries

Step 3: The mean values for each initial class are computed and the sum of squared deviations of class members from the mean values is computed. The total sum of squared deviations (TSSD) is recorded

Step 4: Individual values in each class are then systematically assigned to adjacent classes by adjusting the class boundaries to see if the TSSD can be reduced. This is an iterative process, which ends when improvement in TSSD falls below a threshold level, i.e. when the within class variance is as small as possible and between class variance is as large as possible. True optimization is not assured. The entire process can be optionally repeated from Step 1 or 2 and TSSD values compared

Positioning of breaks may be pre-determined by the classification procedure (e.g. Jenks Natural breaks), but these are often manually adjustable for some of the schemes provided — this is particularly useful if specific values or intervals are preferred, such as whole numbers or some convenient values such as 1000s, or if comparisons are to be made across a series of maps. In some instances options are provided for dealing with zero-valued and/or missing data prior to classification, but if not provided the data should be inspected in advance and subsets selected prior to classification where necessary. Some authors recommend that maps with large numbers of zero or missing data values should have such regions identified in a class of their own.

Multivariate classification and clustering

Harvey (1969, p.346) describes the key steps of multivariate classification as follows:

- quantitative analysis of the inter-relationships among the attributes or among the objects
- transformation or reduction of the correlations to a geometric structure with known properties (usually Euclidean)
- grouping or clustering of the objects or attributes on the basis of the distance measured in this transformed space, and once the classes have been firmly identified
- the development of rules for assigning phenomena to classes

Important additional steps to those listed by Harvey are in-process and post-process analysis of the quality of classification achieved. In-process quality analysis attempts to use multiple

classification approaches (typically *soft classifiers* — see further, below) to evaluate the robustness and strength of the classes that have been defined. Alternatively or additionally, random samples of classified data (e.g. pixels) are taken and compared with the ground truth, by field survey (using new or previously collected data).

Classification of datasets based purely on attribute information (i.e. ignoring spatial relationships) can readily be conducted using general purpose statistical and mathematical toolsets. Some techniques, such as clustering based on multivariate analysis of variance, require that input datasets be Normally distributed, but many methods make no such assumptions. Most statistical and mathematical libraries provide a range of tools to facilitate classification and assignment. These typically include:

- facilities that can be used to reduce the complexity (dimensionality) of the source datasets, such as *factor analysis*, *principal components analysis* and *multidimensional scaling*
- facilities to identify clusters within the dataset, either on a single level for a fixed number of clusters (e.g. *K*-means clustering) or on a hierarchical basis (typically techniques that rely on some form of linkage function, usually based on a function of the simple *n*-dimensional Euclidean distance between points within separate clusters); and
- facilities for optimally assigning new observations to existing classes (e.g. *linear discriminant analysis*)

Before we examine the application of such methods to the analysis of multi-band image data (the main application area for such methods within GIS and remotely-sensed image processing packages), we first summarize the *K*-means and linear discriminant analysis procedures mentioned above.

K-means clustering attempts to partition a multivariate dataset into *K* distinct (non-overlapping) clusters such that points within a cluster are as close as possible in multi-dimensional space, and as far away as possible from points in other clusters. The input dataset can be viewed as a set of objects with an associated set of *n* (typically real-valued) measures or attributes. Each object can then be viewed as a single point or vector $\mathbf{x} = \{x_1, x_2, x_3 \dots, x_n\}$ in *n*-dimensional space. The cluster procedure is then as follows:

- The method starts with a set of *K* initial cluster centers. These might be assigned: (i) as *K* random locations in *n*-space within the observed data ranges; (ii) as *K* uniformly sited locations within the observed data ranges; (iii) as a user-defined set of *K* locations; (iv) by selecting *K* datapoints at random from the observed set as cluster centers; or (v) as a set of *K* locations obtained by clustering a small subset of the data using random or uniform allocation. A minimum separation distance for starting points may be specified
- The distance between every point and each of the *K* means is computed, based on some prior defined metric (often using squared Euclidean, L_2^2 , or City block, L_1 , as these are both fast to compute — see further, Section 4.4.1, [Metrics](#)). The [TerrSet/Idrisi](#) software package, however, uses Euclidean distances and the general expression:

$$d_{ik} = \sum_n \sqrt{|X_{in} - C_{kn}|^2}$$

where d_{ik} defines the distance from the i^{th} pixel in an image to the k^{th} centroid, X_{in} is the n -element (band) vector of the i^{th} pixel, and C_{kn} is the n -element vector of the k^{th} centroid

- Points are assigned to the nearest center (i.e. it is a locally optimizing *greedy heuristic*; see further, Section 7.2.2, [Heuristic and meta-heuristic algorithms](#)). Any entirely empty clusters may be discarded and (optionally) a new start point added

- The location of the center of each cluster is then re-computed based on the set of points allocated to that center, and the previous step is then repeated. This sequence is iterated until no further reassignments occur, or a preset number of iterations have been completed, or no cluster center is moved by more than a pre-specified small amount. The total distance (DSUM) of all points to the centers of the K clusters to which they have been assigned is calculated

- Each point is then re-examined in turn (Phase 2 of the process) and checked to see if DSUM is reduced if the point is assigned to another cluster. If DSUM is reduced the point is reassigned to the cluster that results in the maximum overall reduction

- Stopping criteria may be determined by setting a limit to the number of iterations of the process (e.g. 50), or the level of change in DSUM, or the number or percentage of data points that continue to be re-allocated (e.g. 1%), or a combination of these options

- Special consideration may be given to the re-allocation of data items where the cluster membership is very small (e.g. contain less than 1% of the data). The option to remove low membership clusters and re-allocate their members (i.e. consolidating the clusters to provide a smaller number of larger clusters) is quite common

- The entire process may then be repeated with a different choice of starting points. This may result in an overall reduction in the objective function, DSUM, since the algorithm is not guaranteed to provide an optimal partitioning

- Generic clustering algorithms, such as K -means, may result in many spatially isolated pixels or zones, which may be members of a well-represented class ($\gg 1\%$). In some instances a smoothing filter can consolidate such occurrences with their surrounding zones, but must be applied with care since the resultant classification will knowingly contain mixtures

- It is often also the practice to assign different weights to clusters when finalizing the classes, in accordance with *a priori* interpretation of the resulting outcomes. The benefits of so doing should be set out in a clear and transparent manner in order to aid interpretation of the results by third parties

Linear Discriminant Analysis (LDA), as applied within [TerrSet/Idrisi](#), assumes we have P classes and P linear discriminant functions. These functions, which are rather like linear regression equations, are computed by analysis of training site data comprised of m image bands. Having computed the weights for each of these linear functions, a set of P weighted sums is then calculated for each pixel in the target image and the sum that has the highest value or score, S_i , is taken as the class to which the pixel is to be assigned. The sum is of the form:

$$S_i = c_i + \sum_{j=1}^m w_{i,j} x_j$$

where c_i is a constant for class i , x_j is the observed pixel value in class j , and $w_{i,j}$ is the pre-computed weight applying to band j and class i . Hence with $m=7$ spectral bands there would be a set of 7+1 associated weights for each discriminant function. If there were $P=5$ such functions (i.e. 5 classes), there would be a matrix, $\mathbf{W}$, of 40 weight values, and each pixel would be assigned to one of the 5 classes.

Most GIS packages, such as [TNT/Datum Workstation](#), [TerrSet/Idrisi](#), [GRASS](#) and [ArcGIS Spatial Analyst](#), include a range of multivariate classification procedures. As noted earlier, these are principally for image classification and pixel assignment, as are those provided within the specialized image processing suites, [ERDAS](#) and [ENVI](#). The most commonly provided include K -means, ISODATA, and Maximum Likelihood procedures. More details, with specific reference to multi-band and hyperspectral image analysis, are provided in Sections 4.2.12, [Classification and clustering](#), below. Details of image classification techniques that use artificial neural network (ANN) methods are provided in Section 8.3, [Artificial Neural Networks \(ANN\)](#). Details of many of the traditional methods described may be found in Tou and Gonzales (1974, Chapter 3), Ball and Hall (1965), Richards (1986) and Burrough and McDonnell (1998, 2015).

Multi-band image classification

In this subsection we provide a brief description of many of the multi-band classifier procedures generally available, with special reference to those provided within one of the main raster-focused GIS packages, [TerrSet/Idrisi](#). Other packages, notably [ENVI](#), also provide many of these facilities.

Within GIS and remotely-sensed image processing software multivariate classification methods are almost exclusively applied to image datasets — typically multi-band image files. The objective is to identify distinct areas or features (collections of pixels or cells) such as forest, water, grassland and buildings and assign all occurrences of such features to distinct classes. This is either performed semi-automatically (supervised, requiring training site data) or automatically (unsupervised).

Supervised classification (S) involves the manual analysis of selected images to identify features of interest and assign polygonal regions to user-defined classes. The statistical characteristics of these regions within and across rasters, sometimes described as *signatures*, can then be used to guide subsequent classification. Special toolsets are generally provided for signature development. It may also be necessary to normalize the datasets (to ensure the range of values in each band is similar) and/or to (dimensionally) transform the datasets in some way before proceeding to the formal clustering and assignment processes (for example using techniques such as principal components analysis, PCA, or factor analysis, FA – see further, [CATMOGs 7 and 8](#)).

Unsupervised classification (U) either does not require training data to be specified in advance, or performs a form of automated selection from the input images. With these techniques it is

often helpful to specify more clusters than you would ultimately like or expect and then to combine some of those produced if they seem essentially the same in terms of your own objectives rather than purely computational factors. For example separate clusters may be produced for deciduous and coniferous vegetation cover, but if the objective is to classify areas as woodland without separating these by type, post classification combination may be effective. Most of the more sophisticated raster-based GIS packages provide such facilities, for example [TNT/Datum Workstation](#) (see the [TNT/Datum Workstation](#) Image Classification Guide) and [TerrSet/Idrisi](#) as well as the more generic [ArcGIS](#) Spatial Analyst toolset and the specialized image processing suites, [ERDAS](#) and [ENVI](#). Several of these packages refer to Richards (1986) as a primary source for many of the standard procedures they adopt for classification and interested readers are recommended to refer to this volume for more information.

Each pixel of a multi-band image to be classified can be regarded as being comprised of a vector $\mathbf{x} = \{x_1, x_2, x_3, \dots, x_m\}$ of data values, one per band. Typically the data are real-valued and positive. In so-called remotely-sensed *multi-band image processing* the number of bands is generally not very large, typically 7 or fewer. In such multi-band images it is often the case that most of the useful information is contained in just two or three bands, for example the red, near infra-red and middle infra-red bands, and in some instances these may be used rather than the full set. *Hyperspectral images*, on the other hand, are comprised of a very large number of finely divided spectral bands, from 10s to 100s of such bands. A third type of multi-band image is one in which the images are co-registered but are not all derived from remote-sensing — some of the bands or raster layers may be derived from other forms of survey and/or be computationally derived (e.g. rasters generated by rasterizing zonal, linear or point vector data). An example involving the analysis of such data using artificial neural network (ANN) methods is provided in Section 8.3.3, [Self organizing networks](#).

[Table 4-6](#) summarizes the principal set of so-called *hard classifiers* (H), some of which also provide soft classification as an option. Hard classifiers assign every pixel in the target image to one of n classes, or to a null class (usually coded as 0). *Soft classifiers* (S) assign pixels to classes with a value that indicates their *degree of membership* of that class. This may be expressed as a probability value, p_k , with the sum over all $k=1..P$ classes being 1 (or 100%), or may be a value that is not regarded as a probability and may not sum to 1 (typically summing to less than 1). Display of soft classified images is often as a series of separate layers, one per class, with the pixels being color- or gray-scale coded to indicate their degree of membership.

The output of soft classifiers can be *hardened*, in a variety of ways. One approach is to take the output from a soft classification and generate a set of classification maps that provide the first (most likely/highest scoring), second etc. class assignments as separate displays. Another approach is to utilize ancillary data, such as (buffered) GIS vector data that is overlaid on the classified results. For example, this may assist in identifying classes that are or are not built-up land (i.e. increasing the strength of evidence for specific classes) based on the known distance from roads. Yet another approach is to utilize prior geostatistical analysis of higher resolution remotely-sensed imagery obtained from a smaller area, or field survey data, to provide enhanced guidance for the classification process.

Most of the classification procedures described in [Table 4-6](#) are non-hierarchical, i.e. they classify pixels in the target image into distinct classes that are not grouped into levels. Some

do provide a tree-like structure (e.g. CTA). In this case, conceptually one might imagine high level groups such as Built, Wetland, Forest, Grass, and Rock. Forest, for example, might be divided into deciduous and evergreen. Deciduous might be further divided into species-type subgroups. In Table 4-6 the H/S column identifies whether the classification is hard or soft, with some classifiers providing both options. The S/U column identifies whether the procedure is essentially supervised (S) or unsupervised (U). In addition, and not described in detail here, are *mixture analyses*, in which pixels may be allocated to one or more than one class simultaneously. For example, a pixel might be assigned to two different kinds of woodland or two different kinds of soil type, thus regarded as being of explicitly mixed rather than single class membership. Mixture assignments are usually on an hierarchical basis, e.g. with 3 classes [A], [B] and [C] the 7 possible mixtures would be [A], [B], [C], [A,B], [A,C], [B,C], and [A,B,C]. Clearly there may be a very large number of mixture class combinations, so this approach must be applied with caution.

Image classification facilities — Selected classifiers (Idrisi function shown in CAPS)

A wide range of supervised and unsupervised classifiers are used in GIS and remote-sensing applications, each with different assumptions, strengths, and computational characteristics. **Simple one-pass clustering** is an unsupervised, heuristic method that rapidly forms up to P clusters by assigning each pixel to the nearest existing cluster if within a distance threshold, or creating a new cluster otherwise; although fast, its classification quality is often weak. The **Parallelepiped (PIPED)** classifier is a supervised method that assigns pixels to a class only if their values fall within the lower and upper thresholds of that class on every band; because class “boxes” typically overlap, assignments can be arbitrary, and many pixels may receive no label, so it is generally not recommended. **Minimum Distance to Mean (MINDIST)** is a supervised classifier similar to one-pass clustering but uses pre-computed class means from training data; it assigns each pixel to the nearest class mean but ignores variance, making it fast but less robust than probabilistic approaches. **Maximum Likelihood (MAXLIKE)** improves on MINDIST by incorporating variance–covariance structure under the assumption of normally distributed class signatures; it uses Bayesian posterior probabilities and assigns pixels to the class with the highest likelihood unless all probabilities fall below a user-defined threshold. It requires adequate training data ($\geq 10 \times$ the number of bands) and is strongly affected by training-site homogeneity.

Linear Discriminant Analysis (FISHER) computes linear functions that maximize separation among classes; in stepwise form, new discriminant functions are added sequentially and are orthogonal. Pixels are assigned to the class with the highest discriminant score, and all pixels receive assignments. **K-Nearest Neighbors (KNN)** computes distances from each pixel to all training samples, assigning classes based on the dominant class among the k nearest neighbors (hard classification) or reporting membership proportions (soft classification); speed depends heavily on the size of k . **Histogram clustering (CLUSTER)** is an unsupervised peak-finding method that identifies peaks in 1–7 dimensional histograms and assigns each pixel to the nearest peak; it is fast and often used to generate signatures for supervised classification.

ISOCCLUS (ISODATA/ISOcluster) is an unsupervised iterative method similar to K-means but includes rules for splitting or merging clusters depending on criteria such as cluster size, center proximity, or within-cluster variance; it combines CLUSTER for initial seeds with MAXLIKE for iterative class assignment and can produce high-quality results, though it is

slow. **K-means** itself is a widely used clustering algorithm that requires the number of clusters in advance and is sensitive to initialization; TerrSet/Idrisi offers random partition, random center, and diagonal-spacing strategies. **Fuzzy c-means (FCM)** extends K-means by assigning each pixel weighted membership values across all clusters based on fuzzy rules and prior membership coefficients, providing soft classification.

Other unsupervised similarity measures include **Minimum Distribution Angle (MDA)**, which treats pixel vectors as rays from the origin and classifies based on the angle between a pixel vector and each cluster center, emphasizing proportions rather than magnitude (similar to spectral-angle methods). Neural-network-based classifiers include **Multi-Level Perceptrons (MLP)** and **Self-Organizing Maps (SOM)**, both of which support supervised and unsupervised modes; MLPs are relatively slow whereas SOMs are moderate-speed and often use K-means for initial cluster seeding. **Fuzzy ARTMAP**, based on Adaptive Resonance Theory, supports incremental learning, handles fuzzy input patterns, and adds a MAP layer for supervised classification. Finally, **Classified Tree Analysis (CTA)** uses hierarchical binary splits based on metrics such as entropy, Gini, or gain ratio to recursively partition training data, producing a decision-tree classifier capable of both supervised and soft/hard outputs.

Uncertainty and image processing

Uncertainty is an intrinsic feature of remotely-sensed image data for several reasons, including:

Resolution: the resolution of a remotely-sensed image defines the area represented by a single pixel. If a pixel represents a 100m by 100m region it is almost certain that its contents will be of mixed landcover types, and hence will generate data that reflects this mixture. Specific techniques (generally based on soft classifiers such as [Idrisi's](#) UNMIX, BAYCLASS and FUZCLASS functions) are available for analyzing within-cell spectral variation, often described as sub-pixel classification

Temporal variables: time of day and season affect both the landcover (e.g. growth stages) and illumination patterns (e.g. lighting of slopes, vegetation surfaces, moisture levels) resulting in additional variations within and between image datasets and bands

Spectral response: remote-sensing may not measure the spectral response band or bands on which the particular surface provides maximum response. The response data may therefore be weak or mixed/easily confused with other surface responses

Soft classification procedures can provide an image displaying the *degree of uncertainty* of the classification. For example, for every soft classification image one pixel or set of contiguous pixels (region) may have the same or very similar degree of class membership, hence no one class is clearly favored. This may be contrasted with a hard classification process in which the winning class, i.e. the one with slightly higher probability or degree of membership, would be selected in preference to any other. An uncertainty map illustrates the degree to which one class (from a total of P classes) stands out from the rest for each pixel in the image. Definition of uncertainty could be based on Shannon's information statistic (entropy) statistic or variations thereof, or on an application-specific measure. [TerrSet/Idrisi](#) adopts the latter approach, using an uncertainty measure, U , of the form:

$$U = 1 - \frac{m - s / P}{1 - 1 / P}$$

Here, m is the maximum degree of membership recorded for the pixel in question, and

$$s \leq 1$$

is the sum of the degree of membership for that pixel. Thus, with $P=10$ classes and $m=0.3$ and $s=1$, say, we have $U=1-0.22=0.78$ whilst with $m=0.1$ we have $U=1$ (complete uncertainty) and with $m=1$ $U=0$ (complete certainty).

Hyperspectral image classification

Hyperspectral images are comprised of a large number of finely divided spectral bands, from 10s to 100s of such bands. The following analysis utilizes a hyperspectral dataset produced by the NASA AVIRIS (Airborne Visible/Infrared Imaging Spectrometer) project. As noted previously, hyperspectral imagery is become increasingly available and is essentially the same as multi-band imagery but with far more bands provided. It has particular application in geological work, both terrestrial and for planetary/lunar data analysis.

The dataset we use in this subsection is comprised of 224 contiguous spectral bands in the range 400nm (nanometers) to approximately 2500 nm at 10 nm intervals. The raster image files cover 512x614 cells of 20m squares and relate to a mining region (Cuprite) in western Nevada, USA (hence c.10km by 12km in area). This dataset is available for download via the data section of the AVIRIS website and from the [TNT/Datum Workstation](http://TNT/Datum/Workstation) web site as one of their tutorial datasets. It is discussed in more detail at the USGS spectral lab site: <https://www.usgs.gov/labs/spectroscopy-lab>, from which reference spectra may be obtained. The layers of raster datasets of this type are spatially matched and highly correlated. This means that storage of such data as 224 raster objects would be extremely inefficient and hence of much more compact (compressed) 3D object variant is stored by [TNT/Datum Workstation](http://TNT/Datum/Workstation) which they describe as a *hypercube*.

Figure 4-24 shows two alternative 2D representations of the region, where 3 spectral layers have been chosen to represent the red/green/blue (RGB) components of the image for display purposes. This provides a strong visual impression of the data, but each image is based on just 3 of the 224 layers and depends for its interpretation upon the choice of the three spectral bands and color scheme applied. There are a number of alternative representations for datasets of this type.

The 3D hypercube described above can be displayed and layers selected systematically or as desired (Figure 4-25). This is essentially a visualization tool that enables closer inspection of the data and highlights the correlated nature of the information, but does not provide a direct route to analysis. Several software packages support analysis of this type of dataset, including TerrSet/Idrisi (the HYPERSAM and HYPERMIN facilities, and the Earth Trends Modeler) and the [TNT/Datum Workstation](http://TNT/Datum/Workstation) spectral angle mapping facility, which we have used in the current example. Each pixel of the 3D hypercube has a spectral value for the ordered set of frequencies (i.e. across the depth of the cube). This ordered set can be viewed as a graph or spectrogram and can be compared with reference spectrograms (e.g. ground survey reference

samples) or using laboratory sets obtained from prior research for different surface materials and minerals. Similarity of (normalized) spectral plots to the image spectral plot provides a means of classifying individual pixels and hence the region as a whole.

Laboratory-generated reference spectra are available from various sources, including the USGS spectral analysis laboratory cited above. It should be noted, however, that laboratory-generated reference spectra will differ from remotely-sensed spectra owing to many factors, including non-uniform atmospheric distortions. Special facilities (e.g. the hyperspectral PROFILE facility in [TerrSet/Idrisi](#)) are available to identify and, where necessary, eliminate those bands that are subject to particularly high atmospheric distortion. As noted above, the analysis of hyperspectral images can be undertaken by regarding each cell or pixel as a single point in 224-dimensional space. The recorded value for each spectral band is then the coordinate value for this pixel for the selected dimension. The single point in n -space can be seen as a vector from the spectral origin, $\mathbf{0}$, with the full pixel set (512x614 pixels) being seen as clouds of points in n -space.

The *direction* of these vectors is more important than their *magnitude*, since the latter are affected by average brightness. Simple *spectral angle mapping* (SAM) consists of comparing a known (reference) spectrum, which has its own characteristic m -dimensional vector, with those in the sample image. Pixels that have an angular proximity to the reference vector below a user-specified value are assigned to that class (cf. [Table 4-6](#), minimum distribution angle method). Every pixel in the source hypercube can be compared with the reference vectors in m -space and the angle between the pairs of vectors computed. The resulting set of scalar values can then be mapped as a form of “angular distance” raster (fuzzy, or soft class assignment) or just those values that are below a given angular distance threshold can be mapped (absolute, or hard class assignment).

Boundaries and zone membership

The definition and selection of boundaries is often a difficult, but nevertheless essential element in many areas of spatial analysis. One group of problems where this is especially important is in the analysis of point patterns — sets of points in the plane representing cases of a particular disease, the location of ancient artifacts, tracked locations of individual animals, or a particular species of plant. If the density of points (frequency of occurrence per unit area) is required then the total area under examination needs to be defined. Definition of this area, A , requires identification of a boundary, although there is now a variety of special procedures (known as kernel density methods) available for computation of densities without pre-defined fixed boundaries (see further, Section 4.3.4, [Density, kernels and occupancy](#)).

A further problem relates to the process of modeling the real world with abstracted linear forms (i.e. polylines and polygons). There are a variety of ways of modeling linear forms — mathematical, statistical, fractal, zonal — and each of these models will have implications for analysis. Data recorded in a GIS database, either as vector or grid elements, have already been modeled and to some extent the kinds of analyses that may be carried out are pre-determined. For example, most GIS packages do not provide facilities for analyzing the complexity or “texture” of modeled lines and surfaces, although some, such as [TerrSet/Idrisi](#), do support such analyses by examining grid datasets at various scales and/or using variable sized windows. The [ArcGIS](#) add-in, Hawth’s Tools/GME, provides a *line metrics* tool for generating measures of estimated fractal dimension and sinuosity of stored polylines.

Convex hulls

For point sets in the plane a simple and often useful boundary is the convex hull (Figure 4-27). This is taken as the convex polygon of smallest area that completely encloses the set (it may also be used for line and polygon sets). If the data you are working with consist solely of a single point set then the convex hull of this set encloses all the information you have available, even though you may know that the data extend beyond these limits.

The convex hull may also be useful in the analysis of optimum location and networking problems. For example the center of gravity of a set of points in the plane will always lie inside or on the convex hull of the point set, and the sequence of points which comprises the convex hull form part of the solution to certain network routing problems (systematic minimal tours of a set of locations). However, it is important to note that the convex hull of a set of objects (points, lines etc.) is affected markedly by extreme values, so outliers and/or errors can have a substantial impact on the shape, area, perimeter length and centroid of a convex hull. For example, one or two extreme values can substantially increase the area enclosed by the convex hull. Typically, however, convex hulls under-estimate area for many applications and hence will over-estimate densities if computed using an area measure based upon this hull.

By definition the convex hull includes a number of points as part of its boundary. For some problems this may be satisfactory whilst for others it may not be suitable. For example, when analyzing the distribution of points in a study region it may be preferable to minimize the potential distortions to sample statistics by avoiding points that lie on or close to the convex hull. Creating an inner buffer of the convex hull and analyzing the subset of points lying within the core of the region, with or without consideration of points lying in the buffer zone, may be preferable. Another approach, implemented in the R package [adehabitatHR](#) (mcp function), is to remove a small percentage (e.g. 5%) of the observed points that are farthest from the centroid of the cloud of observations before the process is run. Alternatively the convex hull may be uniformly increased (buffered) in size, for example by a given percentage of area or by a specified distance such as half the mean distance to the nearest-neighbors of enclosed points.

Non-convex hulls

The computation of convex hulls is a fast and useful facility, but does not necessarily match the requirements of the analyst attempting to define a region enclosing a set of objects. We have already noted, for example, that the method is strongly affected by the location of outliers, and as a result may enclose large regions that are of no interest or relevance to the analysis. Whilst the convex hull of a point set in the plane is unique, any number and variety of non-convex hulls may be generated. Such alternative hulls may meet certain pre-defined application-related criteria. Examples might include: the hull must be a polygon; the polygon must be as near convex as possible; the polygon must have the smallest possible area enclosing the set as possible; the polygon must reflect the density of points in the sample; the polygon must assign each point its own nominal region and include all points.

There are three principal methods for generating non-convex polygonal hulls (NCPHs) that meet specified criteria. These are: (a) expansion; (b) contraction; and (c) density contouring. Some of these methods may be implemented using most GIS packages, but the majority require some programming. [TNT/Datum Workstation](#) includes facilities for computing several of the variants described within its Polygon Fitting facility. These algorithms were originally

developed to help identify animal “home ranges” but may be utilized in many areas — see for example, Harvey and Barbour (1965), White and Garrott (1990, pp 145-82, “Home range estimation”), and Mitchell (2006).

Case (a) expansion: in this case the non-convex hull is created by assigning each point a surrounding region and then growing this region until all points are covered and a continuous external hull is obtained. This may be achieved in several ways. One method is to compute Voronoi polygons for all the points and use the external boundary of the (finite) set as the NCPH (see for example [Figure 4-33B](#)). This leaves the problem of how to treat the points that make up the convex hull. An alternative non-polygonal approach is to buffer all the points and increase the buffer width until a single merged zone is formed. This process may highlight the need to consider the benefits of using several sub-regions rather than a single region — for example, in cases where many of the observations lie near the region boundary with a large central region containing few observations. Yet another alternative is to overlay the study region with a rectangular grid and define the NCPH with reference to this (for example, requiring that the grid includes every point and a minimum number of cells surrounding every point). Where necessary interstitial gaps in the grid are grown to ensure a contiguous overall region is achieved without necessarily growing the external boundary.

Case (b) contraction: this case involves reduction of the convex hull according to selected rules. The simplest of these is a systematic area minimization procedure. First the area of the convex hull is computed. Next a point on the convex hull is removed and the new convex hull recomputed. The area of this new region is calculated and stored. The process is repeated for all points on the original convex hull and the point that results in the greatest decrease in area is permanently removed. The procedure then iterates until only a pre-defined percentage of the original points remain (e.g. 90%) or the area has been reduced by a given percentage. The result is a convex hull of an optimized subset of the original points. A second procedure, which results in a NCPH, involves a form of shrinkage of the convex hull around the point set, rather like a plastic bag full of small balls having the air removed. In this method the longest linear segment of the convex hull is selected and replaced with two segments connecting the original two end points via an intermediate point that is the closest to the original line — almost the reverse of point-weeding (Section 4.2.4). The process continues until the overall area enclosed has been reduced by a pre-determined amount, or a number of iterations have been reached — 10 is a typical number to use. Hybridization of this method with the preceding method is possible.

Note that in the majority of contraction and expansion methods the result is either a grid or polygon form, with the latter having straight line edges. Procedures also exist in which the restriction of convexity, connectedness and/or to straight edges is relaxed, e.g. forming so-called *alpha hulls* or *alpha shapes* where concavities and unconnected sets are permitted. Consider a circle or disc of radius $1/\alpha$. When α is small (close to zero) the disc radius is large (infinite at 0). With a large enough radius the disc can be placed over any given point set such that the set is fully enclosed and precisely two points from the set lie on its boundary (Figure 4-28A). These are then typically connected by a straight line, and a complete hull can thus be generated. Figure 4-28B to D illustrate the results obtained by varying α , as explained below.

In Figure 4-28B, with larger α the disc radius is reduced and only a limited number of circles can be drawn that enclose the point set and have a pair of points from the set on their

boundary. The resulting alpha shape excludes many of the original points. Figure 4-28C and Figure 4-28D illustrate the shapes formed as alpha becomes increasingly negative. In this case the complement of the positive disc shapes are used. Some have likened this process in 3D to using an ice-cream scoop of a given radius to extract the 3D solid surrounding the immovable point set — smaller scoops can remove more material until the form becomes increasingly convex and starts to break up. Specialized tools exist for constructing such hulls in 2D and 3D and are of particular application in visualization of environmental and similar datasets. The images shown in Figure 4-28 were generated using the interactive Java applet developed by François Bélair whilst at the Computational Geometry Laboratory of McGill University.

Case (c) density contouring: this last of the three cases involves overlaying the study region with a (fine) grid and then computing an estimated point density at each grid point. There are several ways in which the density can be estimated, but the most widely used, within GIS and related packages, is kernel estimation (discussed in some detail in Section 4.3.4, [Density, kernels and occupancy](#)). Once a density value has been assigned to each intersection or cell, the region taken for the NPCH may be determined by specifying a threshold density and using the grid directly, or by contouring the grid. The final result may then be used directly or converted to polygonal form, depending on the requirement.

Minimum Bounding Rectangles (MBRs)

One difficulty with convex hulls is that they are slightly awkward shapes to work with, and in many cases it is simpler to deal with a boundary that is a regular shape, such as a circle or rectangle. Rectangles that align with the coordinate system and enclose selected features compactly are known as Minimum Bounding Rectangles, or Minimum Enclosing (or Envelope) Rectangles (MBRs or MERs).

In most instances it is very fast to determine whether the MBR for one feature or set of features is completely contained within a study region or within the MBR of another set of features. Identifying where the center of an MBR lies is also very simple and thus they are useful for first-stage identification of simple overlays and membership.

MBRs and convex hulls provide an indication of the limits to the area over which you can reliably interpolate values and many GIS packages take one or other of these boundaries as the limit for procedures such as interpolation, generation of contour lines or the creation of triangulated irregular networks (TINs). Figure 4-29 shows the result of interpolating income per capita data to a fine grid using the census map shown earlier in Figure 4-10 (darker areas have higher per capita income). In this case the [ArcGIS](#) Spatial Analyst facility has been used, and as can be seen the interpolation process extends to the MBR of census tract centers and no further.

We can also see, in this example:

- the interpolation process replaces census tracts with a grid surface of values, comprising around 250x500 cells, which we have classified into 9 groups for display purposes
- the generated surface pattern is closely tied to the designated centroid locations; and
- interpolation has occurred for our rogue locations, including those that should have been screened out as being lakes (no data) *prior* to interpolation, and two small polygons which are classified as land but for which no census data are provided

Another problem with hulls (MBRs, convex polygons or more general hull forms) is that by definition they enclose the points for which you have data, and you may be aware of other points, which could or should be taken into consideration, but which lie outside this region. For example, the nearest tree to one on the boundary may actually be in the adjacent field and not within the pre-defined boundary (Figure 4-30). MBRs and other simple shapes are provided as options for feature enclosure in many GIS packages and in more specialized toolsets such as [Crimestat](#) (which is concerned with point events). Frequently the aim is to select a boundary which is reasonable and meaningful for subsequent analysis, often analyses that involve statistical calculations that rely on knowing the point density in advance. As boundaries are altered, the area, A , is changed and this can have a dramatic influence on the analytical results. This may distort analysis, for example giving poor interpolation results near to the region boundary or under-estimating the average distances between sampled points.

Fuzzy boundaries

A number of GIS packages include special facilities for identifying, selecting and analyzing boundaries. These may be the borders of distinct zones or areas, within which values on one or more attributes are relatively homogeneous, and distinct from those in adjacent areas, or they may be zones of rapid spatial change in more continuously varying attribute data. Amongst the tools available for boundary determination are raster-based GIS products, like [TerrSet/Idrisi](#), and the more specialized family of spatial analysis tools from [Biomedware](#). Indeed [Biomedware](#) has a specific product, Boundaryseer, for boundary detection *and* analysis, using techniques such as overlap and sub-boundary statistical analysis (the pattern analysis software, [PASSaGE](#), provides similar facilities). The problem these products seek to address is that of identifying where and how boundaries between different zones should be drawn and interpreted when the underlying data varies continuously, often without sharp breaks. For example, soil types may be classified by the proportion of sand, clay, organic matter and mineral content. Each of these variables will occur in a mix across a study region and may gradually merge from one recognized type to another.

In order to define boundaries between areas with, for example, different soil types, clear polygon-like boundaries are inappropriate. In such cases a widely used procedure is to allocate all grid cells to a set of k zones or classes based on *fuzzy membership* rather than binary or 0/1 membership. The idea here is to assign a number *between* 0 and 1 to indicate how much membership of the zone or class the cell is to be allocated, where 0 means not a member and 0.5 means the cell's *grade of membership* is 50% (but this is not a probabilistic measure). If a zone is mapped with a fuzzy boundary, the 50% or 0.5 set of cells is sometime regarded as the equivalent to the boundary or delimiter (cross-over point) in a crisp, discrete model situation. Using this notion, crisp polygon boundaries or identified zone edges in raster modeled data may be replaced with a band based on the degree of zone membership. Alternatively grid datasets may be subjected to fuzzy classification procedures whereby each grid cell acquires a membership value for one of k fuzzy classes, and a series of membership maps are generated, one for each class (see further, the discussion on soft classifiers in Section 4.2.12, [Classification and clustering](#)). Subsequent processing on these maps may then be used to locate boundaries. Several fuzzy membership functions (MFs) have been developed to enable the assignment process to be automated and to reflect expert opinion on the nature and extent of transitional zones. Burrough and McDonnell (1998 Ch.11 or 2015 Ch.13) provides an excellent discussion of fuzzy sets and their applications in spatial classification and boundary determination

problems. The GIS product [TerrSet/Idrisi](#), which utilizes some of Burrough's work, provides four principal fuzzy MFs:

Sigmoidal or (double) s-shaped functions, which are produced by a combination of linear and $\cos^2()$ functions in [TerrSet/Idrisi](#)'s case, or as an algebraic expression of the form:

$$m=1/(1+a(z-c)^2)$$

where a is a parameter that varies the width of the function, z is the property being modeled (e.g. proportion of clay) and c is the value of this proportion at the function midpoint (see Figure 4-31; here membership values of >0.5 are regarded as being definitely members of the set A)

J-shaped functions, which are rather like the sigmoidal MFs but with the rounded top sliced off flat over some distance, x (if $x=0$ then the two sides of the J meet at a point). The equation used in this case is of the form:

$$m=1/(1+((z-a)/(a-c))^2)$$

Linear functions, which are like the J-shaped function but with linear sides, like the slope of a pitched roof, and are thus simple to calculate and have a fixed and well-defined extent, and

User-defined functions, which are self-explanatory. In most applications MFs are symmetric, although monotonic increasing or decreasing options are provided in [TerrSet/Idrisi](#)

An alternative to using MFs as spreading functions is to apply a two-stage process: first, use a (fuzzy) classification procedure to assign a membership value ($m_{ik}=0.0-1.0$) to each grid cell, i , for each of k classes. This value is based on how similar the cell attributes are to each of k (pre-selected) attribute types or clusters. These may then be separately mapped, as discussed earlier.

Having generated the membership assignments boundaries are then generated by a second algorithm. Boundaryseer provides three alternative procedures:

- Wombling (a family of procedures named after the author, W H Womble, 1951)
- Confusion Index (CI) measures, which are almost self-explanatory, and
- Classification Entropy (CE) based on information theoretic ideas

Wombling involves examining the gradient of the surface or surfaces under consideration in the immediate neighborhood of each cell or point. Typically, with raster datasets, this process examines the four cells in Rook's or Bishop's move position relative to the target cell or point — boundaries are identified based on the rate of change of a linear surface fitted through these four points, with high or sudden rates of change being the most significant. Wombling methods can be applied to vector and image datasets as well as raster data. Wombling edge-detection is provided in a number of other software products, including the [SAM](#) and [PASSaGE](#) macroecology packages.

The Confusion Index (CI) works on the presumption that if we compute the ratio of the second highest membership value of a cell i , m_{i2} , to the highest, m_{i1} , then any values close to 1 indicate that the cell could realistically be assigned to either class, and hence is likely to be on the boundary between the two classes.

Finally, Classification Entropy, devised by Brown (1998), creates a similar value to the CI measure, again in the range [0,1], but this time using the normalized entropy measure:

where the summation extends over all the k -classes and the measure applies to each of the j cells or locations. This latter kind of measure is used in a number of spatial analysis applications (see further, Section 4.6).

Fuzzy procedures are by no means the only methods for detecting and mapping boundaries. Many other techniques and options exist. For example, the allocation of cells to one of k -classes can be undertaken using purely deterministic (spatially constrained) clustering procedures or probabilistic methods, including Bayesian and belief-based systems. In the latter methods cells are assigned probabilities of membership of the various classes. A fuller description of such methods is provided in the documentation for the Boundaryseer and [TerrSet/Idrisi](#) packages. As is apparent from this discussion, boundary detection is not only a complex and subtle process, but is also closely related to general-purpose classification and clustering methods, spatially-constrained clustering methods and techniques applied in image processing.

As noted earlier, boundaries may also be subject to analysis, for example to test hypotheses such as: “is the overlap observed between these two boundaries significant, and if so in what way, or could it have occurred at random?”; “is the form of a boundary unusual or related to those of neighboring areas?” Tools to answer such questions are provided in Boundaryseer but are otherwise not generally available.

Breaklines and natural boundaries

Natural boundaries, such as coastlines, rivers, lake shores, cliffs, man-made obstacles, geological intrusions and fault lines are widespread and are frequently significant in terms of understanding spatial processes. When conducting spatial analysis it is often important to examine datasets for the existence of such features, and to take explicit account of their impact where appropriate. A large number of structures and arrangements may be encountered, and the standard analytical tools provided within your preferred GIS package may not handle these adequately for your purposes. For example, simply masking out regions, such as lakes, areas of sea, regions where the underlying geology is not igneous etc., is often misleading, if all this procedure does is to hide those areas which are covered. Ideally the analytical procedure applied should be able to take account of the breakline or natural boundary in a meaningful manner. In the case of breaklines there are three different types that tend to be supported, principally in connection with surface analysis and modeling (see further, Chapter 6, [Surface and Field Analysis](#)).

Terminology usage varies here, so we shall use the form provided within [ArcGIS](#) 3D Analyst for convenience:

- Case 1: hard breaklines, which are well-defined surface structures, whose shape and heights are known and fixed, and which mark a significant change of surface continuity — for example, indicating a road, a stream, a dam or a major change of slope which has been surveyed along its length (e.g. a ridge). Such breaklines are usually specified as a set of triples $\{x,y,z\}$ and intermediate values are usually determined by simple linear interpolation
- Case 2: soft breaklines are similar to hard breaklines, but do not imply a change in surface continuity. They indicate that the known values along the breakline should be maintained, for

example when creating a TIN of the surface, but no sharp change in the continuity of the underlying surface is implied

- Case 3: faults are more complex, in that they often have a spatial extent due to displacement and not just a simple linear form

The surface analysis package, [Surfer](#), models breaklines as 3D polylines and faults as 2D polylines or polygons, reflecting their variable form. [ArcGIS](#) models breaklines as 3D polylines and faults as 2D polylines. Software packages treat these types of boundary in very different ways. Cases (1) and (2) take into account all data points provided, although in Case (1) z -values computed for the linear breakline take precedence over points that may lie on the other side of the line from the point being analyzed. Case (3) is quite different, since fault lines and zones are treated as barriers. In [ArcGIS](#) data values beyond a barrier are not included in computations, where the notion of beyond is defined by visibility from the source point; in [Surfer](#) linear barriers are treated as obstacles, which may be skirted around (involving considerable additional distances and extra processing), whilst polygonal faults are treated more like self-contained regions, with interpolation inside and outside the polygonal regions being treated separately.

Tessellations and triangulations

A region may be divided into a set of non-overlapping areas in many ways. As we noted in [Table 1-1](#) a (regular or irregular) tessellation of a plane involves the subdivision of the plane into polygonal tiles that completely cover it. Within GIS regular tiles are almost always either square or rectangular, and form a (continuous) grid structure. Regular triangular and hexagonal grids are also possible in the plane but are rarely implemented in software packages. GIS does make extensive use of irregular triangular tessellations, both for division of plane regions and as an efficient means of representing surfaces. More recently mathematicians have shown that plane-filling tessellations can be constructed that are "soft", in the sense that they consist of regular shapes but without sharp corners (Domokos *et al.*, 2024).

Delaunay Triangulation

Given a set of points in the plane, P , lines may be drawn between these points to create a complete set of non-overlapping triangles (a triangulation) with the outer boundary being the convex hull of the point set. It has long been known that triangulation provides a secure method for locating points on the Earth's surface by field survey. It has also been found that a desirable characteristic of such triangulations is that long thin triangles with very acute internal angles are to be avoided in order to provide the best framework for measurement and analysis.

To ensure the triangulation will have the best chance of meeting the characteristics desired, a construction rule was devised by the mathematician Delaunay. This states that three points form a Delaunay Triangulation if and only if (*iff*) a circle which passes through all three points contains no other points in the set. Figure 4-32, below, illustrates this process for a set of spot height locations in a small area south of Edinburgh in Scotland (GB Ordnance Survey tile NT04). Each location is represented by an (x,y) pair. Many programs support the creation of the triangulation required, for example the `delaunay(x,y)` function in [MATLab](#) where x , and y are $n \times 1$ vectors containing the coordinates of all the points, or (as here) the Grid Data operation in [Surfer](#), using the option to export the triangulation used as a base map. The circles illustrate how the circle circumscribing each triangle has no other points from the set within it.

Worboys and Duckham (2004, pp 202-207) provide a useful discussion of triangulation algorithms.

A number of other observations should be made about this process:

- this triangulation is unique in the sense that it is the only triangulation of the point set that satisfies the rule described above. However, with some point set arrangements (e.g. four points at the corners of a square or rectangle) multiple Delaunay triangulations are possible, hence to this extent it is not unique
- long thin triangles with acute angles do occur using this procedure, although the number and severity of these is minimized compared to other triangulations
- the center of the circumscribing circle (marked with a • here) may lie outside the triangle in question, as is the case with one of the circles illustrated
- this procedure is by no means the only, or necessarily the best method of triangulation for all purposes. For example, when modeling a surface there may be important breaks of surface continuity or surveyed transects which it is highly desirable to include within the dataset, which are not captured by simple Delaunay-type procedures (as noted in Section 4.2.13, [Boundaries and zone membership](#))

If all points included in the Delaunay triangulation of a set P also lie on the convex hull of P the result is known as a furthest point Delaunay triangulation. The set of lines (edges) that belongs to all possible furthest site Delaunay triangulations is known as a *Delaunay diagram*, and has important applications in network analysis (see further, Chapter 7, [Network and Location Analysis](#)).

TINs — Triangulated irregular networks

Regular lattices such as grids provide a very inefficient means of storing details of surface topography. Areas that have similar slope could be better represented and stored using a series of inter-connected triangles forming an irregular mesh or network (a TIN). The nodes or vertices of these triangles can be placed closer together where landscape detail is more complex and further apart where surfaces are simpler. TINs may be designed or derived. Designed TINs arise in engineering and surveying models, whereas derived TINs are typically programmatically generated from grid-based datasets (digital elevation models or DEMs).

Although such TINs could be generated by selecting vertices using key feature extraction and surface modeling, a simpler and faster approach is to systematically divide the entire region into more and more triangles based on simple subdivision rules. For example, if you start with a rectangular region and define vertices at each of the four corners, two triangulations are possible: (i) a line connecting the NE-SW corners creates one pair of triangles; or (ii) a line connecting the NW-SE corners creates a second pair. For each triangle a statistic can be computed, such as the maximum deviation in elevation of the triangle from the original DEM. The arrangement that produces the smallest maximum absolute deviation (a minimax criterion) can then be selected as the preferred triangulation. Each of the two initial triangles can then be subdivided further. For example, a point can be placed at the centroid of each triangle which creates a further three triangles, or 6 in total. The minimax criterion test can be applied for each of the subdivisions of the two original triangles and the process proceeds iteratively until the minimax criterion is less than a pre-defined threshold value (e.g. 5 meters) or the number of

triangles generated has reached a maximum value, also pre-defined. This kind of procedure is used in many GIS packages — for example [TNT/Datum Workstation](#) describes this process as *adaptive densification*. TINs and DEMs are discussed in more detail in Chapter 6 (see for example, [Figure 6-5](#)).

Voronoi/Thiessen polygons

Given a set of points in the plane, there exists an associated set of regions surrounding these points, such that all locations within any given region are closer to one of the points than to any other point. These regions may be regarded as the *dual* of the point set, and are known as proximity polygons, Voronoi polygons or Thiessen regions. These various terms are equivalent, as is the term Dirichlet cell, all being derived from the names of their proponents: Dirichlet (in 1850); Voronoi (in 1909); and Thiessen (in 1912). Detailed explanations and discussion of Voronoi polygons is provided in Boots ([CATMOG 45](#)) and Okabe *et al.* (2000).

[Figure 4-33A](#) shows an example of Voronoi region creation, in this case using [ArcGIS](#). The point set from which the map was generated is also shown and as can be seen, points are frequently close to polygon edges. Indeed, in some applications it may be desirable to exclude points that are closer together than some predefined tolerance level. Note that in this example Voronoi regions at the edges of the sample extend to infinity, and are shown as bounded (e.g. by the MBR of the point set plus 10%). [Figure 4-33B](#) shows the same point set plus the associated Delaunay triangulation once more, created within the [MATLab](#) mathematical modeling package, which (correctly) limits the set of regions generated to those that can be unambiguously computed. [Figure 4-33B](#) enables us to see exactly how Voronoi regions are created. There are two steps: the first is the triangulation process, connecting the point set as described earlier; and the second is bisection of the triangulation lines of adjacent points to define Voronoi polygon boundaries.

There are many reasons an analyst may wish to generate Voronoi polygons from a point set, but a common requirement is to assign values to regions or cells based on the attributes of the point set. The most common assignment is that of the individual point, or nearest-neighbor model. Other options are offered by some GIS packages in much the same way as is applied to grid and image files (in which each cell is surrounded by 4 or 8 immediate neighbors, depending on the model used). Note that the [MATLab](#) implementation does not extend to the MBR or other bounding region (e.g. rectangle) defined by the user. A selection of (a) [MATLab](#) and (b) [GRASS](#) functions that support operations discussed in this section is shown in [Table 4-7](#).

A variety of geometric and spatial analysis functions support operations on point sets, polygons, and triangulations. **Convex hull** tools such as *convhull* and *v.hull* compute the minimum convex polygon enclosing a set of points and can optionally return its area. Delaunay-related operations include *delaunay* and *v.delaunay* for generating Delaunay triangulations, *dsearch* for finding the nearest vertex in a Delaunay network to a given query point, and *tsearch* for identifying the specific Delaunay triangle containing a specified point. Polygon-related functions include *inpolygon*, which tests whether points fall inside, outside, or exactly on the boundary of a polygon; *polyarea* and *area_poly2.c*, which compute polygon area; and *rectint*, which calculates the area of overlap among two or more rectangles. For partitioning space based on proximity, *voronoi* and *v.voronoi* generate Voronoi diagrams corresponding to input point sets.

Assignment facilities provided within [ArcGIS](#) include: local mean value — the cell is assigned the average value of the cell and its direct neighbors; and local modal value — determined by computing the frequency distribution of all adjacent cells in 5 classes and then assigning the cell the class value from this set that is the most common amongst the cell and its immediate neighbors. These assignments represent local smoothing procedures. Local variation rather than values may also be computed. For example:

- local entropy value (equivalent to Shannon’s Landscape Diversity Index, SHDI — see Section 5.3.4, [Landscape Metrics](#)) — in this model the adjacent cells are again grouped into 5 sets (smart quantiles or natural breaks classes) and the number in each class is represented as a proportion of the total, p_i , $i=1-5$, from which Shannon’s entropy statistic is computed ([Table 1-3](#)) and this value is assigned to the cell — values will range from 0 to $\log_2(5)$; and
- local standard deviation and local inter-quartile range — these statistics are computed for the cell and its immediate neighbors and the value assigned to the selected cell

Voronoi polygons may also be generated in grid or raster datasets (perhaps better described as Voronoi patches). Since the definition of a Voronoi polygon in the plane is that it consists of all locations closer to selected points than to any other points, such regions can be generated by a form of spreading algorithm. In this case we start with a number of points (the source points, sometimes called the target points) and spread outwards in all directions at a constant rate, like ripples spreading across the surface of a pond when a stone has been thrown into the middle. The process stops when the ripple (spreading) from another point is encountered. The rectangular grid structure and the number of search directions (typically 8) affect the form of the resulting cells (Figure 4-34). With the raster representation Voronoi patch shape and boundary definition are not as clear cut as in vector operations. Within GIS packages processing activities of this type are usually carried out using techniques described as cost distance, or accumulated cost distance tools (e.g. in [ArcGIS](#) Spatial Analyst). In image processing the procedures adopt a somewhat different set of algorithms, based on scanning rather than spreading, which are known as distance transforms. The latter may be exact (in Euclidean distance terms) or approximate (see further, Section 4.4.3, [Distance Transforms](#)). Furthermore, with both approaches the space across which the scan or spread procedure is conducted may be homogenous (i.e. treated as uniform in all directions and at all points) or non-homogenous, containing areas of varying difficulty to traverse, varying cost, and/or including obstacles.

Okabe and colleagues have developed a software package, [SANET](#), which can be installed as an [ArcGIS](#) add-in for “Spatial Analysis on a Network”. The [SANET](#) software includes a network-based Voronoi tool, very similar to some of the network analysis tools provided in [ArcGIS](#) itself and in network-oriented packages such as [TransCAD](#) and [Cube](#). An illustration of its facilities is shown in [Figure 4-35](#). In this example, showing the Shibuya district of Tokyo, the point set (off-network sites which are Bank premises) have been linked to the nearest network edge, and then *non-directed* network Voronoi regions generated from these locations (new network nodes) using shortest path distances. If one-way streets are taken into account separate inward- and outward-directed Voronoi regions are computed (see Okabe and Satoh, 2008, for more details). In each case these regions derived using unweighted calculations. Weighted calculations, for example taking into account the cost or time of travel across network links results in additional region definitions.

Queries, Computations and Density

Spatial selection and spatial queries

As with many other terms, the expression “spatial query” means different things to different suppliers. In the majority of cases it refers to the process of *selecting* information (features and associated attribute data in tables) by some form of query operation that includes a spatial constraint. For example: SELECT all features WITHIN a specified radius of a specified point; or SELECT all point objects that fall WITHIN a specific polygon. For most (all) GIS packages this selection process can be achieved by a point-and-click visual interface, subject to certain selection rules. In parallel there is the option to execute a single command or series of commands as a script or program, or as a form of extended SQL query. The result of such queries may be a new table and feature set, which can be viewed, analyzed further or saved for additional analysis. Raster-based GIS packages do not generally support spatial query operations, although they may support standard SQL queries on attribute data. The various approaches are best understood by describing a number of specific implementations:

[ArcGIS](#) provides full support for standard SQL SELECT queries, such as: SELECT <all polygons in a given layer> WHERE <attribute table field> EQUALS <value>. It does not appear to support other forms of SQL query, such as MAKE TABLE, UPDATE or DELETE. It supports JOIN and RELATE operations between attribute tables based on specific fields, but it also provides spatial extensions to the JOIN and RELATE process enabling tables to be joined and queried based on their relative locations. Where joins are based on spatial location, data from one set of features are incorporated into the attribute table of the other set of features according to selectable rules. The ArcMAP visual interface provides a “Select by location” facility with selection qualified by a variety of spatial conditions (applicable to point, line and polygon features where meaningful): Intersect; Are within a distance of; Completely contain; Are completely within; Have their center in; Share a line segment with; Touch the boundary of; Are identical to; Are crossed by the outline of; Contain; Are contained by. Each of the options may be further modified by selecting a tolerance level (a buffer — see further Section 4.4.4, [Buffering](#)) around the points, lines or polygons in question.

[Manifold](#) provides support for specific spatial extensions to the standard SQL syntax. This enables a very large range of operations to be carried out, including so-called Action Queries (e.g. queries that CREATE, UPDATE and DELETE table data). Example spatial extensions that result in true or false being returned include: Adjacent, Contains, Intersection, IsArea/IsLine/IsPoint and Touches. These operations may be combined with other spatial operations (such as distance and buffer functions) to select records, and may also be combined with operators that generate spatial objects (e.g. a triangulation or convex hull of a point set) rather than simply selecting or amending existing objects. As such many of [Manifold](#)’s spatial query functions are an extension of its “Transform” facilities, performing geometrical processing on the selected data as well as simply selecting items.

[MapInfo](#), like [ArcGIS](#), provides a range of SQL SELECT facilities and support for spatial joins. Spatial join operators supported include: Contains (*A* contains at least the other object, *B*’s centroid), Contains Entire (*A* contains whole object *B*), Within (as per Contain, but *A* and *B* roles are reversed), Entirely within, and Intersects (*A* and *B* have at least one point in common). These operators enable attribute tables to be JOINed without sharing a common field (i.e. a spatial join only).

Simple calculations

Almost all GIS packages provide a wide range of calculation facilities. By simple calculations we mean arithmetic operations and standard functions applied to: (a) tabular data; (b) within queries and similar operations; and (c) applied to map layers (see further, Section 4.6, [Grid Operations and Map Algebra](#)). As with spatial queries, there are many different implementations of simple calculation operations and these are widely used, readily understandable, and applied as often to raster datasets as to vector data. A number of examples help to clarify the range of facilities provided:

(a) [ArcGIS](#) provides a field level calculator within attribute tables by right-clicking on any field (existing or newly added). This enables basic operations such as +, -, x and / to be performed, plus a range of standard functions such as cos(), exp(), log(), abs() etc. Operations are applied to one or more fields, e.g. $current\ field = (field1 + field2 * 100) / field3$, and this process either updates the values for all records in the table selected or, if a subset have been pre-selected, updates records in just that subset. In addition an “Advanced” option is provided enabling Visual Basic for Application (VBA) commands to be entered and executed. Such commands are often needed in [ArcGIS](#) in order to add or update intrinsic data (as explicit fields in the attribute table) such as the area of a polygon or the (x,y) coordinates of a polygon centroid. [Manifold](#) implements such operations through its extended SQL query facilities and using the Query toolbar facility it provides.

(b) Simple calculations and expressions are extensively used in SQL queries and in association with spatial extensions to standard SQL. Again, the common arithmetic operators are used (extended in some cases to facilitate integer division, modulo arithmetic etc.) combined with comparison operators (e.g. =, >, < etc.) and Boolean operators (e.g. AND, OR, NOT, XOR). Since queries often involve non-numeric fields additional operators are provided to enable the use of strings, dates and times. Note that it is often possible, and sometimes simpler, to perform calculations outside of the GIS framework using standard tools, such as spreadsheet or database programs.

(c) Within raster-based GIS, and in the raster processing facilities of general purpose GIS, there is a very large set of simple calculation facilities. Such facilities are a standard part of the set of operations provided in specifically raster-based GIS packages (e.g. [TerrSet/Idrisi](#), [PCRaster](#)), but are often additional components in general purpose (vector-oriented) GIS. For example in [ArcGIS](#) the majority of such tools are provided in the Spatial Analyst extension whilst in [Manifold](#) they are provided in the Surface Tools extension.

The term *map algebra* is often used to describe operations that combine two or more grid layers according to a simple algebraic expression, e.g. $A + 2 * B + C / 100$ where *A*, *B* and *C* are matching grid layers (Figure 4-36a). This is commonly the case with multi-band [remote sensing](#) data (e.g. 7 bands representing different spectral components of a single image). Map algebra may also be used where the grid resolutions do not match, in which case decisions have to be made (rules applied) on how resolution differences are to be accommodated. The example in Figure 4-36b shows how grid *C*, being lower resolution, determines the resolution of the result grid. The general term for this kind of process is *resampling*.

If distinct raster layers have different resolutions (cell sizes) then one or more must be resampled to ensure their resolutions match. The normal practice is to adjust the layers of finest resolution to match the coarsest, and to use simple nearest-neighbor interpolation to estimate values. By nearest-neighbor is meant that values are assigned based on the position of the center

of the output grid relative to the centers of finer resolution input grids. Grids do not have to consist entirely of square zones, nor do they have to align precisely, but if they do not then resampling and some form of localized interpolation is inevitable.

However, map algebra refers to a far broader set of operations, extending to functions that operate on individual grid cells, blocks of grid cells or entire grid layers (often described as *local*, *focal* or *neighborhood*, *zonal*, and *global* analysis) as we shall see in Section 4.6 and Chapter 6. In this Section we restrict our consideration to operations that apply on a cell-by-cell or point-by-point basis (the term *local* is generally used for such operations).

In [TerrSet/Idrisi](#), for example, pairs of images (grids of pixels) may be combined using the operators: +, -, x, /, ^ (power), maximum, minimum, and cover. These operators combine images *A* and *B* on a pixel by pixel basis, for example adding each pixel in image *B* to each matching pixel in image *A*. Operations such as maximum and minimum identify a value from corresponding positions in image *A* or *B* as appropriate, whilst the cover function creates an image just containing pixels from *A* except where these are zero, in which case the pixels of *B* are taken (showing through the “holes” in *A*). With the exception of + and -, all other operations are non-symmetric, i.e. $A<\text{operator}>B$ is not equal to $B<\text{operator}>A$. An example application of operations of this kind would be the algebraic expression:

$$C=(A-B)/(A+B)$$

where *C* is the output raster, *A* and *B* are input raster maps (e.g. remote sensed images or image bands) of vegetation cover, with pixels in *C* being taken as a vegetation index (VI, with values being in the range -1 to +1). Note that a check is required to ensure that if $A=B=0$ for a pixel location then the calculation gives the result 0.

Many GIS packages allow the creation and saving of macros or scripts that implement regularly used map algebra expressions. For example [TNT/Datum Workstation](#) and [ENVI](#) include numerous pre-built combination formulas of this type. TNT/Datum Workstation holds such formulas as scripts that may be modified or augmented.

Here *A* might be the near infra-red band (Figure 4-37A) from a [remote sensing](#) image and *B* the red band image (Figure 4-37B), with the result being an index raster providing a measure of vegetation vigor (Figure 4-37C). The basic index has the range [0,100], and as elsewhere, checks for division by 0 are required.

[PCRaster](#), a non-commercial package designed specifically for manipulating raster data, provides a rich set of point operations, many of which have been derived from the work of Tomlin (1990). These include a similar set to those described above, i.e. arithmetic, trigonometric and logarithmic functions, coupled with Boolean and relational operators. Point operations are augmented by facilities for missing data detection and substitution, conditional expressions, data lookup and other facilities that mirror those provided in the extended SQL-type command structures described earlier.

Combination operations of this type are not confined to raster datasets. Polygon layers may be combined with raster layers, either by converting the polygon layers to rasters and then proceeding as above (the most common approach in GIS packages), or by direct support for such combinations within the GIS (as is the case with TNT/Datum Workstation for example).

Ratios, indices, normalization, standardization and rate smoothing

The vegetation index (VI) calculation we described in Section 4.3.2, [Simple calculations](#), was generated in a manner that ensured all output pixels (grid cells) would contain entries that lie in the range $[-1, +1]$; an index value of 0 indicates that matching pixel locations have the same value. There are many reasons for changing attribute data values in this manner before mapping and/or as part of the analysis of spatial datasets. One of the principal reasons is that if you create a map showing zones (e.g. counties) for a *spatially extensive* attribute, such as total population, it is very misleading — obviously a single total value (indicated perhaps by a color) does not apply throughout the zone in question. A more correct or meaningful view would be to divide by the area of the zone and show a *spatially intensive* attribute, the density of the population across the zone in this case. In this way zones of different size can be analyzed and mapped in a consistent manner.

The field of geodemographics, which involves the analysis of population demographics by where they live, involves the widespread use of index values — for a full discussion of this subject see Harris *et al.* (2005) and the “Geodemographics Knowledgebase” web site maintained by the UK Market Research Society (www.mrs.org.uk/geodemographics/).

In many instances, especially with data that are organized by area (e.g. census data, data provided by zip/post code, data by local authority or health districts etc.) production of ratios is vital. The aim is to adjust quantitative data that are provided as a simple count (number of people, sheep, cars etc.) or as a continuous variable or *weight*, by a population value for the zone, in a process known as *normalization*. This term, rather confusingly, is used in several other spatial contexts including statistical analysis, in mathematical expressions and in connection with certain topological operations. The context should make clear the form of normalization under consideration. Essentially, in the current context, this process involves dividing the count or weight data by a value (often another field in the attribute table) that will convert the information into a measure of intensity for the zone in question — e.g. cars owned per household, or trees per hectare. This process removes the effects associated with data drawn from zones of unequal area or variable “population” size and makes zone-based data comparable.

Three particular types of normalization are widely used: (i) averages; (ii) percentages; and (iii) densities. In the case of averages, an attribute value in each zone (for example the number of children) is adjusted by dividing by, for example, the number of households in each zone, to provide the average number of children per household, which may then be stored in a new field and/or mapped. In the case of percentages, or conversion to a $[0, 1]$ range, a selected attribute value is divided by the maximum value or (in the case of count data) the total count in the set from which it has been drawn. For example, the number of people in a zone registered as unemployed divided by the total population of working age (e.g. 18-65). The final example is that of densities, where the divisor is the area of the zone (which may be stored as an intrinsic attribute by some software packages rather than a field) in which the specified attribute is found. The result is a set of values representing population per unit area, e.g. dwellings per square kilometer, mature trees per hectare, tons of wheat per acre etc. — all of which may be described as density measures.

The term *standardization* is closely related to the term normalization and is also widely used in analysis. It tends to appear in two separate contexts. The first is as a means of ensuring that data from separate sources are comparable for the problem at hand. For example, suppose that

a measure of the success rate of medical treatments by health district were being studied. For each health district the success rates as a percentage for each type of operation could be analyzed, mapped and reported. However, variations in the demographic makeup of health districts have not been factored into the equation. If there is a very high proportion of elderly people in some districts the success rates may be lower for certain types of treatment than in other districts. Standardization is the process whereby the rates are adjusted to take account of such district variations, for example in age, sex, social deprivation etc. *Direct standardization* may be achieved by taking the district proportion in various age groups and comparing this to the regional or national proportions and adjusting the reported rates by using these values to create an age-weighted rate. *Indirect standardization* involves computing expected values for each district based on the regional or national figures broken down by mix of age/sex/deprivation etc., and comparing these expected values with the actual rates. The latter approach is less susceptible to small sample sizes in the mix of standardizing variables (i.e. when all are combined and coupled with specific treatments in specific districts).

The second use of the term standardization involves adjusting the magnitude of a particular dataset to facilitate comparison or combination with other datasets. Two principal forms of such standardization are commonly applied. The first is z-score standardization (see [Table 1-3](#)). This involves computing the mean and standard deviation of each dataset, and altering all values to a z-score in which the mean is subtracted from the value and the result divided by the standard deviation (which must be non-zero). This generates values that have a mean of zero and a standard deviation of 1. This procedure is widely used in the statistical analysis of datasets. The second approach involves range-based standardization. In this method the minimum and maximum of the data values are computed (thus the *range*=*max-min*), and each value has the min subtracted from it and is then divided by the range. To reduce the effect of extreme values range standardization may omit a number or percentage of the smallest and largest values (e.g. the lowest and highest 10%), giving a form of trimmed range. Trimmed range-based standardization is the method currently applied within the UK Office of National Statistics (ONS) classification of census areas (wards) using census variables such as demographics, housing, socio-economic data and employment.

Another way of looking at the process of dividing a count for a zone, e.g. population, by the zone area is to imagine the zone is divided into very many small squares. Each square will contain a number of people registered as living in that square. The total for the zone is simply the sum of the counts in all the little squares. But this form of representation is exactly that of raster or grid GIS, and hence raster representations generally do not require area-based normalization, since they are already in a suitable form. Of course, the process of generating a grid representation may have involved computation of values for each cell from vector data, either assuming a uniform model or some other (volume preserving model) as discussed in [Section 4.2.10](#). Normalization is used with raster data for a variety of reasons, for example: (i) to change to set of values recorded in a single map layer to a [0,1] or [-1,1] range, e.g. by dividing each cell value, z , by the theoretical or observed maximum absolute value for the entire grid:

$z^* = z/z_{max}$ or using the transformation:

$$z^* = (z - z_{min}) / (z_{max} - z_{min})$$

and (ii) to combine two or more grid layers or images into a single, normalized or “indexed” value (sometimes this process is described as indexed overlay). In this latter case the overlay process may be a weighted sum, which is then normalized by dividing the resulting total by the sum of the weights. These various observations highlight a number of issues:

- the division process needs to have a way of dealing with cases where the divisor is 0
- missing data (as opposed to values that are 0) need to be handled
- if an attribute to be analyzed and/or mapped has already been normalized (e.g. is a percentage, a density, an average or a median) then it should not be further normalized as this will generally provide a meaningless result
- highly variable data can lead to confusing or misleading conclusions. For example, suppose one zone has a population of 10,000 children, and the incidence of childhood leukemia recorded for this zone over a 10 year period is 7 cases. The normalized rate is thus 0.7 per 1000 children. In an adjacent zone with only 500 children one case has been recorded. The rate is thus 2 per 1000 children, apparently far higher. Effects of this type, which are sometimes described as *variance instability*, are commonplace and cause particular problems when trying to determine whether rare illnesses or events occur with unexpectedly high frequencies in particular areas
- the result of normalization should be meaningful and defensible in terms of the problem at hand and the input datasets — using an inappropriate divisor or source data which are of dubious value or relevance will inevitably lead to unacceptable results
- obtaining an appropriate divisor may be problematic — for example if one is looking at the pattern of disease incidence in a particular year, census data (which in the UK are only collected every 10 years) may be an inadequate source of information on the size of the population at risk

GIS packages rarely emphasize the need for normalization. One exception is [ArcGIS](#), where the Properties form for map layers includes a normalization field when quantitative data are to be classified and mapped (Figure 4-38). In this example we are normalizing the number of people who are categorized as owner occupiers (home owners, OWN_OCC) by a field called TOTPOP (total population). The normalization prompt is provided for graduated colors, graduated symbols and proportional symbols, but not for dot density representation as this already provides a spatially intensive representation of the data. The result of normalization in such cases is a simple ratio. Assuming the field being normalized is a subset of the divisor (which it should be) the ratio will not exceed 1.0. Specialized spatial analysis software, like [GeoDa](#), may provide a more extensive and powerful set of pre-defined normalization tools (see further, [Section 5.2.1](#)), although in this case the software does not support built-in user-definable calculation facilities.

Many of [GeoDa](#)'s facilities are designed to address problems associated with simple normalization, or in ESDA terminology, the computation of *raw rates*. The example of OWN_OCC/TOTPOP above would be a raw rate, which could be plotted as a percentile map in order to identify which zones have relatively high or low rates of owner occupancy, unemployment or some other variable of interest. As an illustration of this procedure, Figure 4-39 shows 100 counties in North Carolina, USA where the incidence of sudden infant death

syndrome (SIDS, also known as “cot death”) in the period 1st July 1974 to end June 1978 is mapped. This dataset has been normalized to form a *raw rate* by dividing the number of cases in each zone by the births recorded for the same period (BIR74). The resulting raw rates have been sorted, from the lowest to the highest, and divided into 7 groups (or quantiles), each containing approximately the same number of zones. The uppermost range is comprised of the zones in which the relative incidence or *rate* of SIDS is highest.

However, even raw rates can be misleading, as noted earlier, and various techniques have been devised for analyzing and mapping such data to bring out the real points of interest. One example will suffice to explain the types of procedure available. The ratio $SID74/BIR74$ is normally computed for every zone. However, one could compute this ratio for all zones first, based on the sum of the selected fields. This would give an average ratio or rate for the entire study area. This value could be regarded as the *expected rate* for each zone, E , say. We can then calculate the *expected number* of SIDS cases for each zone by multiplying the average rate E by the relevant population in each zone, i : $E_i = E * BIR74_i$. Now consider the ratio: $R_i = SID74_i / E_i$ — this is a ratio with a value from $[0, n]$. It highlights those areas with higher than expected or lower than expected rates (Figure 4-40), i.e. *excess risk rates*. As can be seen, one zone in Figure 4-40 is picked out in the lower center of the mapped zones as appearing to have an unusually high SIDS rate, being at least 4 times the average for the region as a whole. This technique is one of a series of procedures, many of which are described as *rate smoothing*, that seek to focus attention on interesting data variations rather than artifacts of the way the zones are laid out or their underlying population size. As the [GeoDa](#) documentation states: “Using choropleth maps to represent the spatial distribution of rates or proportions represents a number of challenges. These are primarily due to the inherent variance instability (unequal precision) of rates as estimates for an underlying ‘risk’.” One approach proposed by Anselin to address this issue is their Empirical Bayes (EB) rate smoothing method (described in Anselin *et al.*, 2003 and supported in [GeoDa](#)). However, researchers should be aware that such smoothing is simply one form of ‘reasonable’ adjustment of the dataset that can introduce its own artifacts into subsequent data analyses.

When attempting to develop explanatory models of such data (mortality data, disease incidence data, crime data etc.) it is the widely accepted practice to utilize the expected rate based on overall regional or national statistics as an essential component of the analysis. This issue is discussed further in Section 5.6.4, [Spatial autoregressive and Bayesian modeling](#), where the data illustrated here are re-visited — see also, Cressie and Chan (1989); and Berke (2004).

Single ratios provide the standard form of normalization, as described above, but more complex combinations of attributes, from one or more sets of spatial data, may be combined in a variety of ways in order to generate *index values*. One advantage of such procedures is that index values may be used as a guide for planning policy or as comparative measures, for example as a means of comparing crime rates and clear-up rates between different police forces. As the need to incorporate several factors in the construction of such indices becomes more important, simple weighting and ratio procedures will be found wanting, and more sophisticated methods (such as various forms of cluster analysis) will be required.

Density, kernels and occupancy

In Section 4.3.3, [Ratios, indices, normalization, standardization and rate smoothing](#), we noted that spatially extensive variables, such as total population, should not be plotted directly for

zones under normal circumstances, but standardized in some manner. An exception to this general rule may be made where cartograms rather than conventional maps are produced (see below, [Cartograms](#)). For such maps the areas are distorted in some manner to reflect the (positive) magnitude of an attribute of each region, whilst retaining a sense of the original areal-based map. Densities, with respect to the variable under consideration, thus become uniform in such maps.

For variables that are a subset of a total it is often useful to compute and map these data as a proportion of this total, but for totals themselves we will need to compute a density measure — dividing the total population, N , by the zone area, A . This yields a density value, N/A , for each zone, but assumes that population density is constant through the zone and then may suddenly change to some other value at the zone boundaries. In our discussions on boundaries and on areal interpolation we have already alluded to the inadequacies of simple density calculation, but it remains in widespread use. Conversion of the source dataset to a raster model using some form of intelligent interpolation may well provide a more satisfactory result. A good introduction to mapping density is provided in Mitchell (1999, Chapter 4). Techniques for computing density estimates, including so-called kernel density measures, are discussed in the subsections below.

Point density

If the dataset we have is not zone-based but point- or line-based, alternative methods of determining density are required. The simplest measures apply some form of zoning and count the number of events in each zone, returning us to a zone-based dataset. For example, with data on thefts from cars (TFC events), each event may be geocoded and assigned to a unique zone such as a pre-defined 100mx100m grid (or *quadrat*), or to city blocks, census blocks, or policing districts. The density of TFC events is then simply the event count divided by the area of the associated zone in which they are reported to have occurred.

A difficulty with this approach is that densities will vary substantially depending on how the zoning, grid or quadrat is selected, and on how variable are the size and associated attributes of each zone (e.g. variations in car ownership and usage patterns). Overall density calculation for all zones (e.g. a city-wide figure for the density of TFC events) suffers similar problems, especially with regard to definition of the boundary (such problems are generally known as *edge effects*, and methods to adjust for edge effects are known as *edge correction* or *edge bias correction* methods). Use of such information to compare crime event levels within or across cities is clearly fraught with difficulties. An alternative approach, particularly suited to ecological datasets, is to define zones based on a covariate of interest, for example the underlying geology and terrain slope. Each classified area can then be taken as a separate zone type and densities computed within these (polygonal) boundaries. This type of computation is supported within [spatstat](#).

An alternative approach to density computation for two-dimensional point-sets is based on techniques developed in one-dimensional (univariate) statistical analysis. This is simplest to understand by looking at the example in Figure 4-41, below. We have a set of 5 events marked by crosses. They occur at positions 7, 8, 9, 12 and 14 along the line. We could argue that the point density across the entire 20-unit line length is $5/20=0.25$ points per unit length and assign a value of 0.25 to each section, as shown on the gray line. We might equally well argue that if we divide the overall line into two halves, the density over the first half should be 0.3 per unit

length and over the second, 0.2 per unit length, to reflect the variation in positioning of the 5 events.

There is clearly no single right answer or single method to assign the points to the line's entire length, so the method we choose will depend on the application we are considering. Important observations to notice about this problem include: the length of the line we start with seems to have an important effect on the density values we obtain, and since this may be arbitrary, some method of removing dependence on the line length is desirable; if the line is partitioned into discrete chunks a sudden break in density occurs where the partition boundaries occur, which is often undesirable; depending on the number of partitions and distribution of points, areas may contain zero density, even if this is not the kind of spread we are seeking or regard as meaningful; the line is assumed to be continuous, and allocation of density values to every part is valid; and finally, if we have too many partitions all sections will only contain values of 1 or 0, which is essentially back to where we started from.

These observations can be dealt with by treating each point in the original set as if it was spread over a range, then adding together overlapping zones and checking that the total adds up to the original value. For example, choosing each point and smoothing it over 5 units in a uniform symmetric manner, we obtain the result shown in Figure 4-42. The rows in the diagram show the spreading of each of the 5 original points, with the total row showing the sums (densities) assigned to each unit segment. These add up to 5, as they should, and a chart showing this distribution confirms the pattern of spread. This method still leaves us with some difficulties: there are no density values towards the edges of our linear region; density values still jump abruptly from one value to the next; and values are evenly spread around the initial points, whereas it might be more realistic to have a greater weighting of importance towards the center of each point. All of these concerns can be addressed by selecting a well-defined, smooth and optionally unbounded function, known as a *kernel*, and using this to spread the values. The function often used is a [Normal distribution](#), which is a bell-shaped curve extending to infinity in each direction, but with a finite (unit) area contained underneath the bell (Figure 4-43).

In Figure 4-43, for each point (7,8,9,12 and 14) we have provided a [Normal distribution](#) curve with central value (the mean) at the point in question and with an average spread (standard deviation) of 2 units. We can then add the areas under each of these curves together to obtain a (cumulative) density curve with two peaks (dotted line), and then divide this curve by 5 to adjust the area under the curve back to 1 giving the lower (red) curve shown. When adjusted in this way the values are often described as probability densities, and when extended to two dimensions, the resulting surface is described as a probability density surface, rather than a density surface. We now have a density value for every position along the original line, with smooth transitions between the values, which is exactly what we were trying to achieve.

There still remain some questions: why should we use the [Normal distribution](#)? could we not use almost any unimodal symmetric function with a finite area under it? and why did we choose a value of 2 units for the average spread? The answer to these questions is that the specific selections made are a matter of choice and experience, although in some instances a symmetric distribution with a finite extent (e.g. a box or triangular function) may be regarded as more suitable than one with an infinite possible extent. Figure 4-44 shows a selection of commonly used functions plotted for the same point set, using the [MATLab](#) Statistics Toolbox function `ksdensity()`. As may be seen from examining the various curves shown in Figure 4-44 the exact

form of the kernel function does not tend to have a major impact on the set of density values assigned across the linear segment (or area in 2D applications). Of much greater impact is the choice of the spread parameter, or bandwidth. For further details on these and other functions used for smoothing and density estimation see Bowman and Azzalini (1997) and Silverman (1986).

Two-dimensional kernel density

All of this discussion so far addresses problems in one dimension (univariate density estimation). We now need to extend the process to two dimensions, which turns out to be simply a matter of taking the univariate procedures and adding a second dimension (effectively rotating the kernel function about each point). If we were to use the [Normal distribution](#) again as our kernel function it would have a two-dimensional bell-shaped form over every point (Figure 4-45). As before, we place the kernel function over each point in our study region and calculate the value contributed by that point over a finely drawn grid. The grid resolution does not affect the resulting surface form to any great degree, but if possible should be set to be meaningful within the context of the dataset being analyzed, including any known spatial errors or rounding that may have been applied, and making allowance for any areas that should be omitted from the computations (e.g. industrial zones, water, parks etc. when considering residential-based data). Values for all points at every grid intersection or for every grid cell are then computed and added together to give a composite density surface.

The resulting grid values may be provided as: (i) relative densities — these provide values in events per unit area (i.e. they are adjusted by the grid size, giving a figure as events per square meter or per hectare) — this is the default or only option in many GIS packages, including [ArcGIS](#); (ii) absolute densities — these provide values in terms of events per grid cell, and hence are not adjusted by cell size. The sum of the values across all cells should equal the number of events used in the analysis; (iii) probabilities — as per (ii) but divided by the total number of events. [Crimestat](#) supports all three variants. Computed density values may then be plotted in 2D (e.g. as density contours) or as a 3D surface (as in Figure 4-46). In this latter case the kernel density procedure has been applied to a dataset of reported cases of lung cancer in part of Lancashire, England. This example is discussed further in Section 5.4.4, and fully in Diggle (1990) and in Baddeley *et al.* (2005). The software used in this instance was [Crimestat](#), with a Normal kernel function and average spread (bandwidth) determined from the point pattern itself. Map data were output in ESRI shape file format (SHP) and mapped in [ArcGIS](#) to produce a 2D map and in ASCII grid format for visualization in 3D using [Surfer](#). Other GIS packages support a variety of kernel functions and procedures. [ArcGIS](#) Spatial Analyst provides kernel density estimation for point and line objects, but only supports one kernel function, which it describes as a quadratic kernel (a bounded kernel) but which is often described as an Epanechnikov kernel (see further, [Table 4-8](#)). In general KDE software implementations do not take boundaries or barriers into account (fixed or porous) — typically areas are simply masked off, which can result in inaccuracies when computing probabilities or matching absolute densities to regionalized values.

[MapInfo](#)'s grid analysis add-on package, Vertical Mapper, includes kernel density mapping as do the spatial statistics packages [splancs](#), [spatstat](#) and [adehabitatHR](#), and the crime analysis add-on for [MapInfo](#), Hot Spot Detective (in the latter two cases based on quartic kernels). [TransCAD](#)/Maptitude supports what it describes as density grid creation with the

option of count (simple), quartic, triangular or uniform kernels. [Crimestat](#) supports four alternative kernels to the Normal, all of which have finite extent (i.e. typically are defined to have a value of 0 beyond a specified distance). These are known as the quartic, exponential, triangular and uniform kernels. If kernel density methods are applied in urban areas the use of network kernel density estimation should be considered (see further, below), as Okabe *et al.* (2009) have shown that the use of planar KDEs can lead to erroneous conclusions regarding clustering for point events on networks.

The details of each of the main kernel functions used in various GIS and statistical packages are as shown in [Table 4-8](#). The value at grid location g_j , at a distance d_{ij} from an event point i , is obtained as the sum of individual applications of the kernel function over all event points in the source dataset. The table shows normalized functions, where the distances d_{ij} have been divided by the kernel bandwidth, h , i.e. $t=d_{ij}/h$. Graphs of these functions are shown in [Figure 4-47](#), where each has been normalized such that the area under the graph sums to 1.

Whether the kernel for a particular event point contributes to the value at a grid point depends on: (i) the type of kernel function (bounded or not); (ii) the parameter, k , if applicable, which may be user defined or determined automatically in some instances; and (iii) the bandwidth, h , that is selected (a larger bandwidth spreads the influence of event points over a greater distance, but is also more likely to experience edge effects close to the study region boundaries). Event point sets may be weighted resulting in some event points having greater influence than others.

Bandwidth selection is often more of an art than a science, but it may be subject to formal analysis and estimation, for example by applying kernel density estimation (KDE) procedures to sets of data where actual densities are known. An alternative to fixed bandwidth selection is *adaptive selection*, whereby the user specifies the selection criterion, for example defining the number of event points to include within a circle centered on each event point, and taking the radius of this circle as the bandwidth around that point. A somewhat different form of adaptive density model is provided within the [spatstat](#) software. In this case a random sample of points (a fraction, $f\%$) is taken, and this is used to create a Voronoi tessellation of the sample point set. The density of the remaining points ($1-f\%$) is then computed for each cell of the tessellation. The process is then replicated n times and the average densities of all replicates computed.

Widely used univariate kernel density functions

The Normal (or Gaussian) kernel is unbounded and therefore defined for all values of t , where $t = d_{ij} / h$ and h is the bandwidth. It is the standard kernel used in CrimeStat, and the bandwidth represents the standard deviation, which may be fixed or adaptive. The Quartic (spherical) kernel is bounded and approximates the Normal kernel, with k being a constant. The (Negative) Exponential kernel is optionally bounded, where A is a constant (for example, $A = 3/2$) and k is a parameter (such as $k = 3$); this kernel assigns more weight to the central point than other kernels. The Triangular (conic) kernel is bounded and features a very simple linear decay with distance. The Uniform (flat) kernel is also bounded, with k as a constant, and applies no central weighting, making the function resemble a uniform disk placed over each event point. The Epanechnikov (paraboloid or quadratic) kernel is bounded and serves as an optimal smoothing function for certain statistical applications; it is used as the smoothing function in the Geographical Analysis Machine (GAM/K) and in ArcGIS.

Kernel smoothing, or kernel density estimation methods (KDE methods) of the type described have a variety of applications: exploratory point data analysis; point data smoothing; creation of continuous surfaces from point data in order to combine or compare these with other datasets that are continuous/in raster form; probability distribution estimation; interpolation (although this terminology is confusing and not recommended — [Crimestat](#) is amongst a number of packages that use this terminology, which is essentially incorrect); and hot spot detection.

KDE can also be used in visualizing and analyzing temporal patterns, for example crime events or animal movement at different times of day and over prolonged periods, with the objective of understanding and potentially predicting territories and/or event patterns. In the diagram below 1000+ data recordings from tracking of two panthers (*Puma concolor coryi*) in 2006 have been used to define the core territory or "home range" of the panthers using KDE. The probability envelopes for the home range are set at 99% for the darkest regions diminishing to 50% for the outermost shaded areas. In this case the computations and plot creation used the R package [adehabitatHR](#) with data and code from the test datasets provided via W D Walter's "[Manual of Applied Spatial Ecology](#)". The kernel function in this case is the Bivariate Normal with fixed bandwidth, *href*, defined as a fraction of the average two-dimensional standard deviation of the point sets. Note that the female panther range is much smaller than the male panther range.

KDE can also be applied with more than one point set, for example a set of cases and a set of controls. The output of such "dual" dataset analysis is normally a ratio of the primary set to the secondary set (or ratio of the log transform of the variables), with the objective being to analyze the primary pattern with background effects being removed or minimized. Care must be taken in such cases that the results are not subject to distortion by very low or zero values in the second density surface, and that common bandwidths are used. [Crimestat](#) provides support for 6 alternative dual density outputs: simple ratio; log ratio; difference in densities (two variants, with and without standardization); and sum of densities (again, with and without standardization). Levine (2007, Chapter 8) provides an excellent discussion of the various techniques and options, including alternative methods for bandwidth selection, together with examples from crime analysis, health research, urban development and ecology.

Kernel density for networks

The preceding discussion is not readily applicable to the case of points on networks, such as accidents along roads or thefts in urban areas. In these instances one might attempt to compute the density by applying a two-dimensional kernel procedure as described in the preceding section, overlaying the network, and examining the density patterns along the network routes. However, this approach has been shown to produce biased density estimates (e.g. it suggests variable densities along routes when these are known to be uniform). To resolve this problem kernel density computations in situations where events occur along or adjacent to networks, should be calculated using shortest paths and adjusted kernel functions (the conventional functions described earlier are not suitable without modification. This question has been addressed by Okabe *et al.* (2009) and is available in their [SANET](#) software. They show that a non-continuous fixed bandwidth kernel function, which is unimodal and symmetric, can be used to compute unbiased kernel densities on large networks (e.g. the street network of Kashiwa, Japan, with 35,235 links, 25,146 nodes and 3109 point events). The function used is essentially a modified form of the linear or triangular kernel ([Figure 4-47F](#)), applied to

networks which may divide into multiple segments at junctions, but for which the integral of the kernel function is held at unity. The authors suggest that for problems of the type they have investigated (i.e. dense, urban networks), the bandwidth should be 100-300metres.

Line and intersection densities

Density methods may be applied to linear forms as well as point forms. Some statistics of interest include:

- the number of line segments per unit area (sometimes called the *line frequency*, λ , rather than the density)
- the length of line segments (by type) per unit area. The equation $L_a = L_0 e^{-r/b}$ has been found to provide a good model of the distribution of urban road length with radial distance, r , from many city centers. L_0 =the road density at the city center, which for London, UK is c.19.5 kms/km²; b is a parameter to be fitted from sample data. The inverse of this density measure is the average area associated with a unit of path length, sometimes called the *constant of path maintenance*
- the number of *line intersections* per unit area
- the number of line intersections per unit of length (the latter is known as the *intersection density*, k). This is a particular case of the density of points on lines, which may be subject to kernel density smoothing in the manner discussed earlier in this Section. It can be shown that if λ is the line density (frequency) then for a random network of straight lines there will be an average of $k = \lambda^2/\pi$ intersections per unit length. $K = k/\lambda^2$ is sometimes called the *crossings factor*, and experimental evidence suggests that the random model provides an upper limit to this measure, so K has the range $[0, 1/\pi]$.

All these measures provide information that may be useful for analysis, including inter-regional comparisons. Applications include hydrology, ecology and road traffic analysis. For further discussion of applications in these areas and statistical measures see Haggett and Chorley (1969, Chapter 2.II) and Vaughan (1987, Chapter 3). A number of GIS packages provide facilities for kernel density mapping of lines (i.e. not points on lines – this is covered above) as well as points (e.g. [ArcGIS](#) Spatial Analyst, Density operations). Essentially these work in a similar manner to 2D point operations, spreading the linear form symmetrically according to a specified kernel function. [ArcGIS](#) provides two sets of point density and line density functions (for SIMPLE or KERNEL density operations), one set that results in a grid being generated and the other for use in map algebra environments (i.e. where a grid already exists). SIMPLE density operations generate a set of grid values based on a neighborhood shape (e.g. circular, annular, rectangular, or polygonal) and size (e.g. radius). Similar facilities do not appear to be available with most other mainstream GIS packages. The [GRASS](#) open source package include a vector processing function v.kernel (formerly s.kernel) that now includes linear kernel generation (i.e. a 1D kernel) in addition to its generation of 2D raster grids from vector point sets with a Normal (Gaussian) kernel.

Cartograms

A cartogram is a form of map in which some spatially extensive variable such as population size or gross national product is substituted for land area. The geometry or space of the map is distorted (transformed) in order to convey the information of this alternate variable. The end

result is a map or diagram in which the density of the variable under consideration is uniform throughout the map; variation in values are represented by variations in the size of mapped regions. Cartograms may be created use a variety of algorithms. One simple approach, due to Dorling (1995, [CATMOG](#) 59), is to associate the centroid of each mapped zone with the value of the variable to be mapped, and then to construct a circle around this point with area proportional to the magnitude of the variable. The size of the circles are typically scaled so that the largest is some fraction of a bounding rectangle encompassing all the original zones. Generally these circles will overlap, as illustrated for Zurich cantons in [Figure 4-48A](#). By systematically moving and rescaling the circles overlapping can be minimized whilst retaining approximate contiguity of the mapped regions ([Figure 4-48B](#)).

Dorling (1995, p32) describes the algorithm, for which sample code is provided, as follows:

Each region ... is treated as an object in a gravity model which is repelled by other circles with which it overlaps, but is attracted to circles which were neighboring regions on the original map. Forces akin to forces of gravity are calculated, including velocity and acceleration, and the whole process is acted out as if it were occurring in treacle, to avoid any circles moving too quickly (as can occur when small objects orbit too close to large objects in the original gravity models).

This example illustrates a procedure that seeks to represent the variable ‘population’ by the areas of the circles, whilst retaining the approximate relative positions and contiguity of the communes whilst avoiding overlap. It results in a discontinuous map with no attempt to retain any of the original region shapes. This form of cartogram is supported in several GIS and mapping packages, including [MAPresso](#) (Java-based, now archived) and [GeoDa](#).

If the objective is to create a cartogram that retains full contiguity, then the original area-based map needs to be stretched, or ‘rubber-sheeted’, in some way. Many algorithms have been proposed for such cartograms, commencing with Tobler’s work on map transformations, Tobler (1963). More recently continuous transformation cartograms have been implemented using the rubber-sheeting algorithms of Dougenik *et al.* (1985) and the diffusion-based algorithm of Gastner and Newman (2004) – see [Figure 4-49B](#) (Zurich cantons), [Figure 4-50](#) and [Figure 4-51B](#) (which uses the North Carolina dataset) discussed in several sections of this Guide. Examples and discussion of visualizations that utilize this approach can be found on the World Mapper website (worldmapper.org/): and the Views of the Word website: www.viewsoftheworld.net/

It is important to note that within-zone variation is not represented in this technique, hence representation of highly concentrated datasets (such as population) will provide a very particular view of the dataset. If zoned or gridded data are available within zones it may be useful to consider using such information when generating cartograms.

The Gastner-Newman technique commences with the source map with its varying ‘population’ density and conceptually allows the individual members of this population to diffuse from higher density zones to lower density zones until such time as density is uniform. Zone boundaries are progressively moved during the diffusion process such that the net flow through them is zero, until they reach a stable position. It is important to note that here, as with many software implementations of geospatial algorithms, the final cartograms may differ depending upon the software package used to generate them and the parameters used or selected. For

example, Dorling cartograms are produced very rapidly by [GeoDa](#), with no option for parameter selection, and with some residual overlapping of the circular regions ([Figure 4-51C](#)). If the MapViewer software is chosen with the same dataset the results are markedly different for almost all parameter selections ([Figure 4-51D](#)). Of course, as another alternative, the shape of regions could be retained and their size altered to reflect the mapped variable if they are sufficiently well separated (i.e. a form of ‘explosion’ or multipolygon representation of the original mapped regions – see [Figure 4-51E](#)).

A final cartogram procedure that warrants attention involves identifying a zoning of the original source data in which the population is approximately equal (e.g. census areas, electoral districts). These zones, which will generally be of varying sizes, are then replaced with equal-sized hexagons and re-arranged into a lattice that approximates the source map. This procedure provides a contiguous mapping of the source data, retaining some of the overall form of the conventional map, whilst eliminating most of the variations due to underlying population levels. The Atlas of Mortality in Britain by Shaw *et al.* (2008) uses this approach for all of its maps (see for example, [Figure 4-52](#)) – these show the cause of death for roughly 15 million cases over a 24 year period.

Distance Operations

Fundamental to all spatial analysis is the use of distance calculations. If our model of the world (mapped region under investigation) is regarded as a flat plane (2D Euclidean space) or a perfect sphere, then we can use the simple distance formulas d_E and d_S provided in Section 4.2.1, [Length and area for vector data](#), to compute distances between point pairs. These provide the basic building blocks for many forms of spatial analysis within most GIS packages. However, these two formulas are too limiting for certain forms of analysis and additional distance measures are required. Figure 4-53a shows a simplified diagram of the Earth as a spherical object and as a reference ellipsoid based on the [WGS84 definition](#). Note that the WGS84 ellipsoid is not the same as the WGS84 geoid, the latter being defined by the Earth's gravitational field (EGM84). In some cases distance measures (or metrics) can be computed directly from the coordinates of input point pairs, but in others computation is an incremental process. In fact incremental methods that compute length along a designated path (between a series of closely spaced point pairs) by summing the lengths of path segments, are the most general way of determining distance, i.e. the notions of distance and path are inextricably linked.

Figure 4-53b provides a simple cross-sectional illustration of the kind of issues that can arise when measuring *terrain* distance. In this example we wish to determine the distance separating points *A* and *B*. If *A* and *B* are not too far apart (e.g. less than 10 kms) we could use a high precision laser rangefinder to establish the *slope distance* between *A* and *B*, assuming there is no atmospheric distortion. In practice there will be some distortion and the laser wave path will need to be adjusted in order to provide an estimated slope path distance (this may be carried out automatically by the measuring device). This in turn may require further adjustment if it is to be referenced to a common datum or a level terrain surface. In each case the distance recorded between *A* and *B* will be different. From the diagram (Figure 4-53b) it is clear that none of these distances corresponds to the actual distance across the *terrain surface* along a fixed transect, nor to a distance adjusted or computed to reflect the particular model of the

Earth or region of the globe we are using. In many cases these differences will be small, whilst in other they may be highly significant.

In general, simple Euclidean 2D (incremental) computations are satisfactory for many problems involving relatively smooth terrains extending over an area of perhaps 20km by 20km, even using suitably defined longitude and latitude coordinates (assuming these latter coordinates are not inside the Arctic or Antarctic regions). Closer to the poles the distortion of lines of latitude results in increasing errors in this formula and a variant known as the Polar Coordinate Flat-Earth formula may be used as an improved approximation.

In this formula we are using polar coordinates (ϕ, λ) to represent latitude and longitude expressed in radians, and R = the radius of the terrestrial sphere. With this formula the error is at most 20 meters over 20 kms even at 88degN or S. For example, the formula evaluates the distance between $A(70\text{degN}, 30\text{degN})$ and $B(71\text{degN}, 31\text{degN})$ as 117.6 kms (using the WGS84 ellipsoid), 112.8 kms 10deg further north, and 111.4 kms at 87degN/88degN. One degree separation (ignoring longitude differences) would be 111.3 kms. However, for: (a) larger regions, or problems where *surface distance* is important; (b) problems involving networks; and (c) for problems involving variable costs; alternative procedures are required. Within mainstream GIS packages a range of tools exist to support such analysis, including the provision of support for Euclidean and Spherical distance (but not always ellipsoidal distance), network distance and cost distance (where cost is a generalized concept as we discuss in the subsections that follow).

Most software packages offer at least three categories of distance measure: (i) coordinate-based distance measures in the plane or on a sphere (which may be adjusted to a more close approximation of the Earth's shape); (ii) network-based measures determined by summing the stored length or related measure of a single link or multiple set of links along a specified route (e.g. shortest, quickest); and (iii) polyline-based measures, determined by summing the length of stored straight line segments (sequences of closely spaced plane coordinates) representing a given feature, such as a boundary. The first and third of these distance measures utilize the simple and familiar formulas for distance in uniform 2D and 3D Euclidean space and on the surface of a perfect sphere or spheroid. It is important to check to see if distance computations provided represent true distances (e.g. in miles or kms) since Euclidean computations on raw latitude/longitude values produces incorrect results.

These formulas make two fundamental assumptions: (i) the distance to be measured is to be calculated along the shortest physical path in the selected space — this is defined to be the shortest straight line between the selected points (Euclidean straight lines or great circle arcs on the sphere); and (ii) that the space is completely uniform — there are no variations in terms of direction or location. These assumptions facilitate the use of expressions that only require knowledge of the coordinates of the initial and final locations, and thereby avoid the difficult question of how you actually get from A to B .

In most cases measured terrestrial distances are reduced to a common base, such as the WGS84 reference ellipsoid. Note that the reference ellipsoid may be above or below the geoidal surface, and neither distance corresponds to distance across the landscape. Laser-based measurement is rarely used for distances of 50km or more, and increasingly DGPS measurements of location followed by direct coordinate-based calculations avoid the requirement for the wave path adjustments discussed above.

Whilst coordinate-based formulation is very convenient, it generally means that a GIS will not provide true distances between locations across the physical surface. It is possible for such systems to provide slope distance, and potentially surface distance along a selected transect or profile, but such measures are not provided in most packages. A number of packages do provide the option to specify transects, as lines or polylines, and then to provide a range of information relating to these transects. Examples include the Grid|Slice command procedures in [Surfer](#), the Profiles and Elevations facilities in [Manifold](#) (Surface Tools) and the Profile facility in [ArcGIS](#) 3D Analyst. In addition to path profiles (elevations, lengths and related transect statistics) such paths may be linked with volumetric cut-and-fill analyses (see further Sections 6.2.3, [Directional derivatives](#), and 6.2.6, [Pit filling](#)).

For problems that are not constrained to lie on networks it is possible to derive distances from the surface representation within the GIS. For example this can be achieved using a regular or irregular lattice representation along whose edges a set of links may be accumulated to form a (shortest) path between the origin and destination points. For the subset of problems concerned with existing networks, such questions could be restricted to locations that lie on these networks (or at network nodes) where path link lengths have been previously obtained. Both of these approaches immediately raise a series of additional questions: how accurate is the representation being used and does it distort the resulting path? which path through the network or surface representation should be used (e.g. shortest length, least effort, most scenic)? how are intermediate start and end points to be handled? are selected paths symmetric (in terms of physical distance, time, cost, effort etc.)? what constraints on the path are being applied (e.g. in terms of gradient, curvature, restriction to the existing physical surface etc.)? is the measure returned understandable and meaningful in terms of the requirements? how are multi-point/multi-path problems to be handled? Answers to some of these questions are discussed in Section 4.4.2, [Cost distance](#), but we first discuss the notion of a metric.

Metrics

Introduction

The term metric in connection with spatial analysis normally refers to the form of distance calculation applied, although it has a number of other meanings (e.g. the metric system of weights and measures; in the [Manifold](#) GIS package it means the sequence of coordinates that defines a vector object). Here we stick to its use in connection with distance computation. The essential features of the notion of distance are:

- a finite or infinite set of distinct, definable objects, and
- a rule or set of rules for determining the degree of separation of object pairs (a “measure”).

From a mathematical perspective a metric d_{ij} , denoting a measure of separation between points i and j , should satisfy a number of specific conditions. These conditions describe the properties of a valid distance metric: $d_{ij} > 0$ if $i \neq j$, which represents distinction or separation; $d_{ij} = 0$ if $i = j$, indicating co-location or equivalence; $d_{ij} + d_{jk} \geq d_{ik}$, which expresses the triangle inequality; and $d_{ij} = d_{ji}$, which represents symmetry.

Asymmetry is also a characteristic of a number of telecommunications networks, where some modem standards and some broadband Internet technologies (e.g. ADSL) operate in this way — information is generally received at a faster rate than it can be transmitted. A similar situation

applies in many hybrid broadcast/telecommunications systems, such as satellite, coax or fiber-optic downstream and copper/cable upstream, and in some mobile network architectures. Pure broadcast technologies, such as terrestrial television and radio, are strictly one-way and only facilitate two-way communication via completely separate media (such as post, telephone, Internet or mobile data). This kind of asymmetry also reflects a fundamental human characteristic — our ability to receive and process information is far greater than our ability to respond to it in an active manner.

If the triangularity constraint is not met we have the possibility that

$$d_{ij} + d_{jk} < d_{ik}$$

which offends our notion of “between-ness” and thus of conventional notions of distance and space. For example, this would suggest that the “distance” from London to Madrid plus the “distance” from Madrid to Lima might be less than (cheaper than, faster than) the direct “distance” from London to Lima. A similar pattern can be found with railway and other transport timetables, where express trains and stopping trains in combination may reach a destination faster than a direct stopping train, even though the rail distance traveled may be greater. It may not be possible to draw a time- or space-consistent map of such data, but this does not mean that the measure used is not valuable. Indeed, in such cases non-cartographic representations (e.g. GIS internal representation, to/from matrices) of the data are often more meaningful and useful in optimization and routing problems, and as such are explicitly supported in GIS-T packages such as [TransCAD](#) and [Cube](#).

Terrestrial distances

Most GIS software uses the standard Euclidean formula, d_E , when computing distances between plane coordinate pairs and for calculation of “local” distances, as in the case of the vector buffer operator (see further, Section 4.4.4, [Buffering](#)). It is important to check to see if distance computations provided represent true distances (e.g. in miles or kms) since as noted earlier, Euclidean computations on raw latitude/longitude values produces incorrect results. Some packages, such as [MapInfo](#), provide the option of using spherical distance, d_S , as a default. [Manifold](#) computes distances either using Euclidean or Ellipsoidal methods (supported via its functions `Distance()` and `DistanceEarth()`, and in its Tracker facility).

The [MATLab](#) Mapping Toolbox provides a range of distance and associated mapping functions, which help to illustrate some of the differences that apply on larger scale maps. Figure 4-55 shows a map of the North Atlantic on a Mercator projection, with the great circle path and the line of constant bearing, or *rhumb line* between Boston, Mass. and Bristol, England. The [MATLab](#) function `distance()` will compute the length of each path, taking the latitude and longitude of Boston and Bristol as inputs, and optionally a geoid parameter. The [GRASS](#) module `geodist.c` provides similar functionality. In the example shown the great circle distance is computed as 5105.6 kms based on its default terrestrial radius of 6371 kms; the Haversine formula we gave in Section 4.2.1, [Length and area for vector data](#), produces the same result, whilst the rhumb line distance is 5283.4 kms.

The geoid parameter, rather confusingly named, provides the option to select a model of the Earth which has an ellipsoidal form, specified by a two element vector: (i) the length of the

major semi-axis of the ellipsoid and; (ii) its eccentricity or flattening coefficient. If these parameters are not specified it is assumed that the two semi-axes are equal, and thus provide a spherical or great circle measure. If geoid (ellipsoid) values are provided then an approximation method is used to calculate distances. It first treats the two terrestrial coordinates as if they were Cartesian coordinates in three dimensions and calculates the Euclidean distance between them. This value, which is the chord length of an elliptical section, is then adjusted by a correction factor to produce the final estimate of ellipsoidal distance. In our example this gives a revised distance of 5110.4 kms, a few kilometers longer than the great circle distance, as expected. Finally, yet another alternative is to use the more accurate Vincenty (1975) algorithm to obtain the ellipsoidal distance, which gives a figure of 5119.7 kms for the matching ellipsoid in this example.

Extended Euclidean and L_p -metric distances

Euclidean distance is the most widely used GIS metric. In 2D space (planar or projected space) the Euclidean distance between points $a(x_1, y_1)$ and $b(x_2, y_2)$ is a special case of the n -dimensional Euclidean metric. Within GIS this extended Euclidean metric is used to determine 3D distances (across surfaces and within solid structures) and in multi-dimensional analyses such as clustering.

This formula is processor-intensive to compute for very large number of point pairs and dimensions, mainly due to the square root operation. For some applications, such as K -means clustering (see Section 4.2.12, [Classification and clustering](#)) and line generalization (see Section 4.2.4, [Line Smoothing and point-weeding](#)) the square root is not taken, which breaks its metric behavior, but remains suitable for these specific purposes. In other areas, such as incremental grid operations, approximations to the Euclidean metric are taken to avoid such problems (see further, Section 4.4.2, [Cost distance](#)).

The formula above also arises, with some modifications, in the analysis of statistical datasets. The standard Euclidean metric in n -space takes no account of the average variation of the data points. However, it can be normalized (made scale free) by calculating the variance of the point set components, σ_i^2 , and using these as a divisor:

This is known as the *normalized Euclidean metric* (see also, Section 4.3.3, [Ratios, indices, normalization, standardization and rate smoothing](#)) and is essentially the same as the Euclidean metric if the variances of point set components (the average variation of values or attributes of each dimensional component) are equal. Pre-normalizing the attributes of input datasets (e.g. raster files) and then computing conventional n -dimensional Euclidean distances will produce the same result. This expression is also closely related to a more general measure, introduced by P C Mahalanobis in 1936 and named after him, which is used to determine the similarity of an unknown sample set to a known one. In addition to the variances of the components *Mahalanobis distance* takes into account the covariances that may exist. This more general measure is used in a number of GIS packages in connection with the analysis of [remote sensing](#) data.

For applications that require network distance measures between locations, Euclidean distance is generally inadequate and computing large numbers of (shortest) network distances may be impractical or inappropriate. In order to obtain more useful measures of inter-location distance many authors have suggested that the standard Euclidean distance formula be amended in

various ways. The most frequently used modifications are based on replacing the use of the square and square root operations — i.e. replacing the powers of 2 and 1/2 — with other values, typically p and $1/p$, estimated from sampling the network to be analyzed. We shall refer to such modifications as the family of L_p metrics or p -norms, with L_2 being the standard Euclidean measure. They are amongst a range of alternative metrics supported by the now defunct Locational Analysis package, LOLA, which provides output in [ArcGIS](#) shape file format (SHP) as an option (see further, Section 7.4.1 *et seq.*, [Location problems](#)).

In geographic literature the case $p=1$ is variously called the Manhattan, Taxicab, Rectilinear or City block metric, and is supported in a number of software packages, including the now defunct LOLA project and the [Crimestat](#) package (in the latter instance with both Plane and Spherical computations). With $p=2$ we have the standard Euclidean metric and with $p=\infty$ we have the so-called minimax or Chebyshev metric used in a number of optimization problems. Note that the shape of the “circle” for $p=1$ is a square at a 45 degree angle, and this is the same as the shape for $p=\infty$.

Values for p of approximately 1.7 have been found by many studies to approximate road distances within and between towns, and such estimates may be used in place of direct-line distance (where $p=2$) in problems such as locational analysis and the modeling of travel patterns — see for example, Love *et al.* (1972) and (1988). An alternative is to use a simple *route factor* (typically around 1.3) as a multiplier and to continue using Euclidean distances, but inflated by this route factor. Note that if $p<1$ the triangle inequality no longer holds, i.e. it is possible for the distance (or time) from A to B plus the distance from B to C to be *less than* the distance (or time) from A to C . This characteristic is quite common in transport infrastructures, where one-way systems, timetabled routes and traffic congestion alter the expected paths significantly. Numerous variations on the simple L_p metric model have been studied, including using a mixture of powers for the x - and y -directions (p and q), adding weighting factors for the two directions, including a different power, s , for the root etc. Most of these variations provide modest improvements in estimation of transport route distances at the expense of increased complexity in determination of the parameters. A summary of the possible values for p in the L_p metrics and their interpretation is provided in [Table 4-9](#).

One of the most striking features of the L_p metric diagram (Figure 4-56) is that for every value of p except 2 the “circles” show rotational bias — maximized at 45 degrees. Similar bias exists if other variants of the L_p metric family are plotted. This highlights the fact that a distance value obtained by measurement and application of any of these modified formulas will change if the reference frame is rotated. If this seems a bit obscure, imagine measuring the distance between two points randomly located on a rectangular street network oriented north-south/east-west. In general this will involve finding the length of a route in the north-south direction and adding the length of a route in the east-west direction. If, however, we could ignore the physical network and rotate and translate the pattern of streets about one of the points until the two points both lay along one road (a rotated north-south route), the length measured would be shorter. This comment applies equally to grid files, where distances (and many other measures) are often calculated in an incremental manner which is affected by the grid orientation.

The interpretation of p -values in the L_p distance metric varies across ranges. When $0 < p < 1$, the L_p formula does not yield a true metric because the triangle inequality no longer holds; for instance, constructing minimal spanning trees by progressively connecting nearest neighbors

is not valid. In urban settings, routes may have $p < 1$, indicating complex route structures—for example, a value of $p = 2/3$ implies that effective travel distances are roughly twice as long as direct line estimates would suggest, and the locus of points in this range is concave. When $1 < p < 2$, the resulting distance d is a valid metric that lies between the Manhattan (grid-like) metric and the Euclidean metric, making it suitable for representing more complex city street networks. For $p > 2$, increasing p places greater weight on the largest components of distance within the network. Considering the distance from $(0,0)$ to $(1,1-E)$ where $0 < E < 1$: for $p = 1$, $d = 2 - E$; for $p = 2$ and $E \ll 1$, $d \approx \sqrt{2}\sqrt{1-E}$; and as p continues to increase, $d \rightarrow 1$. In all cases, as $p \rightarrow \infty$, the distance becomes $d = \max(|x_1 - x_2|, |y_1 - y_2|)$.

The above observations on rotational relationships can be used in problem solving—for example, in some cases it may be possible to solve selected problems by applying the sequence of actions:

- rotate the problem coordinate frame
- solve the problem using the metric appropriate to the rotated frame, and then
- rotate the frame back to its original position.

For example, Okabe *et al.* (2000, Section 3.7) show that in order to determine the Voronoi diagram for a set of planar points under the L_∞ metric, the set of points may be rotated by an angle of $+\pi/4$ radians and the regions computed using the L_1 metric, with the resulting diagram rotated back by $-\pi/4$ radians to produce the final solution.

Although the p -metric family provides real value in analytic studies, this lack of rotational invariance is undesirable in the context of distance measurement. Clearly it would be simple for suppliers to include such distance expressions in their GIS packages, but the estimation of the parameters and the interpretation of their results require considerable care. In practice the advent of more powerful hardware and software, and greatly improved network datasets, has meant that many problems can be tackled by explicit network analysis rather than the application of alternative metrics (see further, Chapter 7, [Network and Location Analysis](#)).

Cost distance

The term cost distance is widely used within GIS and in this context has a dual meaning: (i) the notion of an alternative family of distance metrics; and (ii) a procedure for determining least cost paths across continuous surfaces, typically using grid representations

Within GIS the term is distinct from the computation of varying costs or times of travel across a network. The term cost in this connection is also a generic idea, not necessarily implying financial cost, but some composite measure that varies (i.e. is not constant) across the study region, and which needs to be taken into account when computing paths and distances.

An example serves to clarify this notion. Suppose that we wish to construct a section of road between two points, $A(0,2)$ and $B(4,2)$, where costs are constant over the region of interest. In this case the distance is simply 4 units and the path involved is a straight line in the plane (Figure 4-57, bold line). Construction costs will simply be $4k$ units, where k is some constant (the cost per unit length of path construction). Now suppose that construction costs also depend on the cost of land acquisition, and that land costs vary in our study region in a simple manner—greater to the “north” of our region than to the “south”. In this situation it is cheaper to

construct our road with a curve in it, taking advantage of the fact that land costs are lower to the south (nearer the x -axis in our model).

If costs increase in a precisely linear manner, the optimal (least cost) path will have a smooth curve form of the kind shown by the thinner line (a shape like a chain hanging between two posts). This case turns out to be solvable mathematically, as are cases where costs vary in radially symmetric manner (e.g. around a city center), whereas almost every other example of varying costs cannot be solved precisely and we need to resort to computational procedures to find the least cost path.

Accumulated cost surfaces and least cost paths

Within GIS packages solutions to such problems are found using a family of algorithms known as Accumulated Cost Surface (ACS) methods — for a fuller discussion of ACS methods and representational accuracy see Douglas (1994) and Eastman (1989). The ACS procedure is applied to raster datasets, in which the primary input surface is a complete grid of generalized costs, i.e. every cell is assigned an absolute or relative cost measure, where costs must be (positive) ratio-scaled variables. In Figure 4-58 we show a hypothetical cost grid for the same example as shown in Figure 4-57. Values increase linearly away from the x -axis, from 0.5 to 2.5. This essentially a much simplified representation of the function $f(x,y)=y$, so $f(x,y)=0$ when $y=0$ for all x and increases steadily so at $y=2$ $f(x,y)=2$. In practice cost functions should rarely if ever have zero-valued sections, so this example should be considered illustrative!

Providing a gridded representation of the cost surface highlights the importance of using a sufficiently fine representation of the surface, since the solution path must be computed by adding up the costs along alternative paths. If the grid resolution is quite coarse, as in Figure 4-58, the solution path will not have the curved form of the optimal path, but will simply be a very blocky stepped line of cells approximating the curve. Amending the grid resolution (halving the cell size to 0.25 units) as shown in Figure 4-59A improves both the cost surface representation and the possible solution paths that are considered.

If we look at a path in Figure 4-58 that starts at $A(0,2)$ and goes diagonally down towards $C(2,0)$ and then back up to $B(4,2)$ we have cell costs of $2+1.5+1+.5+.5+1+1.5+2=10$ units (times cell size of 0.5 gives a cost distance of 5 units). But this path is longer than the sum of the steps — all the diagonal steps are $\sqrt{2}$ (1.414... units) longer than horizontal and vertical steps, so the total cost must be increased for such steps by this factor, giving a final cost distance of 6.85 units. This is less than the cost of 8 units that would be incurred for a straight line path from A to B . But can we be sure this is the best route possible for this problem and is the calculation “correct”?

ACS provides a systematic procedure for answering this question. With this algorithm we start at A and sum up the costs in all directions, step by step, choosing the least cost increment at each step. Eventually, when we arrive at B , we will have the overall least cost of getting to B , and if we have kept track of our route (the steps we have taken) we will have the least cost path from A to B . Thus ACS is a form of spread algorithm, with values increasing in all directions depending on the grid resolution, cost surface and algorithmic implementation. Almost all GIS implementations of ACS use immediate neighbors in calculating values (i.e. queen’s moves, to the 8 adjacent cells) but some facilitate calculation to neighbors further away. [GRASS](#), for

example, offers the option in the `r.cost` function, of using knight's move neighbors (one step up and one diagonal) in its calculations. [Distance transform](#) methods utilize a much wider range of neighbors, and hence are not restricted to 8-directional cost surfaces and path alignments, and thus do not exhibit the common directional bias and computational errors of ACS distance models.

As we start to consider the ACS idea more closely we see that a number of additional issues need to be addressed. The first is that in a GIS-enabled grid structure cells (or more specifically cell centers) are frequently used as references rather than grid lines. Thus our starting point A cannot be $(0,2)$ exactly but must be a cell center and our end points will also all be cell centers. For example, see Figure 4-59B where $A(0,2)$ is approximated on the grid by the point $A(0.125, 1.875)$. For the best results we should estimate the value of the cost function at cell centers if possible, rather than at cell edges or grid lines. In the example we are using we have ignored this latter factor, which will often be the case since finer detail for within cell estimation may not be available (although could be obtained by interpolation).

A second issue is that if we take grid cell centers as our reference point, the start point will be half way across the first cell, and the next point will be half way across an adjacent cell, so it can be argued that the cell values (costs) should be allocated 50:50 over this one step, and likewise for steps 2, 3 etc. This can be seen in Figure 4-59B where the starting point for A is shown as a selected cell center and the step to A' in the diagonally adjacent cell lies 50% in a cell with cost 2 and 50% in a cell with cost 1.75. This 50:50 division of costs is the normal approach used in GIS packages. The averaged cost from A to A' is then $(2+1.75)/2=3.75/2$ and thus the incremental cost distance is $\sqrt{2} \times 0.25 \times 3.75/2$ units (adjusting for the diagonal path using $\sqrt{2}$ and for the cell size of 0.25 units in this example).

We now have all the basic elements we need in order to compute accumulated costs and identify the least cost path from A to B . Using our example cost grid, with a cell size of 0.25 units, we need to carry out the following operations using a typical GIS:

- Create a source grid that contains a 0 in the cell in which our starting point A lies, and some other value (e.g. -1, 1, 9999) for all other grid cells. This can be done with a text editor and converted to a GIS compatible grid format if necessary
- Create a cost grid in the same manner, exactly the same size and the same resolution as the source grid, but containing the costs associated with each cell position (i.e. similar to the cells in Figure 4-59a)
- Run the cost distance facility within the GIS on these two input grids, requesting the output as a cost distance (ACS) grid, and also requesting (if necessary) that a direction or tracking dataset is generated that can be subsequently used to identify least cost paths
- Finally, to create the least cost paths, a target grid is required, which contains the target or destination points (again as 0s or a similar cell identifier for the target points). The least cost path facility in the GIS is then run using the target grid, ACS grid and tracking grid

A sample result for our test problem is shown in Figure 4-60. Each grid cell has been separately colored and a series of additional destination points (cells) have been included in the least cost path finding stage. The least cost path from A to B , corresponding to our original problem, hugs the bottom of the diagram as it seeks to take advantage of the low cost first row (cells with a

value of only 0.25 units). Routes to other cells may share common paths for parts of their route and may not be unique, as indicated by the two routes to the cell diagonally below point *A* to the right. If the cost surface had been flatter, for example $f(x,y)=y/4+1$, the ACS model with the cell resolution illustrated would fail to select the correct path, but would show a horizontal path from A to B. To obtain a more accurate estimate of the correct path and cost distance a much finer grid would be required. For this latter example the true path is a gentle curve with a minimum value at around $y=1.65$ and total cost of 5.88 units.

This simple example provides a flavor of cost distance computation, but this procedure (or family of procedures) has far more to offer. The first and perhaps most important point to note is that the source and target features do not need to be single points. They can be multiple distinct points (cells), lines (linear groups of cells), or regions (clusters of adjacent cells). In all such cases the ACS procedure computes the lowest total distance to a target cell from its nearest source cell. If requested the GIS may also produce an allocation raster, which identifies which source each cell in the grid is closest to (typically by giving each cell a unique integer identifier). Obstacles (e.g. lakes, cliffs etc.) may be included in these models by setting cells in the cost raster to very high values or by the GIS explicitly supporting unavailable (masked) cells (e.g. in [TerrSet/Idrisi](#), coding barrier cells as -1).

The input cost raster may be, and often is, a composite grid created from a number of other (normalized) input grids by simple map algebraic operations. The final cost grid may also be normalized to provide an indication for each cell of the relative cost, difficulty or *friction* of crossing that cell. Figure 4-61 illustrates the result of such a process, which has been applied to determining possible alternative routes for a main road in part of South-East England. The map shows the existing road (the A259T) from Hastings in the south-west to Brenzett in the north-east. Three alternative routes, identified here by arrows, were generated by ACS methods. These routes were generated from cost surfaces that included different weighted values for a series of inputs: land-use (grade and type of land); SSSIs (Sites of special scientific interest); areas of designated Ancient Woodland; the slope of the land across the study area (derived from a DEM); existing (buffered) built-up areas (many of which are conservation areas) and ancient monuments; National Trust land; and existing (buffered) road and rail routes. The final route alternatives were then computed from the composite ACS surface and smoothed to reduce the effects of the gridding of all the datasets and output paths. This information was then provided as input to a public consultation exercise.

This is a relatively simple example of a general class of problems categorized as corridor planning — selection of alternative routes for roads, rail lines, power lines or other purposes frequently requires the involvement of many parties, and formal multi-criteria decision support systems, such as [AHP](#) and ANP, have been extensively used in this field — see, for example, Azis (1990), Bailey (2003) and Piantanakulchai (2005).

The above examples are based on the procedures utilized in [ArcGIS](#), Spatial Analyst. Many other packages offer similar facilities. For example, [TerrSet/Idrisi](#) offers two alternative algorithms: one, called the *pushbroom* algorithm, is similar to [distance transform](#) procedures, and is fast but not suitable for use with more complex problems (for example truly implementing barriers); a second [TerrSet/Idrisi](#) procedure, *costgrow*, is much slower but does facilitate complex friction patterns and fixed barriers. [TerrSet/Idrisi](#) also allows for anisotropy in the cost modeling process, with some directions (e.g. downhill) being preferable to others

(e.g. uphill). [SAGA](#), [GRASS](#) and [PCRaster](#) provide similar functions, with inputs optionally including a friction surface (with relative friction values and absolute barriers) and a physical surface, with movements restricted to uphill, downhill and either stopping at or crossing flat areas. Such procedures are clearly targeted at hydrological and related studies, so whilst being well suited to these environments are not readily generalized. Many such programs, particularly those with an environmental sciences and [remote sensing](#) background, perform the path finding stage using a form of steepest descent procedure (often described as a *drainage* procedure) applied to the ACS. This approach is not guaranteed to provide the correct shortest path — tracked paths should be used. This is illustrated in [Figure 4-62](#), where local Euclidean grid distances and 50:50 splitting of costs between the cells shaded in gray has been used to generate the ACS grid, Figure 4-62B. The steepest path from the target cell back to the source cell (Figure 4-62C) is not the least cost path, as its total accumulated cost is 13.726. The correct least cost path is shown in Figure 4-62D. ACS procedures are subject to a number of other problems, for example: becoming trapped in localized plateaus or pits; inadequate handling of multiple equivalent paths; or failing when input and/or output datasets are specified to operate in integer rather than float mode.

A number of GIS packages incorporate extensions to the standard cost distance/ACS model. For example, [ArcGIS](#) provides a variant of the ACS procedure which it describes as Path Distance. This is essentially the same as ACS but includes options such as inclusion of surface distance (using a surface raster or DEM), and vertical and horizontal movement rasters that are utilized in adjusting the optimum path computation. [TNT/Datum Workstation](#) also includes a path analysis procedure that takes into account the physical surface form, but the authors are cautious about its optimality. [TerrSet/Idrisi](#) provides a similar facility, Disperse, designed to provide a simple force-based model of anisotropic object dispersal ([SAGA](#) provides comparable facilities). In this case the inputs are a set of four grid files: (i) the location of the target and source features (non-zero entries are targets, zeros are sources); (ii) and (iii) two grids, one containing the magnitude and the other containing the direction (azimuth) of the anisotropic force field, i.e. a vector field modeled using a pair of grids; and (iv) an output grid. The notion of the magnitude of the force field in this implementation is as the inverse of a friction field, F , $\text{force} = 1/F$ where F in this case is taken to be greater than 1. α is an angular difference between the angle being considered and the stated friction angle, and k is a parameter ($k=2$ is the default). The process may be modeled as a series of steps (time periods) by pre-specifying a maximum dispersal distance at each stage. *The anisotropic dispersal model is either user specified or utilizes an inbuilt cosine function that modifies the stated friction field $F' = F * f.\alpha$* is an angular difference between the angle being considered and the stated friction angle, and k is a parameter ($k=2$ is the default). The process may be modeled as a series of steps (time periods) by pre-specifying a maximum dispersal distance at each stage.

Distance Transforms

In the image processing world a procedure similar to Accumulated Cost Surface (ACS) has been developed in connection with work on pattern recognition. The procedure is known as a Distance Transform (DT), and is typically implemented on binary images in a scanning manner rather than the spreading approach of ACS. Extremely efficient algorithms have been developed to generate exact Euclidean DTs and excellent approximations to exact Euclidean DTs (EDTs) on large images. Some of these procedures have even been implemented within

hardware chipsets for real-time applications. GIS vendors do not generally utilize DT methods, but they are included in many image processing toolsets and packages, including the [MATLab](#) Image Processing Toolkit: e.g. see the `bwdist()` function, which provides a variety of metrics for black and white images, including City Block and exact Euclidean. In a series of publications, de Smith (2004, 2006a) has shown how Distance Transforms can be applied to solve complex optimization problems in the geospatial domain. The technique has been applied to a range of practical optimization problems by a number of authors, including selection of optimal routes in road, rail and pipeline engineering projects, some of which are described later in this section.

Using DTs, distances over the image or grid are calculated in an incremental manner based entirely on the distance to *nearby* (not necessarily *adjacent*) cells. The standard algorithm involves a two-pass scan of a square or rectangular lattice: a forward scan from top left (NW corner) to bottom right (SE corner), and then a backwards scan from bottom right to top left (i.e. SE→NW corner). The scan commences by placing the forward scan mask over the first available cell in the grid to be transformed – in a 3x3 scan mask ($M \times M$ cells, where $M=3$) this will be grid cell (row 2, col 2).

A basic DT 3x3 mask is illustrated in Figure 4-63, where 5 values are used based on the two-value (3,4) integer mask. (3,4) is useful as it is extremely fast to process, and dividing through by 3 at the end yields direct steps of 1 and diagonal steps of 1.33... giving a reasonable approximation to *local* Euclidean measure (1.414...) and within 6% of exact *global* Euclidean measure. Typically the underlying grid to be transformed is an array in which unreached cells are initially assigned a large number (e.g. 999) and target or source cells are assigned a value of 0. Cells that act as barriers or restricted areas (e.g. buildings) can be coded with even larger numbers (e.g. 9999), in which case they are never altered. On completion of the two-pass scan each cell in the resulting lattice will contain the distance to the *nearest point* in the set of source points. If barriers are included they must be in blocks of at least $(M-1) \times (M-1)$ cells, and accessible areas should consist of at least $(M+1) \times (M+1)$ cells. In addition in this case two scan pairs: (NW→SE/SE→NW) and (SW→NE/NE→SW), with one or more iterations, are required to ensure full scanning of the grid. On completion of the scan each cell in the resulting lattice will contain the distance to the *nearest point* in the set of source points. The pushbroom algorithm described by Eastman (1989) and used in [TerrSet/Idrisi](#) for their cost distance modeling is very similar to this approach.

The central function in this algorithm is of the form: $d0 = \min\{d + D(i), d0\}$, where $d0$ is the current value at the central point (0) of the template, $D(i)$ is the local distance to the i^{th} element of the template, and d is the current value at (r,c) . The basic algorithm involves $O(Mn^2)$ computations where n is the maximum dimension of the grid. Details of DT procedures are provided in the references cited, and code samples in [MATLab](#) and [Python](#) are available from the author. A selection of 3x3 masks is shown in Table 4-10.

These are known as *chamfer metrics* because the locus of the metric generates a figure similar to a cross-section of a piece of wood with chamfered or beveled edges. The generic form of such masks is defined by the two unique mask values (a,b), e.g. (2,1) or (3,4). The maximum error percentages shown in Table 4-10 provide the *global* maximum errors for the metric as compared with global Euclidean measure. The use of 3x3 local metrics is fast and simple, but

results in directional bias in the resulting DT. These show the familiar 4- or 8-direction patterns seen in many GIS processes that rely on 3x3 neighborhoods (see, for example, Figure 4-34).

This problem can be considerably reduced by extending the neighborhood to a 5x5 mask ($M=5$), in which case 3 values define the mask: (a,b,c) as shown in Figure 4-64. Amongst the best integer values for this mask size are the triple (5,7,11), dividing all cell values by 5 at the end to give (1,1.4,2.2). The best fractional values are (0.9866, 1.4141, 2.2062). Use of the integer values above ensures that every grid cell distance computed (i.e. *global* values) is within 2% of the exact Euclidean distance and within 1.36% for the fractional masks. This compares with global errors of up to 7.6% with a pure *local* Euclidean metric.

his shows a uniform region with a single target point in the center (originally coded as 0, with all other grid cells coded 999). This figure shows the *best* non-exact DT possible using a fractional-valued 5x5 scanning template. The exact Euclidean distance transform (EDT) of this uniform space yields a perfectly circular set of filled contours around the central point. The vector equivalent would be a multi-ring buffer operation. When applied to multiple points (e.g. as part of a point-pattern analysis exercise) the set of computed distances, $F()$, is known as the *empty space function*.

If a simple barrier is introduced, e.g. representing a river with a bridge point, shortest paths appear *diffracted* through the gap (Figure 4-65B). In this example the target point has been set as (2,50). With a complex barrier, as in Figure 4-65C, a single extra iteration has been used to ensure full convergence. In all cases shortest paths to target points can be drawn as orthogonal to the distance bands, but are best constructed using tracking of the selected steps, as described further in the examples below. Although the DT approach may seem somewhat divorced from practical GIS analysis, DT procedures have been extended and developed by: Ratti *et al.* (architecture and the built environment; 2001, 2004); de Smith (transport routes; 2004, 2006), Ikonen and Toivanen (image processing; 2005); Kristinsson *et al.* (geothermal pipelines, 2005); and Wei Li *et al.* (railways in mountainous regions, 2017) to provide a new and fast family of ACS-like procedures for solving many complex grid-related distance problems.

The central function in the DT procedure in this instance is of the form:

$$d0 = \min\{d + D(i) * \text{COST}(r,c), d0\}$$

where $d0$ is the current value at the central point (0) of the mask, $D(i)$ is the local distance to the i^{th} element of the mask, d is the current value at (r,c) and $\text{COST}(r,c)$ is the (averaged) cost function at row r , column c . As can be seen, the velocity field distorts the distance bands. Shortest (quickest) paths across this transformed surface may be plotted by following routes to or from the source point in any given direction, remaining at right angles (orthogonal) to the equal time contours, or by tracking of the computed quickest paths. Hence the quickest route from the start point due South of the center to locations North of the city center would curve around (East or West) the center to avoid congestion (low traffic speeds). In this particular example these curve form log spirals (Wardrop, 1969). Angel and Hyman applied models of this type to the analysis of traffic flows in Manchester, UK. Distance transform methods enable such models to be calibrated against known analytical results and then applied to more realistic complex velocity and cost fields, for example to examine the possible effects of introducing central zone congestion charging.

A second example of this procedure is shown in Figure 4-67. This shows a section of the Notting Hill area of central London, with the annual Carnival route shown in white and areas in dark gray being masked out (buildings etc.). Colors/grayscale indicate increasing distances (in meters) from the Carnival route along nearby streets.

Although this example has been created by a DT scanning procedure the result is sometimes described as a “burning fire” analogy, with the fire spreading from the source locations (in this case a route) outwards along adjacent streets and squares (for true fire spread models see www.fire.org and Stratton, 2004). The same techniques can be applied in public open spaces and even within building complexes such as shopping malls, galleries and airports. This illustrates the ability of DT methods to operate in what might be described as complex friction environments.

Distance transforms are very flexible in their ability to incorporate problem-specific constraints. For example, in a typical ACS procedure variable slopes are modeled by regarding steeper slopes as higher cost, rather than relating the spreading process to slope directly (other than some uphill/downhill constrained implementations). The selection of a route for a road, railway or pipeline is affected by slope, but primarily in the direction of the path rather than the maximum slope of the physical surface (although this does have a bearing on construction cost).

Figure 4-68 illustrates the application of a gradient-constrained DT (GCDT) to the selection of road routes through a hilly landscape. The most northerly two routes (upper right in the diagram) were constrained to a 1:12 and 1:11 gradient (before engineering grading) whilst the most southerly two represent routes found using less restrictive 1:9 and 1:8 gradient constraints.

This technique has been applied to the problem of mountain railway alignment optimization (Wei Li, Hao Pu, Paul Schonfeld *et al.*, 2017). As the authors explain

"Various complex constraints must be satisfied in railway alignment design for topographically complex mountainous regions, where the natural terrain gradient between the start and end points greatly exceeds the maximum allowed design gradient. The alignment design for such environments is so challenging that existing methodologies have great difficulties in automatically generating viable railway alignment alternatives. We solve this problem with an alignment optimization methodology combining the bidirectional distance transform and a genetic algorithm. This methodology consists of a two-stage search that first identifies promising corridors within a broad area and then optimizes alignment alternatives within those corridors.... This approach has been employed by the China Railway Eryuan Engineering Group and has been applied in designing the Sichuan-Tibet, Tongdao-Quanzhou, Zhengzhou-Guiyang, and Ganluo-Yaan railways as well as a railway across South America in Peru and Brazil. "

In the case study area illustrated below (Figure 4-69A, B), an area of extremely mountainous terrain covering over 900 sq km is shown. This forms part of the Sichuan-Tibet railway design project. The blue (dark) line shows the Distance Transform optimized path, which differs markedly from the manually aligned path chosen by specialists in the field, and which surprisingly crosses itself. The reason for this unexpected behavior is that the constrained path does not lie on the terrain surface, but almost entirely under the surface (in tunnels) or above the terrain surface via bridges. Where the path crosses itself there is a large difference in the

vertical location of the relevant sections of the path, which occur because of the gradient constraints applied. The final optimized path, generated using genetic algorithm (GA) techniques with the DT path as a starting point, is shown in Figure 4-69b. It represents an estimated saving of approximately \$3million per annum over the best manually aligned path.

With minor modifications the GCDT procedure described above has been utilized with a high resolution (10meter) DEM to locate alternative pipeline routes for the Hellisheiði 80MW geothermal power station in Iceland (Kristinsson, Jonsson and Jonsdottir, 2005). In this case additional constraints were applied giving the maximum and minimum heights permitted for the route and differential directional constraints that permitted limited flow uphill, which is possible for these kinds of two-phase pipelines (Figure 4-70A,B).

Network distance

Network distance, as the name suggests, is the distance (or cost, time etc.) involved in traversing a network. In most instances the term refers to the lowest value distance measure between pairs or lists of points (generally network nodes or points upstream and/or downstream in hydrological applications). Typically the network is a logical collection of linked polylines, joined at network nodes (or vertices), and possessing a network topology (sometimes called a geometry). Network links and nodes may have additional explicit attributes, such as type of link, weights applied to links and nodes, rules governing behavior of flows (e.g. link directionality, permitted turns, barriers) etc., and intrinsic attributes (e.g. position and segment lengths). When network distances are computed by a GIS the procedure takes into account the various explicit and intrinsic attributes of the network and the rules applied for the problem selected. Thus network distance will vary in meaning according to the problem at hand. A variety of hybrid concepts exist, which may be modeled within some GIS packages, but generally only in a limited manner. These include multi-modal networks (thus seeking routes that may combine road, rail and bus for example), and mixed network/cost distance situations, in which sources or targets may not lie on the network, but for which sections of optimal routes (or network building programme) need to utilize existing network infrastructures.

Most GIS packages have not historically included extensive networking tools as part of their standard offering, but may provide limited functionality in this area (e.g. [ArcGIS](#) Utility Network Analyst and Linear Referencing tools). This situation has changed in recent years: [Manifold](#) does include a range of network analysis tools via its Business Tools extension; [ArcGIS](#) V9 and later editions ship with the new Network Analyst option as part of the main distribution kit; [TNT/Datum Workstation](#) includes a network analysis module with a wide range of facilities; and the open source GIS, [GRASS](#), includes a range of network analysis tools in its core modules for vector processing, with functions such as v.net.path generating shortest paths and outputting associated attribute data such as distances and times in linked attribute tables. A rather different tool, [SANET](#), is specifically designed to carry out certain types of spatial analysis on a network (e.g. statistical analysis), and is provided as an [ArcGIS](#) add-in. [TransCAD](#) and [Cube](#) also differ from most GIS products as they are essentially “GIS for transportation” (GIS-T) offerings. [TransCAD](#) is built on a general purpose GIS engine, Mapititude and [Cube](#) is built on [ArcGIS](#). Figure 4-71 illustrates the use of network distance within the [TransCAD](#) shortest path facility, utilizing least distance (heavy line shown in blue/dark gray) and least time (line shown in green/light gray) with a single intermediate node. Such algorithms are widely used in web-based GIS providing routing and traffic advice

systems that are now pervasive on internet and app-based mapping systems. Output from these systems is in a variety of forms, including tabulation of section-by-section network distance/time and cumulative distance and/or time.

Depending on the GIS package, for large networks with potentially complex routing it is often advisable to compute link lengths as an explicit field or table and work with this data rather than expecting the GIS to re-compute such information dynamically from an intrinsic field. For more details see Chapter 7.

Buffering

Buffering is the process of creating one or more zones around selected features, within a pre-specified distance (almost always Euclidean distance) from these features. The resulting buffer is an area object or series of area objects (typically polygons) that may be automatically merged (dissolved) where they overlap. In many cases the buffer regions are created from a combination of straight lines and circular arcs centered on the vertices of the convex hull. In Figure 4-72A simple symmetric buffers generated using the [GRASS](#) v.buffer operation are shown, whilst in Figure 4-72B the result of applying a 100metre buffer to sections of road network is shown, without buffer segment merging (selected segment highlighted). Buffers may be applied to:

- point, line or polygon objects, or to cells in a grid dataset
- they may be symmetric, i.e. spreading outwards from lines and polygon boundaries in both directions (left and right or inside and out), or
- they may be asymmetric, spreading inwards or outwards only (see for example the buffering used in polygon skeletonization, [Figure 4-14](#))
- buffers may also be applied differentially, for example larger buffers applied to major roads and smaller buffers applied to minor roads, as part of a pollution analysis exercise (air-borne, noise, vibration etc.)
- buffers may be applied incrementally, progressively increasing and/or reducing in size

Although the idea of buffering appears very straightforward, and has many applications, different GIS packages implement the facility in a variety of ways. In [ArcGIS](#) extensive support is provided for buffer creation of selected features, with facilities to select buffering: to the left, right or both sides of polylines and polygons; to select whether linear feature buffers are terminated with rounded or squared-off ends; and to select features which may have their buffered regions combined (dissolved together) and optionally a selected attribute (field) to be used for determining whether dissolve operations should be performed. An additional buffer tool is provided to generate *multiple buffer rings* in a single operation. [MapInfo](#) provides a similar facility, which it calls “Concentric ring buffers”.

Figure 4-72 Simple buffering

[Manifold](#)’s principal buffering facility is invoked via the Transform toolbar, and the standard Buffers operation when applied to a polygonal area provides a buffer that extends outside the polygon plus the entire polygon internal area, effectively growing the entire polygon. To limit the buffer zone to the boundary of a polygon the Border buffer zone operation must be used (Figure 4-73B). An additional operation, Inner buffer zone is provided which selects the inner

part of a polygon, comprising all areas at least the buffer distance from the boundary (Figure 4-73C). Repeated buffering operations may be applied, either to the original feature or to the resulting buffered entity itself. Buffer features may also be modified, e.g. by clipping operations or intersection, to generate new or modified zones, as required. The Buffers operation in [Manifold](#) creates distinct buffer zones (Figure 4-74A), whilst an alternative function, ‘Common buffers’ creates merged zones (Figure 4-74B).

Once a buffer has been created, or as part of the creation process, a new map layer containing the buffer polygons will be generated. This map layer will have an associated attribute table, and if not already generated, attributes such as polygon areas can be computed. The new layer may, in turn be combined with other layers (e.g. via an Intersect overlay) to analyze attribute data within the buffer zone.

Raster buffering

Buffering may also be applied in raster GIS, whereby cells are classified according to their distance from one or more target cells (grid distance, Euclidean distance or a more complex, cost-related distance). For example, in [TerrSet/Idrisi](#) by default target cells are coded as 1, buffer cells as 2 and other cells as 0, and the buffer width to apply is specified in reference system units (based on true Euclidean distances or grid-based cost distances). The process of generating a buffer in this environment is very similar to performing a standard image processing task, that of generating a distance transform of the (binary) input image, and then selecting from the resulting output image all those cells that have entries (distance values) less than or equal to the buffer width specified. These selected cells are then re-classified (RECLASS in [TerrSet/Idrisi](#)) as required (e.g. to a value of 2; see also, Section 4.4.2, [Cost distance](#)).

Hybrid buffering

Buffer operations within vector GIS generate regions, and distances within those regions meet the overall buffering criterion set. The use of multiple buffering does enable a degree of gradation within such buffer regions, but for more detail some form of hybrid buffering may be more appropriate. In this approach either hybrid buffering is directly supported by the GIS tool or the relevant vector objects will require conversion to grid format and then grid buffering applied. The latter may be achieved using cost distance (ACS) methods or distance transforms (DTs), as described in Section 4.4.2, [Cost distance](#).

Network buffering

A simple buffer operation applied to a vector map of roads will create a set of merged or unmerged regions either side of the selected roads. An alternative form of buffering for such networks is a time or distance buffer from a point source or sources with respect to the network. In the case of roads, this form of buffer generates travel-distance or drive-time zones. The [Manifold](#) GIS Business Tools option provides for such buffering. In their model drive times are computed to all network points in one or more time zones, plus a distance buffer (speed related) for off-road travel (e.g. walking). This is a very processor intensive operation. [TransCAD](#) and [ArcGIS](#) Network Analyst provide similar functionality via their “drive time” analysis facilities (see, for example, [Figure 7-18](#)). [Basemap](#) provide more advanced toolsets that incorporate multi-modal travel and timetables.

Distance decay models

Almost by definition, spatial modeling makes extensive use of distance measures. In many instances a suitable metric is selected and used directly, but for many other situations some generalized function of distance is used. Typically these models involve the assumption that the strength of relationships between locations, or the effect one location has upon another, diminishes as separation increases. There are a wide range of GIS analysis techniques that make use of this concept — we describe a number of these briefly below, and address them separately in a number of the later chapters of this Guide. There are, of course, many situations in which such models are not appropriate, such as in the context of telecommunications links and in situations where time-distance computations are more relevant (e.g. single- or multi-modal network travel modeling, discussed further in Section 7.1 *et seq.*, [Introduction to Network and Location Analysis](#)).

The most widely used distance decay models are those in which distance is introduced as an inverse function to some power, typically 1 or 2. Thus the value of some variable of interest, z , at location j , z_j , might be modeled as some function, $f()$, of attribute values, z_i , associated with other locations, i , weighted by the inverse of the distance separating locations i and j , d_{ij} raised to a power, β :

$$z_j = \frac{f(\{z_i\})}{d_{ij}^\beta}, \beta \geq 0$$

The exponent, β , has the effect of reducing the influence of other locations as the distance to these increases. With $\beta=0$ distance has no effect, whilst with $\beta=1$ the impact is linear. Values of $\beta \gg 1$ rapidly diminish the contribution to the expression from locations that are more remote.

One of the commonest applications of this kind of model is in surface interpolation. Most GIS and surface analysis packages provide support for such operations (see further, Section 6.6.1, [Deterministic Interpolation Methods](#)). However, inverse distance weighting (IDW), as it is known, is used in many other situations. An example is the study by Draper *et al.* (2005) referred to in Section 3.1, [Spatial analysis as a process](#). The distance of cancer cases and controls from overhead power lines was calculated and the inverse and inverse squared distances were then used as weights in modeling the observations. The idea behind this approach was to take into account the fact that electromagnetic radiation intensity diminishes with distance from the source.

Inverse distance modeling is one of the main techniques used in the fields of transportation, travel demand modeling and trade area analysis. The propensity to travel between pairs of locations is often assumed to be related to the separation between the locations and some measure of the size of the source and destination zones. The basic model is derived from analogies with the force of gravity:

$$I_{ij} = \alpha \frac{M_i M_j}{d_{ij}^\beta}$$

Here the interaction (e.g. number of trips) between zones i and j is taken to be some measure of the mass, M , at each location, divided by the distance between the zones. As before, α and β are parameters to be estimated. The general form of such ‘spatial interaction’ models is:

$$T_{ij} = A_i B_j O_i D_j f(d_{ij})$$

where T_{ij} is the number of trips between zones i and j , O_i is a measure of the size of the origin location i (e.g. the total number of trips to commuters in zone i), D_j is a measure of the size of the destination zone (e.g. the total number of work places in zone j), and $f(d_{ij})$ is some appropriate function of the distance or degree of separation of zones i and j . A_i and B_j are balancing factors of the form:

$$A_i = 1 / \sum_j B_j D_j f(d_{ij}), \quad B_j = 1 / \sum_i A_i O_i f(d_{ij})$$

These latter expressions are designed to ensure that the modeled row and column totals add up to the total number of trips from and to each zone (i.e. a so-called doubly-constrained model, which assumes that the totals are correct and not subject to sampling error). Note that by taking logarithms the model is transformed into a (weighted) linear sum, and is often analyzed in this form. In this case the weights are powers applied to the origin, destination and distance variables that must be estimated from sample data.

For a fuller discussion and background to such modeling see: [CATMOG 2](#) and [CATMOG 4](#); Roy (2004) who provides a very complete book on the subject; Miller and Shaw (2001, Ch.8); Wilson (1967);, and Fischer (2006, chapter 3, reprinted from a paper of 2000). Fischer’s paper describes both the basic spatial interaction model outlined above and alternative methods for model calibration, including least squares, maximum likelihood, and in Parts II and IV of the same volume, neural network methods. Common variants of this type of model are the origin-constrained and destination-constrained versions, in which case either the A_i or the B_j are determined by balancing, but not both. In the case of trade area modeling, gravity models or Huff models (which are origin constrained) are often used. The former typically use inverse distance power models coupled with measures of location attraction, such as retail square footage. Typical models for $f(d_{ij})$ include: simple inverse distance models (as above); negative exponential models; and combined or Gamma models. All three types (plus others) are supported by [TransCAD](#) as impedance functions within its trip distribution modeling facility. [Crimestat](#) incorporates similar functions within its trip distribution modeling component. The statistical models within the [BUGS](#) project provide for a generalized powered exponential family of distance decay models of the form:

$$f(d) = e^{-(\varphi d)^k}, k \in [0,2], \varphi > 0$$

where φ is the principal decay parameter relating the decay pattern to the effective range of correlations that are meaningful. Note that the parameter k acts as a smoothing factor and with $k=2$ this is essentially a Gaussian distance decay function. Geographically Weighted Regression ([GWR](#)) utilizes similar decay functions — the authors describe these as kernel functions — see further, Section 5.6.3, [Geographically Weighted Regression \(GWR\)](#), this time of the form:

$$f(d) = e^{-d^2 / 2h^2}, \text{ or } f(d) = e^{-d/h}, \text{ or}$$

$$f(d) = \left(1 - \frac{d^2}{h^2}\right)^2, d < r, f(d) = 0 \text{ otherwise}$$

In these functions the parameter, h , is also known as the *bandwidth*. A small bandwidth results in very rapid distance decay, whereas a larger value will result in a smoother weighting scheme. The Gaussian model is an unbounded function, and as an alternative [GeoBUGS](#) provides the following bounded or disk model of distance decay:

$$f(d) = \frac{2}{\pi} \left[\cos^{-1} \left(\frac{d}{\alpha} \right) - \left\{ \left(\frac{d}{\alpha} \right) \left(1 - \left(\frac{d}{\alpha} \right)^2 \right) \right\}^{1/2} \right], d < \alpha$$

With $f(d)=0$ otherwise. This model decays slightly more rapidly than a straight line over the interval $[0,\alpha)$, especially over the initial part of the interval. The parameter, α , here has a similar role to that of the bandwidth parameter in the [GWR](#) expressions above.

All such models may suffer from problems associated with clustering — for example where a number of alternative (competing) destinations are equidistant from an origin zone, but may be located either in approximately the same direction or in very different directions. Such clustering has implications for modeling, since competition and even interaction *between* destination zones is inadequately represented. Most GIS and related software packages do not provide direct support for analysis of such factors, and it is up to the user to specify and implement an appropriate modeling framework to address these issues.

These are all models based on a mathematical function of distance. In addition to such models other models or measurements of impedance are extensively used, typically in the form of origin-destination (O-D) matrices. The latter may be derived from measurements or from alternative distance-related models such as intervening opportunities, see Stouffler (1940), or lagged adjacencies (e.g. see the spatial weights modeling options in [GeoDa](#), and Section 5.5.2, [Global spatial autocorrelation](#)). Distance, d , in this instance is a free-space or network-

related measure of impedance to interaction (e.g. network distance, time, cost or adjacency). The most widely implemented distance impedance expressions are of the form:

$$\alpha / d^\beta \text{ or } \alpha e^{-\beta d} \text{ and } \alpha e^{-\gamma d} / d^\beta$$

where α , β and γ are parameters to be selected or estimated. Figure 4-75 and Figure 4-76 illustrate the first two of these models, in each case standardized with an α value to provide a value of 10 at $d=0.1$ units and a range of values for β . The Gamma model provides curves that are intermediate between the two sets shown, depending on the specific values of γ and β chosen or fitted from sample data. Pure inverse distance models (and the Gamma model) have a (serious) problem when distances are small, since the expression will tend to infinity as d tends to 0. A partial solution to this problem is to include a small distance adjustment to such formulas, such that $d=d+\varepsilon$ where $\varepsilon>0$, or to exclude distances that are below a given threshold from the inverse distance calculations (possibly defining a fixed value for the function in such cases). These approaches stabilize computations but are essentially arbitrary. The exponential model has the advantage that it tends to 1 as d tends to 0, so may be preferable if observations are known to be very closely placed or even coincident.

Models of this kind may be constrained in various ways: by limiting the maximum range to be included in computations; by limiting the maximum number of locations to be considered; by restricting the range of values the parameters may take; and by explicitly taking account of barriers (e.g. inaccessible regions). More sophisticated analysis would take into account aspects of the dynamics and uncertainty associated with the interactions or flows being modeled. Somewhat different models of distance decay are incorporated into radial basis interpolation and geostatistical analysis (see further, Sections 6.6.4, [Radial basis and spline functions](#), and 6.7, [Geostatistical Interpolation Methods](#)).

The analysis of data variation by distance or distance bands (e.g. variograms and correlogram analysis) is designed to identify overall patterns of variation in a measured variable with linear (Euclidean) distance. In such instances a maximum distance or range is identified if possible, and the pattern of variation in the measured variable is modeled to reflect the diminishing associations observed as distance increases. Here the models effectively treat distance as an independent variable and the observations (generally averaged by distance band) as a form of dependent variable.

Directional Operations

Directional analysis is perhaps less well known and well used than other forms of spatial analysis. It primarily deals with the analysis of lines, points sets and surface orientation, sometimes from different time periods, examining patterns in order to identify and utilize information on specific directional trends. Researchers in the fields of geology, sedimentology and geomorphology utilize directional analysis to examine fault lines and fracture systems, glacial striations, particulate distributions and lithography patterns; in ecology it is used to study patterns of wildlife movement and plant dispersal; in hydrology and other forms of flow analysis directional considerations are paramount; in wildfire management directional analysis is used to predict local (30metre-100metre) wind flows as a function of local topography; and directional analysis can be used when studying patterns of disease incidence or the distribution of particular crime events.

If a particular dataset exhibits no discernible variation in direction across a study region it is described as *isotropic*. Whilst such situations may be regarded as unusual in the real world, it is frequently the starting assumption for many forms of analysis. Variations that have a particular directional bias are known as *anisotropic*. In some areas of GIS analysis, for example geostatistics, examination of anisotropy is a fundamental part of the modeling and prediction process (see further, [Section 6.7](#)). For an introduction that includes a discussion of hypothesis testing for directional data, and a review of applications of these techniques to spatial datasets, see Gaile and Burt (1980) [CATMOG](#) 25. It should be noted that when comparing and analyzing directional datasets computed with projected (plane) coordinate systems, there may be introduced directional bias. Comparisons should be made between datasets that have been generated using the same projections and approximate scale, or by utilizing spherical coordinates where necessary. For small areas (under 100kmx100km) the problems for most applications should be minor.

Directional analysis of linear datasets

The formulas used by Oriana follow those provided in Fisher (1993) and Mardia and Jupp (1999). Measures are provided which compare the observed distribution to a uniform distribution and/or the distribution due to von Mises that is the circular equivalent of the [Normal distribution](#). Graphically the [von Mises distribution](#) with a mean at π is essentially identical to the [Normal distribution](#) with the x -axis being in the range $[0, 2\pi)$. The general form of the [von Mises distribution](#) is:

$$p(x) = \frac{e^{\kappa \cos(x-\alpha)}}{2\pi I_0(\kappa)}$$

where α is the mean direction of the distribution in the range $[0, 2\pi)$, and $\kappa > 0$ is a shape parameter known as the concentration (effectively equivalent to the standard deviation). $I_0(\kappa)$ is the modified Bessel function of the first kind, of order 0 (see Section 1.4.2, [Statistical measures and related formulas](#)). The circular variance of this distribution is:

$$\sigma^2 = 1 - \frac{I_1(\kappa)}{I_0(\kappa)}$$

This modified Bessel function is relatively straightforward to compute for integer orders, although it requires summation of an infinite series:

$$I_0(\kappa) = \sum_{i=0}^{\infty} \frac{(-1)^i (\kappa/2)^{2i}}{(i!)^2} \quad \text{and} \quad I_1(\kappa) = \frac{\kappa}{2} \sum_{i=0}^{\infty} \frac{(-1)^i (\kappa/2)^{2i+1}}{i!(i+1)!}$$

Several types of spatial dataset consist of multiple components that incorporate directional information. For example: wind direction and speed over time for a specific location or set of locations; single grid files of derived gradient data (slope, or gradient magnitude and aspect, or

direction of maximum gradient — see further, Section 6.2, [Surface Geometry](#)); dual grid files containing either direction and magnitude data (polar information) in separate grids, or Cartesian data (x and y components) in each grid, for example simple optimal path tracking data (see further, Section 4.4.2, [Cost distance](#)). Plotting such data may utilize a variety of display methods.

For temporal data a variant of the rose diagram can be utilized that provides stacked histogram functionality in the radial direction in addition to the circular histogram facility provided by a standard rose diagram. The [Grapher](#) program, available from the same providers as [Surfer](#), is an example of a package that supports such displays, as do [Oriana](#) and [Matplotlib](#), but many statistical graphing packages provide similar functionality. This kind of diagram is illustrated in Figure 4-78, where the radial extent shows the frequency of measured wind speeds (hourly intervals) separated into speed categories in knots (with zero speed values removed). The black arrow shows the mean vector (direction and speed) for the entire dataset of almost 4500 records.

In some instances smoothed contour plots of wind direction are created rather than stacked histograms. This enables multiple plots to be superimposed, as is the case for selected sites in the [London Air Quality](#) network. In this case the plots show pollution levels, by type, as the magnitude and average wind direction over a selected time period as the directional component.

Directional analysis of point datasets

A somewhat different form of directional analysis is used within many packages to summarize the distribution of point sets around a mean center. The average variation in the distance of points around the mean can be viewed as a circle or set of circles at a set of standard distances (like standard deviations in univariate statistics). In addition, separate variations in x - and y -coordinate values may be used to generate a standard distance ellipse, with major and minor axes reflecting the directional variation of the point pattern.

Figure 4-79 illustrates this for the location of churches on Romney Marsh in SE England. The two ellipses are set at one and two standard deviations respectively. Determination of the standard deviations in x and y are not made using the original data, but using transformed coordinates selected by minimizing the squared deviations from the point set to transformed (rotated) x and y axes. This is similar to the process of computing a linear regression on the (x,y) coordinate pairs and using the slope of the best fit line in angular measure to calculate the adjusted standard deviations.

The formula for the angle of (clockwise) rotation of the y -axis is somewhat lengthy:

$$\theta = \tan^{-1} \left(\frac{\sum (x_i - \bar{x})^2 - \sum (y_i - \bar{y})^2 + \sqrt{C}}{2 \sum (x_i - \bar{x})(y_i - \bar{y})} \right)$$

where

$$C = \left(\sum (x_i - \bar{x})^2 - \sum (y_i - \bar{y})^2 \right)^2 + 4 \sum ((x_i - \bar{x})(y_i - \bar{y}))^2$$

Using this value the two standard deviations can then be computed in much the same way as for conventional standard deviations, but with the degrees of freedom in this case being $n-2$, where n is the number of points in the sample (for n large the divisor will be close to n , as per the common formulation of standard distance).

The formulas cited are those used within [Crimestat](#), which have been adjusted to ensure the ellipse axes are the correct length and the formula is consistent if the standard deviations in x - and y - are equal:

$$SD_x = \sqrt{\frac{2 \sum ((x_i - \bar{x}) \cos \theta - (y_i - \bar{y}) \sin \theta)^2}{n - 2}}$$

$$SD_y = \sqrt{\frac{2 \sum ((x_i - \bar{x}) \sin \theta - (y_i - \bar{y}) \cos \theta)^2}{n - 2}}$$

Software packages may provide information on the rotational angle and other attributes of the standard deviational ellipse. For example, [Crimestat](#) includes the angle of rotation and the individual standard deviation values, together with the area of the standard deviational ellipse (which is simply $A = SD_x SD_y \pi$; the lengths of the two ellipse axes are $2SD_x$ and $2SD_y$). For the data illustrated in Figure 4-79 the clockwise rotation of the y -axis is 53.2 degrees and the ratio of major to minor axis length, or *coefficient of eccentricity*, is 1.76.

[Crimestat](#) includes directional data in another technique, Correlated Walk Analysis (CWA). CWA is an analytical procedure based on examining a sequence of events (e.g. crimes, animal movements) in terms of location, bearing (direction from a given point) and time lapse. Its objective is to predict where subsequent events, e.g. burglaries, may occur. Random walks, of various types (see further below), may be used as part of such analysis. For more details of these methods and their applications see the [Crimestat](#) manual, Chapters 4 and 9, and the more recent paper by [Demšar et al. \(2021\)](#).

An essential part of such analyses is the examination of patterns in the distribution of point data as the distance and direction of examination is varied. A simple Correlated Random Walk (CRW) simulation facility was provided with Hawth's Tools/GME for [ArcGIS](#) that enabled model paths to be generated and saved as features ([Figure 4-80](#)). This facility highlights the impact of model parameters on the resulting path form (in this example using CRW as the basic model, but other models such as pure random/Brownian motion and so-called persistent random walks are also supported).

Graphs that plot the average difference in values for pairs of points in a region, over given blocks of distance, optionally grouped into different blocks of directions, are known as correlograms or variograms. If the region is isotropic there will be no significant differences between the variograms drawn for one direction (e.g. 0-30 degrees) or from another (e.g. 60-90 degrees), whereas with significant anisotropy the shapes of these variograms will differ. Hence if modeling is to utilize such information, different models or model parameters may be

required to accommodate these directional differences (these issues are covered in greater detail in Section 6.7.1, [Core concepts in Geostatistics](#)).

Directional analysis of surfaces

There are many applications of directional analysis relating to surfaces. These include: analyzing and mapping surface *aspect* (e.g. the `d.rast.arrow` function in [GRASS](#), the aspect generation tools of [Landserf](#), [SAGA](#), [Whitebox](#), [ArcGIS](#) and [TNT/Datum Workstation](#)). Aspect is defined as the *direction* of the surface gradient (as opposed to the *magnitude* of the gradient — see Figure 4-81); analyzing and modeling flow directions and *friction* as part of hydrological and similar studies (e.g. local drainage directions such as the `lddcreate` function in [PCRaster](#), the Flow function in [TerrSet/Idrisi](#), and the FlowDirection function in the [ArcGIS](#) Spatial Analyst hydrology toolset); and analyses relating to the lighting of surfaces, as part of solar energy analysis, lighting for visualization, and for visibility analysis (e.g. in the [ArcGIS](#) 3D Analyst and Spatial Analyst toolsets). Each of these topics is discussed in some detail in Chapter 6, [Surface and Field Analysis](#).

In general the representation of direction in such instances is strongly affected by the underlying model or representation of the surface — for example, with grid (raster) surfaces there is often directional bias as a result of the limited range of directions available in immediate neighborhoods (e.g. only 8 directions in a 3x3 neighborhood). For vector-like grid data (single or dual-grid source data) a vector plot may be informative. An example of such a plot is shown in Figure 4-81 where gradient vectors have been plotted using [Surfer](#) for a sample grid file of the Mt St Helens volcano. In this case color/grayscale has been used to indicate vectors of greater magnitude (from white through blue to red/darker gray). Similar mapping of simulated wind directions is performed by software tools such as WindWizard and WindNinja designed for use in conjunction with wildfire modeling and management (Figure 4-82). In this example wind flow has been modeled for the OS NT04 DEM (Pentland Hills area) with 100 meter output grid resolution and input direction of 230 degrees and speed 50 mph. Output here is displayed using Google Earth (.KML file).

Grid Operations and Map Algebra

In this Section we provide an introduction to grid analysis methods, including an overview of the kind of facilities available in raster analysis packages and toolsets. Many of the facilities relate to operations on surfaces (e.g. computation of gradient and aspect) and field-like datasets (e.g. interpolation and prediction techniques). These topics are covered in detail in Chapter 6, [Surface and Field Analysis](#).

Operations on single and multiple grids

perations on single and multiple grids

Grid or raster data files typically have a single attribute associated with each cell position. Datasets of this type can be thought of as triples (x,y,z) where x and y are the cell coordinates and z the attribute values. z -values might be: a color code in the case of image data; a binary value, for example in the case of converted vector data (without additional attributes); a cell classification number, such as the type of vegetation cover associated with that location; or a measured or calculated data value, such as an interpolated figure for soil heavy metal content in parts per million (ppm), the height of a surface in meters, or the estimated average level of

noise from a given source type (e.g. roads). In all instances the set of values observed in a single grid of size M-rows and N-columns is limited, either by the coding or classification procedure adopted, or by the fact that ultimately only MxN values can possibly be recorded.

This set of values may be summarized and analyzed using basic univariate statistics (non-spatial analysis) and many GIS packages provide a wide number of simple single-grid measures. These include many of the basic statistics described in [Table 1-3](#), including the arithmetic mean, max, min, count, sum, variety, minority, majority, range, mid-range, variance and standard deviation. Each of these operations applies to the set of individual cells of a single grid and may be described as local statistics.

Simple statistical operations may be applied to groups of cells, for example the immediate 8 cell neighborhood (or window) of each cell or to individual cells across multiple grids (see also, Section 5.3, [Grid-based Statistics and Metrics](#)). Where regular neighborhoods of cells are involved such operations are termed focal, or if the neighborhoods are defined using pre-defined (irregular) zones, then the operations are described as zonal.

There are a large number of basic grid operations supported for image and general raster files. These include local, focal and zonal operations depending on the scope of the operation. Such operations may be applied to a single grid, or to a number of input grids, depending on the operation in question. The set of possible operations of this type are often referred to as *map algebra*. Originally this term was introduced by Tomlin (1983) as the process of map combination for co-registered layers with rasters of identical size and resolution. Combinations involved arithmetic and Boolean operations. A typical example would be the addition of separate grids for an area, where each grid cell represented the level of noise, L , due a particular source such as A: road, B: aircraft, measured or computed in the same manner. The total noise level for each cell would then be computed as the sum of the individual grid datasets, i.e. $A+B+C+...$, as described in Section 4.3.2, [Simple calculations](#), and [Figure 4-36A](#). This is how grids such as that illustrated in [Figure 3-4](#) (Noise map, Augsburg) have been created. However the term map algebra is now used more widely by many suppliers. For example, [ArcGIS](#) describes the set of all operations performed using its Spatial Analyst option as “map algebra”. More specifically it divides such functions into five main categories:

- Local functions, which include mathematical and statistical functions, reclassification, and selection operations
- Focal functions, which provide tools for neighborhood analysis
- Zonal functions, which provide tools for zonal analysis and the calculation of zonal statistics
- Global functions, which provide tools for full raster layer or raster dataset analysis, for example the generation of Cost Distance rasters
- Application functions, specific to tasks such as hydrology and geometric transformation

Similar facilities are provided in many of the mainstream raster-based GIS packages, such as [PCRaster](#), [TNT/Datum Workstation](#) and [TerrSet/Idrisi](#), and in the raster-processing toolsets of more generic packages such as [GRASS](#). Many of these are described in greater detail in Chapter 6.

The map algebra concept is capable of being extended in several ways. One example is extension to explicitly handle the time domain coupled with 1, 2 or 3-dimensional spatial datasets. Mennis *et al.* (2005A, 2005B) have developed such a model and implemented a set of Java classes for handling what they describe as *multidimensional map algebra* (mma). Performance issues are one of the principal barriers to widespread use of such concepts, but as improved data structures and processing architectures emerge for spatio-temporal datasets, concepts of this kind are likely to be more widely applied. A second example of extension to map algebra ideas is *map calculus*, put forward by Haklay (2004, 2006). Haklay proposes retaining the functional form of surfaces to be combined, wherever possible, and performing operations on these forms rather than the resultant grids. For example, instead of generating a high resolution grid from a point dataset using some interpolation function, $F()$, one could retain the functional form $F()$ and only generate the resulting grid as an output when required. This has the potential to provide a more manageable approach to spatio-temporal modeling than the *mma* procedure in some instances.

Linear spatial filtering

The term spatial filtering is principally associated with digital image processing, although such methods may be applied to almost any type of grid or image. The term is also used, in a related manner, in the area of spatial statistics (see further, Section 5.6.5, [Spatial filtering models](#)).

Most GIS packages provide simple grid filtering functions, whilst those that have a bias towards image processing and grid file handling (e.g. [TNT/Datum Workstation](#), [Surfer](#), [TerrSet/Idrisi](#)) tend to provide a broader range of functions. The most commonly provided functions are so-called low-pass and high-pass spatial filters. These are focal functions whose operation is determined by a kernel or neighborhood of $N \times N$ cells around each pixel or grid position. Grid cells “covered” by the kernel are multiplied by the matching kernel entry and then the weighted average is calculated and assigned as the value for the central cell, G . A 3×3 symmetric kernel might look something like:

$$\begin{bmatrix} a & a & a \\ a & b & a \\ a & a & a \end{bmatrix}$$

or perhaps

$$\begin{bmatrix} c & a & c \\ a & b & a \\ c & a & c \end{bmatrix}$$

In the first of these examples a and b are positive numbers, typically integers. If $a=b=1$ then the kernel provides a simple averaging or smoothing operation, whereas if $a=1$ and $b=4$ more

weight would be applied to the grid cell being filtered. In both cases the weighted average is divided by the sum of the kernel elements. No divisor is applied in cases where the kernel sums to 0 or 1. Filters of this type are sometimes described as *low-pass spatial filters*. For example, consider the filter:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

and grid section

$$\begin{bmatrix} 98 & 176 & 200 \\ 90 & 187 & 188 \\ 124 & 170 & 175 \end{bmatrix}$$

The value at the central grid position $G(2,2)=187$ is replaced by the weighted average of itself (assigned a weight of 4) and the surrounding 8 cells (each assigned a weight of 1). The adjusted value is 164 after rounding. In the second example if a , b and c are positive with $b>a>c$ the kernel is described as Gaussian, being symmetric but center-weighted, for example:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

With the same data grid as above the filtered value is $G=162$. More generally, if $\{C_i\}$ is the set of coefficients of an $m=N \times N$ kernel matrix and $\{P_i\}$ is the set of source grid values within this kernel neighborhood, then the filtered grid value, G , is defined as:

$$G = \sum_{i=1}^m C_i P_i \bigg/ \sum_{i=1}^m C_i + B$$

The extra term, B , here is often set to 0. It is a bias factor that increases or decreases the resulting filter value by a fixed amount.

High-pass spatial filters emphasize the difference between the central point of the kernel and the values in its immediate neighborhood. Typically the entries in such kernels have a mix of positive and negative values, but the entries still add up to 1:

$$\begin{bmatrix} -a & -a & -a \\ -a & +b & -a \\ -a & -a & -a \end{bmatrix}$$

Values are often used to provide image sharpening prior to further processing. [Table 4-11](#) provides examples and a summary of typical linear spatial filters used in GIS and image analysis. Basic linear image filtering operations fall into a number of categories:

- Sharpening — for which the kernel matrix elements sum to 1 and the matrix is symmetric. Matrix elements are a mix of positive, negative and zero entries
- Blurring — also known as a smoothing, averaging or low-pass filtering, since this reduces or removes extreme values. In this case the kernel matrix elements sum to >1, entries are normally all positive and the matrix is symmetric
- Edge detection — for which the kernel matrix elements sum to 0 and the matrix exhibits some symmetry. As with sharpening, matrix elements are a mix of positive, negative and zero entries
- Embossing — for which the kernel matrix elements sum to 0 as per edge detection, but the matrix is asymmetric. It is usual to compute the embossed version of an image on a grayscale version, which is obtained by finding the average value of the bands, e.g. red (r), green (g) and blue (b) components, as $x=(r+g+b)/3$ for each pixel, and assigning the resulting values in the range [0,255] to a gray scale (where 0=black and 255=white)

A number of general characteristics of this type of filtering should be noted:

- if the range of values permitted within the grid are limited (e.g. RGB values in the range [0,255]) then computed values that fall outside of this range must be truncated to the range limits. This generates some computational (as opposed to user defined) bias in the processing
- multi-band image data filters are applied to each band separately. Hence a sharpening filter applied to an RGB image, for example, would be applied to the R, G and B components
- filtering may be conducted more than once (single pass or multi-pass). Each pass re-applies the same filter or filters to the target grid or image (layer or band)
- user-defined kernels are widely supported, both in image processing packages and within GIS software. Users may be able to select a pre-defined filter and modify this to suit their requirements, or to create their own $N \times N$ kernel
- all of the examples described involve simple linear operations, essentially weighted averaging (sometimes described as *linear convolution*). Non-linear filtering is also widely available and is described in more detail in [Section 4.6.3](#)

It should be noted that at image or grid edges it is not possible to apply the kernel symmetrically. This effect is more pronounced for larger kernels, but applies in every instance. There are many options for resolving such effects. Common procedures include: leaving the edge pixels unaltered; blanking edge pixels (which causes the grid to shrink); using an asymmetric (i.e. partial) kernel, which can generate artifacts; and mirroring, where grid values inside the kernel are used as surrogates for the missing values outside the kernel.

In addition to filters that operate in the spatial domain, pure frequency domain filters may also be applied. These operate on the set of grid values within an image without reference to the spatial pattern. Frequency domain filtering (for example Fast Fourier Transforms or FFTs) is provided in some GIS packages such as [TNT/Datum Workstation](#) and [TerrSet/Idrisi](#), but such procedures are not described in this Guide. FFTs are used, for example, to removing striping effects from remotely-sensed imagery.

Linear spatial filters can be categorized into several types based on their purpose and kernel design. Low-pass symmetric filters include averaging filters, which act as smoothing, noise-reduction, or blurring filters (focal mean), and Gaussian filters, which also perform smoothing and noise reduction but use a focal weighted mean. High-pass symmetric filters include sharpening filters, which remove mean effects and enhance sharpness using focal sum operations; these provide limited edge detection, and although their entries typically sum to 1, larger variants such as 3×3 or 7×7 Laplacian kernels may be more complex and may sum to values greater than 1. Gradient (asymmetric) filters are used for edge detection and can be applied either singly or in a two-pass process to highlight vertical and horizontal edges; when used together they form Gradient or Order-1 derivative filters. These kernels typically have parameters such as $a = 1$ and $b = 1$ or $b = 2$, with entries summing to 0. A notable variant is the Roberts method, which uses a modified structure with row 3 entirely zero and column 1 zero for the first pass, and column 3 zero for the second pass, effectively forming a 2×2 filter. Embossing filters are edge-detecting kernels that enhance edges in a chosen compass direction to create an embossed effect, with the example given illustrating a north-east kernel. Directional filters compute simple gradients in one of eight compass directions, with east and north directional derivatives shown in the first two examples. The third example corresponds to Horn's method for computing the north-south component of the derivative ((dz/dy)), as described in Section 6.1.3, and is comparable to edge-detection filtering. A non-linear variation of this approach involves computing all eight directional values and selecting the maximum absolute value as the output.

Non-linear spatial filtering

With linear filtering the value at each grid position, G , is altered using a simple kernel as a weights matrix. With non-linear filtering kernels are not generally used. Instead the value at G is determined by some procedure other than weighted averaging. For example, G may be determined by a statistical measure (other than the mean) based on an $N \times N$ neighborhood of G . $N \times N$ is typically 5×5 or greater, and almost any basic statistic (as listed in [Table 1-3](#)) may be used. Typical examples might be the minimum value in the neighborhood, or the cell value that is most common (occurs in the majority) within the neighborhood. Other measures commonly supported included variance, standard deviation, coefficient of variation and inter-quartile range. In each case the computations are based on the defined neighborhood.

[ArcGIS](#) describes such operations as Neighborhood operations involving Focal or Block statistics, with neighborhoods defined more generally than in typical filtering software (e.g. as rectangles, rings or wedges). Most other packages stick with the image processing terminology of filtering and square or rectangular neighborhoods. [ArcGIS](#) also provides a special function to handle the Majority case (MajorityFilter) with integer-valued grids that incorporates an additional spatial contiguity constraint. The rule applied is that either half or at least half of the cell values in the 3x3 neighborhood must be the same as the value at *G*, and these must satisfy a rook's move (4-way) or queen's move (8-way) contiguity constraint. Multiple passes of this filter can be made and will stabilize to a fixed pattern.

Several non-linear filters examine local deviations in grid values. In addition to standard statistical measures, median deviation and threshold averaging may be applied (see also Section 6.7.2, [Kriging interpolation](#), subsection on median-polishing). In median deviation filtering the value at *G* is replaced by *G-B*, where *B* is the local median. This emphasizes outliers. With thresholding averaging the value at *G* is retained if it is within a defined absolute value of the local mean (calculated excluding the value at *G* itself), otherwise it is replaced with this local average. The effect is to remove outliers. This is a form of “noise reduction”, a procedure that is supported in many image processing packages and in some GIS software. [TNT/Datum Workstation](#), for example, provides 6 noise reduction filters, all of which utilize local statistics to adjust the resulting dataset or image. These are principally based on the local median or mode value (variously computed), but also include a form of trimmed mean (the Olympic filter) in which 1 or more values at the upper and lower end of the local ranked set of values is dropped and the mean of the subset calculated as the selected output value.

Simple hillshading may be achieved using a form of non-linear filter. In this case horizontal, vertical or diagonal difference filtering is applied. The procedure subtracts all the values in row 1 say, from row 2 and assigns a value of -1, 0 or 1 for example to the outcome, depending on the sign of the result. A color-graded map of the output grid provides the hillshading effect.

Erosion and dilation

Erosion and dilation, in the context of grid operations, refer to morphological operations applied to binary and grayscale images. These are terms commonly used within image processing but only arise in a small number of GIS packages in this context. Erosion involves the removal (alteration) of pixels at the edges of regions, for example changing binary 1 values to 0, whilst dilation is the reverse process with regions growing out from their boundaries. Buffering is a form of dilation.

These two processes are often carried out using a form of kernel known as a structural element. A structural element is an *N*x*N* kernel with entries classified according to a binary scheme, typically as 0 or 1. If all entries are coded 1 then the structural element is a solid square block, the center of which is laid over each pixel in the source image in turn. Pixels that are coded as 1 in the structural element and extend beyond the boundary of a shape in the source image result in that element being extended or dilated. The shape of the structural element may vary, for example as a vertical bar, horizontal bar, cross shape or a user-defined pattern.

To illustrate this process we have taken a small section of a USGS DEM file and then applied dilation and erosion processes to this source. The results are shown in Figure 4-83A-C. If dilation is followed by erosion the process is described as a *Closing* operation, whilst Erosion

followed by dilation is known as *Opening*. These processes are not symmetric, and thus are generally not reversible. Opening eliminates small and thinner features, resulting in smoother edged regions, whilst Closing also smooths shapes but makes thin narrow features larger and eliminates small holes and narrow gaps. These changes are the result of deliberate morphological modification of the source data files. However, similar effects may be observed as a result of transformation operations performed manually or automatically for other purposes — for example resampling (changing resolution), rotation, multiple overlay, or map algebra operations.

Spatial Autocorrelation

The term spatial autocorrelation owes its origins to work in a related field, [time series analysis](#) (TSA), and in turn to the notion of correlation in univariate statistics. Since spatial autocorrelation follows these concepts so closely we will introduce the subject using these more familiar origins. Readers may also wish to refer to Cliff and Ord (1981), Goodchild (1986, [CATMOG](#) 47), or Fortin and Dale (2008). Throughout this discussion it is important to recognize that correlation may *suggest*, but does not *imply*, [causation](#); it is extremely easy to make erroneous assumptions regarding cause-effect relationships from correlation data, but such results should be examined in great detail before suggesting any causation.

Global Spatial Autocorrelation

The procedures adopted for analyzing patterns of *spatial* autocorrelation depend on the type of data available. There is considerable difference between:

- a) a set of 100 values obtained for a 10x10 grid of (100mx100m) squares which covers a 1000mx1000m region;
- b) a set of values obtained for 100 contiguous but arbitrarily shaped polygons which again cover the same region;
- c) a set of nominal values (classes) for 100 contiguous but arbitrarily shaped polygons which again cover the same region; and
- d) a set of data values obtained from 100 arbitrarily distributed sample points in our study region

Each case warrants a slightly different approach, but each utilizes the notion of proximity bands as a means of imposing some form of serial behavior to the data. The idea is then to examine the correlation between areas or points at given levels of separation, to obtain a similar measure to that used within time series analysis. Initially we examine the case of nominal-valued data, which helps identify the key concepts involved and focuses our attention on a set of questions regarding such analysis that we examine in more detail in the subsections that follow.

Join counts and the analysis of nominal-valued spatial data

Consider a regular grid of cells that completely covers a sampled region with a value associated with each grid square. The set of values in this case could be binary, e.g. presence/absence; or they could be classified into one of k classes (see Dacey, 1968, for a full discussion of this case); or k classes could be re-classified back into two classes, one class from k and the remaining $k-1$ classes. Typically this latter kind of data are nominal-valued, i.e. the cell or zone values can be regarded as categories, such as woodland, grassland, sandy desert etc. rather than ordered values.

If we assume the data are binary, perhaps representing the presence or absence of a particular species of insect or plant variety in sample quadrats, a range of possible patterns might be observed ([Figure 5-25A-D](#)). In [Figure 5-25A](#) all the observed values for presence are in one half of the 6x6 grid (strong positive autocorrelation), whilst in [Figure 5-25B](#) they are perfectly evenly distributed (strong negative autocorrelation). [Figure 5-25C](#) gives a particular case of a random pattern (of which there are many). In each case 50% of the cells show presence, but this value could easily be 10% or any other value depending on the data being studied. [Figure 5-25D](#) is an example of a real-world dataset showing the presence or absence of desert holly in a 16x16 cell sample region. Note that in each instance our comments refer to autocorrelation at the scale of the cell, i.e. between cells; within cells autocorrelation is typically strongly positive.

One way of analyzing these patterns is to ask “what is the chance that a particular pattern could occur at random?” In each of the three sample patterns shown we can look at the spatial equivalent of one time step or lag, the patterns observed at one cell step, i.e. adjacent cells. If steps are restricted to rook’s moves we can count the number of instances of 1-1, 0-0, 1-0 and 0-1 occurring, and compare these to the number we might expect if the pattern was random. These numbers are referred to as *join counts* (sometimes erroneously described as ‘joint counts’). For smaller regions edge effects will be significant, so calculations need to be adjusted to reflect the fact that along the borders not all of the four directions are possible. For example, only two adjacent cells exist for the four corner positions, and only three adjacencies for other border cells.

Row and column totals for the adjacencies, or joins, are shown in [Figure 5-26](#) with the overall total being $120/2=60$ joins. For our patterns, with 50% occupancy, we might expect 15 of these to be 1-1 joins, 15 to be 0-0 joins and the remaining 30 to be 0-1 or 1-0 joins. We can count up the number of each type of join and compare this to our expected values to judge how special (significant) or not our patterns are. In [Figure 5-25A](#) there are 27 1-1 joins, 27 0-0 joins and only 6 0-1 or 1-0 joins — this seems very unlikely, and is indeed most unlikely. Similar calculations can be undertaken for cases B and C, and as expected for case B all 60 joins are of type 1-0 or 0-1, which is again extremely unlikely to occur by chance. Case C has 35 1-0/0-1 joins compared with perhaps 30 expected, with 1-1 joins being 13 and 0-0 being 12, as against perhaps 15 in each case.

A test for the significance of the results we have observed can be produced using a Normal or *z*-transform of the data. In practice three separate *z*-transforms are needed, one for the 1-1 case, one for 0-0, and one for 0-1 and 1-0. These transforms evaluate expressions of the form $z=(O-E)/SD$ where *O* is the observed number of joins of a given type, *E* is the expected number based on a random model, and *SD* is the expected standard deviation. The procedure and formulas can be implemented in software scripts for use within mainstream GIS packages or data may be externally analyzed using specialized software and then the results mapped within a GIS; for example, using the Rookcase Excel add-in produced by Sawada (1999). Results for the four examples given in [Figure 5-25A-D](#) generated using the Rookcase add-in are shown in [Table 5-8](#) (“B” and “W” here refer to Black and White, rather than 1 or 0; “Rand.” refers to the randomization model, which is discussed further below; and # means the number (of joins).

Two features of the results in Table 5-8 should be noted: (i) the expected number of joins is not 30,15,15 as we suggested earlier, but 30.86, 14.57,14.57 these being adjusted values to take account of what is known as non-free sampling (i.e. sampling without replacement); and (ii) the z -statistic in Table 5-8B shows a large positive value for BW joins, and large negative values for BB and WW joins (absolute values of >1.96 would be significant at the 5% probability level). This mixed pattern can be confusing, especially if one or two of the z -scores shows a significant value whilst others do not, as in Table 5-8D.

The join counts method has been utilized in a variety of application areas including: ecological data analysis; to analyze patterns of voting (for example voting for one of two political parties or individuals); for analysis of land-use (e.g. developed or undeveloped land); and for examination of the distribution of rural settlements. However, the complexity of computing the theoretical means and variances in more realistic spatial models, the difficulty of interpreting the multiple z -scores, coupled with the availability of alternative approaches to analysis, have meant that join count procedures are not widely used and are rarely implemented in mainstream GIS packages. Support for this kind of analysis is provided in the *joincount* functions of the R-Project spatial package, *spdep* and within the [PASSaGE](#) software. Metrics based on these concepts are provided in [Fragstats](#) and similar packages (see further, Section 5.3.3, [Quadrat analysis of grid datasets](#), and Section 5.3.4, [Landscape Metrics](#)), for example in the computation of Connection and Contagion indices.

The standard form of the join counts statistical analysis makes a number of assumptions regarding the observation data. These include (under free sampling) that the distribution of BB and WW joins is asymptotically Normal (Gaussian) and that the data are first-order homogeneous. To clarify this latter point, we use an example from Kabos and Csillag (2002) who have investigated relaxing this assumption. They give the example of two simulated images with same B/W distribution (50%B and 50%W), but with different first-order effects. Their first pattern was simulated with first-order homogeneity. i.e., p_b , the probability of any cell being black was 0.5 for all cells (Figure 5-27 left hand image). Their second pattern was simulated by setting $p_b=0.6$ for half of the image and $p_b=0.4$ for the other half, so the overall probability p_b remained at 0.5 but was no longer homogeneously distributed across the image (Figure 5-27, right hand image). They found that the join count statistics (JCS) using $p_b=0.5$ accepted the null hypothesis of spatial randomness for the first image, but rejected it for the second. Using unmodified JCS, they have thus demonstrated that we would erroneously (but understandably) conclude that there is significant spatial autocorrelation in the second image.

Additional questions might reasonably be asked about this analytical procedure that point the way to more powerful extensions and developments of these ideas. Key questions, which we discuss below include:

- how sensitive is this technique (and, of course, many other techniques) to the particular size of grid cell used and the number of cells?
- if there are k classes rather than just two, how might this kind of data be analyzed?
- why choose rook's moves — why not permit queen's moves, which would extend the idea of contiguity but increase the complexity of computation?

- why restrict the notion of contiguity to directly adjacent cells (a lag of 1) — why not examine longer range effects, i.e. higher order spatial lags? Indeed, with separate analyses for different lags or distance bands one could produce a form of *correlogram* of autocorrelation effects
- why should every cell have the same level of importance or weight — why not permit cells to have differing weights, depending on the kind of model of spatial association we are considering?
- should the analysis be restricted to regular grids, or could it be extended to irregular grids, polygonal regions and even pure point data?
- is a single statistic for a whole area meaningful, or should separate statistics be computed for sub-regions or individual cells and then plotted, giving a Local Indicator of Spatial Autocorrelation (LISA) measure, which would highlight clustering effects?
- if the data in the study cells are integer or real-valued, why restrict analysis (lose valuable information) by ignoring the *levels* in regions?
- how should we determine the cell probabilities, e.g. p_b ? Is it sufficient to always derive the probability estimate from the observed proportions of cells of type B, or are there other approaches that might be more appropriate?

We briefly comment on the first two of these questions in the discussion below. The remaining questions in the list, notably those relating to cases where the data values in the grid, lattice or point-set are integer or real-valued, are discussed in the subsections that follow.

The first point raised above can be examined directly by reviewing our example in [Figure 5-25D](#). If the grid had been sampled at half the frequency in each direction there would be $8 \times 8 = 64$ cells rather than $16 \times 16 = 256$ cells, and in this instance only one new larger cell contains no data, i.e. a 0, all others showing a desert holly plant present ([Figure 5-25A](#), empty 2×2 cell shown in white/all zeros). So we immediately see that such techniques may be highly sensitive to grid resolution, or, with irregular lattices, to the particular density, shape and connectivity of individual areas.

Furthermore, with large grids, e.g. 128×128 cells, the z-scores may become huge, raising doubts over the sensitivity and interpretation of the technique as the resolution of sampling or the size of the area covered is increased; in [Figure 5-28B](#) there are 16384 cells and all three z-scores are over 160. A critical question here is whether the data in such cases reflect the result of direct observation, or whether they represent results from re-sampling or similar computational procedures. In the latter instance test scores may not be valid. At the other extreme scores may be suspect where the number of cells or regions is small (less than 30) or the category selected only occurs in a small percentage of areas.

The second question deals with the k -class case. There are several ways of dealing with this situation. Perhaps the first question to ask is whether the classes are genuinely nominal-valued (e.g. distinct species of tree); or whether the classes have been created from a set of underlying data that are themselves, integer or real-valued. In the latter instance it is sensible to examine the underlying data separately using more powerful techniques, such as Moran's I (see below). It is also possible to tackle each of the k classes in turn, treating these as B=Black say, and regarding all other classes as W=White, and proceeding as described earlier. Alternatively one can examine all the k classes simultaneously by identifying all instances of the same class being

found in adjacent areas. This pattern of occurrences can then be compared with the expected pattern assuming the zones are labeled in a spatially random manner, and/or using a pseudo-probability distribution generated using random permutations of the known values.

Support for this form of ‘multi-colored map’ analysis is to be found in the R-Project package, *spdep*, *joincount* functions and the join counts option of the [PASSaGE](#) software. Note that for some of the analyses performed it is assumed that the spatial weights matrix, $\mathbf{W}$, is symmetric, or can be adjusted to be symmetric; this excludes the use of some, more complex, neighborhood relations.

Moran I and Geary C

Many of the remaining questions from the preceding subsection relating to join count procedures can be addressed by considering a more general case than presence/absence data, with a region that is not a regular lattice. We start with a simple incomplete rectangular lattice, shown in Figure 5-29 as an arrangement of 10 cells with real-valued entries. Note that although this is shown with rectangular edges it is topologically equivalent to any set of planar regions that exhibit the same adjacency pattern, however irregular in form.

Instead of displaying this data as an irregular lattice, it can be recast as a table of (x,y,z) triples, where rows are numbered 1,2... from the top and columns are likewise numbered 1,2... from the left, as shown in Table 5-10. This coordinate based description of the data immediately suggests a generalization that may be applied to point datasets, with the (x,y) pairs providing the coordinate information. These (x,y) pairs could be well-defined points, or grid cells, or even polygon centroids.

This tabular form loses information, in that cell adjacencies are not explicitly captured. Figure 5-30 provides this missing adjacency information for the sample lattice as a binary matrix using rook’s move definitions. In this example cells in the original dataset are numbered from 1 to 10, starting with the first cell being in row 2 column 1 of the source data and continuing down the rows and across the columns.

This matrix can be viewed as a specific example of a spatial weights matrix, $W = \{w_{ij}\}$, which indicates the elements of computations that are to be included or excluded; in this example, $w_{ij} = 1$ if two cells are adjacent (joined) and $w_{ij} = 0$ otherwise. The matrix is symmetric and binary, but these features are not required, as one could instead assign rook’s-move adjacencies a weight of 0.9 and bishop’s-move (diagonal) connections a weight of 0.1, or, more generally, if the source dataset is polygonal (such as well-defined agricultural tracts), adjacency weights may be chosen to reflect the relative lengths of shared boundaries. With a set of values $\{z_i\}$ and a set of weights $\{w_{ij}\}$, we seek a function $f()$ that combines this information and satisfies the criteria that $f(z_i, z_j) > 0$ when both values are positive or both negative, $f(z_i, z_j) < 0$ when the pair contains one positive and one negative value, $f(z_i, z_j)$ is large when both values are large, and the adjacency structure in W is reflected in the computation. A simple way to meet these criteria is to multiply zone values together, optionally adjusting for the overall mean and incorporating adjacency weights, giving expressions that resemble covariance; when weights are binary they simply identify which elements are included or excluded. To obtain a covariance value, this sum is divided by the sum of all weights used, and standardizing this expression by dividing by the variance of the data yields the Moran’s I index, which typically ranges from approximately +1 (complete positive spatial autocorrelation) to approximately -1

(complete negative spatial autocorrelation). The bounds vary with the configuration of weights, and Moran's I is the spatial analogue of the correlation coefficient r , similar to the unweighted time-series autocorrelation coefficient r_1 . If the observations z_i are spatially independent Normal random variables, the expected value of Moran's I is given by its standard expression, which approaches 0 for large n . The variance of I under these conditions is also defined, and the variance under randomization (i.e., redistribution without replacement) is more complex but provided in many software packages; from expected values and variances, z-scores and significance levels can be computed. The computation of Moran's I for the sample dataset in Figure 5-29 begins by forming a 10×10 matrix of variance- and covariance-like quantities (Figure 5-31A), with the 10 values and their deviations from the overall mean, 1.022, shown along the margins; the diagonal contains variance-like terms and the off-diagonal elements contain covariance-like terms. The weight-adjusted matrix (Figure 5-31B) retains only off-diagonal terms for which adjacencies occur, summing to 16.19, while the diagonal variance-like values sum to 196.68. The final adjustment multiplies by $1/p$, where p is the number of cells (10) divided by the sum of all weights (26), resulting in Moran's $I = \frac{10 \times 16.19}{26 \times 196.68} = 0.0317$. A value close to 0 indicates very little spatial autocorrelation, which is confirmed in this example.

If we rearrange the cell values the index value will change accordingly. For example, if the negative values in our example are swapped with the positive values in column 3, so that column 2 is entirely positive and column 3 entirely negative (Figure 5-32) the index value computed is $I=0.26$ (see Figure 5-37 for the significance test of this index value).

Figure 5-32 Revised source data

	+4.55	-5.15
+2.24	+5.54	-4.39
+3.10	+9.02	-2.09
	+0.46	-3.06

Moran's I is one of the most frequently implemented statistics within GIS packages, including [ArcGIS](#) Spatial Statistics Toolbox, the AUTOCORR function in [TerrSet/Idrisi](#), and in most other spatial analytical tools. This is in part due to its ease of computation across a range of datasets and weightings, and because of its relationship to well-understood measures used in univariate statistics and time series analysis. It is also readily extended in a number of ways that increase its application and usefulness. These extensions include: adjustment for differences in underlying population size; treatment of spatial point data utilizing distance bands rather like time lags; and computation of local Moran I values rather than a single whole map or global Moran I value. We examine each of these in turn.

Oden (1995) proposed an adjustment to Moran's I for zonal data where population size data (e.g. population at risk) is available. This statistic is supported within Clusterseer, but is not generally provided in other packages. Under this adjusted version the null hypothesis is that measured rates (e.g. disease rates, crime rates) in adjacent areas are independent, with the spatial variation observed simply reflecting the spatial variation in the underlying population. The computation of this statistic is quite involved, but essentially utilizes the difference

between the proportion of cases or incidents in a given zone (i.e. as a proportion of the total cases in the study region) and the expected proportion in the population at large (again, computed as the population in zone i /total population at risk). The expected value for the adjusted statistic is similar to that for the standard statistic, being $E(I_{pop}) = -1/(N-1)$ where N is now the total population at risk. This statistic is not independent of population size, but can be readily standardized by dividing the computed value by the overall incidence rate for the study region.

We now turn to treatment of spatial point data rather than zoned or lattice data. Suppose that instead of zones we had a set of points (x,y) with associated data values, z (which could be values assigned to zone centroids). We could then compute:

$$I(h) = N(h) \frac{\sum_i \sum_j z_i z_j}{\sum_i z_i^2}$$

where $N(h)$ is the number of points in a distance band of width d and mean distance of h ; z_i is the (standardized) value at point i ; and z_j is the (standardized) value at a point j distance h from i (in practice within a distance band from $h-0.5d$ to $h+0.5d$). The summations apply for all point pairs in the selected band. Hence the adjacency matrix $\mathbf{W}$ is replaced by distance, h , and by using a number of bands a set of I values can be computed and plotted against h . This form of graph or diagram showing correlation information, as noted earlier, is known as a *correlogram*. To be meaningful the width of distance bands, h , should be greater than the minimum inter-point distance in the dataset, and the initial band should commence at $0.5d$ in order to give a better representation to values near the origin.

For example, the GS+ demonstration dataset comprises 132 separate spatial measurements of lead and aluminum levels in soil samples. An extract of the first 10 records is shown in [Figure 5-33A](#) with missing values being blank entries (coded on input in this case as -99). Note that missing values are not the same as zero values, and recorded zero values may actually be a reflection of the limitations of the measuring device rather than a true zero value. In such instances it may appropriate to adjust zero values to the minimum recordable value. Figure 5-33B shows a map of the point set for the selected z -variable (Pb, lead), with separate symbols indicating the four quartiles of the overall dataset that each point lies within (the triangular shape is the upper quartile, the square is the lower quartile).

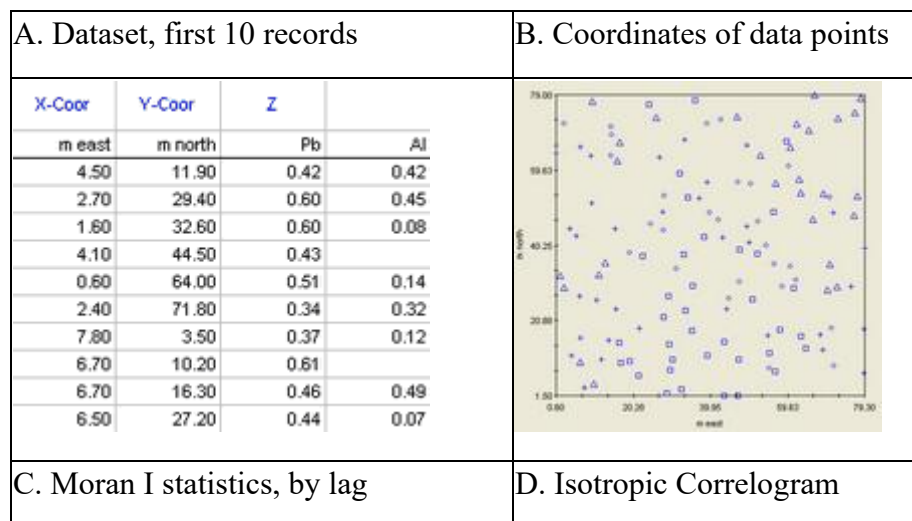
There are well over 10,000 possible point pairings in this sample dataset, but when grouped into distance bands only a subset of pairings is considered (Figure 5-33C). For example, in the first band, lag class 1, there are 226 point pairs, the average distance between pairs is 5.77 units, the bandwidth, h , is fixed for all lags at 8.21 units (defaulted to the maximum pairwise distance/10), and the Moran I value is 0.4179. This value decreases for the first few lags and then becomes closer to 0 and somewhat random. These values may be plotted, as shown in Figure 5-33D. Spatial autocorrelation (positive) as measured by this function, appears to be present for separation distances up to around 20-25 meters.

Each square on the graph in Figure 5-33D represents an index value generated from a large number of point pairs, and the individual contributions to the covariance can be plotted as a cloud of (in this case) 226 values (Figure 5-34). For example, the largest value in this cloud (see arrow) has a value of 0.4276 and is derived from points 100 and 115, whose separation distance is 6.44 units, and whose data values are 0.17 and 1.25 respectively. Now $(0.17-1.25)^2$ is not 0.4276, so we must look rather more carefully at what GS+ is doing with the data. The input dataset, like many such samples, is strongly left skewed — there are many small values recorded and only a few larger ones, with 1.25 being the maximum z-value. By default GS+ automatically transforms the data to make it more Normal, using the log transform $z^*=\ln(z+1)$. This gives the values:

$$z^*_{100}=0.157 \text{ and } z^*_{115}=0.811 \text{ and thus } (0.157-0.811)^2=0.4276$$

The correlogram shown is isotropic, i.e. it has been calculated for all points, irrespective of direction. Spatial patterns may well exhibit directional bias and hence it may be of interest to divide the circular lag regions into sectors and investigate the correlograms for each sector. Due to symmetry only half the possible bearings need to be considered. Taking a sector window of, say, 45 degrees (i.e. $X \pm 22.5$ degrees) results in a set of 4 possible sectors to be analyzed. The choice of initial bearing, X , can be selected by reference to the problem at hand, or by examining the dataset to locate the orientation of the principal axis of the standard deviational ellipse. Software packages often identify this for the user and then generate a set of anisotropic correlograms to enable the data to be examined sector-by-sector. If all are broadly similar it is likely that directional effects are small. Again, looking at this sample dataset, the principal axis is at 130 degrees, and the four anisotropic correlograms do not differ greatly.

Figure 5-33 Sample dataset and Moran I analysis



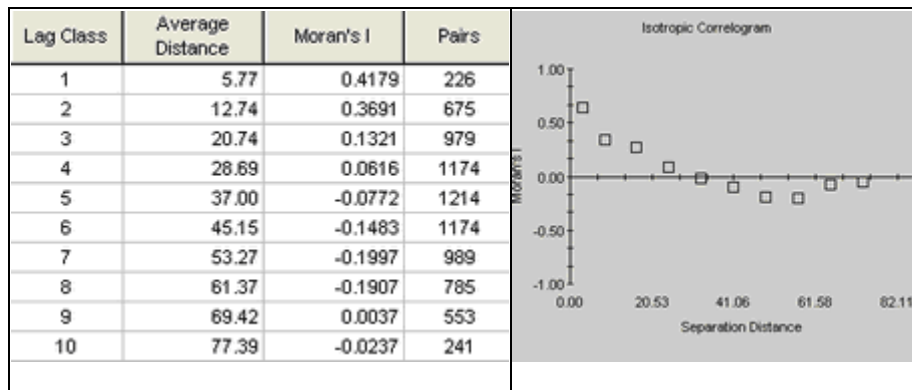
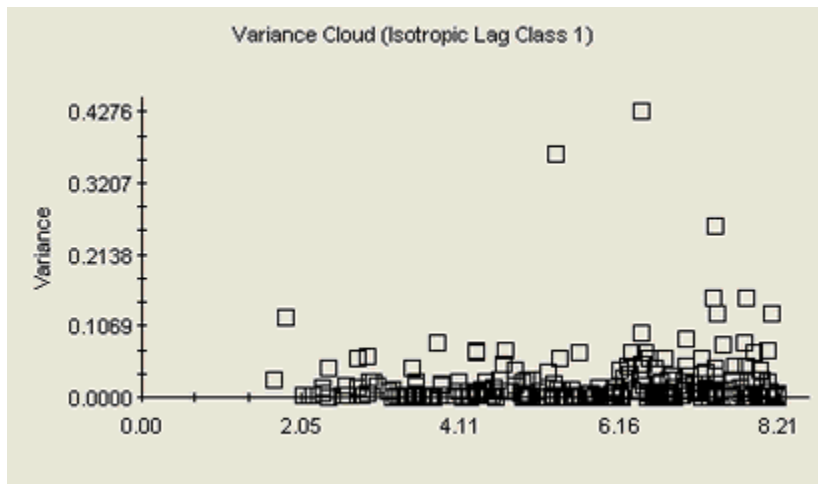


Figure 5-34 Moran I (co)variance cloud, lag 1



One alternative to the Moran I form of computation involves returning to the original set of criteria for a suitable function. Instead of computing the product sum:

$$S1 = \sum \sum z_i z_j$$

We could equally well have computed the sum of squared differences:

$$S2 = \sum \sum (z_i - z_j)^2$$

Or the absolute differences:

$$S3 = \sum \sum |z_i - z_j|$$

since these have similar desirable properties.

These expressions once again can be standardized by dividing by the variance and adjusted for the ratio of zones or point pairs sampled. If the expression $S2$ above is used the resulting measure is known as the Contiguity ratio or Geary's C ratio:

$$C = \frac{1}{p} \frac{\sum \sum w_{ij} (z_i - z_j)^2}{\sum (z_i - \bar{z})^2}, \text{ where } p = 2 \frac{\sum \sum w_{ij}}{n-1}$$

This expression may be compared to the formula for Moran I provided earlier. Unfortunately this ratio is not particularly easy to interpret: its values vary around 1, which indicates no spatial autocorrelation; to values as low as 0, which indicate positive autocorrelation; and values greater than 1 which indicate negative autocorrelation. Perhaps for this reason it is rarely implemented in GIS packages in this form, but expression *S2* turns out to be extremely useful in what is known as semivariance analysis as we shall see in our discussion on geostatistics (see Section 6.7, [Geostatistical Interpolation Methods](#)).

Due to the inconvenience of working with absolute values, expression *S3* is rarely used, but it is applied in semivariance analysis and is less sensitive to extreme values than computations based on *S2*. In this case the correlograms are known as Madograms — for more details see Deutsch and Journel (1992) and Goovaerts (1997).

Weighting models and lags

Our discussion on correlograms highlights the way in which distance bands can be used to generate data series in a manner similar to the lags used in time series analysis and Ripley's K function. Weighting matrices that incorporate values other than 0 and 1 were introduced by Cliff and Ord (1973). One way in which distance-based weighting is applied is to use the actual distances, rather than distance bands, as the means by which data pairs are “ordered”. The usual approach (for example that adopted within the [Crimestat](#) package) is to use inverse distance, so that pairs that are further apart contribute less to the overall index value than those that are close together. The distance weighted Moran's I is then given by:

$$I = \frac{1}{p} \frac{\sum \sum [(z_i - \bar{z})(z_j - \bar{z})] / (d_{ij} + \varepsilon)}{\sum (z_i - \bar{z})^2}$$

where

$$p = \frac{1}{n} \sum \sum 1 / (d_{ij} + \varepsilon)$$

The ε here represents an increment applied to the distance to ensure that when d_{ij} is small, results are not overly distorted or overflow. Other software packages implement a range of such distance weightings. For example, [ArcGIS](#) includes: inverse distance; inverse distance squared; and specified distance bands amongst its options.

With datasets that represents zones rather than points, as in our earlier examples, GIS tools may be used to map the data and compute spatial weights. Amongst the most extensive facilities

available at present for conducting such calculations are provided by the [GeoDa](#) and [ArcGIS](#) packages, so we will use these to illustrate the range of facilities that have been implemented in readily available software.

Simple rook's move and queen's move contiguity weights are supported within [GeoDa](#) for polygonal regions. Typically these will be first order contiguity values, but 2nd or higher order weights can also be generated. Secondly, distance-based weights can be generated based on polygon centroid locations or using a defined (x,y) pair falling within the selected polygon. This requires a threshold value to be set, and all points (representing polygons) within this distance band are included within the weighting (as per [ArcGIS](#) distance bands). [ArcGIS](#) also supports a combination of fixed distance bands and distance decay, which it calls *zone of indifference weighting*. Distance computations in both [GeoDa](#) and [ArcGIS](#) are based on projected (Euclidean) distance, plus a second option: in [GeoDa](#)'s case unprojected (approximate arc distance); and in the case of [ArcGIS](#), city block calculations (L_1 metric).

A further method, based on the number of nearest-neighbors, is also supported in [GeoDa](#). This requires the number of neighbors to be included as a parameter, and then varies the distances to reflect the value set (i.e. an adaptive method). The [GeoDa](#) code uses the Approximate Nearest Neighbor (ANN) software library developed at the University of Maryland to determine such neighborhood relations.

The advantage of this nearest-neighbors approach is that the spatial weights for tightly packed small zones will be more similar to those of larger zones within the same study region, which tends not be the case with pure distance-based selection. Finally, a user-defined spatial weights matrix may be defined and read into either of these two packages — of course this requires very careful construction to ensure the weights matrix reflects the topology and weighting patterns intended. Neither of these packages explicitly supports weights based on shared boundary lengths. With initial “point set data” weights could be determined by construction of Voronoi polygons and computing the lengths of shared boundaries, but facilities to perform such operations directly are not provided within existing GIS and related software. Information from the Delaunay triangulation and/or Voronoi polygon creation could be used to generate a spatial weights file, which [ArcGIS](#) and [GeoDa](#), for example could import. In this case, and by modifying or augmenting a program-generated standard weights file, weights with closer application-specific meaning may be created, for example based on measured interactions or flows between the sampled regions.

[TerrSet/Idrisi](#) supports simple autocorrelation computations for raster datasets using Moran's I. Its weights matrix options include rook's move and king's move (we would call this queen's move), with weights of 1 and 0.7071... for the direct and diagonal moves respectively. A form of support for higher order lags is indirectly supported in [TerrSet/Idrisi](#) by resampling the grid.

Local indicators of spatial association (LISA)

If we look more closely at the Moran I index we see that it can be disaggregated to provide a series of local indices. We start with the standard formulation we provided earlier:

$$I = \frac{1}{p} \frac{\sum_i \sum_j w_{ij} (z_i - \bar{z})(z_j - \bar{z})}{\sum_i (z_i - \bar{z})^2}, \text{ where } p = \sum_i \sum_j w_{ij} / n$$

Examining the computation of this “global” index provided earlier in [Figure 5-31B](#), we can see that each row in the computation contributes to the overall sum ([Figure 5-35A](#)), with the covariance component being 16.19 and overall Moran I value 0.03167. Each row item could be standardized by the overall sum of squared deviations and adjusted by the number of cells, n , as shown. The total of these local contributions divided by the total number of joins, $\sum \sum w_{ij}$, gives the overall or Global Moran I value. These individual components, or Local Indicators of Spatial Association (LISA), can be mapped and tested for significance to provide an indication of clustering patterns within the study region ([Figure 5-36](#)).

The calculations require some adjustment in order to provide Global Moran I and LISA values in their commonly used form. The first adjustment is to the weights matrix. It is usual to standardize the row totals in the weights matrix to sum to 1. One advantage of row standardization is that for each row (location) the set of weights corresponds to a form of weighted average of the neighbors that are included, which also enables different models to be readily compared. Row standardization may also provide a computationally more suitable weights matrix. This procedure alters the Global Moran I value when rows have differing totals, which is the typical situation. For the example above the effect is to increase the computed Moran I to 0.0736. The second adjustment involves using $(n-1)$ rather than n as the row multiplier, and then the sum is re-adjusted by this factor to produce the Global index value. These changes are illustrated in [Figure 5-35B](#) and [Figure 5-36](#) and correspond to the computations performed within [GeoDa](#), R and SpaceStat (Anselin, personal communication). [ArcGIS](#) offers the option to have no weights matrix standardization, row standardization (dividing rows by row totals) or total weights standardization (dividing all cells by the overall cell total).

The Geary C contiguity index may also be disaggregated to produce a LISA measure, in the same manner as for the Moran I. Another measure, known as the “G” statistic, due to Getis and Ord (1992, 1995), may also be used in this way. The latter is supported within [ArcGIS](#) as both a global index and in disaggregated form for identifying local clustering patterns, which [ArcGIS](#) considers this as a form of hot-spot analysis. The [ArcGIS](#) implementation of the G statistic facilitates the use of simple fixed Euclidean distance bands (threshold distances) within binary weights (as described below), but also permits the use of the Manhattan metric and a variety of alternative models of contiguity such as: inverse distance; inverse distance squared; so-called Zones of Indifference (a combination of inverse distance and threshold distance); and user-defined spatial weights matrices. Unlike the Moran I and Geary C statistics, the Getis and Ord G statistic identifies the degree to which high or low values cluster together.

The standard global form of the Getis and Ord G statistic is:

$$G = \frac{\sum_i \sum_j w_{ij} z_i z_j}{\sum_i \sum_j z_i z_j}, i \neq j$$

whilst the local variant is

$$G_i = \frac{\sum_j w_{ij} z_j}{\sum_j z_j}, i \neq j$$

The weights matrix, $\mathbf{W}$, is typically defined as a symmetric binary contiguity matrix, with contiguity determined by a (generalized) distance threshold, d . A variant of this latter statistic, which is regarded as of greater use in hot-spot analysis, permits $i=j$ in the expression (so includes i in the summations, with $w_{ii}=1$ rather than 0 in the binary weights contiguity model) and is known as the G_i^* statistic rather than G_i — Note that these local statistics are not meaningful if there are no events or values in the set of cells or zones under examination.

The expected value of G under Complete Spatial Randomness (CSR) is:

$$E(G) = \frac{\sum_i \sum_j w_{ij}}{n(n-1)} = \frac{W}{n(n-1)}, i \neq j$$

where n is the number of points or zones in the study area and W is the sum of the weights, which for binary weights is simply the number of pairs within the distance threshold, d . The equivalent expected value for the local variant is:

$$E(G_i) = \frac{\sum_j w_{ij}}{(n-1)} = \frac{W_i}{(n-1)}$$

where n is the number of points or zones within the threshold distance, d , for point/region i . Closed formulas for the variances have been obtained but are very lengthy. They facilitate the

construction of z-scores and hence significance testing of the global and local versions of the G statistics. For the local measure both computed values and z-scores may be useful to map.

The Rookcase add-in provides support for all three LISA measures with variable lag distance and number of lags. Further discussion of these methods is provided in several other sections of this Guide: in Section 5.2, [Exploratory Spatial Data Analysis](#) (ESDA); in Section 5.6.5, [Spatial filtering models](#); and in Sections 4.6, [Grid Operations and Map Algebra](#) and Chapter 6, [Surface and Field Analysis](#), relating to grid files, since there are parallels between several of the concepts applied.

Significance tests for autocorrelation indices

As noted in the preceding sections the various global and local spatial autocorrelation coefficients discussed can be tested for statistical significance under two, rather different, model assumptions. The first is the classical statistical assumption of Normality, whereby it is assumed that the observed value of the coefficient is the result of the set $\{z_i\}$ of values being independent and identically distributed drawings from a [Normal distribution](#), implying that variances are constant across the region. The second model is one of randomization, whereby the observed pattern of the set $\{z_i\}$ of values is assumed to be just one realization from all possible random permutations of the observed values across all the zones.

Both models have important weaknesses, for example as a result of underlying population size variation and lack of homogeneity of probabilities, but are widely implemented in software packages to provide estimates of the significance of observed results. In the case of the randomization model many software packages generate a set of N random permutations of the input data, where N is user specified. For each simulation run index values are computed and the set of such values used to provide a pseudo-probability distribution for the given problem, against which the observed value can be compared. A z-transform of the coefficients under Normality or Randomization assumptions is distributed approximately $N(0,1)$, hence this may be compared to percentage points of the [Normal distribution](#) to identify particularly high or low values.

The Rookcase add-in for Excel computes both values, for the global Moran I and Geary C , and the summary results in [Figure 5-37](#) illustrate this for the revised test dataset shown in [Figure 5-32](#). Note that these results are calculated using overall weight adjustment rather than row-wise adjustment. The results suggest the observed index value of 0.26 could reasonably be expected to have occurred by chance. With the local variant of Moran's I the same procedures as for the Global statistic can be adopted.

[Crimestat](#) provides the option to compute z-scores for this statistic but notes that it may be very slow to calculate, depending on the number of zones. [GeoDa](#) uses random permutations to obtain an estimated mean and standard deviation, which may then be used to create an estimated z-score.

The local variants of these statistics utilize values in individual cells or zones over and over again. This process means that the values obtained for adjacent neighbors will not be independent, which breaches the requirements for significance testing, a problem exacerbated by the fact that such cells are also likely to exhibit spatial autocorrelation. There are no generally-agreed solutions to these problems, but one approach is to regard each test as one of a multiple set. In classical statistics multiple tests of *independent* sets (the members of which

are not correlated) are made using a reduced or corrected significance level, α . The most basic of these adjustments (known as a Bonferroni correction, after its original protagonist) is to use a significance level defined by α/n , where n is the number of tests carried out, or in the present case, the number of zones or features being tested. If n is very large the use of such corrections results in extremely conservative confidence levels that may be inappropriate. [Monte Carlo](#) randomization techniques that seek to estimate the sampling distribution may be more appropriate in such cases.

Spatial Regression

Regression analysis is the term used to describe a family of methods that seek to model the relationship between one (or more) *dependent* or *response* variables and a number of *independent* or *predictor* variables. Spatial regression methods are similar, but take explicit account of the spatial structure of data, in particular the lack of independence that typically exists between measurements made at nearby locations. Initially we describe the basic models applied in ordinary regression and the application of this kind of technique to trend surface analysis. We then examine three different approaches to addressing the complexities that arise when attempting to model spatial datasets using regression techniques. The first of these involves varying the model parameters spatially using a technique known as Geographically Weighted regression ([GWR](#)). The second seeks to analyze the pattern and degree to which the sampled data are spatially correlated (or *autocorrelated*), and to use this information in order to construct regression models that recognize this structure and include it within their designs. The third approach attempts to apply simple filters or differencing to the data to remove the spatial variation and then model the filtered data.

Regression analysis is the term used to describe a family of methods that seek to model the relationship between one (or more) *dependent* or *response* variables and a number of *independent* or *predictor* variables. A typical (linear) relationship might be of the form:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots, \text{ or}$$

$$y = \mathbf{x}\boldsymbol{\beta}$$

where $\boldsymbol{\beta}$ is a column vector of p parameters to be determined and $\mathbf{x}$ is a row vector of independent variables with a 1 in the first column. It should be noted that the expression shown below is also described as linear since it remains linear in the coefficients, $\boldsymbol{\beta}$, even if it is not linear in the predictor variables, $\mathbf{x}$, e.g.:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_2 + \beta_4 x_1 x_2 + \dots$$

If $\mathbf{y}=\{y_i\}$ is a set of n observations or measurements of the response variable, with corresponding recording of matching values for the set of independent variables, then a series of linear equations of the type shown above can be constructed. However, typically the number of observations will be greater than the number of coefficients (the system is said to be *over-determined*). One approach to this problem is to seek a best fit solution, where best fit is taken to mean a solution for the vector $\boldsymbol{\beta}$ that minimizes the difference between the fitted model and observed values at these data points. In practice the procedure most widely applied seeks to

minimize the sum of squared differences, hence the term *Least Squares*. Each linear expression requires an additional "error" component to allow for the degree of fit of the expression to the particular data values. If we denote these error components by ε_i , the general form of the expression above now becomes:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \dots + \varepsilon_i, \text{ or}$$

$$\mathbf{y} = \mathbf{XB} + \boldsymbol{\varepsilon}$$

where $\mathbf{X}$ is now a matrix with 1's in column 1, and $\boldsymbol{\varepsilon}$ is a vector of errors that in conceptual terms is assumed to represent the effects of unobserved variables and measurement errors in the observations. Typically we assume that the expected value of these errors is zero, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, and the variance, $E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T) = \sigma^2 \mathbf{I}$, is constant. Here $\mathbf{0}$ is a column vector of zeros and $\mathbf{I}$ is the identity matrix.

This set of n equations in p unknowns ($n > p$) is typically solved for the vector $\boldsymbol{\beta}$ by seeking to minimize the sum of the squared error terms, $\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T$ (*Ordinary Least Squares*, or *OLS*). The solution for the coefficients using this approach is obtained from the matrix expression:

$$\hat{\mathbf{B}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\text{var}(\hat{\mathbf{B}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

where the hat (^) symbol denotes concern with sample estimates of population parameters and the sample estimates of the population error ($\boldsymbol{\varepsilon}$) terms are described as 'residuals'.

The variance for such models is usually estimated from the residuals of the fitted model, using the expression:

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p}$$

Methods such as these may be used in the context of spatial description and mapping, as part of a process of data exploration, or for explanatory and/or predictive purposes.

The broad principles of regression analysis and modeling, as described in basic statistics texts, also apply to spatial datasets. In a spatial context one is seeking to model the variation in some spatially distributed dependent variable, for example house prices in a city, from a set of independent variables such as size of property, age of property, distance from transport facilities, proportion of green space in the region etc. (see further Table 5-11, *hedonic*

regression). Spatial coordinates might be explicitly or implicitly incorporated into such models, for example including the coordinates of the centroid of a census tract or event location into the modeling process, or by taking account of the pattern of adjacency of neighboring zones.

Similar methods are also widely supported in raster-based packages, with the dependent variable being taken as one grid from a set of matching layers or bands, and the independent variables being taken as the matching grids. Grid cell entries are treated as the sample values (responses) and the measured predictor values. In most packages this technique is used to aid grid combination, but others suggest that it may be used for predictive purposes (e.g. predicting crop yield from a series of measured grids that provide details of soil structure and chemical makeup). This kind of modeling is generally unsafe due to the high degree of spatial autocorrelation between nearby cells in each grid (possibly generated by interpolation or resampling).

Ideally in regression analysis the form of the chosen model should be as simple as possible, both in terms of the expression employed and the number of independent variables included — these are sometimes referred to as the *simplicity* and *parsimony* objectives. In addition, the proportion of the variation in the dependent variable(s), y , explained by the independent variables, x , should be as high as possible, and the correlation between separate independent variables should be as low as possible. These criteria form part of a *strength of evidence* objective. If some, or all, of the x are highly correlated (typically having a correlation coefficient above 0.8) the model almost certainly contains redundant information and may be described as being *over-specified*. If a strong relationship exists between selected independent variables and is also broadly linear, the variables are said to exhibit *multi-collinearity*. There are a number of standard techniques for reducing multi-collinearity, including: applying the so-called centering transform (deducting the mean of the relevant independent variable from each measured value); increasing the sample size (especially if samples are quite small); removing the most inter-correlated variable; and combining inter-correlated variables into a new single composite variable (e.g. using principal components analysis, PCA, or other data reduction techniques). The latter procedures can be viewed as forms of model re-specification.

With higher-order spatial trend analyses, multi-collinearity is to be expected since various combinations of the coordinates are themselves the independent variables. Technically, imperfect multi-collinearity does not result in the OLS assumptions being violated, but does lead to large standard errors for the coefficients involved, also referred to as *variance inflation* (VI). This limits their usefulness. As a rule of thumb, if the measure known as the Multi-collinearity Condition Number, MCN, is greater than 30 this problem is quite serious and should be addressed. The MCN is derived from the ratio of the largest to the smallest eigenvalue of the matrix $\mathbf{X}^T\mathbf{X}$. Generally the square root of this ratio is reported, but it is important to check this for any specific software package (if no square root is taken values greater than 1000 are taken as significant).

When fitting a model using OLS it is generally assumed that the errors (residuals for sample points) are identical and independently distributed (denoted *iid*). If the spread of errors is not constant, for example if in some parts of the study area the residuals are much more variable than in others these errors are said to exhibit *heteroskedasticity*. If the latter is the case (which may be detected using a range of standard tests) the estimated variance under OLS will be biased, either being too large or too small (it is not generally possible to know which direction

the bias takes). As a result the specification of confidence intervals and the application of standard statistical significance tests (e.g. F tests) will not be reliable.

Many (most) spatial datasets exhibit patterns of data and/or residuals in which neighboring areas have similar values (*positive spatial autocorrelation*) and hence violate the core assumptions of standard regression models. One result of this feature of spatial data is to greatly over-estimate the effective sample size, and hence the degrees of freedom that may be indicated (for example in multiple grid regressions) will be stated as being far higher than they should be. One possible (partial) solution to this problem, in the case of grids, is to take (stratified or equalized) random samples from the response grid and apply regression to the subset of grid cells that correspond to these samples in the predictor grids. This approach is sometimes (rather confusingly) referred to as resampling.

A common additional requirement for standard regression modeling, as noted above, is that the distribution of the errors is not only identical but also Normal, i.e.

$$\epsilon \sim N(0, \sigma^2 I)$$

If the distribution of the errors is Normal and the regression relationship is linear, this equates to the response variable being Normally distributed, and vice versa, i.e.

$$y \sim N(X\beta, \sigma^2 I), \text{ or setting the vector } \mu = X\beta$$

$$y \sim N(\mu, \sigma^2 I)$$

The assumption of Normality permits the analytical derivation of confidence intervals for the model parameters and significance tests on their values (against the null hypothesis that they are zero). In order to satisfy the Normality conditions required in many models it is common to transform (continuous valued) response data using Log, Square Root or Box-Cox functions ([Table 1-3](#)) prior to analysis where this can be shown to improve the approximation to the Normal. It is also advisable to examine and, where appropriate, remove outliers even if these are not errors in the dataset but represent valid data that would otherwise distort the modeling process. Exclusion must be made explicit and separate investigation of any excluded data carried out. An alternative to exclusion is to weight observations according to some rule. Spatial declustering (see Section 5.1.2, [Spatial sampling](#)) is an example of such weighting applied to spatial datasets.

If spatial autocorrelation has been detected (see Section 5.5.1, [Autocorrelation, time series and spatial analysis](#), *et seq.*) a more reasonable assumption is:

$$y \sim N(\mu, C)$$

where **C** is a positive definite covariance matrix. Typically **C** would be defined in such a way as to ensure that locations that are close to each other are represented with a high value for the modeled covariance, whilst places further apart have lower covariance. This may be achieved using some form of distance-decay function or, for zoned data, using a contiguity-based spatial weights matrix. This kind of arrangement is similar to *Generalized Least Squares* (GLS), which allows for so-called second-order effects via the inclusion of a covariance matrix. The standard form of expression for GLS is:

$$y = X\beta + u$$

where $\mathbf{u}$ is a vector of random variables (errors) with mean 0 and variance-covariance matrix $\mathbf{C}$. The GLS solution for $\boldsymbol{\beta}$ is then:

$$\hat{\mathbf{B}} = (\mathbf{X}^T \mathbf{C}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{C}^{-1} \mathbf{y}$$

$$\text{var}(\hat{\mathbf{B}}) = (\mathbf{X}^T \mathbf{C}^{-1} \mathbf{X})^{-1}$$

In spatial analysis this kind of GLS model is often utilized, with the matrix $\mathbf{C}$ modeled using some form of distance or interaction function. Note that $\mathbf{C}$ must be invertible, which places some restrictions on the possible forms it may take. Simple weighted least squares (WLS) is a variant of GLS in which the matrix $\mathbf{C}$ is diagonal and the diagonal values are not equal (hence the errors are heteroskedastic). In this case the weights are normally set to the inverse of the variance, thereby giving less weight to observations which have higher variance.

In addition to regression on dependent variables that take continuous values, similar methods have been developed for binary and count or rate data. In the case of binary data the data typically are transformed using the logit function (see [Table 1-3](#)). The central assumption of this model is that the probability, p , of the dependent variable, y , taking the value 1 (rather than 0) follows the logistic curve, i.e.

$$p = \text{prob} \langle y = 1 | \mathbf{x} \rangle = \frac{e^{\mathbf{B}\mathbf{x}}}{1 + e^{\mathbf{B}\mathbf{x}}}$$

This model can be linearized using the logit transform, $q = \ln(p/(1-p))$. The regression model then becomes

$$q_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \dots + \varepsilon_i, \text{ or}$$

$$\mathbf{q} = \mathbf{X}\mathbf{B} + \boldsymbol{\varepsilon}$$

As with standard linear regression the errors in this model are assumed to be *iid* and the predictor variables are assumed not to be co-linear. The parameters in such models are determined using non-linear (iterative) optimization methods, involving maximization of the likelihood function. With count or rate data Poisson regression may be applicable. The model is similar in form to the binary logit model, but in this case the response variable, y , is count (integer) data (hence non-continuous). The basic model in this case is of the form:

$$y = A e^{\mathbf{B}\mathbf{x}}$$

where A is an (optional) offset or population-based value (e.g. the underlying population at risk in each modeled zone). This model may be linearized, as per the logit model, but this time by simply applying a log transform: $q=\ln(y)$. The regression model then becomes:

$$q_i = (\ln(A) + \beta_0) + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \dots + \varepsilon_i, \text{ or}$$

$$\mathbf{q} = \mathbf{XB} + \boldsymbol{\varepsilon}$$

There are many variants of linear and non-linear regression analysis and a wealth of associated terms applied in this field — a number of these are summarized in [Table 5-11](#). In some instances the techniques are not always described as regression analysis *per se*, but share many underlying statistical assumptions and associated tests.

Choosing between alternative models presents a particular challenge. Each additional parameter may improve the fit of the model to the sample dataset (or data set), but at the cost of increasing model complexity. Model building should be guided by the principal of *parsimony* — that is the simplest model that is fit for the purpose for which it is built. Overly complicated models frequently involve a loss of information regarding the nature of key relationships. Drawing on this *information content* view of modeling, an information theoretic statistic is often used as a guide to model selection. The most common statistic now used is known as the Akaike Information Criterion (AIC) and is often computed in its small-sample corrected version (AICc) since this is asymptotic to the standard version. The measures are defined as:

$$AIC = -2\ln(L) + 2k$$

$$AICc = -2\ln(L) + 2k \left(\frac{n}{n - k - 1} \right)$$

where n is the sample size, k is the number of parameters used in the model, and L is the likelihood function. It is recommended that the corrected version be used unless $n/k > 40$. If the errors are assumed to be Normally distributed then the standard expression can be re-written as:

$$AIC = n(\ln(2\pi\sigma^2) + 1) + 2k$$

The AIC measure is utilized in several of the software packages described in this Section and elsewhere in this Guide (e.g. in the point pattern analysis software, [spatstat](#)). For example [GeoDa](#) uses the version above in OLS estimation, whilst [Geographically Weighted Regression \(GWR\)](#) uses a variant of the version corrected for small sample sizes. Both packages use appropriate alternative expressions in cases where the likelihood functions are different from the Normal model. A wide range of regression models, including OLS and [GWR](#), and several autocorrelation models (see below) are supported by the [SAM](#) package

(which also supports AIC-based model selection). Although [SAM](#) has been designed for and by macro ecologists, it can equally well be applied in other fields of spatial research.

A second information theoretic measure, which places greater weight on the number of parameters used, is often also provided. This is known as the Bayesian Information Criterion or BIC, or sometimes as the Swartz Criterion. It has the simpler form:

$$BIC = -2\ln(L) + k\ln(n)$$

Several forms of regression models are commonly used in spatial and statistical analysis. Simple linear regression involves a single approximately continuous response (dependent) variable and one or more predictor variables related by an expression that is linear in its coefficients, though not necessarily linear in the predictor variables, and shows similarities to weighted averages and geostatistical models. When multiple predictor variables are all quantitative, the model is referred to as multiple regression. Multivariate regression involves more than one response variable, and when multiple predictors are also present, the model is termed multivariate multiple regression. Simultaneous autoregressive (SAR) models, also known as spatial autoregression, incorporate adjustments for spatial autocorrelation, and many variants of SAR have been developed. Conditional autoregressive (CAR) models are similar in purpose but differ from SAR models in how the inverse covariance matrix is specified; in CAR models, the expected value of the response variable is conditional on recorded values at all other locations. Logistic regression applies when the response variable is binary (1/0 or Yes/No), typically using the logit transform, with linear regression performed on the transformed data; extensions exist for multi-category responses, whether ordered or unordered. Poisson regression applies when the response variable represents counts (such as crime incidents or disease cases) or standardized rates and assumes a Poisson distribution whose expected value depends on predictor variables, usually modeled as having a linear relationship with the log of the mean; datasets with many zeros may pose challenges. Ecological regression refers not to ecology but to regression applied to data aggregated into zones, such as census or administrative units, and relates to issues such as the ecological fallacy, which concerns the difficulty of making inferences about individuals based on aggregated data. Hedonic regression, used mainly in economics and real estate analysis, estimates demand or prices as a combination of component attributes treated as independent variables. Analysis of variance is used when all predictors are qualitative or grouped into a small number of categories and is often applied to assess the significance of alternative regression models under Normality assumptions for error distributions. Analysis of covariance is used when predictors include both qualitative and quantitative variables and is widely used in spatial modeling to examine covariance of observations at pairs of locations.

Simple regression and trend surface modeling

Simple regression techniques have been widely applied in spatial analysis for a very long time. The general form of model employed can be represented by the expression:

$$y = f(x_1, x_2, w)$$

where: y is the (possibly transformed) observed value of some continuous dependent variable y at location (x_1, x_2) ; $f()$ is a general function to be defined, but often a linear expression in the coefficients; and $\mathbf{w}$ is a vector of attributes measured at (x_1, x_2) . For example, if $f()$ is simply modeled as a linear function of the coordinates we can write

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Model performance can be assessed using a number of statistics that are discussed in standard statistical text books. One such statistic, which will be referred to in our subsequent discussion, is the correlation coefficient squared or [R² statistic](#). This statistic records the proportion of variation in the data that is explained by the model. A modification this statistic, adjusted R^2 , takes into account the complexity of the model in terms of the number of variables that are specified.

With a set of n observations made at different locations in the study area, $y_i(x_{1i}, x_{2i})$, a series of n equations can be set up and solved for the $p=3$ unknown parameters using OLS. This is simply a linear trend or best fit linear surface through the point set, $\{y_i\}$. A set of differences between the observed values and the best fit surface can then be computed:

$$\varepsilon_i = \hat{y}_i - y_i$$

where the $\hat{y}$ values (the estimated values) are obtained by evaluating the trend surface at each point i . Assuming the value of n is reasonably large and provides a representative coverage of the complete set of locations in the study area, these residuals may be treated as samples from a continuous surface and mapped using simple contouring or filled contouring procedures. If, on the other hand, the sample points represent centroids of zones (pre-defined regions or generated zones, e.g. Voronoi polygons) the visualization of residuals is generally applied to these zones as a choropleth map. Plotting the residuals using a combination of the fitted surface and stick/point display in 3D may also be helpful but is less common within GIS packages.

Whichever approach is adopted the fitting of a trend surface and mapping of residuals is a form of [Exploratory Spatial Data Analysis](#) (ESDA) and is supported in many GIS and related packages. Within GIS packages such residual maps may be overlaid on topographic or thematic maps with a pre-specified transparency (e.g. 50%) enabling spatial associations between large positive or negative residuals to be visually compared to potential explanatory factors.

With higher order polynomials or alternative functions (e.g. trigonometric series) the fit of the trend surface to the sample data *may* be improved (the sum of the squared residuals reduced) but the overall utility of the procedure may well become questionable. This may occur for many reasons including the increased difficulty in interpreting the fitted surface (e.g. in terms of process), and increased volatility of the surface close to and immediately beyond the convex hull of the sample points and/or in sparsely sampled regions. Higher order polynomial trend surfaces are usually limited to second order equations, although in some applications 3rd order or above have been shown to be useful — for example, Lichstein *et al.* (2002) fit a full 3rd order polynomial trend equation to the spatial coordinates of bird species sightings. They then proceed to analyze the distribution of bird species by modeling habitat + trend (large scale

spatial pattern) + spatial autocorrelation (small scale spatial pattern) in order to identify separate factors that might be used to explain observed distribution patterns by species. Note that any analysis of trends in this way will result in a fitted surface whose form is heavily dependent upon the set of specific locations at which measured values of y are available and the scale factors operating for the problem being examined. Frequently data locations are not the result of selections made by the analyst but are pre-determined survey locations that may reflect convenience for data gathering (e.g. proximity to a forest path, access to quarry sites, location of wells) rather than an ideal sampling frame for analytic purposes.

The most common measure of the quality of fit of a trend surface or similar regression is the *coefficient of determination*, which is the squared coefficient of (multiple) correlation, R^2 . This is simply a function of the squared residuals with standardization being achieved using the sum of squared deviations of observations from their overall mean:

$$R^2 = 1 - \frac{\sum \varepsilon_i^2}{\sum (y_i - \bar{y})^2}, R^2 \in [0, 1]$$

Under appropriate conditions, e.g. $\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$ the significance of this coefficient can be evaluated in a meaningful way, with values close to 1 generally being highly significant. Frequently, however, the statistical significance of the measure cannot be determined due to distribution conditions not being met, and the value obtained should be seen as no more than an indicator of goodness of fit.

Trend surface and residuals analysis is essentially a form of non-statistical ESDA — it is not necessary to state how significant the fit of the model is or what confidence limits might be placed around the values of the estimated coefficients and the fitted surface. For inferential analysis the standard pre-conditions that apply are extremely unlikely to be met in the context of spatial analysis. Conditions that may apply for statistical purposes include:

- the set $\{y_i\}$ is comprised of independent (uncorrelated) observable random variables. This set does not have to be drawn from a [Normal distribution](#) but it is often helpful or necessary that they are at least asymptotically Normal (possibly after transformation)
- the set $\{\varepsilon_i\}$ is comprised of independently and identically distributed (*iid*) unobservable random variables with mean 0 and constant variance, σ^2 , where σ^2 is not a function of β or x
- the set $\{\varepsilon_i\}$ is Normally distributed — this condition is not a pre-requisite but is a requirement for inference on finite sets (as previously noted, this condition is directly connected to the first bullet point above)
- the model applied is appropriate, complete and global. This assumption incorporates the assumption that the independent variables, x , are themselves uncorrelated, and the parameters β are global constants

Tests for the Normality of the observations and the residuals are commonplace (the first and third conditions above), and a variety of tests for heteroskedasticity (resulting in failure to meet the second condition) have been implemented in a several GIS-related packages. Note that

heteroskedasticity may or may not be associated with positive spatial autocorrelation. As discussed in Section 5.5.2, [Global spatial autocorrelation](#), testing for (global) spatial autocorrelation using the Moran I statistic is a useful starting point for such analyses. A spatial weights matrix, $\mathbf{W}$, is required in order to generate the Moran I statistic. As shown earlier this statistic has the form:

$$I = \frac{1}{p} \frac{\sum \sum w_{ij} (y_i - \bar{y})(y_j - \bar{y})}{\sum (y_i - \bar{y})^2}, \text{ where}$$

$$p = \sum \sum w_{ij} / n$$

This measure is the spatial equivalent of the standard product moment correlation coefficient. The weighting terms, w_{ij} , are defined either from patterns of zone adjacency or using some form of distance-decay function. A value of Moran's $I=1$ indicates perfect positive spatial autocorrelation, whilst a value of I very close to 0 indicates no spatial autocorrelation. A value such as $I=0.2$ indicates some positive spatial autocorrelation, although this may or may not be significant — with large spatial datasets small positive I -values are very often technically identified as significant, although the value of this interpretation may not be great.

Assuming that significant spatial autocorrelation is identified several options exist for continued analysis using regression methods:

- proceed with the analysis ignoring the observed spatial autocorrelation (particularly if it is small) accepting that the drawing of confidence intervals and determination of significance levels will not be possible, or if computed may be misleading
- drop the requirement that the parameters β are global constants and allow these to vary with location. Effectively this involves fitting a series of regression surfaces to the dataset in such a way as to ensure a continuous prediction surface is created. One procedure for achieving this objective is known as [Geographical Weighted Regression](#) (GWR); or
- include additional elements into the regression model that take explicit account of the observed pattern of spatial autocorrelation, as described in Section 5.6.4, [Spatial autoregressive and Bayesian modeling](#)

Geographically Weighted Regression (GWR)

GWR is the term introduced by Fotheringham, Charlton and Brunsdon (1997, 2002) to describe a family of regression models in which the coefficients, β , are allowed to vary spatially. GWR uses the coordinates of each sample point or zone centroid, t_i , as a target point for a form of spatially weighted least squares regression (for some models the target points can be separately defined, e.g. as grid intersection points, rather than observed data points). The result is a model of the form:

$$\mathbf{y} = \mathbf{XB}(\mathbf{t}) + \boldsymbol{\varepsilon}$$

The coefficients $\beta(t)$ are determined by examining the set of points within a well-defined neighborhood of each of the sample points. This neighborhood is essentially a circle, radius r , around each data point (anisotropic modeling is not currently supported). However, if r is treated as a fixed value in which all points are regarded as of equal importance it could include every point (for r large) or alternatively no other points (for r very small). Instead of using a fixed value for r it is replaced by a distance-decay function, $f(d)$, as described earlier (Section 4.4.5, [Distance decay models](#)). This function may be finite or infinite, much as with kernel density estimation (Section 4.3.4, [Density, kernels and occupancy](#)). The functions utilized in the GWR software package are of the form:

$$f(d) = e^{-d^2/2h^2}, \text{ or } f(d) = e^{-d/h}, \text{ or}$$

$$f(d) = \left(1 - \frac{d^2}{h^2}\right)^2, d < r, f(d) = 0 \text{ otherwise}$$

In these functions the parameter, h , also known as the *bandwidth*, is the key factor determining the way in which the weighting schemes operate. A small bandwidth results in very rapid distance decay, whereas a larger value will result in a smoother weighting scheme. This parameter may be defined manually or alternatively by some form of adaptive method such as cross-validation minimization, e.g. jackknifing — see further, Section 6.7.2, [Kriging interpolation](#), and Efron (1982) and Efron and Tibshirani (1997), or minimization of the Akaike Information Criterion (AIC). The use of a kernel function also raises the possibility of generating additional descriptive statistics, as have been described earlier.

Using a selected kernel function and bandwidth, h , a diagonal weighting matrix, $\mathbf{W}(t)$, may be defined for every sample point, t , with off-diagonal elements being 0. The parameters $\beta(t)$ for this point can then be determined using the standard solution for weighted least squares regression:

$$\hat{\mathbf{B}}(t) = \left(\mathbf{X}^T \mathbf{W}(t) \mathbf{X}\right)^{-1} \mathbf{X}^T \mathbf{W}(t) \mathbf{y}$$

or, letting

$$\mathbf{D} = \left(\mathbf{X}^T \mathbf{W}(t) \mathbf{X}\right)^{-1} \mathbf{X}^T \mathbf{W}(t)$$

$$\hat{\mathbf{B}}(t) = \mathbf{D} \mathbf{y} \text{ and } \text{var}(\hat{\mathbf{B}}(t)) = \mathbf{D} \mathbf{D}^T \sigma^2$$

The standard errors of the parameter estimates can be computed as the square root of these variances and used in t -tests to obtain estimates of the significance of the individual components. In this model the variance component, σ^2 , is defined by the normalized residual sum of squares (RSS) divided by the degrees of freedom. The latter are defined by the number of parameters, p , in a global model, or the *effective number of parameters* in the GWR model. This value is approximated by the authors as the trace of a matrix $\mathbf{S}$, $\text{tr}(\mathbf{S})$ (the sum of the diagonal elements of $\mathbf{S}$) defined by the relation:

$$\hat{\mathbf{y}} = \mathbf{S}\mathbf{y}$$

A set of such equations is solved for all points, t . The fit of the model may be examined in the usual manner, although it is to be expected that the fit in terms of variance explanation will almost always be an improvement over global methods, if only because there are far more parameters fitted to the dataset. For this reason comparisons should be made on additional criteria, for example the AIC measure which takes account of the model complexity. As with conventional regression, the modeled surface and (standardized) residuals may be mapped for exploratory purposes, but additionally the parameters $\boldsymbol{\beta}(t)$ and their estimated standard errors may also be mapped since these also vary spatially. Within GWR standardized residuals are determined as the sum of squared residuals, $\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T$, divided by the degrees of freedom, $n-\text{tr}(\mathbf{S})$. The authors recommend examining any values for these residuals >3 .

To illustrate this process we shall use an example dataset comprising educational attainment by county in the state of Georgia, USA (the dependent or response variable, [Table 5-12](#)). This dataset lists the percentage of University graduates by county together with a range of social data that might act as independent variables to be used in predicting the dependent variable. The data have been assigned to a set of 159 point locations (county centroids) and show an overall average of 19% of the population recorded as being graduates, with an average per county of 10.9% (i.e. not population weighted). The range by county is from 4.2% to 37.5%. [Table 5-12](#) shows the predictor variables and the global regression parameters estimated by OLS, which collectively account for around 63% of the variance. Also shown are the GWR parameter estimates, expressed as a range of values that have been computed. In the diagnostics section of this table note the drop in residual sum of squares, the increase in the adjusted R^2 , and a modest fall in the AICc statistic. The authors suggest that a fall of 3 or more in the AICc value warrants examination as demonstrating a meaningful improvement in model fit. Note that differences in the method of calculation of the AIC statistic can easily result in differences of greater than 3, so caution is required when comparing alternative software packages on this measure. As mentioned above, the standardized residuals from the GWR predictions can be mapped in order to identify any prediction outliers. [Figure 5-38A](#) shows this mapped dataset, with the dark gray/dark blue and red counties illustrated being those with the highest and lowest deviations. These counties may then be examined to try and ascertain if there are any special characteristics of these cases that might explain the large residuals. By definition the GWR modeled parameters falling within the range shown in [Table 5-12](#) include a value for every county. Hence each parameter can also be mapped, as shown in [Figure 5-38B](#). In this case the map highlights a distinct pattern of variation, with higher values in the north and lower values in the south.

As Fotheringham *et al.* (2005) have noted: “In some instances ... it is difficult to justify why some relationships should be allowed to vary spatially. In others, empirical results may suggest that some relationships are stationary over space while others vary significantly. In these instances, ‘mixed’ GWR models, where some relationships are allowed to vary spatially while others are held constant, would seem to be more appropriate.” For such cases the authors are planning to release a Version 4 of the GWR software which will provide specific support for mixed models.

The same kind of GWR analysis can be carried out on count data, using Poisson regression (GWPR), and on binary data, using Logistic regression (GWLR). The GWR V3 program supports both models. The standard Poisson and Logistic regression models described in Table 5-11 are utilized, but with the coefficients $\beta(t)$ varying with location, t , as before. As an example, GWPR has been applied to counts of disease incidence amongst a particular age/sex grouping, recorded by health district. In such models an *offset* value is applied to the model based on a matching count variable, such as the total number of people in that district in the selected age/sex cohort. For example, Nakaya *et al.* (2005) applied GWPR to mortality rates in Tokyo. The dependent variable in this instance was based on the standardized mortality ratio (SMR) for each of 262 municipality zones. The SMR is defined as the observed number of deaths, O_i , in a specified time period (e.g. 1990) in a given zone i , divided by the expected number, E_i , for that zone based on national or regional mortality rates (i.e. by demographic grouping). In the GWPR model the O_i became the response variable and the E_i provided the offset values. The study was able to examine relationships between the dependent variable and a number of independent variables at the local level, highlighting particular relationships that global models may not have identified. In fact the best model the authors were able to produce included a mix of global and local parameter estimates, with the proportion of older people (64+) and of house-owners taken as globals and the proportion of professional and technical people and the proportion of unemployed people being allowed to vary regionally.

For Logistic GWR one might have true presence/absence data, or recoded continuous data based on some critical threshold value. For example, with the Georgia dataset the dependent variable could be recoded as 1 if the percentage of graduates is above the state average and 0 otherwise. This is rather an artificial example, but recoding of this type is often applied in decision-making — for example coding land as contaminated (1) if, say, the average measured cadmium level in the soil exceeds a certain number of parts per million (ppm) and not contaminated (0) with respect to this trace element if below this threshold level.

Both Poisson and Logistic GWR require model fitting using a technique known as iteratively re-weighted least squares (IRLS). The analysis is carried out in much the same manner as previously described, but the computation of the Akaike Information Criterion (AIC) and AICc differs from the OLS expressions.

The ready availability of GWR software, supporting Gaussian, Poisson and Logistic models, together with a companion book and materials, has resulted in an upsurge in interest in the technique. This includes its consideration by spatial econometricians, medical statisticians and ecologists amongst others (GWR support has recently been included in the [R-Spatial](#) collection as `spgwr` (see also `simplegwr`) and in the [SAM](#) ecology package, as well as in the latest versions of ESRI's [ArcGIS](#)). It has the attraction of accepting the non-stationarity of most spatial datasets (see further, Section 6.7.1, [Core concepts in Geostatistics](#)), and proceeding to create

models with improved information characteristics and amenable to further exploratory analysis. For large-scale problems the processing overheads of GWR may become prohibitive, but the technique is well-suited to parallel or grid-enabled processing, as has been demonstrated by Harris *et al.*, 2006 and others.

Spatial autoregressive modeling

Many of the techniques that are briefly described in this final subsection originate from time series analysis and were subsequently developed from the mid-1950s within the discipline known as spatial statistics. They have been applied and substantially extended in the last 25+ years, notably by econometricians, geographers and medical statisticians. Additional disciplines that have made extensive use of these techniques include the actuarial, ecological, environmental and biomedical sciences. Detailed discussion of the methods and underlying theory may be found in Cressie (1993), Bailey and Gatrell (1995), Anselin (1988), Anselin and Bera (1998), Anselin (2002) and Haining (2003). Haining (2003) discusses the various forms of model that may be appropriate for the statistical analysis of spatial data. He notes the choice of a particular modeling approach often reflects the data types, preferences and experience within the disciplines involved, with social sciences, earth sciences and medical researchers all tending to focus on different approaches. In the last few years, notably since 2023, autoregressive models have been combined in regenerative AI systems to produce some remarkable results, in fields as diverse as natural language processing, weather forecasting and protein modeling.

"Autoregressive AI refers to a class of models that generate data (like text, images, or audio) by predicting future values based on past values in a sequence. These models work by assuming that the current output depends on the previous outputs. They process data one step at a time, using the previous step's output as input to predict the next step." This definition was generated by the LLM AI service [ChatGPT](#) - see further, Chapter 9

Spatial autoregressive procedures have been implemented in: [SpaceStat](#); S-Plus; the [R-Spatial](#) project; [MATLab](#) Spatial Statistics Toolbox (Pace *et al.*); [WinBUGS](#); [SAM](#); [PySal](#); and [GeoDa](#) packages, amongst others. A number of these have been specifically developed to deal with large (and often sparse) matrix difficulties that arise with detailed regional and national datasets.

A pure spatial autoregressive model simply consists of a spatially lagged version of the dependent variable, y :

$$y = \rho Wy + \epsilon$$

As can be seen this is similar to a standard linear regression model where the first term is constructed from a predefined n by n spatial weighting matrix, W , applied to the observed variable, y , together with a spatial autoregression parameter, ρ , which typically has to be estimated from the data. Anselin (2008, p257) describes spatial lag models as “a formal representation of the equilibrium outcome of processes of social and spatial interaction”. Essentially a spatial lag model is expressing the notion that the value of a variable at a given location is related to the values of the same variable measured at nearby locations, reflecting some kind of interaction effect. The spatial weights matrix, W , is almost always standardized such that its rows sum to 1, hence it is effectively including a weighted average of neighboring

values into the regression equation. Note that in this case $\mathbf{W}$ is not necessarily symmetric (compare this with the symmetry requirements on conditional autoregressive, or CAR, models that are discussed later in this section).

For an individual observation the basic spatial lagged autoregression equation is simply:

$$y_i = \rho \sum_j w_{ij} y_j + \varepsilon_i$$

Note the similarity of this model to a series of simultaneous equations (hence the description of such models as simultaneous autoregressive, or SAR, models). The model can also be compared with a simple time series autoregressive model, from which it owes its origins:

$$y_t = \rho y_{t-1} + \varepsilon_t$$

Since the dependent variable, $\mathbf{y}$, appears on both sides of the expression:

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \boldsymbol{\varepsilon}$$

it can be re-arranged to give:

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1} \boldsymbol{\varepsilon}$$

from which we can obtain an expression for the variance of $\mathbf{y}$ as:

$$\text{var}(\mathbf{y}) = E(\mathbf{y} \mathbf{y}^T) = (\mathbf{I} - \rho \mathbf{W})^{-1} E(\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}^T) (\mathbf{I} - \rho \mathbf{W}^T)^{-1}$$

hence

$$\text{var}(\mathbf{y}) = E(\mathbf{y} \mathbf{y}^T) = (\mathbf{I} - \rho \mathbf{W})^{-1} \mathbf{C} (\mathbf{I} - \rho \mathbf{W}^T)^{-1}$$

where $\mathbf{C}$ is the variance-covariance matrix.

This derivation has made no distributional assumptions regarding the response variable or the errors. In this model the spatial weights matrix, $\mathbf{W}$, is effectively raised to the power 1, so only first-order neighbors are included in the autoregressive function, and for this reason this model is therefore sometimes described as having a ‘first-order’ specification.

Although widely used, Wall (2004) has pointed out significant weaknesses in the interpretation of the spatial structure of commonly applied SAR and CAR spatial weighting schemes. These models were originally designed for use on infinite regular lattices, rather than finite irregular lattices (which have edges and variable numbers of neighbors per zone) and in this latter context are less well behaved. She recommends considering use of geostatistical models as an alternative or additional approach when analyzing lattice (zonal) datasets, since in this approach the covariance function is modeled directly. Of course, the use of geostatistical

methods also have their weaknesses in this context, notably the need to use zone centroids or similar arbitrary points rather than the zones themselves when constructing the experimental variogram (see further, Sections 6.7.1, [Core concepts in Geostatistics](#)).

If we add additional predictor variables, $\mathbf{x}$, to the pure spatially lagged autoregression model described earlier we have a *mixed regressive spatial autoregressive* model (*mrsa*):

$$\mathbf{y} = \mathbf{XB} + \rho\mathbf{W}\mathbf{y} + \boldsymbol{\varepsilon}$$

As can be seen this is the same as a standard linear regression model with the addition of the autoregressive component (the [SAM](#) software refers to this formulation as a ‘lagged response model’). As before we can re-arrange this expression to solve for $\mathbf{y}$:

$$\mathbf{y} = (\mathbf{I} - \rho\mathbf{W})^{-1}\mathbf{XB} + (\mathbf{I} - \rho\mathbf{W})^{-1}\boldsymbol{\varepsilon}$$

The design of this kind of mixed model incorporates spatial autocorrelation together with the influence of other (aspatial) predictor variables. The objective of this revised approach is to obtain a significant improvement over a standard OLS model. The level of improvement will depend on how well the revised model represents or explains the source data, and to an extent this will vary depending on the detailed form of the weighting matrix, $\mathbf{W}$.

Theoretical analyses have shown that this type of model can be derived from a variety of different processes, including direct processes such as spatial diffusion, certain forms of spatial interaction (including spillover and gravity or potential-type process models), and indirect processes such as resource distribution. This lack of a well-defined link between process and form is commonplace in spatial analysis, and is well-documented in fields such as point set clustering and fractal analysis. That is also applies here, in spatial regression modeling, should come as no surprise.

A second approach to SAR modeling is known as the spatial error model. This model is applied when there appears to be significant spatial autocorrelation, but tests for spatial lag effects do not suggest that inclusion of the latter would provide a significant improvement. A decision diagram for selecting the appropriate model based on a set of additional diagnostics (Lagrangian multiplier test statistics) is included in [GeoDa](#) (see the tutorial materials for a discussion of their use). The spatial error model (from [GeoDa](#)) is defined as:

$$\mathbf{y} = \mathbf{XB} + \boldsymbol{\varepsilon}, \text{ where } \boldsymbol{\varepsilon} = \lambda\mathbf{W}\boldsymbol{\varepsilon} + \mathbf{u}$$

Hence the basic model is as per the standard linear model, but now the error term is assumed to be made up of a spatially weighted error vector, $\lambda\mathbf{W}\boldsymbol{\varepsilon}$, and a vector of *iid* errors, $\mathbf{u}$. We can re-arrange the expression for $\boldsymbol{\varepsilon}$, above, to obtain:

$$\boldsymbol{\varepsilon} = (\mathbf{I} - \lambda\mathbf{W})^{-1}\mathbf{u}$$

hence the error variance-covariance matrix, $\mathbf{C}$, in this case is given by:

$$E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T) = \sigma^2 \left[(\mathbf{I} - \lambda \mathbf{W})^T (\mathbf{I} - \lambda \mathbf{W}) \right]^{-1}$$

The Georgia, USA educational attainment dataset used in Section 5.6.3, [Geographically Weighted Regression \(GWR\)](#) and in [Table 5-12](#) can be analyzed in a similar manner using spatial autoregressive methods. If this is conducted within [GeoDa](#) the OLS results match those within GWR (although the AIC values differ slightly owing to the differences in the detailed expressions applied). However, to apply a spatial autoregressive model a spatial weights matrix is required. In the following example we have set the spatial weights to be defined by simple first-order rook's move contiguity (adjacent edges), and then examined the [GeoDa](#) diagnostics to determine which form of regression model seems most appropriate to apply. In this instance the spatial error model was identified as the most appropriate and the regression re-run using this model. The results are summarized in Table 5-13, which is simply an extended version of the [Table 5-12](#), including the new autoregressive parameter estimates.

Although the RSS value in Table 5-13 is not as low as with GWR, the model is intrinsically far simpler and enables a more global view of the relationship between variables. There is an argument for utilizing both global regression and GWR approaches when analyzing datasets of this type, since they provide different perspectives on the data, and different insights into the use of such data for predictive purposes. The spatial error model applied in this example was defined above as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \text{ where } \boldsymbol{\varepsilon} = \lambda \mathbf{W}\boldsymbol{\varepsilon} + \mathbf{u}$$

Observing that:

$$\boldsymbol{\varepsilon} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$$

we have:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \lambda \mathbf{W}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \mathbf{u}, \text{ or}$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \lambda \mathbf{W}\mathbf{y} - \lambda \mathbf{W}\mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

Hence this expression models the dependent variable $\mathbf{y}$ as a combination of a general (global) linear trend component, $\mathbf{X}\boldsymbol{\beta}$, plus a pure spatial autocorrelation component, $\lambda \mathbf{W}\mathbf{y}$, plus a (negative) neighboring trend or predictor component, $\lambda \mathbf{W}\mathbf{X}\boldsymbol{\beta}$, plus a set of *iid* random errors, $\mathbf{u}$ (the [SAM](#) software refers to this formulation as a ‘lagged predictor model’)

Comparing this to the *mrsa* model above:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \rho \mathbf{W}\mathbf{y} + \boldsymbol{\varepsilon}$$

we see that the spatial error model can be viewed as a form of mixed spatial lag model with an additional spatial component, the neighboring trend, $\lambda \mathbf{W}\mathbf{X}\boldsymbol{\beta}$.

These types of model can be generalized still further (Haining, 2003, p355), for example as:

$$\mathbf{y} = \alpha \mathbf{1} + \mathbf{X}\mathbf{B} + \rho \mathbf{W}\mathbf{y} + \mathbf{W}\mathbf{X}\mathbf{\delta} + (\mathbf{I} - \rho \mathbf{W})^{-1} \mathbf{\epsilon}$$

where the scalars α , ρ , and ϕ , and the vectors $\mathbf{\beta}$ and $\mathbf{\delta}$ are all parameters to be estimated, and the final term represents spatial autocorrelation on the errors. Clearly one could proceed from the generalized model to the particular, or *vice versa*. Likewise one could progressively increase or decrease the set of explanatory variables in the model.

Given the considerable complexity of spatial phenomena, Haining suggests a data-driven approach to statistical modeling, which can be seen as fitting comfortably within the Data and Analysis components of the PPDAC framework described in Section 3.1, [Spatial analysis as a process](#), and [Figure 3-3](#). His approach commences with ESDA, proceeds to model specification for the current data, and then progresses to an iterative cycle of selection and implementation of parameter estimation, assessment of model fit and re-specification where necessary.

Conditional autoregressive and Bayesian modeling

A somewhat different conceptual model, which in practice may produce similar results, is known as conditional autoregressive modeling (CAR). The essential idea here is that the probability of values estimated at any given location are conditional on the level of neighboring values. The standard or ‘proper’ CAR model for the expectation of a specific observation, y_i , is of the form:

$$E(y_i | all y_{j \neq i}) = \mu_i + \rho \sum_{j \neq i} w_{ij} (y_j - \mu_j)$$

where μ_i is the expected value at i , and ρ is a spatial autocorrelation parameter that determines the size and nature (positive or negative) of the spatial neighborhood effect. The summation term in this expression is simply the weighted sum of the mean adjusted values at all other locations j — this may or may not be a reasonable assumption for a particular problem under consideration.

In the standard CAR model spatial weights are often computed using some form of distance decay function. The range of this function may be unbounded or set to a value beyond which the weights are taken as 0. This range might be determined from some *a priori* knowledge relating to the problem at hand, or perhaps estimated from a semivariogram or correlogram (see further, Section 6.7.1, [Core concepts in Geostatistics](#)).

In the CAR model the covariance matrix is of the form:

$$\text{var}(\mathbf{y}) = (\mathbf{I} - \rho \mathbf{W})^{-1} \mathbf{M}$$

and if the conditional variances of $\mathbf{y}$ are assumed constant this simplifies to:

$$\text{var}(\mathbf{y}) = (\mathbf{I} - \rho \mathbf{W})^{-1} \mathbf{I} \sigma^2$$

Requirements on the specification of the weighting matrix, $\mathbf{W}$, and conditional variance matrix, $\mathbf{M}$, include: (i) $\mathbf{M}$ is an n by n diagonal matrix with $m_{ii} > 0$; (ii) to ensure symmetry of the variance-covariance matrix $w_{ij}m_{ji} = w_{ji}m_{ij}$; and (iii) $0 < \rho < \rho_{\max}$ (typically) where $\rho_{\max}$ is determined from the largest eigenvalues of $\mathbf{M}^{-1/2} \mathbf{W} \mathbf{M}^{-1/2}$.

In the study by Lichstein *et al.* (2002), cited in Section 5.6.2, [Simple regression and trend surface modeling](#), they chose to use CAR following the recommendation of Cressie (1993) and because they felt it to be more appropriate for their study. They found no real difference in the results obtained with the CAR model from those achieved using SAR modeling of the type described in the previous section. Wall (2004) likewise found little difference between CAR and SAR models in her analysis of educational data across USA states.

A range of CAR models are supported by the GeoBUGS extension to the [WinBUGS](#) package. This software is specifically designed to support Bayesian rather than frequentist statistical modeling, and uses computationally intensive techniques (Markov Chain Monte Carlo or MCMC simulation with Gibbs sampling) to obtain the fitted parameter estimates and confidence intervals. An [ArcGIS](#) tool (Adjacency for WinBUGS) is available from the USGS to generate the spatial adjacency matrix required for [WinBUGS](#) CAR models. The application in this case is for modeling and mapping avian abundance, especially for migratory bird species whose conservation is of concern.

Haining (2003) discusses the use of such Bayesian models, in which additional (prior) information (for example, national or regional crime survey data) is used to strengthen the modeling process and reduce bias in local estimates. The Bayesian approach treats the unknown parameters (e.g. the vector $\boldsymbol{\beta}$) as a set of random variables, just like the data, to which may be associated prior distributions. The prior guesses for these parameters (possibly ‘borrowed’ from spatially adjacent and/or regional or national information) are then combined with the likelihood of the observed data to obtain posterior distributions for the parameters, from which inferential analysis proceeds. Essentially this provides a broader range of modeling approaches than pure (classical) frequentist analysis, and has been shown to result in substantial improvements over using simple rate data such as standard mortality ratios (SMRs); see for example, Yasui *et al.* (2000) for a fuller discussion of this question. These kinds of model have mostly been applied in epidemiological studies, for both mapping and modeling purposes, but have also been applied to other forms of spatial data (e.g. see Li *et al.*, 2007).

In the so-called *proper* CAR model ([WinBUGS](#) function *car.proper*) the variance-covariance matrix is positive definite. The example values given in the [WinBUGS](#) manual for $\mathbf{M}$ and $\mathbf{W}$ based on expected counts, E_i , are of the form:

$$m_{ii} = 1/E_i$$

$$w_{ij} = (E_j/E_i)^{1/2} \text{ for neighboring areas } i,j \text{ or}$$

$$w_{ij} = 0 \text{ otherwise}$$

This particular example relates to the Sudden Infant Death Syndrome (SIDS) data described in Section 4.3.3, [Ratios, indices, normalization, standardization and rate smoothing](#), and in

Cressie and Chan (1989) and more recently revisited by Berke (2004). Here the definition of *neighboring area* was not based on adjacency but on distance between county seats ($d < 30$ miles), a value determined from an examination of an experimental variogram (an estimate of a variogram based on sample data). The specific model applied in this case was actually of the form:

$$w_{ij} = \left[C(k) d_{ij}^{-k} \right] (E_j / E_i)^{1/2}$$

where the term in square brackets is a (Euclidean) distance decay function, with k selected as 0, 1 or 2, and $C(k)$ is a constant of proportionality to ensure results are easily compared across different values of k . In this study the authors chose $k=1$ as this provided the best results when considered from a likelihood perspective, hence their weights were of the form:

$$w_{ij} = (E_j / E_i)^{1/2} 1 / d_{ij}$$

Edge effects in this model are quite significant, since over a third of counties lie on the State boundary and clearly States do not represent closed systems for many (most) applications.

In this example the ‘proper’ (or autoGaussian) model fitted for this dataset was not applied to the full raw dataset, but to a Freeman-Tukey variance-stabilizing square root transform of the data (see [Table 1-3](#) and Freeman and Tukey, 1950) with Anson County omitted as an outlier. This county is the one picked up as an outlier in Figure 4-40, the “Excess risk rate map for SIDS data”.

Cressie and Chan had looked for non-spatial explanatory variables based on population density, percentage urban, number of hospital beds per 100,000 population, median family income and non-white live-birth rate. They then extended their analysis to include spatial patterns, but even after doing so could not adequately explain the observed variations in the data for this period, or for the subsequent 5 year period. It remains the case that the causes of SIDS are not fully understood, but medical research has shown that the placement of very young children on their back when sleeping, the use of pacifiers (dummies) and avoidance of overheating, all help to reduce the risks involved substantially. It is reasonable to suggest that the spatial variations observed and their changes over time might have been, in part, a reflection of cultural and social factors (such as advice given to mothers by local medical staff). These factors were not explicitly picked up by the non-spatial explanatory variables. Although such factors may be related to race-specific customs, it is likely that the spatial variations observed and modeled *may* have reflected variations in these advisory and behavioral factors. Certainly it would have warranted a very close examination of such factors in counties with unusually high and low death rates in each time period.

An *intrinsic* version of the CAR model (IAR or ICAR) is also supported, in which the variance-covariance matrix is not positive definite, but is semi-definite ([WinBUGS](#) functions *car.normal* and the robust variant *car.ll*). The intrinsic version (applied initially in an image processing context) is based on pairwise differences between the observed values (similar to the computations used in variogram analysis, from which it originates — see

Matheron (1973) and Künsch (1987) for a detailed mathematical treatment) and is now a more popular choice of CAR model for many researchers. Intrinsic models are a generalization of the standard conditional autoregressive models to support certain types of non-stationarity. The example values given in the [WinBUGS](#) manual for $\mathbf{M}$ and $\mathbf{W}$ for the intrinsic CAR model, based on Besag *et al.*, (1991) and Besag and Kooperberg (1995) are of the form:

$m_{ii}=1/n_i$ where n_i is the number of areas adjacent to i , and $w_{ij}=1$ for neighboring areas (i not equal to j) or $w_{ij}=0$ otherwise

The use of simple 1/0 weighting schemes for CAR models is not really appropriate for finite irregular lattices, and frequently a row-adjusted scheme of the form $\mathbf{W}^*=\{w_{ij}^*\}$ is used, where $w_{ij}^*=w_{ij}/w_{i+}$ (often written within this field as w_{ij}/w_{i+}). Hence the expected conditional means, for example, refer to an average rather than a summation. The symmetry requirement for CAR models cited earlier, i.e. $w_{ij}m_{ji}=w_{ji}m_{ij}$ implies that the conditional variances should be proportional to $1/w_{i+}$.

Having fitted the chosen model to the sample data, the residuals may be examined by mapping and/or by using the Moran I correlogram, $I(h)$, to identify any remaining patterns. If the residuals appear to show little or no spatial pattern it supports the view that the fitted model provides a good representation of the observed spatial patterns. However, as noted earlier, different models with fundamentally different interpretations may provide equally good fits to the data, hence drawing inferences from such models is difficult. Detailed examinations of the likely processes that apply for the particular dataset under consideration are vital for such analyses.

In the examples cited in this subsection, the response variable, y , has been assumed to be continuous. As with GWR, autoregressive models have been developed to handle discrete and binary data, for example autoLogistic and autoPoisson models — see Haining (2003, Chapters 9 and 10) for more details. Haining (2003, p.367 *et seq*) provides examples of the use of [WinBUGS](#) for Bayesian autoregressive modeling of burglaries in Sheffield, UK, by ward (Binomial logistic model) and children excluded from school (Poisson model). He includes sample code and data for these examples, together with maps of the results and provisional interpretations.

Griffith (2005) tested six alternative spatial regression models using data on cases of West Nile Virus (WNV) in the USA, by State (for mapping of cases see Figure 5-39 below and the US Disease Maps website).

The tests applied by Griffith included SAR, CAR and spatial filtering specifications (see further, Section 5.6.5, [Spatial filtering models](#)). He concluded that none of the models provided an ideal specification for the observed data, but a number of general lessons could be learnt to assist analysts in determining the best approach to use in disease map modeling:

- switching between alternative model specifications should yield similar intercept values – if markedly different values are obtained an analyst should be suspicious and ascertain why
- non-Normal data (such as the WNV cases) are best described with non-Normal probability models

- it appears from the tests on WNV data that [Spatial filtering models](#) can be used to explore the nature of spatial autocorrelation effects (positive and negative effects) more quickly and effectively than SAR and CAR models

Spatial filtering models

Conceptually the spatial lag models and spatial error models are substantially different, yet the results from their use on a given dataset may be similar. Furthermore, both are related to a process known as *spatial differencing* which is a form of *spatial filtering*. This is similar to simple differencing methods applied in time series analysis, but instead of analyzing values at each time slot, y_t , one examines the differences between sequential time slots (i.e. first differences), $y_t - y_{t-1}$. In the spatial context the expression:

$$\mathbf{y} = \mathbf{XB} + \boldsymbol{\varepsilon}$$

can be written with a first-order spatially autocorrelated difference as:

$$\mathbf{y} - \rho \mathbf{W}\mathbf{y} = \mathbf{XB} - \rho \mathbf{W}\mathbf{XB} + \boldsymbol{\varepsilon}, \text{ or}$$

$$\mathbf{y}(\mathbf{I} - \rho \mathbf{W}) = (\mathbf{I} - \rho \mathbf{W})\mathbf{XB} + \boldsymbol{\varepsilon}, \text{ hence}$$

$$\mathbf{y} = \mathbf{XB} + (\mathbf{I} - \rho \mathbf{W})^{-1} \boldsymbol{\varepsilon}$$

Thus a model with a spatially autoregressive error term is a form of standard regression equation (OLS) on spatially filtered data. The term:

$$(\mathbf{I} - \rho \mathbf{W})^{-1}$$

is the spatial filter in this instance. Similarly the mixed spatial lag model, *mrsa*, described earlier:

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{XB} + \boldsymbol{\varepsilon}$$

can be written with a first-order spatially autocorrelated difference as:

$$\mathbf{y} - \rho \mathbf{W}\mathbf{y} = \mathbf{XB} + \boldsymbol{\varepsilon}, \text{ or}$$

$$\mathbf{y}(\mathbf{I} - \rho \mathbf{W}) = \mathbf{XB} + \boldsymbol{\varepsilon}$$

Hence, once again the model can be seen as a form of spatial filter where in this case only the dependent variable is filtered.

A somewhat different approach has been pioneered by Getis and Griffith (2002). They apply spatial filtering to potential independent variables in a regression equation using one of two procedures: (i) using the G_i statistic described in Section 5.5.3, [Local indicators of spatial](#)

[association \(LISA\)](#), to transform the source data (Getis method); or (ii) decomposition of the Moran I statistic into orthogonal and uncorrelated map pattern components (Griffith). The approach of Getis is the most straightforward to describe and apply, since it involves a simple transformation of the original data values.

Let $G_i(d)$ be the local Getis and Ord spatial autocorrelation statistic applying to observation (point) i :

where d is the estimated range of observed spatial autocorrelation. This expression is the *observed* value of the statistic for location i . The *expected* value of the statistic for location i is of the form $W_i/(n-1)$. The filtering transformation of the observed data values x_i applied is then.

where W_i is the sum of the binary weights, w_{ij} , within a distance d of i , and n is the number of observations. The original dataset can then be viewed as being comprised of a spatial component, $x_{sp}=\{x_i-x_i^*\}$, and a non-spatial (filtered) component, $x_f=\{x_i^*\}$. These two components are then used in standard linear OLS as dependent variables and analysis proceeds as described earlier. Hence for each filtered dependent variable there will be two additional terms in the regression equation.

Getis and Griffith (2002) applied unfiltered OLS, standard spatial autocorrelation, Getis filtered OLS and Griffith filtered OLS to sample US state expenditure data (observation taken as applying to state centroids). For OLS, standard spatial autocorrelation and the Getis filtering the coefficient of determination was found to be 75%, whilst with the Griffith method a slightly higher value of 80% was obtained. The authors conclude that both filtering methods provide a more appropriate specification of the regression model, facilitating meaningful interpretation of all of the regression coefficients, with the advantage of not requiring formal specification of an appropriate spatial autocorrelation model. Overall the Griffith approach appears somewhat more powerful, but the Getis approach is simpler to apply, although it does require that the filtered variables are positive and have a natural origin (i.e. are positive ratio-scale variables).

Big Data, AI and Geospatial Analysis

Perhaps one of the mostly hotly debated topics in recent years has been the question of "GIS and Big Data". Much of the discussion has been about the data: huge volumes of 2D and 3D spatial data and spatio-temporal data (4D) are now being collected and stored; so how they can be accessed? and how can we map and interpret massive datasets in an effective manner? Less attention has been paid to questions regarding the analysis of Big Data, although this has risen up the agenda in recent times. Examples include the use of density analysis to represent map request events, with Esri demonstrating that (given sufficient resources) they can process and analyze large numbers of datapoint events using kernel density techniques within a very short timeframe; data filtering (to extract subsets of data that are of particular interest); and data mining (broader than simple filtering). Increasingly a combination of massive-scale spatial databases, cloud computing resources, and AI-assisted geographic data analysis (GeoAI), is enabling organizations to interpret and map and utilize critical spatial data on a scale not previously thought possible. Services from [Microsoft](#), [IBM](#), [Esri](#), [Google](#), [Ecopia](#) and many others are driving these developments, with the inclusion of AI-driven analytical tools to facilitate automated processing (e.g. see the [Esri GeoAI Toolbox](#)) and autonomous transport

systems. However, these new services and automated smart tools raise new issues of confidence, reliability, interpretation and auditability.

Several organizations provide notable GeoAI services and applications across different domains. IBM and NASA collaborate on projects that help estimate climate-related risks to crops, buildings, and other infrastructure; assess and monitor forests for carbon-offset programs; and develop predictive models that support enterprise strategies for climate-change mitigation and adaptation, as described at <https://research.ibm.com/blog/geospatial-models-nasa-ai>. Esri applies GeoAI in urban planning and management through the use of Digital Twins; for example, during a major blizzard in January 2024, the city of Vilnius cleared most streets within four hours using a combination of AI, location technology, and aerial imagery, detailed at <https://www.esri.com/about/newsroom/blog/vilnius-uses-drones-ai-to-monitor-city-services/>. Ecopia uses AI to digitize geospatial imagery and rapidly generate high-definition vector maps at a level of accuracy comparable to a trained GIS professional; they have produced 3D land cover data for most U.S. cities and are working, alongside other companies, toward creating a digital twin of the entire Earth, with more information at <https://www.ecopiatech.com/data>. Google Earth Engine and Global Forest Watch contribute to global forestry monitoring, with Global Forest Watch offering an online platform that provides near real-time data and tools to track forest change worldwide, accessible at <https://earthengine.google.com/> and <https://www.globalforestwatch.org/>.

For real-time data, sequential analysis has also been successfully applied; in this case the data are received as a stream and are used to build up a dynamic map or to cumulatively generate statistical values that may be mapped and/or used to trigger events or alarms. To this extent the analysis is similar to that conducted on smaller datasets, but with data and processing architectures that are specifically designed to cope with the data volumes involved and with a focus on data exploration as a key mechanism for discovery. Special software architectures have been developed to handle these very large datasets, including facilities for statistical analysis. Amongst these are the [Apache Hadoop](#), [Storm](#) and [Accumulo](#) projects and associated Spark, Scala and Breeze software, and visualization and data discovery tools such as Tableau, Gephi and Qlik.

Early in the development of these novel systems and services, Miller and Goodchild (2014) argued that considerable care is required when working with Big Data — significant issues arise in terms of the data (the four Vs): the sheer Volume of data; Velocity of data arrival and associated timestamps of the data; the Variety of data available and the way in which this is selected (e.g. self selection); and the Validity of such data. A presentation by Prof Mike Goodchild of some of the key elements of the Big Data debate are provided in the Resources page of the website and can be accessed directly at: <https://www.spatialanalysisonline.com/PPTS/BigData.pdf> — this presentation should be viewed alongside the article by [Miller and Goodchild](#), as the latter provides a fuller explanation of the main ideas covered. Chapter 17 of [Longley et al.](#) (2015) is also recommended reading as this addresses the broader questions regarding information gathering and decision-making.

In an article at the time entitled "[Big Data: Are we making a big mistake?](#)", journalist, economist and statistician [Tim Harford](#) also addressed these issues and more, highlighting some of the less obvious issues posed by Big Data. Perhaps primary amongst these is the bias that characterize many such datasets. Such biases may be subtle and difficult to identify and

impossible to manage. For example, almost all Internet-related Big Data is intrinsically biased in favor of those who have access to and utilize the Internet most, with associated demographic and geographic bias built-in. The same applies for specific services, such as Google, Twitter, Facebook, TikTok, mobile phone networks, opt-in online surveys, opt-in emails — the examples are many and varied, but the problems are much the same as those familiar to statisticians for over a century. Big Data does not imply good data or unbiased data; moreover Big Data presents other problems — it is all too easy to focus on the data exploration and pattern discovery, identifying correlations that may well be spurious as a result of the sheer volume of data and the number of events and variables measured. With enough data and enough comparisons, statistically significant findings are inevitable, but that does not necessarily provide any real insight, understanding, or identification of causal relationships. There is also the added issue of pre-processing of data by automated systems that may amend records and even create records where gaps may have been detected, using advanced machine learning (ML) or GeoAI software, in ways that are unclear and often undetectable. Of course there are many important and interesting datasets where the collection and storage is far more systematic, less subject to bias, recording variables in a direct manner, with 'complete' and 'clean' records. Such [Research-ready data \(RRD\)](#) are curated and managed well and tend to be those collected by agencies and research organizations who supplement the data with metadata and quality assurance information (see also, the section discussing [RRD](#) in Chapter 3).

An important part of the geospatial analysis research agenda is to devise methods of triangulating conventional 'framework' data sources — such as censuses, topographic databases or national address lists — with Big Data sources in order that the source and operation of bias might be identified prior to analysis and, better still, accommodated if at all possible. Yet such procedures to identify the veracity of Big Data are often confounded by their 24/7 velocity and scale, and potential undocumented modifications to the source data. For instance, social media data are created throughout the day at workplaces, residences and leisure destinations, and so it is usually implausible to seek to reconcile them to the size and compositions of night time residential populations as recorded in censuses. As Harford concludes: "Big Data has arrived, but big insights have not. The challenge now is to solve new problems and gain new answers — without making the same old statistical mistakes on a grander scale than ever."

Big Data Research

The academic community has been enticed by the potential to harness Big Data to predict real-world phenomena. It appears that no longer is research restricted by an inability to efficiently collect and process large quantities of data. In many settings Big Data offers comparable statistics to those traditionally collected by governments as officially released datasets, whilst in other settings they offer insights into phenomena that were not previously recorded. Efforts to gain access to new forms of data for research purposes are also motivated by an increasingly acknowledged necessity to extend the breadth of data sources available for research, since traditional forms of data are costly to collect and, in many cases, suffer from decreasing response rates (Pearson, 2015). In many contexts, working with Big Data is a necessity. For population data, in particular, the traditional gold-standard sources of data are in decline around the world. Most developed countries are cutting back on their long-form censuses or replacing them altogether with administrative data (see Shearmur, 2010). In very important respects, Big Data can offer far greater spatial granularity, attribute detail and frequency of refresh. But these

compelling advantages of greater content accrue without any guarantees that the coverage of the entire population of interest is known — in many Big Data applications reported in the literature, the population of interest may not even be defined! The analyst community may need to become increasingly reconciled to use of new data sources, and may certainly benefit from their richness; however, data analysts must also undertake increasing amounts of due diligence in order to establish the provenance of such sources.

There is a growing amount of interest in harnessing Big Data beyond their primary applications since they offer the potential to provide fresh insights into a wide range of phenomena across space and time. Furthermore, they offer large volumes of data and may have a good coverage of particular populations or spaces. In many settings, the insights that Big Data may be able to offer could not be generated by traditional data sources without being very costly. See, for example, Canzian and Musolesi's work on mental health using mobility data generated from mobile phones (2015), Shelton *et al.*'s study of religion using the geoweb (2012), and more recently, Dong *et al.* (2024) "Defining a city — delineating urban areas using cell-phone data". Big Data can present an opportunity to break away from a reliance on data of low temporal frequency to new forms of data that are generated continuously with far more precise temporal and spatial attributes.

Many datasets are generated in real-time by citizens using handheld technologies. Resultant data sources include mobile phone call records, WiFi usage and georeferenced social media posts. The velocity of their data generation allows very large volumes of geospatial information to be uploaded every day — this is a crucial step away from the slow production of many forms of geographic data and the delay between data collection and dissemination. Furthermore, the unrestricted spatial and temporal nature of data collection allows social research to be unshackled from an exclusive fixation on residential level data. For instance, Figure 9-1 shows the spatial distribution of Tweeters who submitted data between 8 am and 9 am on weekdays in 2013 in Central London.

It can be observed that a large proportion of these users were using transport routes, many of which could have been commuting to work and therefore could be indicative of the spatiotemporal distribution of the population at large to a certain extent. Such information could not be acquired for large numbers of individuals from traditional data sources. But the characteristics and views of journeying population segments may not be representative of sedentary ones, and some transport modes are more conducive to social media use than others.

Representation is a fundamental part of scientific knowledge discovery and is crucial for advancing our understanding of the world. All data should be thought of as a partial and selective representation of the totality of real-world occurrences and interactions, and the basis to selection should be understood. Data on places, people or activities are usually compressed into a set of digitally recorded variables in order to be stored efficiently. Therefore, representation is constrained by the scale and scope of each dataset, and the extent to which it depicts the totality of all real-world phenomena of interest. Most of what we know about the population and their activities has historically been dependent on generalizations and descriptions derived from very limited data. The skill of the scientist has been in devising sample designs that may found representations upon very sparse samples that nevertheless facilitate robust and defensible generalizations.

In some fields of geography we are used to being data-rich. For instance, the Landsat programme was producing remotely sensed images in the early 1970s at such a velocity that the data were extremely challenging to handle. With improvements in technology, the supply of data on the physical environment has increased and improved in precision. However, data on people and their activities has historically been scarce. Now, many big datasets are generated from human actions, directly or indirectly, and have countless applications for understanding the real-world. Yet, new forms of Big Data are fundamentally distinct from traditional datasets. Often, in human focused applications, their primary objectives are to make recordings of events and characteristics to assist administrative functions and there is little consideration of coverage as this is extremely costly. While on the one hand this means data are captured about what people actually do, rather than what they say they do in surveys, these actions will influence both who is represented in the data and how trends are manifested. Academics and others have little or no influence over the data collection process and many of the administrative processes of datasets remain hidden.

The diffusion of data collection into everyday activities has drastically enhanced the scope of data analysis. Firstly, in many cases, Big Data may represent a near-total coverage of specific populations, activities and places. Secondly, Big Data has enabled spatial analysis to shift large-scale analysis from aggregate data to individuals. For example, travel card data generated in most sizeable cities in the developed world represents the vast majority of journeys that occur on public transport across the city, and each journey is recorded individually with origin, destination, and times appended to the recording. Moreover, some datasets also record the precise locations for individual records through GPS technologies. Individual-level data enables analysis to separate itself from issues associated with small area aggregations (e.g. the ecological fallacy and the modifiable areal unit problem discussed in section 4.2). Furthermore, often individual records can be linked to unique identifiers (such as accounts, neighbourhoods, etc.) enabling the conflation of large volumes of longitudinal data. It has also been possible to harness new technologies to track anonymized individuals across time and space in order to improve understanding of urban systems and structures. For example, mobile phone app data can be used to estimate where and when a customer boards and departs buses in order to improve understandings of public transport use (Stockdale *et al.*, 2015), or to identify the meaningful structure of modern cities (Figure 9-2, Dong *et al.*, 2024):

These characteristics have led Big Data to change how we discover information from data. Previously, much geographic research and analysis was grounded in theory partly due to the poor availability of data (by today's standards). Indeed many theories of spatial analysis are generalizations built on assumptions in order to estimate trends for situations where there are no or limited data. For example, spatial interaction models have been widely used to estimate movements in a broad range of applications. In the context of retail geography, their use is primarily to estimate the likelihood of persons from each spatial unit visiting each of their local retail destinations. Essentially, the choice to visit a particular destination is based on convenience (inverse distance) and attractiveness (sometimes simply the retail floor space). However, store loyalty data can tell us the addresses of very large samples of patrons, as well as the times and dates when they visit stores, and when and whether they order products and services for home delivery. While one of the main criticisms of positivistic research is that models assume all individuals act rationally, through discrete data on populations researchers can also represent those that seemingly act irrationally. Indeed many actions will not conform

to the generalizations that models are built from. Thus, large volumes of data have enabled us to mine trends and create predictive models with high levels of success. Likewise, data-driven AI forecasting is becoming much faster and more accurate than traditional physics-based models in complex fields like weather forecasting and fluid-flow modeling (see further, the [NAS AI 2024 Symposium](#) presentations, and the Google Deep Minds [Gencast](#) and [Graphcast](#) projects).

As will be described later in this section, Big Data are not devoid of uncertainty especially when they are used to represent the real-world. Therefore, empiricist approaches may succumb to fallacies when the data collection procedures are not thoroughly understood. By letting data speak for themselves researchers neglect key influential geographic concepts pioneered by Geographic Information Science. Instead, data-driven approaches are limited to the identification and description of events and processes, rather than understanding why they exist. Theory is still needed to support critical research as Big Data remain only partially representative and research therefore may be founded upon data that are rich and detailed, but nevertheless not fit for purpose. It is important to think of most Big Data sources as a by-product, or ‘exhaust’ from a process that does not have re-use of data for research purposes at its heart. For example, a store loyalty card profile is a by-product of one or more transactions entered into by a consumer: it records the effectiveness of a company in delivering a service and how it makes a profit. It does not, however, record the entirety of that individual’s consumption, and provides only indicators of a broader lifestyle that requires supplementary data and/or validation from other sources for more broad-based research. As such, there should be no over-reach of the data beyond the phenomena that the data may reasonably be purported to represent. This was demonstrated by the infamous over-predictions of Google Flu Trends - see for example the article by Declan Butler (2013) in Nature News.

Types of Big Data

In this section we briefly discuss Big Data sourced from human interactions (e.g. social media platforms), process-mediated data, and finally machine generated data (e.g. from automated event and activity tracking).

Human sourced data

Human-sourced data is integral to a wide range of services, some of which enable the dissemination of Open Data. For example, Internet enabled handheld devices have supported the growth of various Volunteered Geographic Information (VGI) services. A popular example is OpenStreetMap (OSM), which is a widely used crowd-sourced map of the world and geodata portal (www.openstreetmap.org). Like other information sharing services (such as Wikipedia), the service uses a series of social hierarchies of moderators in order to maintain the quality of its information and sieve out redundant records (Goodchild, 2013). An advantage of VGI over officially produced counterparts is transparency and the velocity of data production. It is easy for lay users to update data and report misinformation, a process that is otherwise cumbersome with governments and businesses that often lay out lengthy internal administrative procedures. However, there are, of course, data quality issues that may emerge due to potential weaknesses in vetting the large quantities of data that are produced daily.

One very prevalent Big Data source is social media. Social media have become an integral component of modern societies. It is estimated that just almost two-thirds of the world’s

population are currently social media users (Statista, 2024). There are several different types of social media including social network services, video sharing services, messaging services and information sharing services, most of which are accessed via online platforms. Consequently, social media have also become a popular source of data on the human condition for the academic community, largely due to the volume and velocity of the data and the rich social information they provide. Furthermore, many social network services such as Twitter/X and Flickr enable users to include coordinates from their mobile devices when uploading content, thereby enabling the study of geospatial phenomena.

As users engage with the online services through handheld devices, so they leave geographic and temporal footprints of their activity in the real-world within social media data (Blanford *et al.*, 2015). It is, therefore, possible that large quantities of social media data can be highly informative of general human activity and, thus, they have many potentially useful applications in geospatial analysis. Taking Twitter/X data as an example, many studies have sought to use the geosocial data to predict real-world trends, thus treating the users as sensors (Haklay, 2013). This has included estimating the spread of influenza (Lamb *et al.*, 2013), predicting customer catchments for retail centers (Lloyd and Cheshire, 2017), and tracking natural hazards (Guan and Chen, 2014). The data can also be re-purposed as indicators of urban activity by using new techniques in text mining to categorize Tweets and identify routine spatial and temporal patterns. For instance, Figure 9-3 displays the relative density of Tweets about education in inner London: here, most areas of high concentration coincide with the locations of university campuses. Although social media data are curated by companies, they are distinct from most other commercial datasets in that the bulk of the data are fashioned by members of the public. The data typically record digitized media and can take many forms. Social network data are unlike other forms of Big Data as there is little control or regulation over what is produced; thus it is inherently difficult to harvest objective research (see Tinati *et al.*, 2014).

Process-Mediated data

Process-mediated data describes the classes of data that are primarily generated through the transactions of businesses and governments. Such data is said to be “found” rather than “made”, as they were typically generated to support commercial or administrative functions rather than to represent some aspects of the real world in research (Connelly *et al.*, 2016). Through new technologies, businesses and institutions now capture data at a much higher velocities. Some example data sources are described below.

Government Administrative Data

Administrative data are routinely collected by public institutions in the form of registrations, transactions and records. These include data generated by welfare, taxation, licensing and electoral registration, as well as records maintained by healthcare and education institutions. Most administrative data sources aim to acquire complete coverage of specific populations; and in some cases, the data are collected from transactions which are legal requirements, making them more suitable for geodemographic representations. For instance, governments usually retain records on 100% of vehicles and legally employed workers for taxation purposes.

Administrative data systems have historically been integral to the development of spatial data. It was for administrative purposes that the UK postcode and US ZIP code systems were built, and through them, Census geographies were carefully designed to assist dissemination at a fine

level without breaching guidelines on disclosure. Due to common interests, much of the data collected by governments correspond with social surveys historically collected by statistics authorities. Consequently, on both sides of the Atlantic, governments are becoming increasingly interested in using administrative data to bolster census statistics. Administrative data has the advantage of being routinely captured and regularly updated, enabling large-scale longitudinal analysis of high quality.

In addition, the data can contribute new information that was not previously available in the public domain. For example, despite considerable interest in social class and wealth from social scientists and users of geodemographic classifications, there are still no granular data on income in the UK. Previous research has found value in harnessing government data to investigate life-chances and criminality (Britton *et al.*, 2015). In addition, while previous geographic research on deprivation has included car ownership as a proxy for social standing due to its collection in censuses (for example the Townsend Index of Deprivation, Townsend *et al.*, 1988), research on car registration data from the UK's Driver and Vehicle Licensing Agency found car characteristics (such as age and vehicle model) to be strongly associated with socio-economics, especially in large cities. Figure 9-4 demonstrates that there is a distinctive geography to the registered locations of new cars.

Of course, the extent and quality of administrative data collection varies around the world. Some countries maintain detailed population registers in order to maintain better data on citizens. Such registers enable different government institutions to link their data in order to improve service delivery. Unfortunately, in most countries, administrative data are collected by multiple different organizations making it difficult to link records, although there have been efforts to overcome this using multivariate linkage approaches.

Consumer Data

An increasing share of data held on people and their actions is generated by commercial organizations in order to assist their activities. These include data created by retailers, utility providers, transport providers and banking and financial services. Typically, they require transactions of some kind to generate data although they often also retain account information. For example, retailers record the times and the locations of transactions. Some large retailers and almost all online retailers also maintain customer accounts. Sometimes, specially devised loyalty programmes (such as supermarket loyalty cards) have been implemented specifically to understand trends in consumer behavior and to target promotions. This enables retailers to retain valuable longitudinal data often including the location of shopping trips and customer residences. Loyalty databases and website shopping accounts empower retailers to focus their data collection and promotions in order to target individual customers.

A popular anecdote about the predictive capabilities of retail data describes how Target (an American discount store) predicted a teenage customer in their loyalty database was pregnant (Duhigg, 2012). Target used their databases to personalize the coupons and advertisements they mail to their customers. In this case, a father complained to the store that his daughter had been sent promotions for baby products and maternity clothing. His concern was that the retailer was implementing marketing techniques that might encourage teenagers to have children. However, a few days later the father contacted Target again to apologize — he had discovered that his daughter had been pregnant for some time. It is not uncommon for private companies to use empiricist methodologies in order to automate tailored service provision at the individual level.

In this case, the high dimensionality and granularity of the data made it feasible to make associations at the individual level.

Some large retailers may achieve a very large coverage of the population. For example, Tesco's Clubcard in the UK had achieved a coverage of 15 million members in 2015 (over 30% of the adult population). However, engagement with the programme will vary considerably as many users may become disengaged with using loyalty schemes over time or may routinely avoid using cards when making small purchases. Yet, it is obvious, that this data will have merits over traditional data sources that will struggle to acquire detailed information on consumption for large shares of the population. Retailers are now able to grasp data on the consumer behavior of groups who are traditionally more averse to participating in market research programmes (Manyika *et al.*, 2011). Retail data, therefore, can be of considerable value to geospatial population research. The core use of geodemographic classifications in industry is to predict lifestyles across space, particularly consumer behavior (Harris *et al.*, 2005). Consumption habits may have broader connotations for understanding segregation in lifestyle choices. Indeed, equivalent data from traditional sources suffer from common limitations of costs, respondent errors, sample biases, and perhaps most importantly, small sample sizes.

Machine- Generated Data

Machines and sensors are routinely used to capture events, measure consumption or record the physical world. The Internet, and technological innovation more generally, has enabled sensors to capture a wide variety of data in a non-disruptive way. Examples include fixed sensors which are in place for security or monitoring purposes, mobile sensors which use GPS technology to capture spatial and temporal trends, and satellite-based remote sensing. One of the reasons that we are able to gather such vast quantities of data at a high velocity is that we have decreased the role of humans in data generation. It is now easy to generate accurate and precise spatial information from satellite and mobile devices (including such technologies as portable GPS devices, aerial LIDAR surveys and the use of drones). In the past georeferencing was expensive, time consuming and far more complicated to implement - this has all changed over the last 25 years. Together, machines generate several different types of new forms of data that can be used to assist decision making in real-time. For instance, "Smart Cities" amalgamate real-time data from a wide range of sensors including CCTV, weather monitors and traffic measuring sensors, in addition to other types of Big Data (such as social network data).

An example of fixed sensors are machines which are used to monitor energy usage. In the UK energy companies have moved towards the installation of smart meters for every property. These devices record energy consumption in real-time enabling the collection of energy trends for addresses at a very high temporal frequency. While the primary objective of the data is to improve the efficiency of energy usage and to provide greater transparency of billing, they can also reveal interesting geographic trends. For instance, Dugmore (2010) suggests that as occupied households use utilities daily, unoccupied dwellings could be identified by pooling utilities companies' data. The high temporal resolution of the data enables us to segment households based on energy usage throughout the day. Research has found smart meter data to be an effective predictor of household characteristics and a viable alternative to the Census for some measures. For instance, Samson (2014) clustered the standardized average daily energy profiles of smart meters from deprived postcodes. The research identified that most energy

profiles fit one of four trends (Figure 9-5). Each of the trends were found to be associated with households of different life stages.

There are also sensors that are unfettered from physical locations. Satellite imagery has historically been a common source of geographic data and there has often been more data than researchers can handle. However, there are now new forms of data that have benefited from GPS technologies. One such example is fitness apps that use mobile devices to record routes taken during exercises (Figure 9-6). These types of apps (and indeed many others) record detailed tracks of sensor movements and can be informative of real-world travel activities. Typically, the apps record locations at very regular intervals so that journeys can be mapped and attributes such as elevation and speed can be appended. The proliferation of smart mobile phones and other GPS-enabled hand-held devices has vastly increased the supply of geospatial data on the population. It is not uncommon for such data to be re-purposed to estimate traffic congestion and the popularity of restaurants and stores.

Challenges of Big Data

Unfortunately, despite the enthusiasm for new forms of Big Data and the more recent, AI-revolution, there has been relatively limited critical research into their validity for geospatial research on real-world phenomena. This is foremost because the provenance of big datasets is often unknown and it may be difficult to link records to alternative databases. In addition, the unstructured and unregulated nature of some datasets has deterred some from using them at all. Whilst Big Data are useful for countless applications, it is important to be cognizant of their limitations. This section describes the core fundamental challenges to harnessing Big Data to form geospatial representations. It is important to consider that these challenges may not apply to each data source to the same extent. Each big dataset will have its own merits and deficiencies that researchers should seek to explore.

Access

Data are unlike other assets in that they are difficult to collect but easy to copy and disseminate. However, only a very small proportion of big datasets are released in the public domain due to their strategic importance, commercial value and potentially disclosive attributes. Breaches of privacy can have serious adverse consequences on the public opinion of businesses and governments, so there are considerable efforts to safeguard data. The events surrounding the misuse of Facebook profile data by an academic and Cambridge Analytica only serve to highlight how fraught with problems issues of access and privacy can be. This inevitably has had a detrimental impact on Big Data research as commercial organizations that produce geospatial Big Data products often value efficiency and customer satisfaction over other interests. Companies are also less likely to share their methodologies and results thus hampering the advancement of research, methodologies and theoretical insights.

Where data access is granted, researchers and analysts are usually at the mercy of their data providers. It is not uncommon for elements of the data collection or sampling process to be kept private due to commercial sensitivities. Many companies that opt to share samples of their data disseminate them via Application Programming Interfaces (API). However, when using such channels the coverage of the data are usually uncertain (Boyd and Crawford, 2012). Taking the example of Twitter/X, little is known about how the free data feeds are sampled and feeds often drop in velocity for short intervals of time. Furthermore, data owners may set

conditions on data access and use, and sometimes these limitations may be contrary to the interests of researchers.

For these reasons there needs to be greater support at all levels for data sharing and safe data practices in order to facilitate research and the maximize the full potential of Big Data. In 2008, the OECD launched a call in support of open data aimed at national governments. In response, some governments have launched open data initiatives to disseminate administrative data which may be of value to research and the general public (for instance data.gov in the USA, data.gov.uk in the UK and aurin.org.au in Australia). Example datasets that have been made open source in the UK include Land Registry Price Paid data (the location and value of almost every domestic property sale) and Domestic Energy Performance Certificate (the energy performance certificates for over 15 million properties).

Commercial organizations are generally under less pressures to share their data. Therefore, with or without legislative intervention, institutions are required to provide safe environments through which commercial data can be accessed by researchers without violating data protection protocols. One such institution, is the UK [Consumer Data Research Centre](https://www.consumerdataresearchcentre.org) (CDRC, based at University College London) which specializes in large consumer datasets managing data from a wide range of providers.

Ethics

Whilst concerns about the use of Big Data have been raised by many parties, ethics are frequently overlooked by practitioners who have been enticed by new and innovative forms of insight (Puschmann and Burgess, 2014). The majority of Big Data production and analysis currently occur within the private and government sectors. Given that data and algorithms are commercial assets, their methodologies are often trade or national secrets and thus not subject to independent scrutiny and critique. Data on people are extremely sensitive, both to the people they relate to and to the organizations that own the data. In 1995, an edited collection of papers entitled "Ground Truth" was published to bring attention to the social implications of geographic information systems and geographic databases of people (Pickles, 1995). All of the chapters in this seminal collection express concern, to differing degrees, over the impacts of GIS, some of which focused on the negative connotations geographic research can have on society. However, the ethical concerns have in fact been exacerbated in recent years owing to the proliferation of data which are now analyzed at the individual level and the often opaque nature of automated data analysis systems, including GeoAI software, services and datasets.

The collection of personal data is integral to the personalization of services (as demonstrated by the Target loyalty programme described earlier). However, the scope and use of Big Data can easily result in "dataveillance", a process through which individuals' details are routinely collected and stored creating detailed databases about people and their behavior (Dodge and Kitchin, 2007). It is therefore often possible to identify individuals and their behaviors from pooling different sources of Big Data. In addition, some data present very detailed footprints of user locations making it possible for analysts to distinguish personal characteristics about individuals (including their place of residence and other personal details). Therefore, when data outputs are publicly disseminated, efforts are undertaken to ensure that data are not disclosive of individuals. Usually, this entails aggregating the data into predetermined spatial units and/or anonymizing records. However, this means that in many cases, researchers cannot take full advantage of the fine granularity of Big Data and must succumb to the challenges of ecological

fallacy. It is therefore challenging to secure privacy in some datasets whilst retaining their quality and precision.

Data protection is becoming a more contentious issue and this is also being reflected by new legislation around the world such as the EU General Data Protection Regulation (GDPR). Once access to data has been granted, the researcher needs to ensure they are compliant with data protection procedures and also have a clear ethical steer. A useful resource is the British Academy and Royal Society's 2017 report entitled "[Data management and use: Governance in the 21st Century](#)" (Royal Society, 2017), which summarizes the ethical issues that arise from data in today's society and how the issues might be mitigated.

Data Quality

Data quality issues are paramount to big datasets and are one of the major criticisms that have deterred many academics. Errors can arise from all parts of the data generation and amalgamation processes, from measurement error to adjustment errors (see the total error paradigm in Groves *et al.*, 2011). These errors can then contaminate other data when records are linked and are inherently difficult to identify. Furthermore, the volume, velocity and commercialization of datasets prevent any efficient means of validating records.

Small sets of data are typically created using procedures that are transparent and robust. However, most forms of Big Data are not subject to standard quality controls and they come from numerous different sources of varying quality and reliability. That said, most datasets never enter the public realm so there is little knowledge about how rigorous the collection and data cleaning procedures are. Within the commercial sector companies may focus on quantity over quality and emphasis is placed on efficiency in order to maximize profits. It is, therefore, unsurprising that Big Data may contain an unknown proportion of noise generated by errors and inaccuracies.

While most forms of Big Data are generated by machines that have reduced the elements of human error (especially with regards to temporal and georeferencing components), there are cases where machines can be inaccurate. For example, GPS coordinates are known to be less accurate in so-called urban canyons. Yet, as seen above, there is still a large share of Big Data that are human-sourced. It is not uncommon for errors (accidentally or deliberately) to occur in VGI. For example, the quality of OSM data has been found to vary due to the skills and interests of contributors (Haklay, 2010). Furthermore, business and administrative data could also include errors that arise due to mistakes during processing. For example, the falsification of administrative records could lead to changes in policy and allocations of funding (Connelly et al, 2016).

Data quality also often deteriorates over-time as places and events are not static. For example, administrative and consumer databases regularly retain the incorrect residential addresses of adults for individuals who have moved. It is, therefore, important that each record is timestamped to improve the transparency of databases. It is also useful to synthesize data from multiple sources to reinforce confidence in observations and measurements and to identify outliers.

Repurposing Data

Often Big Data are used in research despite the fact that they may have been collected for completely separate reasons — this is particularly challenging when attempting to predict human actions. For example, research has found social media data to be of very variable utility for predicting election outcomes. Whilst pollsters can ask the all-important question about voting intentions, researchers attempting to glean political predictions from social media must acquire a plethora of messages and then attempt to contextualize them. Big Data fundamentally represent phenomena on the population and their behaviors that is distinct from traditional social science datasets. There are advantages in this, for instance surveys are prone to misreporting (e.g. because of respondent recall errors) whereas in many instances Big Data are accurate due to the automation of data collection of observed behavior. However, this is not to say that all data must, therefore, be valid and free of constraints. Big Data are not neutral and may be prone to misrepresentations in the context of geospatial research.

Whilst it is reasonable to assume that Big Data are reliable proxies for real-world activities or conditions, in making these assumptions it is important that we take account of the possible implications that the means of data collection may have on representations. Activities which generate data may be constrained by time and geography, they may also be unevenly distributed across the population (as shall be discussed later in this section). In addition, people may engage in the activities for different reasons and it is not possible to obtain information on their motivations. For instance, social media contain an unregulated and unknown proportion of individuals' beliefs, opinions and activities, together with much less reliable and deliberately misleading information. In addition, some data are generated by automated programs (bots). Moreover, the proportion of individuals that actively engage with social media is also unknown. It is, therefore, a challenge to link trends on social media to events occurring in the real world.

When generating inferences about the real-world from Big Data, it is possible to generate misrepresentative indicators. For instance, mobile phone data can leave traces of users' mobility at a relatively fine spatial and temporal resolution. However, inferences of travel and user behavior from these footprints beyond their presence could be difficult. For instance, it may be reasonable to assume that it is possible to make estimations on how crowds move from A to B, but for how many journeys would such models be accurate? Unfortunately, it is not often possible to conclusively validate models built from Big Data beyond links to aggregate datasets, small sample surveys or alternative big datasets with similar limitations.

Demographic Bias

Sampling has always been a fundamental component of geographic research. In order to devise nomothetic (law-based) knowledge, hypotheses need to be built from representative samples. Bias typically arises due to one or more of three reasons: the selection procedure is not random; the sampling frame does not account for all groups in society; and/or some groups are impossible to reach. In the case of Big Data, they are often a by-product of actions and therefore are not random nor scientifically sampled at all, and some groups may be excluded. Instead of building representations from carefully constructed sampling frames, researchers working with Big Data acquire data and then must attempt to work out who they truly represent.

One very early failure of a big dataset was when the Literary Digest's postal poll regarding the presidential election received roughly 2.4 million returns in 1936. With the aim of achieving as large a sample as possible, the magazine sought datasets which contained the names and addresses of millions of adults, these primarily comprised vehicle registration lists and telephone directories, and in total over 10 million ballots were posted. However, despite receiving an impressive number of responses, the poll incorrectly predicted that Landon would beat Roosevelt. Their data sources are now understood to have produced biased samples that were likely to be of a higher socio-economic status. The rates of both automobile and telephone ownership were much lower amongst poorer adults.

Generally, in both academia and the commercial sector, large sample sizes are valued as more favorable where they represent a greater proportion of the "population" of interest. Indeed, Poisson's law of large numbers entails that with larger samples one is more likely to acquire an observed mean closer to the theoretical mean of a given phenomenon. Despite the promises of Big Data that they represent complete populations ("N = All"), the majority of datasets unfortunately fail to capture every citizen. Data may represent all users of a given service, but does everyone in the population use that service? It is not unusual for a minority of the population to produce the majority of the data (Crampton *et al.*, 2013). Big Data are devoid of robust sampling frames meticulously designed to reduce sample errors and sample biases. In addition, the primary objectives of most big datasets are not to acquire complete coverage of the population at large, thus data are prone to representing particular subsets of the population that engage with the various activities that generate data. Moreover, it is very difficult to filter out incorrect information, often government and commercial bodies rely on the individual correcting their records themselves.

Whilst all representations are necessarily partial, incompleteness in scientific surveys can be considered both systematic and understood. A disadvantage of Big Data is that the data are often of unknown provenance. It is not uncommon for big datasets to contain no demographic variables at all making it inherently difficult to estimate their representativeness of the population at large. This is usually because asking participants to volunteer additional information may discourage them from using the particular service. We are, therefore, still reliant on censuses for good quality framework geodemographic data available at fine geographic scale, despite the fact they are collected very infrequently and population estimates may need to accommodate systematic non-response.

Data linkage is often required to ascribe demographic characteristics to Big Data, bolstering their utility and reducing bias. Linkage techniques in geodemographics generally fall into one of three categories: a) linkage to individual-level data, b) linkage to small area population statistics or c) inferences based on other personal variables in the data. Individual data linkage is the only means of acquiring accurate demographic variables without succumbing to issues of ecological fallacy. Unfortunately, individual-level demographic data are very rarely available to researchers due to their confidentiality restrictions. However, such linkage has been made feasible in some countries where individual-level data are routinely collected and anonymized at vast scales, for example, medical research in Sweden.

Another means of estimating the demographic bias of geographic datasets is through data linkage to small area official statistics. However, in these instances, validity is limited by the quality of the official population statistics and is hampered by issues associated with spatial

aggregations : for example, scale and zoning effects (Openshaw and Taylor, 1979) and the ecological fallacy (Openshaw, 1984). While some have taken advantage of microsimulation procedures in order to retain the focus of analysis at the individual level, they are still haphazard estimations based on assumptions.

An additional, but often overlooked means of linking demographic characteristics to Big Data is through linkage to datasets that record the statistics of personal attributes which are often recorded in large databases. A prominent example of such attributes is names. Names appear in many different big datasets that require users to register accounts (such as social media or store loyalty programmes). Both surnames and forenames are likely to be distributed unevenly across a wide range of geodemographic domains. For example, most European names are gender-specific. Consequently, there is value in devising demographic databases on forenames in order to provide typical statistics on age and gender that can be ascribed to other data. Figure 9-7 demonstrates this by comparing UK Census data to estimated demographics from the names detected in a 2011 population register that has near complete coverage of the adult population. Although, of course, not every individual will share the modal characteristics of all bearers of the same name.

Spatial and Temporal Coverage

In addition to representing biased subsets of the population, data are usually partial representations of phenomena across space and time. Activity of some kind is usually required to generate data, and these are bounded by geography and time. Even when machines are used to automatically collect data, their geographic placement may not be uniform and may be influenced by the demographic and other biases described earlier. Most commonly the responsibility of recording data is handed to the people the data pertain to. Therefore, their behavior will heavily influence the data they generate and they may also be influenced by unobserved external factors. Failure to account for any geotemporal biases may lead to the misrepresentation of real-world processes.

Some new forms of data can be generated anywhere and at any time due to the use of new technologies such as hand-held and remote sensing devices. However, despite the presumed freedom of these data collection activities, they will still have some spatial and temporal restrictions. People cannot be assumed to actively contribute data randomly or at regular intervals – unless data are machine generated of course. Taking georeferenced social media data as an example, the action of uploading a georeferenced post may be more common in tourist or social settings than in workplaces. And indeed, some factors such as time of day/night or poor data communications will inhibit the ability to upload content. Therefore, distribution of georeferenced social media does not represent the distribution of people (and activities) across time and space, despite there being strong associations in some settings.

At the international level, Taylor and Overton's first law of geographical information still stands true today at a general level, viz: poorer nations are also poorer in good quality data (Taylor and Overton, 1991). There is a digital divide in a range of different types of geographical data. For example, the quality and precision of Open Street Map is far better in developed parts of the world where there are more contributors and there is generally better connectivity. In addition, social media usage is generally lower in developing nations. Although Internet connectivity is improving globally, developing nations and those with lower

information equality still lag behind. This is unfortunate, as the developing parts of the world are often in the greatest need of good quality geographic information.

Unstructured Data

Geographers have long attempted to make sense of intangible concepts. Non-quantifiable concepts are well suited to ethnographic researchers who are able to collect very detailed observations and derive conclusions based on shared experiences. However, it is a more novel challenge to represent intangible concepts from large quantities of data in research, as quantitative data typically represent phenomena as categorical, ordinal or interval, so that they can be analyzed numerically and efficiently.

One of the challenges with many types of Big Data is that they not only represent intangible concepts but that they are also not numerically structured making it difficult to undertake quantitative analysis. Techniques are, therefore, required to manipulate the data. Taking the example of social media posts that are textual, often research has focused on isolating a series of keywords and then using filtered searches to generate relative frequencies across time and space. This has enabled researchers to use social media posts to track real-world events such as flu epidemics (Lampos *et al.*, 2010) or earthquakes (Sakaki *et al.*, 2010). In these approaches, each case is a binary datum; either a relevant term is present or it is absent (1 or 0).

Another set of techniques attempt to group words or entire social media messages in order to produce categories. By modeling the co-occurrence of particular words across large samples of documents, it is possible to create associations between words, and also, associations between posts. Topic modeling techniques have provided a popular means of analyzing large documents, such as blog posts, speeches or articles; however, their utility on social media is questionable due to the short length of documents and unstandardized use of language. Despite this, it is feasible to devise topic classifications from short social media posts if appropriate data cleaning steps are applied first. For instance, Figure 9-8 displays the relative frequency of geotagged Tweets from Inner London that were allocated to 20 key topic groups by hour of day. Some of the temporal patterns could be indicative of activities, for example the Food and Drink group shows peaks at mid-day (lunch time) and in the evening, as would be expected.

By converting the data into more reductive forms, there is an inherent loss of information. It is likely that qualitative analysis on a small sample of data may reveal entirely different trends and perspectives. In addition, quantitative analysis of large amounts of data tends to ignore the contextual elements of each datum. While it may be possible to group messages based on textual similarity, it is not possible to return meaning in a numerical form. Users describing the same topic may be tackling it from entirely different viewpoints with very different agendas. In the case of tracking flu from Twitter/X messages, it is easy to confuse messages reporting symptoms with those that are merely expressing awareness of the illness.

Data Linkage

Although rich in detail, most big datasets still only offer partial representations and typically only pertain to a very limited number of variables. Therefore, as discussed earlier, data linkage is required to improve the coverage of data and enhance their predictive capabilities (Harris *et al.*, 2005). Combining Big Data maximizes their value and enables critical research to ask questions about distinctive phenomena that may be extrinsically associated with each other.

Having discovered the value of linked data, some countries have introduced data infrastructures which enable the linkage of individual records to facilitate research and public service delivery. However, it is not feasible to link most data from commercial organizations given that few efforts have been made to standardize how they collect information on individuals.

Yet even where consumer data are more widely available, challenges to effective concatenation and conflation remain, not least because of the ways that different data are structured by the organizations that collect them. In short, data linkage is plagued by uncertainty and the potential for propagating errors. There have been countless efforts to link big datasets, often using probabilistic linkage techniques. Such techniques cannot match every case and may often incorrectly match records. However, without individual-level linkage data either remain siloed or trends can be observed only through aggregations and associations with aggregated reference data.

Often unique identifiers are particular to each dataset and so linkage is reliant on other unique elements such as addresses and names. However, both of these can be input and structured inconsistently. Whilst postcode and zip code systems were designed to assist the identification of addresses, in the UK, for example, a single postcode covers 15 properties on average, often considerably more in urban areas. In addition, postcodes may change or be recorded incorrectly. Research has previously attempted to build a longitudinal database of the UK adult population from 20 separate annual population registers. It was not possible to link roughly half of the addresses using a simple string match due to inconsistencies in formatting and spellings. Therefore, a bespoke matching algorithm had to be devised. The same research also found that person names could be recorded inconsistently, thus hampering linkage. However, using specially devised heuristics, over 100,000 women each year were identified to have probably changed name following a marriage for instance. Indeed, with an understanding of how identifiers could be recorded differently, it is possible to tailor linkage algorithms to maximize match rates. Indeed, it was even possible to estimate the origin and destination of over 750,000 domestic migrants by looking at subsets of households that joined and left addresses with unique combinations of names (Figure 9-9).

Tools and Skills

The volume, velocity and variety of Big Data have imposed considerable challenges on geospatial data analytics. Whilst there have been advancements in computer memory, processing power and databases, the majority of conventional analytics packages struggle to handle very large datasets. This is especially true for geographic information systems that also often need to account for complex issues such as spatial dependence and spatial non-stationarity. Here more spatial data can equate to exponential increases in computing required to undertake spatial operations. The complexity and the extensiveness of some spatial operations (and indeed various multidimensional modeling techniques) means that it is not always feasible to employ parallel computing to increase their efficiencies. Indeed, most spatial statistics techniques were originally devised to work on a finite number of points. This situation is changing rapidly for some forms of Big Data, notably as a result of the development of AI-assisted spatial data services. Examples include: the production of 2D and 3D Digital Worlds in vector formats; and the application of advanced AI techniques to autonomous vehicle usage in the [Austin Texas trials](#).

Visualization (including mapping) remains an integral component of quantitative analysis. Carefully selected techniques are required to convey trends at the expense of noise in Big Data. Too much detail can make key patterns unobservable. In addition, scale is still important, zooming in too far can obscure local trends. Therefore, despite the detail and precision of Big Data, visual outputs must remain reductive in order to support decision making. Whilst developments in static image-based representations have been minimal, there has been a growth in the implementation of web-enabled services that let users explore the data and focus on key areas of interest.

Data on people and places are also generated faster than they can be critically explored and understood. It is also the case that most techniques in spatial analysis were built to work on static and pre-existing databases which are therefore treated as static and timeless entities. However, a large share of Big Data are generated at high velocities and there are interests in analyzing them in real-time. Furthermore, the handling of such data is challenging, for instance indexing techniques that were designed to make very large datasets accessible can become inefficient once data goes over their original capacity of extension. In response to this, bespoke software packages and services have been developed to harness real-time geosocial information in order to detect real-world trends. There are also new spatial data streaming algorithms which were primarily used to estimate environmental data (e.g. weather, sea surface temperatures) from new flows of Big Data. However, often data are statistics that are generated for data in intervals, thus the formation of intervals become a fundamental part of the analysis.

Data may also come in a variety of different types, and may require extensive reformatting in order to enable analysis of them. Geospatial data generally falls into three forms: raster, vector and graph, and there are established practices for working with all three. However, as observed in the [Unstructured Data](#) section above, many Big Data sources generate data types that are not conventional for quantitative analysis. While this has not changed how spatial elements of data are standardized, the appended data now vary considerably. These include textual data, images and even video feeds. Therefore, new techniques primarily from the field of computer science have been devised to harvest information from new forms of data.

In response to the technical challenges of working with Big Data, researchers have to become more comfortable working with coding and scripting languages to employ techniques such as parallel computing. Indeed, advanced access to many big datasets (such as Twitter/X and Microsoft's [Planetary Computer service](#)) are typically granted through APIs, and so they are primarily accessible to only those with programming skills. New techniques from computer science and statistics have allowed researchers to generate never-before seen insights (such as those from machine learning and AI) from data that are otherwise very difficult to analyze. For instance, there are now a plethora of text-mining tools that were developed to harvest information from large quantities of textual records. By creating a pipeline that blends such methods with GIS it is possible to identify previously unexplored geospatial phenomena. Such techniques have also supported the growth of geographic data into wider disciplines.

Conclusions

Geospatial analysis is fundamentally changing in the wake of the exaflood of new sources and forms of Big Data, and the accompanying development of machine learning and other AI methods to analyze them. From a geospatial perspective, new data sources and their appropriate governance offer finer spatial and temporal resolution – and hence data volume. But the

absence of research design in their creation means that new kinds of data may disrupt established approaches to representation of social and physical systems. For example, data-based weather forecasting may already be faster, cheaper, and more accurate than physics-based forecasting. Similarly, cities exist because of interactions, and cellphone information about connectivity between individuals and locations is changing many of our concepts of how cities function.

Widespread re-purposing of new data sources challenges the implied view that GIS software can remain agnostic with respect to purpose. Failure to undertake data engineering and due diligence in assessing the quality and provenance of Big Data creates dangers of unreliable and unethical use. Analysts must anticipate the principle of *caveat emptor* when using data of undocumented provenance acquired from third parties. Prospectively it may be possible to render GIS software more ethical – for example by flagging workflows that open the possibility of ecological fallacy, for example, or by enhancing the use of metadata to flag uses for which the data are known to be unfit. In the absence of Big Data that are engineered to predefined research ready data standards, this raises much broader questions about assigning the responsibility for assembling functions into workflows largely to the user. It also speaks to the impossibility of educating all GIS users to be aware of every ethical issue.

Machine learning is becoming integral to data driven geospatial analysis in an era of science which is dominated by complexity, since there are seemingly no simple scientific truths left to be discovered. Yet the results of machine learning have been shown to be non-replicable and unreproducible, black box rather than transparent, incapable of dealing with uncertainty, and not at all consistent with traditional science. Machine learning in important respects akin to curve-fitting, with the scale of its adoption tantamount to abandonment of many of the core organizing principles of science. Explanation, discovery and understanding are all increasingly problematic in a world dominated by machine learning.